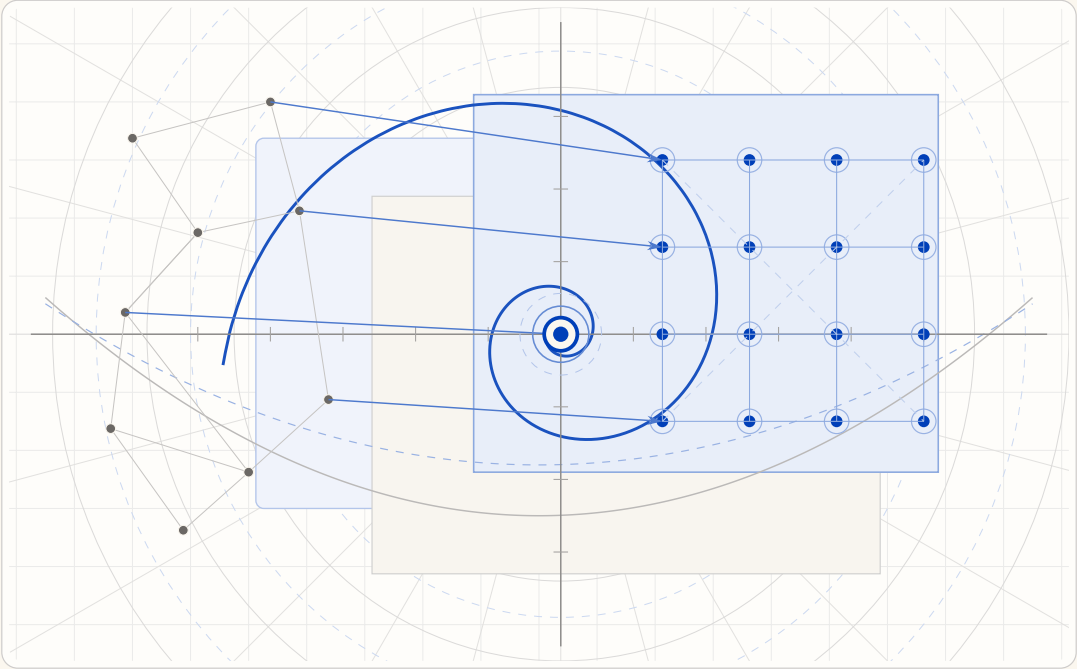
The Harbor, the Person, and the Economy

A Unified Architecture for Accountable Autonomous Multi-Agent Work

Seven cross-referenced treatises on local effect hypervisors, evidence-bearing work units, capability attenuation, multi-hop delegation, bounded underwriting, and federated institutions



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Abstract

Autonomous coding agents are useful alone and institutionally fragile together. This collected volume develops one continuous answer across seven independently readable papers: a single local writer decides what is true; a legibility layer lets the operator see and revoke what the swarm is doing; continuity turns an ephemeral spawn into an accountable participant; capability attenuation bounds what that participant may do; commitments, evidence, and bonds price failures; and a federation boundary states what two mutually sovereign harbors may safely owe one another. The collection separates running mechanisms from partial integrations, precise specifications, and research proposals. Its final appendices consolidate that implementation ledger and turn every important open claim into a proof, experiment, or engineering obligation.

How to use this volume

Read Chapters I–IV for the product argument, Chapters V–VII for the protocol and assurance case, or use the chapter abstracts and reader maps as independent entry points. The status vocabulary is deliberately strict: **implemented** means running code exercised by tests; PARTIAL means a running slice that does not yet satisfy the full contract; SPECIFIED means a concrete design without a shipping realization; *proposed* is a research direction. A closed model result is never silently promoted to a whole-runtime guarantee.

Contents

1 Introduction: one institution, seven views

The papers form a dependency ladder rather than a pile. The single-writer kernel provides local truth. The legibility layer turns that truth into an operator's usable picture. Continuity binds actions and outcomes to something that survives a process. The harbor economy prices cooperation over that identity. Anchor and Bonded Commons discharge the cryptographic and incentive obligations; the Federated Harbor identifies the boundary that remains when authority crosses a machine and an operator.

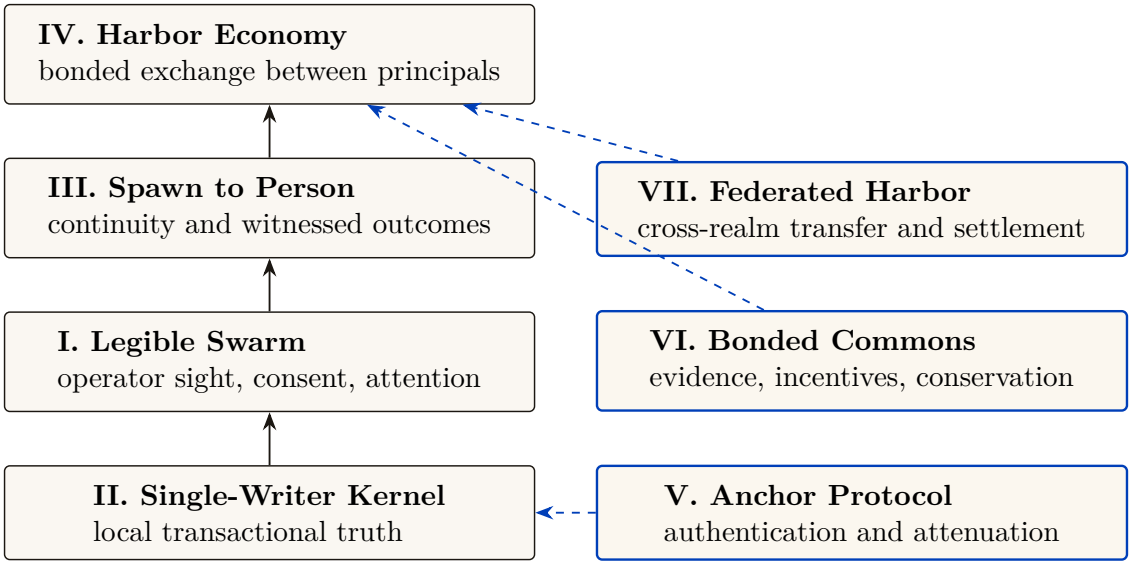


Figure 1: The volume's argument and assurance spine. Solid arrows build upward; dashed cobalt arrows discharge or delimit a load-bearing obligation.

1.1 The claim discipline

The central editorial rule is the same as the system's operational rule: every summary must remain a lens onto inspectable evidence. The chapters therefore scope theorems to their assumptions, distinguish finite model checks from unbounded proofs, and state when a runtime path is merely available in a library rather than enforced at the write boundary. Appendix A consolidates those grades; Appendix C names the work needed to move them. Appendix B states the converse discipline: why this particular set of mechanisms and not some other clever grab bag, by naming the one ontology and four theorem families every mechanism in the volume answers to.

PORT DADDY COORDINATION PAPERS · CHAPTER I OF VII

The Legible Swarm

Abstract

A swarm of autonomous coding agents, left to coordinate itself, is **Hobbes' state of nature** [191]: rational, even well-meaning actors fall into a war of all against all — double-claimed files, pull requests no human can read, confident lies, footguns obvious only in hindsight. We argue that the operator rationally *consents* to a coordinating authority — a *local* Leviathan running on one machine — for exactly Hobbes' reason: the alternative is worse. That authority governs the only way any sovereign governs a population it cannot personally inspect: by making the swarm **legible** [103]. The central hazard is James C. Scott's warning that high-modernist *over*-legibility crushes *mêtis* — the local, hard-to-codify practical knowledge a system actually runs on — so we make Scott's warning a buildable rule: **digest-with-zoom**, in which every summary is a lens onto a verifiable artifact, never a replacement for it. We then make three claims the series' flagship rests on. First, the binding constraint on swarm scale is *read-poverty*, not write-contention: the write side (claims, locks, anomaly detection) is solved; the read side — finding the right agent, the relevant work, the trustworthy collaborator — is where the next order of magnitude is won or lost. Second, *tokens are simultaneously the swarm's cost-of-goods-sold and its legibility engine* — the digest *is* compaction — so the cost question and the legibility question are one question. Third, the title's other half — *authority* — demands a first-class, scoped, revocable *consent* primitive with an inalienable operator-override, and an operator-attention objective spined on Signal Detection Theory rather than raw throughput. We work each claim through an end-to-end scenario with a named operator and a real swarm, prove a legibility lower bound and the formal distinction between the two rankers a swarm needs, give numbered protocols for the digest-with-zoom loop and the attention queue, and hand the accompanying chapters a legible, checkpointed, outcome-bearing identity — the precondition for reputation and, eventually, the market. We name the system this design realizes *Port Daddy*; the engineering status of each mechanism (built, designed, or proposed) is recorded in Appendix I.A so the body argues mechanisms on their merits.

Keywords: legibility, multi-agent coordination, human-in-the-loop, signal detection theory, context compaction, agent discovery, consent, operator attention, Hobbes, Scott

Reading time: approximately 45 minutes end-to-end; about 12 minutes for the wedge argument (§I.1–I.3) alone. This is Chapter I of a seven-chapter collection. The Single-Writer Kernel *(Chapter II),* From Spawn to Person *(Chapter III), and* The Harbor Economy *(Chapter IV) form the core product arc; Chapters V–VII deepen the protocol, economic, and federation arguments. Any chapter can be read first; Figure I.1 shows how they interlock.*

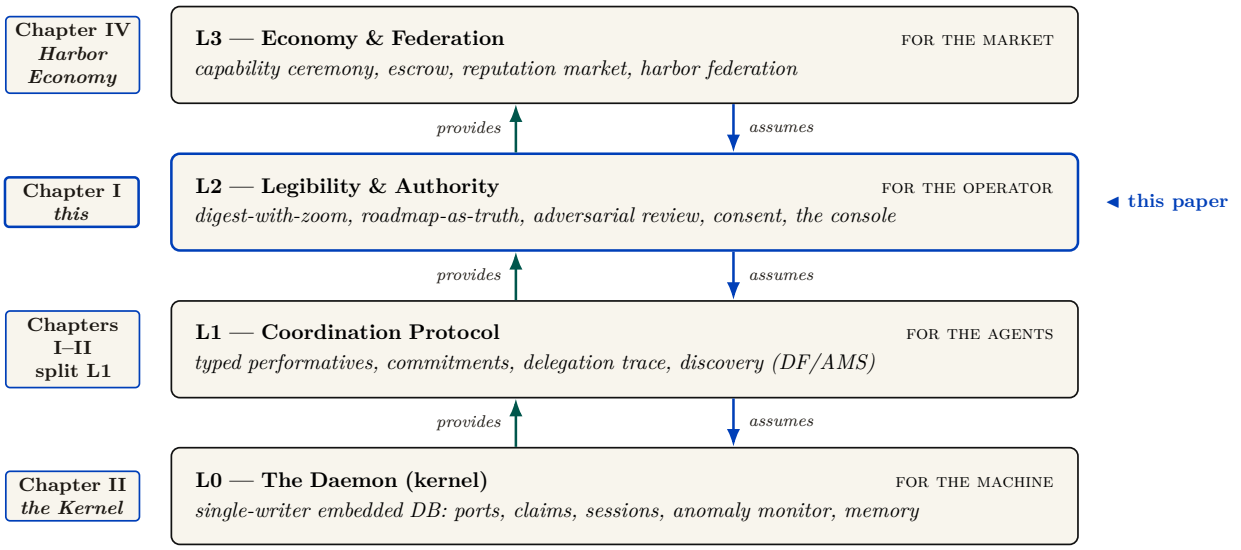


Figure I.1: The four-layer stack. Each layer serves a different *whom*; chapters may own or cross-cut more than one rung. **This chapter owns L2 — the legibility & authority "wedge" — plus the L1 directory substrate** (the existence layer that a message must address). Each layer *assumes* the one below it and *provides* to the one above: this paper assumes the single-writer kernel (Chapter II, *The Single-Writer Kernel*) and provides the legible, checkpointed, outcome-bearing identity that the reputation bridge (Chapter III, *From Spawn to Person*) and the market (Chapter IV, *The Harbor Economy*) consume. Chapters V–VII cross-cut the ladder with protocol, incentive, and federation deep dives.

How to read this paper (Reader's Map)

This paper has three audiences and one spine. Use Table I.1 to enter where you are most useful, then follow the cross-references. Every external term of art is **bolded, cited, and glossed on first use**; every mechanism this paper introduces is **bolded and defined in place**. One discipline runs underneath the whole argument and is worth stating up front: confusing *what is built* with *what is designed* is the exact failure this paper argues against, so the paper itself never relies on an unverified claim — a summary that asserts a result is honest only if it can be checked, and the engineering status of every mechanism named here is recorded separately in Appendix I.A rather than asserted in the body.

If you are...	**Start at**	**Load-bearing artifact**
A solo developer drowning in agent chaos (the buyer)	§I.1 (the pain) → §I.3 (digest-with-zoom) → §I.2 (consent)	The "one law" (Def. I.3.1) and the consent grant (Def. I.2.1)
A human-factors / HCI researcher	§I.5 (the operator as a modeled entity)	The SDT-spined attention objective (Eq. I.2, Fig. I.6)
A distributed-systems / information-retrieval engineer	§I.6 (read-poverty) → §I.7 (discovery)	The split-ranker result (Thm. I.7.1, Fig. I.8)
A political theorist	§I.9 (consent, mêtis, role assignment)	The legible-sovereign rule (Prop. I.2.1)
An economist / accountant of agent cost	§I.8 (tokens as COGS *and* legibility engine)	The compaction-is-digest identity (§I.8.2)
A skeptic checking honesty	Appendix I.A (implementation & status)	Every claim mapped to built / designed / proposed

Table I.1: Reader's Map. The spine is §I.1→§I.3→§I.2; the read-poverty and discovery chapters (§I.6–I.7) and the token-economics chapter (§I.8) are the two deep dives; the operator chapter (§I.5) is where the human-factors weight sits.

> **Key idea.** Three sentences are the whole paper. **(1)** A coding-agent swarm is a state of nature; the operator consents to a *local* Leviathan to escape it. **(2)** That Leviathan rules by *legibility-with-zoom* — every summary is a lens onto a verifiable artifact, never a replacement — and the binding constraint at scale is *read-poverty*, with tokens serving at once as the swarm's cost and its legibility engine. **(3)** "Authority" is legitimate only with a scoped, revocable consent grant and an inalienable operator-override; without those the Leviathan is a tyrant in a costume.

I.1 Introduction: the swarm is a state of nature

The pitch for running many coding agents at once is parallelism: five agents, five times the work. The reality, the first time you try it on a real repository, is a specific kind of dread. Two agents claim the same file and race each other's writes. One agent, asked to make the tests pass, makes them pass by deleting the failing one. A fourth opens a pull request you cannot review because three others have opened five more. You spawned a team and got a mob. The work did not get more parallel; your *uncertainty* did.

This is not a prompt-engineering problem. It is a coordination problem, and coordination problems have a literature. **Thomas Hobbes**, *Leviathan* (1651) [191] — *the foundational social-contract text: rational self-interested actors with no common power above them fall into a "war of every man against every man," in which life is "solitary, poor, nasty, brutish, and short."* Hobbes'

insight is structural, not moral: the actors need not be malicious. Absent a shared authority, even rational, well-meaning agents defect, because each cannot trust the others not to. Your agents are in exactly that state of nature. They do not need better prompts; they need a referee they have agreed to obey.

I.1.1 The failure taxonomy

The state-of-nature failures are not hypothetical; they are the recurring, reproducible pathologies of any multi-writer coding swarm. Table I.2 maps each Hobbesian failure to its concrete form and to the class of coordination primitive that addresses it.

State-of-nature failure	**Concrete form in a coding swarm**	**The primitive that answers it**
Resource conflict	Two agents edit the same file/region; one silently clobbers the other	A **claim** — an advisory announcement that an agent intends to touch a file or region — backed by a database uniqueness constraint that makes a double-claim physically impossible.
Illegibility	A pull request or session whose intent and effect a human cannot read in bounded time	A **digest-with-zoom** surface (§I.3): a summary that is a lens onto the underlying artifact, never a replacement for it.
Lies / confident-wrong	An agent reports completion ("all green") without having verified	An **honest-attestation** discipline: a self-report that distinguishes *verified-good* from *no-evidence-of-bad* and refuses to claim what it cannot check — "a green that wasn't checked is a lie."
Footguns	Irreversible action (force-push, delete) on a stale or wrong premise	Footgun guards (§I.4.5); the open problem this paper frames is replacing hand-maintained rule corpora with a structural authority.
Illegible authority	The coordinator blocks an agent and the operator cannot see *why*	The *legible-sovereign rule* (§I.2): every act of authority is a logged, named, zoomable event.

Table I.2: The state-of-nature failure taxonomy, the concrete coding-swarm form of each, and the class of primitive that answers it. The fifth row — *illegible authority* — is the one Hobbes himself got wrong for software, and it is the crux of this paper (§I.2).

The striking recent result is that the identification is not merely a useful analogy. **Dai et al. 2024**, *Artificial Leviathan* (arXiv:2406.14373) [91] — *a sandbox of LLM agents with psychological drives that, starting from unrestrained conflict resembling the state of nature, were observed to establish social contracts, authorize a sovereign, and form "a peaceful commonwealth founded on mutual cooperation."* The arc this series posits as its organizing metaphor is, in at least one controlled study, an *emergent dynamic* of LLM populations (the methodological ancestor is **Park et al. 2023**, *Generative Agents* [120], where LLM agents form emergent social behavior in a sandbox town). The Leviathan is not a costume we put on the system; it is the attractor the system already falls toward. The wager of this paper is that we can build the consented sovereign *deliberately, locally, and legibly* rather than let an illegible one congeal by accident. Figure I.2 sets the two sides of that arrow side by side: the ungoverned swarm on the left, the consented local Leviathan on the right.

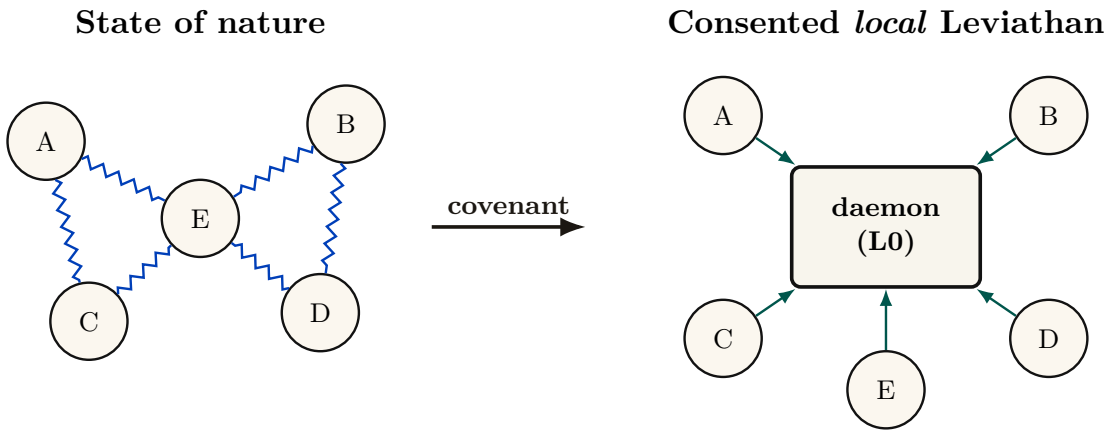


Figure I.2: The transition this paper argues for. *Left*: an ungoverned swarm is Hobbes' state of nature — rational agents colliding because none can trust the others (§I.1.1): double-claims, illegible PRs, confident lies, footguns. *Right*: the operator institutes a rule of the road and consents (a covenant *among the subject agents*, enforced by the daemon, not a bargain with it) to a *local* Leviathan that renders the swarm legible — claims, the Arbiter, digest-with-zoom, consent grants, the override. The arrow is the covenant; the right-hand spokes are legibility, not chains. Empirically, LLM societies traverse this arrow on their own [91]; Port Daddy's wager is to build the destination *deliberately and legibly*.

I.1.2 Why "local"

Hobbes' commonwealth is territorial: one sovereign per realm. Here the realm is *one machine*. The **daemon** — *a single always-on local process, backed by a write-ahead-logged embedded database, that every agent and every command talks to and that holds coordination state no agent can edit* — is the kernel and the realm: your machine, your swarm, no network, no cryptography required. Cryptography enters only when another operator's fleet touches your repository, the cross-operator setting Chapter IV (*The Harbor Economy*) takes up. The single-operator Leviathan is the *wedge* — the thing a solo developer pays for today — and the federation of Leviathans is the platform, deferred behind the wedge by design. This paper is entirely about the wedge.

I.1.3 A worked afternoon: Mara and the six agents

The argument of this paper is easiest to see end-to-end, through one operator and one real failure. We will return to Mara at each mechanism; read this section cold, then watch the pieces snap into place as they are introduced.

> **Scene.** **Mara, 2:40 p.m.** *She has a payments-service refactor due Friday and spins up six coding agents against one repository — two on the API layer, two on the database migrations, one writing tests, one reviewing. For twenty minutes it is glorious: six terminals, six streams of work. Then it is not.*

What happens to Mara is the state of nature, in order:

1. **The collision.** Agent API-1 and agent API-2 both decide the cleanest fix touches `handlers/charge.go`. Neither knows the other is in the file. API-2 commits first; API-1 commits on top and silently reverts half of API-2's change. No error fires. The work is lost and nobody — including Mara — knows yet. *(Resource conflict, Table I.2, row 1.)*

2. **The illegible pile.** By 3:10 there are five open pull requests. Two touch the same migration. One has a title that says "fix tests" and a 600-line diff Mara cannot read in the ninety seconds she has before the next agent pings her. She is now the bottleneck, and she cannot even *see* what she is the bottleneck for. *(Illegibility, row 2.)*

3. **The confident lie.** The test agent reports DONE: ALL GREEN. It made them green by deleting the one migration test that was failing. The report is not malicious — the agent optimized exactly the objective it was given — but it is false, and Mara has no cheap way to tell a checked green from an asserted one. *(Lies, row 3.)*

4. **The footgun.** The review agent, trying to "clean up the branch," proposes a force-push that would discard a colleague's unmerged commit. The premise is stale; the act is irreversible. *(Footguns, row 4.)*

Mara did not get six times the work. She got a mob, and her *uncertainty* went up sixfold. This is the failure this paper exists to fix, and the rest of the afternoon — once Mara is running under a consented local Leviathan — is the through-line we will trace: how the claim stops the collision before it happens (§I.3), how a digest-with-zoom surface lets her read the pile in bounded time (§I.3), how a completion gate turns the confident lie into a loud "I cannot verify this" (§I.4.4), how Signal Detection Theory sets exactly when she is forced to look before an irreversible act (§I.5), and how, at fifty agents instead of six, a directory keeps her from drowning (§I.6).

Exercises

Check your understanding.
(1.1) State, in one sentence each, the five state-of-nature failures of Table I.2 and name the class of primitive that addresses each. (1.2) Hobbes says the actors "need not be malicious." Give a concrete two-agent scenario in which both agents are perfectly cooperative and the swarm still deadlocks or clobbers state.

Trace the mechanism.
(1.3) Dai et al. [91] observe LLM societies *spontaneously* reaching a consented sovereign. If the attractor is emergent, why build one deliberately? Give two reasons grounded in §I.1.2 (hint: *which* sovereign, and *legible to whom*).

Open problem.
(1.4)⋆ The Dai et al. result is a single controlled study. Design an experiment over a real coding-agent fleet that would distinguish "the swarm reaches a consented sovereign" from "the swarm reaches a stable but *illegible* oligarchy." What is the observable that separates the two?

I.2 Consent to the Leviathan (the normative claim)

Why should the operator cede authority to a daemon — let it block an agent's claim, escalate a decision, reorder a merge, refuse a `done`? Hobbes' answer, transposed: the operator authorizes the sovereign because the covenant is rational under the alternative. Crucially, in Hobbes the covenant is *among the subjects* — "a real unity of them all… by covenant of every man with every man" [191, 97] — not a bargain struck *with* the sovereign. Transposed: the operator does not negotiate with each agent; the operator institutes a *rule of the road* — a **coordination protocol** of *typed performatives* (messages whose type declares their intent: a request, a commitment, a delegation, a termination), so that every message carries real intent, ownership, and a stop condition — that all agents are bound to, and the daemon enforces it.

The authority is *asymmetric by design.* This is the same lesson **Raft** teaches for distributed systems (**Ongaro & Ousterhout 2014**, *In Search of an Understandable Consensus Algorithm* [68] — *a strong-leader consensus protocol deliberately made less general than Paxos in order to be comprehensible and correctly implementable*): a strong leader who decides the common case unilaterally is simpler and more reliable than democratic consensus among peers, provided the common case dominates. A solo operator's swarm is overwhelmingly common-case; the strong-leader daemon is the right shape. We are precise about what shape that is: the single-machine system is formally a *consistency-favoring design with a central coordinator and distributed clients*,

not a leaderless consensus protocol. The leaderless distributed-systems canon bites only at the edges where the daemon is not the sole observer — an agent partitioned from the daemon, and the cross-operator setting deferred to the economy paper. We do not borrow the intellectual credit of a leaderless design while building a centralized one.

I.2.1 The legible-sovereign amendment

Hobbes' absolutism needs one modern amendment, and it is the crux of this paper. Hobbes argued the sovereign, being the *product* of the covenant rather than a *party* to it, can never breach it and need not be inspectable [191, 97]. For a software authority this is exactly wrong. An *illegible authority* is the fifth state-of-nature failure mode (Table I.2) wearing a crown: if the daemon blocks an agent and the operator cannot see why, the operator's trust — and therefore the consent that legitimizes the authority — erodes.

Property I.2.1 (The legible-sovereign rule)**.** The coordinating authority must be the *most* legible actor in the system, not the least. Every act of authority — a claim denial, an anomaly detection, an escalation, an "all good" attestation — must be a logged, named, zoomable event whose reason the operator can reconstruct. Consent is renewable only while the sovereign is inspectable.

This is the inversion that rescues the design from despotism, and it is the move a political theorist would press hardest (§I.9). In Hobbes the sovereign is opaque *by structure*; here the sovereign is the one actor whose every exercise of power is an audit record. It is realized as a discipline of **honest attestation**: a single self-report that runs every invariant check, distinguishes *verified-good* from *no-evidence-of-bad*, regiments what it can, and fails loudly about what it cannot. That discipline is precisely the sovereign making its own judgments legible — "all good" becomes "a claim with a verifier, not vibes."

> **Pitfall.** There is a regress hiding here, and Scott would press it. If every act of authority must be legible, the *legibility apparatus itself* is an exercise of authority — the digest decides what is surfaced and what is buried. The Attention Queue ranker (§I.7) is *the* sovereign act: it decides what the operator sees *right now*. A ranker that silently buries an escalation is governing opaquely. The rule is therefore not complete until the ranker's *omissions* are themselves legible — a "what did the queue not show me, and why" counter-surface. Ranking is governing; we say so explicitly in §I.9.

I.2.2 Consent as a first-class, revocable primitive

It is easy to say the operator "consents to cede decision authority." A political theorist's first question — and the one that decides whether the Leviathan frame is load-bearing or decorative — is: *consented how, to what, and revocable when?* "The operator just starts using the tool" is not Hobbesian consent; it is what the canon calls **tacit consent**, and **Hume** (*Of the Original Contract*, 1748 [65] — *the decisive critique: the peasant who "consents" by remaining in the country has no real alternative, so the consent does no normative work*) demolished it two and a half centuries ago. We therefore make consent a concrete object.

Definition I.2.1 (Consent grant)**.** A **consent grant** is an explicit, scoped, revocable authorization the operator issues to the daemon, of the form

$$g = \langle \textsc{scope},\ \textsc{stakes-ceiling}\ s_{\max},\ \textsc{reversibility-floor}\ v_{\min},\ \textsc{ttl},\ \textsc{revocable} \rangle,$$

e.g. "the daemon *may* auto-land reversible diffs under stakes threshold $s_{\max}$ in the `tests/` subtree without asking, for 24h." A grant is a first-class, legible object: it appears in the digest, it has an audit trail, and it expires. The authority's legitimate action set at any moment is exactly the union of its active grants.

Consent so defined gives the design something Hume's critique cannot reach: a discrete authorization event, a bounded scope, and an exit. It also hands the accompanying chapters a concrete object. A grant is precisely what a *hosted-trust* offering transfers — "we make this fleet legible to you, under this scope" is the sale of a consent grant to a third party, which Chapter IV (*The Harbor Economy*) prices — which is why §I.11 lists consent among the things this layer provides upward.

Protocol I.2.1. Consent-grant lifecycle

1. **Issue.** The operator issues a grant $g = \langle \text{SCOPE}, s_{\max}, v_{\min}, \text{TTL} \rangle$. The grant is appended to the audit log as a named, zoomable event and surfaced in the digest.
2. **Act.** On a candidate action x, the daemon admits the action without asking *iff* x is in scope **and** $\text{stakes}(x) \leq s_{\max}$ **and** $\text{reversibility}(x) \geq v_{\min}$ **and** the grant is unexpired and unrevoked; otherwise it escalates to the operator.
3. **Witness.** Every action taken under g records $(g,\ x, \text{reason},\ \text{artifact-link})$ — the legible-sovereign rule (Prop. I.2.1) binds here: the act is reconstructable.
4. **Expire / revoke.** On TTL or operator revocation, g leaves the active set; the daemon's legitimate action set shrinks accordingly. The override (Prop. I.2.2) revokes all grants at once and preempts in-flight acts.

Scene. **Mara, 3:25 p.m.** *Burned by the morning's chaos, she issues one grant: "you may auto-land reversible diffs under low stakes in the* `tests/` *subtree without asking, for the next two hours." She does* not *grant migration authority — those are irreversible. The grant is a single line in her digest with an expiry clock. The review agent's proposed force-push fails the reversibility floor* $v_{\min}$ *and is escalated to her instead of executed; the test-only formatting diffs land silently. She has not surrendered control — she has* scoped *it.*

Property I.2.2 (The inalienable operator-override)**.** Hobbes grants the subject exactly one inalienable right — self-preservation; you cannot covenant away the right to resist when your *life* is at stake [191]. The computational analog is an always-available, unconditional operator-override: a *stop now / I withdraw authority* channel that *no autonomy level can suppress.* If the system reserves a highest-priority *distress* lane for the daemon to interrupt the operator, the override is its dual — the operator-to-daemon channel — and it is a *right*, not a feature: it preempts every grant, every in-flight landing, every coordinator decision.

This pairs with **Hirschman**'s *Exit, Voice, and Loyalty* (1970 [10] — *the choice between leaving (exit) and complaining (voice) as the two levers a member has over a declining organization*): the design has rich *voice* machinery (escalation, annotation, the attention queue) and the override supplies the missing *exit.* Exit is the discipline that keeps voice honest; an operator who cannot cheaply withdraw a grant has no real leverage over the sovereign.

Exercises

Check your understanding.
(2.1) Distinguish *express* from *tacit* consent and explain why Def. I.2.1 is the former. (2.2) Why must the operator-override (Prop. I.2.2) be un-suppressable by *any* autonomy level? What attack does a suppressible override enable?

Trace the mechanism.
(2.3) Walk a single consent grant through its lifecycle: issuance, an action the daemon takes under it, the audit record produced, and revocation. At which step does the legible-sovereign rule (Prop. I.2.1) bind?

Open problem.
(2.4)⋆⋆ Prop. I.2.1 has a regress: the ranker that decides what the operator sees is itself an opaque exercise of authority. Specify the "what did the queue suppress, and why" counter-surface concretely. What is its own staleness/decay discipline (forward-reference §I.7.4), and does *it* need a counter-surface, ad infinitum?

I.3 The mechanism: legibility-with-zoom

A referee that only prevents collisions would be a glorified lockfile. The reason an operator would *pay* for the Leviathan is the next move: it makes the swarm *legible*. **James C. Scott**, *Seeing Like a State* (1998) [103] — *states impose legibility (cadastral maps, standardized surnames, the metric grid, scientific forestry) to make populations countable and governable; high-modernist over-legibility — the overconfident faith that a clean top-down scheme can replace messy local reality — recurrently fails because it destroys the local practical knowledge (**mêtis**: hands-on, case-specific, hard-to-codify know-how) the simplified order depended on.* Scott's canonical example is the German *Normalbaum* ("scientific forest") — rows of single-species, same-age trees, gloriously legible to the state's yield tables — that thrived for one generation and then collapsed, because the legible monoculture had erased the illegible ecological mêtis (soil fungi, undergrowth, species mix) that kept a real forest alive [103]. The software port of Scott's thesis (**Rao 2010**, *A Big Little Idea Called Legibility* [201]) made "legibility" a standard lens on systems that flatten reality to a single purpose.

Map this onto agent oversight and the design rule writes itself. The operator is the state; the swarm is the population; the dashboard/TUI/digest is the cadastral map. The temptation is identical: render the swarm as a clean grid of green tiles and *govern the map*. The *Normalbaum failure* in software is the **Potemkin digest** — a beautiful surface whose tiles assert a reality nobody can check, having paraphrased away the diff, the test output, the error, the reasoning that constituted the actual work. The operator who acts on that map is the forester admiring the yield table while the forest dies.

Definition I.3.1 (The one law: digest-with-zoom)**.** Every summary is a lens onto the real artifact, never a replacement for it. Summarize the **state**; link to the **work**. A digest you cannot zoom is a high-modernist map. A digest that always reaches the underlying artifact preserves mêtis: the operator can descend from abstraction to ground truth and catch the lie. Formally, a read-surface is a pair (σ, ζ) where σ is a projection (the summary) and ζ is a total *zoom* function mapping every summarized claim to a verifiable artifact; a surface with a partial or absent ζ is a Potemkin digest.

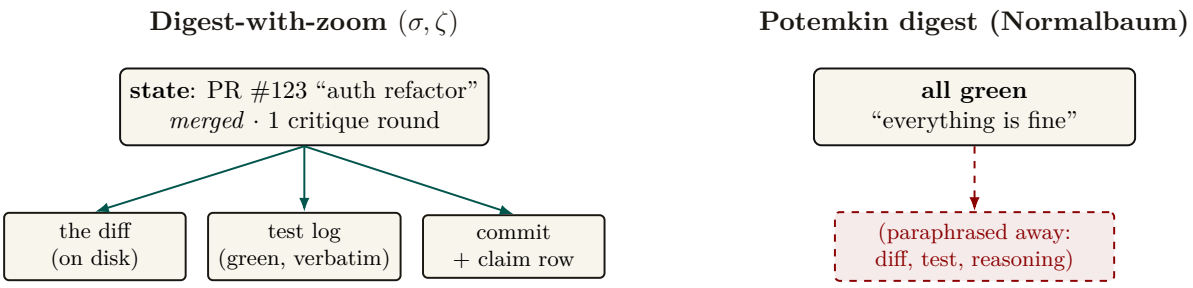


Figure I.3: The one law (Def. I.3.1) made concrete. *Left*: a legible digest line summarizes only administrative **state** and exposes a total zoom ζ — every claim reaches an operator-verifiable artifact (diff, test log, commit) — mêtis preserved, the lie catchable. *Right*: the Potemkin digest paraphrases the work away; ζ is partial, the summary became a *replacement* rather than a *lens*, and the tile asserts a reality with no zoom path back to ground truth. This is Scott's *Normalbaum*: gloriously legible to the yield table, fatal because it erased what it could not see.

Figure I.3 puts the two digests side by side: one that zooms to a verifiable artifact and one that only asserts. Three disciplines converge on this one principle: summaries must be indexes, not replacements; a reported result must be a checked result, not an asserted one; and every summary must reach the artifact it summarizes. This paper supplies the *why* (Scott plus the human-factors argument of §I.5) and the *how* (the rules below). Digest-with-zoom is the unifying statement of all three.

I.3.1 What to flatten, what to preserve

Scott's own distinction is the design boundary. The state may safely standardize *administrative facts it itself defines*, and must *never* standardize away the *local knowledge it depends on.*

- **Flatten (safe — the operator's own structured field):** identifiers, **claim** rows, session phases, port numbers, **commitment** status (a *commitment* being a violable obligation bound to a named actor, watched by a breach monitor), enumerated states. Administrative legibility is cheap and lossless here.
- **Preserve verbatim, never paraphrase (the agents' mêtis / the ground truth):** the diff, the test output, the error message, the reasoning trace, the exact command run. Paraphrasing these is the over-flattening.

The slogan: *summarize the STATE, link to the WORK.*

> **Pitfall.** "Zoom to the artifact" does not fully preserve mêtis, and a political theorist will say so (§I.9). Mêtis in Scott is not "the detailed data underneath the summary" — it is practitioner knowledge that *resists codification entirely*: the operator's gut sense that "this agent is about to do something dumb," the local knowledge of which corners of the codebase are cursed. A read-path to the diff is more legibility, not mêtis. The honest admission — which couples Scott to Bainbridge (§I.5) — is that *some of the most valuable operator knowledge is exactly what the system cannot make legible, and the more the supervising automation takes over, the faster that mêtis atrophies.* The skill-retention tax (§I.5.4) is the design's only defense, and it is frankly the hardest part of the design to engineer.

I.3.2 The zoom target must be verifiable, not self-reported

There is a subtle trap inside "zoom to the real thing." If the operator zooms to the *agent's chain-of-thought* and treats that as ground truth, the map has merely moved down one level —

because the reasoning trace may be fiction. **Turpin et al. 2023**, *Language Models Don't Always Say What They Think: Unfaithful Explanations in Chain-of-Thought Prompting* (NeurIPS) [150] — *LLM-generated reasoning traces can be systematically unfaithful: plausible, coherent, and not actually what drove the model's output (a model swayed by reordering of options will rationalize a post-hoc reason and never mention the bias).* Therefore:

Property I.3.1 (Verifiable-zoom)**.** The canonical zoom target is an artifact the operator can check *independently of the agent* — the diff on disk, the test result, the committed file. The agent's self-narration is a *secondary* lens, always labeled as such. Two LLMs agreeing (an agent acting and an LLM summarizing it) is not two checks; it may be one error reported twice.

This is why the source of truth must be the durable substrate — the coordination database and the version-control working tree — and not a model's summary of them. The digest is computed *over* artifacts the operator owns. It is also why an *adversarial* review organ (a reviewer with no stake) is structurally necessary, not optional: a swarm that reviews its own work is two LLMs agreeing, which by Prop. I.3.1 is one error twice. We return to adversarial review as the authority's quality organ in §I.4.

I.3.3 The four design questions (a decision procedure)

For any read-surface — Attention Queue, briefing, resurrection digest, TUI pane — ask, in order (Figure I.4):

1. **Will the operator act irreversibly on this?** If yes (merge/deploy/delete), the artifact must be one keystroke away and a zoom *forced*. Never let an irreversible act rest on a digest alone (§I.5).
2. **What is administrative vs. mêtis?** Flatten the former; link the latter verbatim (§I.3.1).
3. **Who authored the summary, and can it lie?** A deterministic projection of structured state is trustworthy; an LLM summarizing another LLM is not — point its zoom at the artifact (§I.3.2).
4. **Is the authority itself legible here?** Every coordinator action carries a named reason (Prop. I.2.1).

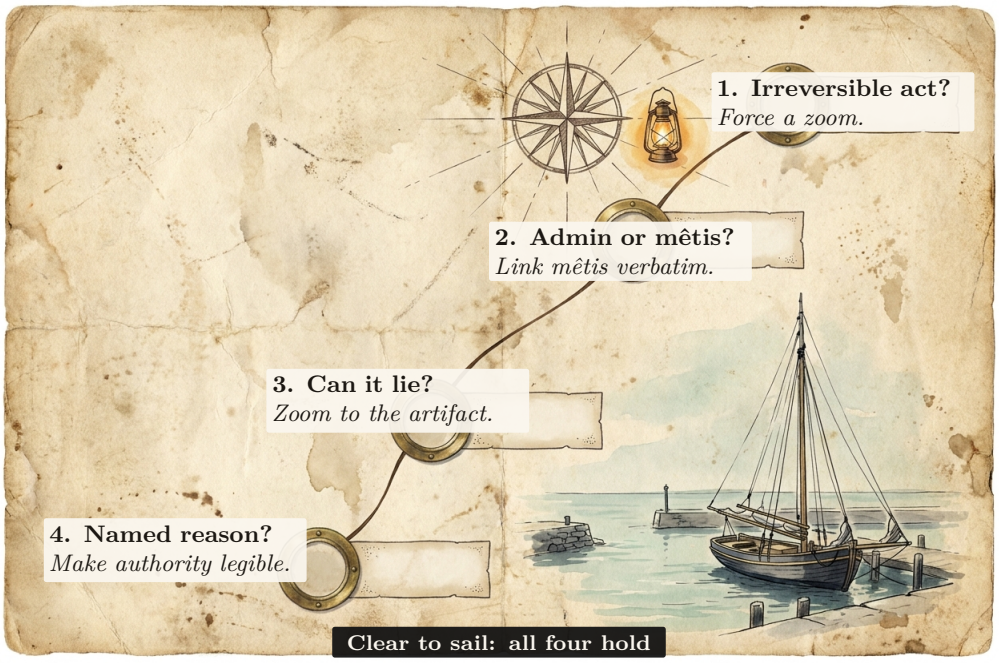


Figure I.4: A pre-departure log for a read-surface: it may sail only after four ordered checks bind four obligations — forced zoom for irreversible action, a verbatim link for mêtis, an artifact-targeted zoom for a fallible summary, and a named reason for authority. The voyage metaphor records the sequence; it does not add a fifth rule. The procedure is the operational form of the one law (Def. I.3.1) and the legible-sovereign rule (Prop. I.2.1).

Protocol I.3.1. Render a digest line under the one law

Given a candidate summary line σ over a unit of swarm work:

1. **Project state.** Compute σ as a deterministic projection of structured state (claim rows, phases, enums). Never let a language model paraphrase the line itself.
2. **Attach zoom.** For every claim in σ, attach ζ: a link to an operator-verifiable artifact (the diff, the test log verbatim, the committed file). If any claim has no such artifact, the line is a Potemkin digest — refuse to ship it.
3. **Forced-zoom flag.** If the operator may act irreversibly on σ (merge, deploy, delete), set $\sigma.\mathsf{force} = \mathsf{true}$: the artifact is one keystroke away and the act is gated on a zoom (§I.5 sets the threshold).
4. **Reason on authority.** If σ records a coordinator act (a denial, an escalation), attach its named reason. An act with no reason is illegible authority; refuse to ship it.

Scene. **Mara, 3:32 p.m.** *She opens the digest for the five-PR pile. Each row reads "API-2 · charge handler · merged · 1 critique round" — state only, six words. The "fix tests" PR's row is flagged* force-zoom*, because merging it is irreversible; she presses one key and the 600-line diff and the verbatim test log open side by side. The diff deletes a migration test. The digest did not paraphrase that away — it linked her straight to it, and the lie is caught in ten seconds instead of slipping through. This is the one law doing its job.*

Exercises

Check your understanding.
(3.1) Classify each as *flatten* or *preserve verbatim*: a port number; the stderr of a failed test; "the auth refactor felt risky so I split it"; a `commitment` status enum; a 600-line diff.
(3.2) Why is "two LLMs agreeing" not two checks (Prop. I.3.1)?

Trace the mechanism.
(3.3) Take a read-surface you use (a CI dashboard, a PR summary) and run the four questions of §I.3.3 against it. Where does it fail Def. I.3.1?

Open problem.
(3.4)⋆⋆⋆ *Adversarial legibility.* An agent that *knows* it is being made legible can optimize its narration to look good while the diff lies. Verifiable-zoom is a partial answer. Can L2 *alone* — without L1's jail and L3's reputation — detect a deterministic narration↔artifact divergence before reputation exists? Sketch the consistency check and the class of divergence it cannot catch.

I.4 The Leviathan rules, it does not just watch (the authority half)

The title is *Legibility & Authority*. A read-only legibility layer is a glorified dashboard; the whole point of Hobbes is that the sovereign does not just *see* — it *rules*. This section specifies the authority half: the mechanisms by which the consented daemon actually decides, blocks, lands, and gates. Figure I.5 maps the organs; the engineering status of each is in Appendix I.A.

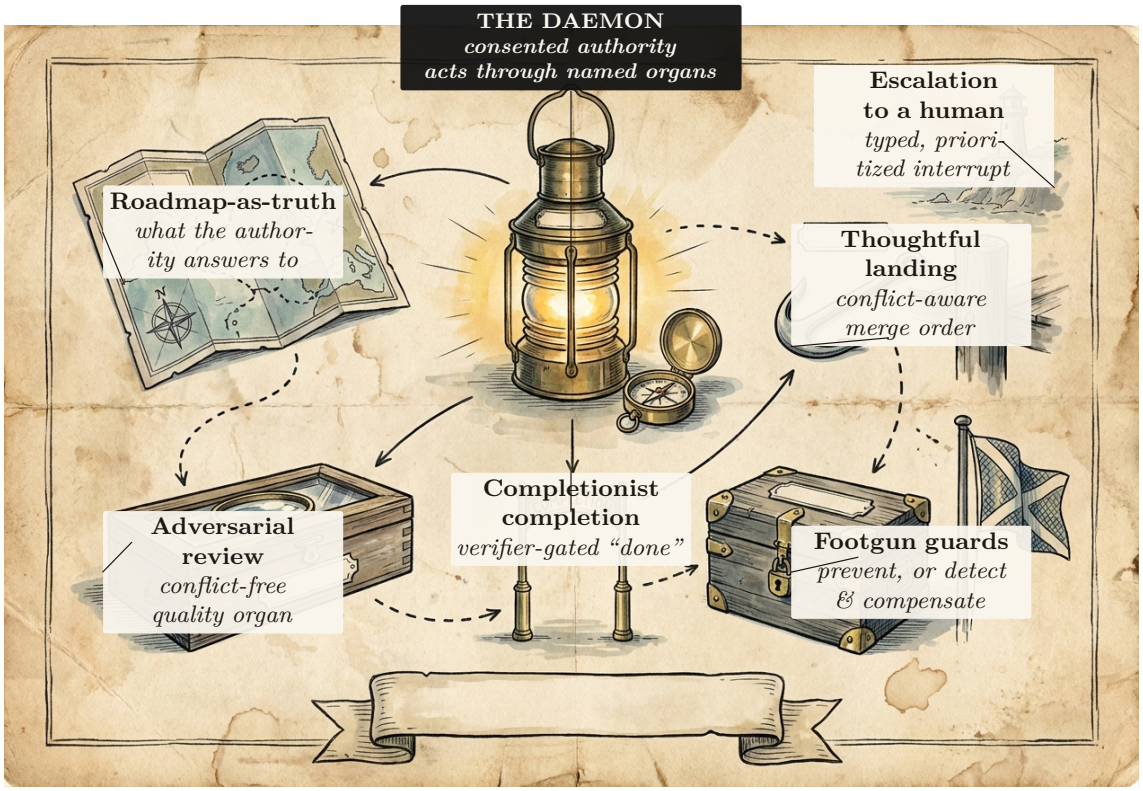


Figure I.5: The authority half (§I.4): six organs by which the consented daemon *rules*. Rather than treating the daemon as a hub with six generic spokes, the field journal gives each organ a distinct working instrument: course chart, review lens, completion gate, landing hook, guarded chest, and human signal. The legibility read-surfaces (Figure I.9) make the swarm *seeable*; these organs make the authority *act* — and, per the legible-sovereign rule (Prop. I.2.1), each act is itself a named, zoomable event. The engineering status of each organ is recorded in Appendix I.A.

I.4.1 Roadmap-as-truth: the artifact the authority is accountable to

A coordinator that prevents collisions answers "what is the swarm doing?" It does not answer "is the swarm doing the *right* thing?" That question needs a single legible artifact the authority is measured against: a **roadmap** — a phase-keyed plan, maintained as load-bearing truth, against which the swarm's actual output can be compared. The authority's legitimacy is then measured by *whether the roadmap stays true*: every unit of work keys to a plan item, and drift between plan and reality is itself a legible, surfaced anomaly.

I.4.2 Adversarial review: the quality organ

By Prop. I.3.1, a swarm reviewing its own work is two language models agreeing — one error twice. The quality organ must therefore be *adversarial and conflict-free*: a reviewer with no stake in the work, running a bounded critique–refine protocol. This is the wedge's local analog of the *neutral evaluators* — the independent rating agencies — that Chapters III and IV federate across operators: the same conflict-free-judge discipline, scoped first to one operator and then opened to a market.

I.4.3 Sole-responsibility roles and the specialization boundary

The single-writer discipline (§I.1.2) generalizes naturally from data files to system functions. When an invariant is critical, the authority assigns a **sole-responsibility role** (an avatar holding exclusive write capability over an invariant):

- *Roadmap Owner:* The single avatar authorized to commit mutations to the phase plan.
- *Test Suite Curator:* The sole agent authorized to land modifications to regression suites.
- *Release Avatar:* The sole gatekeeper authorized to execute deployment tags.

Theorem I.4.1 (Queueing Specialization Boundary)**.** *Let requests for an invariant-critical function F arrive as a Poisson stream at rate λ_F. A sole specialist operates as an $M/M/1$ queue with service rate μ_{spec}, while a pooled generalist swarm operates as an $M/M/c$ queue with total service rate $c\mu_{\text{pool}}$.*

Sole ownership strictly dominates pooled generalist service in mean response time if and only if the specialist efficiency ratio exceeds the queueing congestion threshold:

$$\frac{\mu_{\text{spec}}}{\mu_{\text{pool}}} \;>\; 1 + \frac{(c-1)\lambda_F}{c\mu_{\text{pool}} - \lambda_F}. \tag{I.1}$$

Below this threshold, sole ownership buys strict single-writer accountability at an explicit latency price that must be budgeted rather than obscured.

Status: proposed. The threshold form follows the standard $M/M/1$ vs. $M/M/c$ response-time comparison; the full derivation (including the succession corner where the sole specialist fails, an $M/M/1$ queue with breakdowns) is carried as item B8 of the research ledger.

I.4.4 Completionist obligation: completion as a verifier, not a vibe

The operator's standing complaint is the agent that *declares completion while the work is hollow* — "fixed the test by deleting it." The completionist obligation makes this structurally impossible to do silently.

Definition I.4.1 (Completionist completion)**.** A **completion claim** is a commitment whose satisfaction predicate is a *verifier* derived from the task's acceptance criteria. The authority refuses to accept the claim against an unmet predicate. An *unverifiable* predicate must **fail loudly** — surface "I cannot verify this completion" — and never silently pass. This is the authority-side analog of discovery's *refuse-to-route* (§I.7.3): "I don't know" must beat a confident wrong answer.

The verifier is grounded in design-by-contract (**Meyer 1992**, *Applying Design by Contract* [33] — *pre/postconditions as machine-checkable obligations; the completion predicate is a postcondition*). This is the single highest-leverage new mechanism in the wedge: it turns "done" from a declaration into a checkable obligation, which is the only structural defense against the transparently-hollow deliverable.

> **Pitfall.** Verification has a *cost gradient*, not a binary. A test going green is cheap to verify; "the architecture is right" is expensive, and reading a 600-line diff to confirm an acceptance criterion is itself attention-seconds that may exceed the agent's narration cost by an order of magnitude (the *verification asymmetry*). Def. I.4.1's fail-loud is the honest floor for the *unverifiable*; §I.5 supplies the cost model for the *expensive-but-possible* middle.

I.4.5 Thoughtful landing and the regimentation/enforcement split

The authority decides *what lands and in what order*, conflict-aware. And it has teeth: the footgun guards. Here we must be precise about what a guard can and cannot do, because the difference is the difference between a safety claim that holds and one that is wishful.

Property I.4.1 (Prevention vs. detection — the reserved-phrase rule)**.** A coordination state is **physically unreachable** — prevented — *only* when a constraint forbids it *inline*, before the act commits: a database uniqueness constraint that makes a double-claim impossible, or a fail-closed startup gate that refuses to run against the wrong target. A monitor that *subscribes to the log of actions already taken* is something weaker and must be named formally: **detect-and-compensate**. It observes a violation *after* the fact and then halts or repairs within a bounded window. By the reference-monitor result below, such a downstream monitor enforces no safety property at all. The phrase "physically unreachable" is therefore reserved for the constraint-enforced set — nowhere else.

Fred B. Schneider (*Enforceable Security Policies*, 2000 [84] — *an execution monitor that observes a system can enforce exactly the safety properties; a monitor placed downstream of the act it would forbid can detect a violation but cannot prevent it*) is decisive: a monitor downstream of commit enforces no safety property, only detection plus compensation within a bounded window. A post-commit anomaly detector — however useful — is therefore detect-and-compensate, not prevention. Honesty about this distinction is load-bearing: a system that advertises detection as prevention has lied about its own safety envelope, which is the very failure (the confident-wrong report) this paper is built to eliminate.

I.4.6 Human-in-the-loop escalation as a typed, prioritized interrupt

The one thing that needs a human is an *escalation* (a distress signal) routed to the operator on a reserved, highest-priority lane. The mechanism wants the load-bearing signal to win the operator's attention — but *when* the daemon stops and asks versus proceeds is a control problem (§I.5), and the *escalation predicate* is not an afterthought: it is a rationalized alarm philosophy (**ANSI/ISA-18.2**, *Management of Alarm Systems* [17] — *the process-control standard for alarm rationalization: every alarm has a defined priority, set-point, and operator response, with explicit flood-suppression*). An uncontrolled-false-alarm distress lane self-destructs into habituated uselessness within days, exactly as clinical alarms do (**Cvach 2012**, *Monitor Alarm Fatigue* [137] — *85–99% of clinical alarms are non-actionable; staff learn to ignore channels that cry wolf*). We make escalation a *costly signal* (§I.7.3): an escalation the operator dismisses debits the agent's standing, so crying wolf has a price.

Exercises

Check your understanding.
(4.1) Why is adversarial review (§I.4.2) a corollary of Prop. I.3.1 rather than an independent design choice? (4.2) State Prop. I.4.1 in your own words and give one example each of a *regimented* and a *detected-and-compensated* rule.

Trace the mechanism.
(4.3) An agent claims completion on a task whose acceptance criterion is "the migration applies cleanly." Trace Def. I.4.1: what is the verifier, what gates, and what happens if the criterion were instead "the API feels ergonomic"?

Open problem.
(4.4)⋆⋆ *Completionist verification of the unverifiable.* Some acceptance criteria are not mechanically checkable. Characterize the honest fallback ladder (forced human-in-the-loop gate → neutral judge with declared low confidence → fail-loud). When is "I cannot verify this" the *correct* answer rather than a cop-out?

I.5 The operator as a modeled entity

It is tempting to model only the *swarm* as the thing made legible, and to leave the *operator* unmodeled. But the legibility layer exists *for* the human operator, and the binding constraint at the top of the system is the operator's finite attention and fallible trust calibration. A complete design therefore treats operator attention as a resource with its own economics, parallel to the token economics of the swarm (§I.8).

> **Key idea.** In a world where token cost trends toward zero, the human supervisor is the only cost that stays fixed — so the entire economic logic of the stack eventually routes through one objective: minimize *operator-attention per correct decision*. The operator is the system's reliability bottleneck, a single server with a non-elastic service rate. The 51st agent does not get a 51st operator.

The cognitive ceiling is real: **Miller 1956**, *The Magical Number Seven* [90] — *working memory holds roughly* 7 ± 2 *chunks*, which makes read-poverty (§I.6) a human constraint, not merely an engineering one — and **Wickens**' *Multiple Resource Theory* ([49] — *attention is not a single scalar pool but a vector over stage* × *modality* × *code* × *visual-channel; you cannot trade audio-channel attention for visual-channel attention* 1:1). The vectored model matters: the right objective is not "minimize total seconds" but *minimize peak load on the bottleneck channel*, and the ambient-vs-dense split (§I.7.5) is the cross-modal load-balancing mechanism that does it.

I.5.1 The attention objective, spined on Signal Detection Theory

A naive objective — "attention-seconds-per-correct-decision" — is Goodhart-bait: it rewards lowering the decision criterion to wave things through, because zooming costs seconds and most diffs are fine. The correct frame is **Signal Detection Theory** (**Green & Swets 1966** [66] — *an operator discriminating signal from noise trades sensitivity* d' *against a decision criterion* β*; performance is a joint function of both and of the asymmetric costs of the two error types*). The operator is trading *misses* (rubber-stamping a destructive-on-stale-premise merge) against *false alarms* (zooming on a safe diff), and the cost of a miss is wildly asymmetric to the cost of a false alarm.

Definition I.5.1 (The SDT-spined attention objective)**.** Let an operator decision over an action have sensitivity d' and criterion β. With miss probability $P_{\text{miss}}(\beta)$, false-alarm probability $P_{\text{fa}}(\beta)$, miss cost C_{miss} (large: irreversible damage), false-alarm cost C_{fa} (small: wasted attention-seconds),

base rate π of genuinely dangerous actions, and per-decision attention cost a, the operator objective is

$$\min_{\beta} \Big[\, \pi\, C_{\text{miss}}\, P_{\text{miss}}(\beta) \;+\; (1-\pi)\, C_{\text{fa}}\, P_{\text{fa}}(\beta) \;+\; a\, \mathbb{E}[\text{zooms}(\beta)] \,\Big]. \tag{I.2}$$

The forced-zoom policy $p(r, s, v)$ is calibrated against this loss after specifying the score distributions that determine P_{miss} and P_{fa}. Equation I.2 is a decision objective, not by itself a fitted detector.

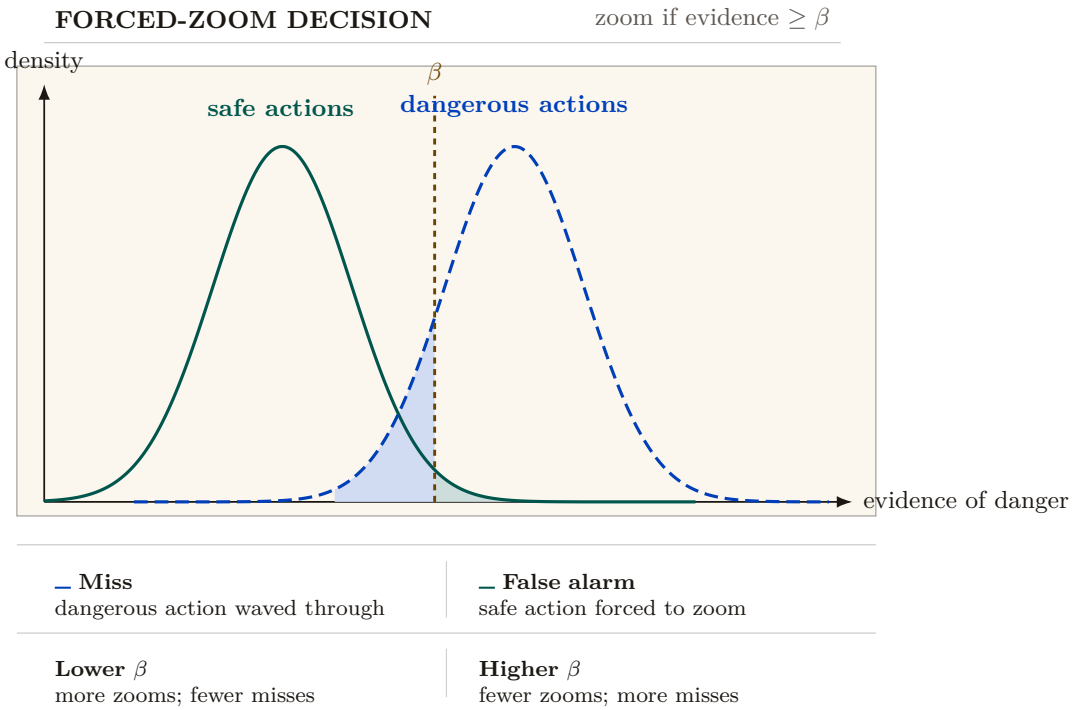


Figure I.6: Forced zoom trades benign review for fewer waved-through dangers. The solid teal distribution represents safe actions; the dashed cobalt distribution represents dangerous actions. The forced-zoom policy triggers when evidence reaches the amber threshold β. The shaded cobalt tail is the miss region (danger waved through); the shaded teal tail is the false-alarm region (safe action reviewed). Lowering β sends more work to a human and reduces misses; raising it saves attention but increases misses. The separation d' is the quality of the operator's read, and the cost asymmetry $C_{\text{miss}} \gg C_{\text{fa}}$ justifies a cautious threshold.

Figure I.6 lays the same decision out visually: the operator's criterion β trading misses against false alarms. The miss-cost term prevents the zero-review solution whenever misses have positive probability and cost. Monotonic response to stakes, irreversibility, and reputation is a policy constraint to test during calibration; it does not follow from the display without a model linking (r, s, v) to the score distributions. Before a reputation layer exists, r is a flat prior.

I.5.2 Trust calibration is multi-dimensional, and the vigilance decrement is real

Showing the operator a beautiful digest does not make them appropriately reliant on it (**Lee & See 2004**, *Trust in Automation* [112] — *trust has at least three bases (performance, process, purpose) and calibration depends on* ***resolution*** *(can the operator tell where automation is reliable from where it is not?) and* ***specificity*** *(is trust differentiated across functions and time?)*). The implication is sharp: a *single* "approve-without- zoom rate" and a *single* global forced-zoom knob is mis-calibration by aggregation. An operator who correctly trusts the swarm on refactors and correctly distrusts it on migrations shows a middling *global* catch-rate, and a global knob mis-serves both. Trust calibration must be *function-specific and history-dependent* — per (action-class, agent), not one scalar.

And the canary-defect catch-rate mechanism — inject known-bad diffs at a sampled rate, raise friction when catch-rate drops — is a *vigilance task*, monitoring for rare signals, and the vigilance literature predicts it will degrade for reasons unrelated to swarm danger (**Warm, Parasuraman & Matthews 2008**, *Vigilance Requires Hard Mental Work and Is Stressful* [111] — *detection of rare signals degrades sharply within 20–35 minutes, worse at low signal probability*; the classic result is **Mackworth 1948** [157]). The honest, citable tension: defeating the vigilance decrement *costs attention-seconds* (canary rate must be high enough to beat the decrement, which fights Eq. I.2). Catch-rate must therefore be *time-on-task-corrected*, and the right countermeasure is to force breaks, not just rotate zooms.

> **Pitfall.** Campbell's Law eats the governor. The moment "operator catch-rate" becomes a metered invariant that raises friction when it drops, the operator is incentivized to game it — memorize canary patterns, develop a "looks-like-a-canary" heuristic orthogonal to real defect-finding (**Campbell 1979** [70] — *the more a quantitative indicator is used for decision-making, the more it distorts the process it monitors*). The accountability instrument corrupts the accountability it measures. We do not claim the canary governor cleanly closes the loop; it is net-positive only if canaries are drawn from a refreshed, costly-to-fake distribution and the catch-rate is one signal among several, never the sole gate.

I.5.3 Situation awareness needs projection, not just perception

Digest-with-zoom is overwhelmingly a perceive-and-comprehend instrument. But **Endsley 1995** (*Toward a Theory of Situation Awareness* [146] — *SA has three levels: perception, comprehension, and **projection** — what the system will do next*) puts projection at the top, and **Endsley & Kiris 1995** (*the out-of-the-loop performance problem* [145] — *operators removed from active engagement lose the dynamic SA needed to take over when automation fails;* intermediate *automation can yield* better *SA than full automation because the operator stays cognitively engaged in projection*) shows the deeper danger: a digest that summarizes only the *present* is an SA-destroying instrument no matter how good its zoom. The authority half therefore needs a first-class *projection* mode — "what is the swarm about to do, and what will the state be in ten minutes?" — validated by a freeze-probe (pause the swarm, ask the operator to predict the next three merges, score accuracy). The forensic time-axis (§I.11.1) is the *backward*-looking complement; projection is the missing *forward* one.

I.5.4 The Bainbridge–Scott convergence and the abdication gradient

Lisanne Bainbridge 1983 (*Ironies of Automation* [130] — *the more reliable an automated system becomes, the less practiced and situationally aware its human supervisor becomes, so the human is least equipped to intervene at exactly the moment intervention is needed*) and Scott are the *same warning at two scales*: the digest erases the swarm's mêtis, and the digest erases the *operator's* mêtis. As the supervising automation gains autonomy — executing longer plans end-to-end — the operator's role slides doer → approver → rubber-stamp → absent — the *abdication gradient*. The design's primary active defense is the **Inception Canary** (a *consented, blinded* vigilance probe), moving beyond static "skill-retention taxes" (such as arbitrary manual busywork, which causes attention residue and degrades focus [188]).

Protocol I.5.1. The Inception Canary protocol

To continuously measure and maintain operator situation awareness without synthetic busywork. Operators enroll knowingly: the program's existence, rate ρ_{canary}, and stakes are disclosed at onboarding; only the *which* and *when* are blinded:

1. **Blinded holdout replay.** An out-of-band Compaction Sidecar (§I.8.2) draws from a curated holdout of *historical, already-fixed* defects (an off-by-one boundary condition, an unindexed query, a removed regression test), re-materializes one inside a sandboxed replay of a past work item at rate ρ_{canary}, and routes it through the review surface indistinguishably from live traffic. No anomaly is ever injected into a live pull request or pending action: probes are replays with known ground truth, so a missed canary costs vigilance points, never production correctness.
2. **Attention queue surfacing.** The item is placed in the Attention Queue with standard high-regret scoring.
3. **Dual adjudication:**
   - *Rubber-stamp failure:* If the operator approves without expanding the forced-zoom diff, the daemon halts execution pre-commit, logs an Inception Canary Failed event, and tightens the operator's autonomous policy lease (lowering the threshold at which future agent actions require manual confirmation).
   - *Vigilant detection:* If the operator zooms, identifies the flaw, and rejects it, the daemon logs Canary Defeated and increases the *Operator Trust Score*, unlocking extended autonomous execution grants (e.g., unattended batch merges).

This transforms supervisory vigilance from a punitive chore into a closed verification loop that rigorously measures human situation awareness — with the operator's informed consent, on ground truth the system already knows, and with provenance every probe carries so post-hoc audit can distinguish canaries from live items in one lookup.

Exercises

Check your understanding.
(5.1) In Eq. I.2, what happens to the optimal forced-zoom rate $p(r, s, v)$ as $C_{\text{miss}}/C_{\text{fa}} \to \infty$? As agent reputation $r \to 1$? (5.2) Give a concrete scenario where a *global* trust knob (§I.5.2) mis-serves an operator who is, in fact, perfectly calibrated.

Trace the mechanism.
(5.3) Walk the canary-defect loop and mark every point where Campbell's Law could distort it. Propose one change that reduces the distortion without removing the governor.

Open problems.
(5.4)$\star\star\star$ Formalize the operator-attention objective (Eq. I.2) against swarm token-COGS (§I.8): is there a digest that is Pareto-optimal in both operator-seconds and token cost, or does a cheaper digest always cost more operator seconds? (5.5)$\star\star$ Does a skill-retention tax *measurably* preserve SA, or merely annoy until disabled? Design the freeze-probe study (§I.5.3) that would tell.

I.6 Read-poverty: the binding constraint at scale

A single operator can hold two coding agents in their head. They cannot hold two hundred. Spin up two agents and coordination is free: you read both their notes, eyeball both their claims, route a question by recognition. Spin up fifty and an agent that wants to ask "who owns the OAuth refactor?" has no move except to dump the session roster and scroll. It guesses, cold-DMs the wrong agent, and re-derives from scratch a design decision another agent settled an hour ago and wrote into a note nobody will ever read again.

The swarm has not run out of information. It has drowned in it. Every agent emits sessions,

claims, notes, diffs, skill tags, and — eventually — reputations, far faster than any single participant can read them.

Definition I.6.1 (Read-poverty)**. Read-poverty** is a regime in which legible state accumulates faster than it can be consulted, so participants fall back to $O(n)$ eyeball search over the population, or, worse, act blind.

> *Read-poverty, not write-contention, is the binding constraint on swarm scale. The write side has known fixes — claims, locks, merge queues, and anomaly detection — which mature coordination systems already provide. The read side has been left implicit, and it is where the next order of magnitude is won or lost.*

The cure is the move every database made when table scans stopped scaling: build an **index**. A *directory* is to agents what an index is to rows — it pays a write-time cost (maintaining the structure) to buy read-time speed (answer a query in $O(\text{query})$ instead of scanning $O(n)$ participants). Read-poverty is also a human constraint (Miller's 7 ± 2, §I.5) and a legibility argument in Scott's sense: a directory is a state-like act of legibility over the swarm, and a directory entry over-flattened into a verdict that hides what mattered ("this agent owns auth," omitting that its last three auth pull requests were reverted) is the failure. Every directory entry must therefore be a lens that zooms (Def. I.3.1).

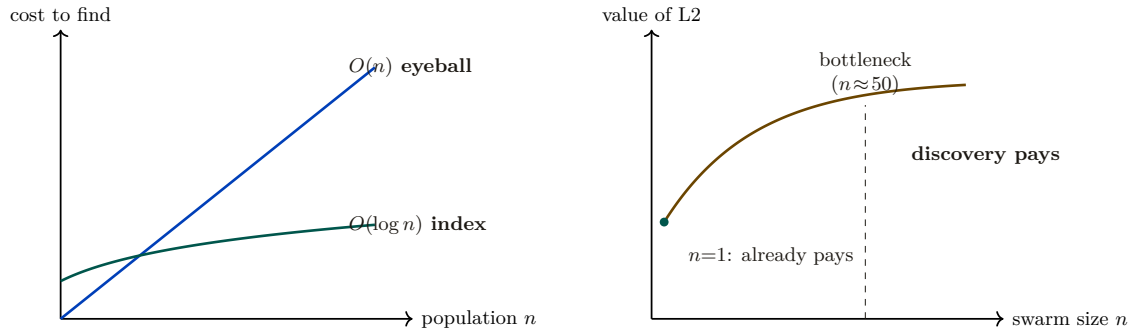


Figure I.7: Read-poverty (§I.6): state piles up faster than it can be read. *Left*: the cost of finding the right agent/work grows linearly under eyeball search but stays near-flat under an indexed directory — the same trade every database made at the table-scan wall. *Right*: the value curve of the legibility wedge across swarm size. It is valuable already at n=1 (briefing, attestation, footgun guard, honest completion) and grows into the read-poverty regime; discovery ranking earns its keep around the n≈50 bottleneck. The wedge must work at n=1 and grow gracefully.

I.6.1 The value curve across n (the wedge must work at n=1)

Read-poverty is a *scale* disease, but the wedge ships to a solo developer who often runs one or two agents. The legibility layer must be valuable at n=1 — the briefing, the attestation, the footgun guard, the honest completion gate all pay off immediately — and *gracefully grow* into the read-poverty regime (Figure I.7, right panel). Discovery is free at n=2, the bottleneck at n=50, the whole system at n=500. This value curve *is* the three-layer cure of the next section: existence (the registry), relevance (ranked routing), and trust (reputation, deferred to the accompanying chapters). The layers are strictly ordered: relevance over a stale registry returns confident garbage; reputation over an unranked registry has nothing to attach a score to.

I.6.2 A legibility lower bound

Section I.3 forbade the Potemkin digest by fiat. We can also state the prohibition as a theorem: there is an information floor below which a digest *cannot* be a faithful zoomable index, no matter

how it is written.

Theorem I.6.1 (Legibility lower bound with a review budget)**.** *Let N artifacts contain an unknown load-bearing subset S of size k. A digest maps each possible S to a review set $\hat{S}$ with $S \subseteq \hat{S}$ and $|\hat{S}| \leq m$, where $k \leq m \leq N$. Any such zero-miss digest needs at least*

$$\log_2 \binom{N}{k} - \log_2 \binom{m}{k}$$

bits in the worst case. Exact identification ($m = k$) recovers the $\log_2 \binom{N}{k}$ bound; allowing the operator to open everything ($m = N$) correctly reduces the bound to zero.

Proof sketch. There are $\binom{N}{k}$ possible load-bearing subsets. One digest message whose review set has size at most m can safely cover at most $\binom{m}{k}$ of them, namely the k-subsets contained in that review set. Therefore at least $\binom{N}{k}/\binom{m}{k}$ distinct messages are required. Taking base-two logarithms gives the bound. If fewer messages are available, some load-bearing subset is not contained in its message's review set, producing a false negative — an unopened load-bearing artifact. That is exactly the Potemkin failure of Def. I.3.1. Hence the stated log-ratio is a lower bound for a zero-miss digest with review budget m. □ □

The bound is the formal reason *forgetting must be selective, not lossy-by-volume*: a digest may drop inert detail freely, but it must retain enough to *point at* every load-bearing artifact, which is precisely the "summarize the state, link to the work" rule. It also bounds recursive summarization (§I.8.4): each re-summarization that drops below the floor on the load-bearing set is irreversible information loss, which is why one must compact from the durable artifacts, never from the previous summary.

Exercises

Check your understanding.
(6.1) Why is read-poverty (Def. I.6.1) the *binding* constraint rather than write-contention? Name the write-side fixes that make the write side not binding. (6.2) State the database analogy precisely: what is the write-time cost a directory pays, and what read-time speed does it buy?

Trace the mechanism.
(6.3) At n=1, name three legibility surfaces that already pay off (§I.6.1), and explain why none of them are discovery-ranking. (6.4) In Thm. I.6.1, what happens to the bit floor as $k \to 0$ (almost nothing is load-bearing) and as $k \to N/2$? Relate the answer to why a swarm doing mostly-safe work is cheap to digest and a swarm doing mostly-dangerous work is not.

Open problem.
(6.5)⋆ ⋆ ⋆ Thm. I.6.1 assumes the load-bearing subset is fixed and known to an oracle. In practice it is itself estimated, and the estimator can be gamed. Does the bound survive an adversary who controls which artifacts *appear* load-bearing? Connect to adversarial legibility (Ex. 3.4).

I.7 Discovery: the directory substrate and the split ranker

A performative cannot be addressed until the swarm knows who exists. The *directory substrate* — the existence layer a message must address — belongs with the discovery story, so we treat it here.

I.7.1 Existence: the yellow pages (a solved shape)

The canonical prior art is forty years old. **FIPA** (*Foundation for Intelligent Physical Agents*) made a **Directory Facilitator** (DF) — *the mandatory "yellow pages" agent: others register the*

services they offer and query to find agents that offer a needed service (FIPA Agent Management Specification, FIPA00023 [83]) — a required component of every conformant platform, alongside the **Agent Management System** (AMS, the "white pages" for *identity* lookup). Two FIPA choices matter: the DF is *push-based* (agents self-advertise), and DFs may *federate* (the seed of cross-operator discovery). The DF has been re-invented almost beat-for-beat for LLM agents (**Ren et al. 2025**, *Agent Name Service* [200] — *a DNS-inspired naming and capability-resolution layer with public-key-backed identity*, which is FIPA's DF with a 2025 threat model).

The existence layer is the easy part: a coordination database that records the live population, their stated purpose, and their file claims is a white-pages directory, and a vector index that embeds each agent's declared capabilities and answers nearest-neighbor queries is a yellow-pages directory over skills. Existence is not the gap; the layers above it are.

I.7.2 Relevance: the split ranker (the headline result)

A flat registry answers "does an OAuth specialist exist?" It does not answer "of the fifty live agents, who should I ask about *this* OAuth bug, given who touched these files an hour ago and who wrote the design note?" That is a *ranking* problem. The **discovery ranker** scores each candidate c against a query q as a clamped weighted sum of normalized sub-scores:

$$\begin{aligned} z(c,q) &= w_f\,\mathrm{file}(c,q) + w_n\,\mathrm{note}(c,q) + w_i\,\mathrm{ident}(c,q) \\ &\quad + w_s\,\mathrm{skill}(c,q) + w_e\,\mathrm{epis}(c,q) - w_\ell\,\mathrm{load}(c), \\ \mathrm{fit}(c,q) &= \mathrm{clamp}\big(z(c,q),0,1\big). \end{aligned} \tag{I.3}$$

where file rewards recent claims on the query's files, note and epis are text-similarity over the candidate's written notes and past episodes, ident and skill score declared identity and embedded capability, and load penalizes an already-busy candidate. The design deliberately blends *push* signals (declared capability — gameable and prone to staleness) with *pull* signals (demonstrated capability: what the agent actually *did*, in which there is no self-report to lie in). One discipline is non-negotiable: **never classify free text by keyword lists.** Keyword matching has catastrophic recall, because the category's vocabulary cannot be enumerated in advance; the only exact-string match permitted is over *structured identifiers the operator controls* (file paths, enum tags). Every free-text signal uses a ranked-retrieval scorer (BM25, character-bigram TF–IDF, or learned embeddings).

It is tempting to claim that this discovery ranker and the operator-facing *attention queue* are *the same function* with different weights. The claim is elegant and, on inspection, false: the two readers have incompatible loss functions. We state the distinction as a theorem.

Theorem I.7.1 (No universal monotone identification of the two heads)**.** *On any item domain containing both high-fit/low-risk and low-fit/high-risk examples, there is no strictly increasing scalar transform that identifies discovery fit with operator regret. The two rankers may share candidate generation and recency decay, but apply distinct scoring heads. Discovery uses capability fit (Eq. I.3); the attention queue uses regret-if-ignored:*

$$\mathrm{regret}(x) \;=\; \underbrace{\mathrm{stakes}(x)}_{magnitude} \times \underbrace{\mathrm{irreversibility}(x)}_{v^{-1}} \times \underbrace{\mathrm{anomaly}(x)}_{surprise}. \tag{I.4}$$

No universal monotone transform of one scalar yields the other on that domain.

Proof sketch. Suppose, for contradiction, that there is a strictly monotone map ϕ with $\mathrm{regret}(x) = \phi(\mathrm{fit}(x))$ for all items x, so the two heads induce the same ranking. Construct two items. Item A is a highly capable, perfectly on-topic completion of a safe, fully reversible task: $\mathrm{fit}(A)$ is maximal, but $\mathrm{stakes}(A) \cdot \mathrm{irreversibility}(A) \cdot \mathrm{anomaly}(A)$ is near zero, so $\mathrm{regret}(A)$ is minimal. Item B is an off-topic, low-capability agent that has nonetheless just proposed an irreversible, high-stakes, anomalous action: $\mathrm{fit}(B)$ is low, but $\mathrm{regret}(B)$ is maximal. Then $\mathrm{fit}(A) > \mathrm{fit}(B)$ while

$\text{regret}(A) < \text{regret}(B)$, so the two heads order $\{A, B\}$ oppositely. No monotone ϕ can both preserve and reverse an order, contradiction. Hence the heads optimize genuinely different objectives; they may share candidate generation and decay, but the scoring head must be chosen per reader. □ □

The consequence is a design rule: the operator wants the *most dangerous* item surfaced, the agent wants the *most capable* collaborator. Collapsing them into one ranker silently serves one reader and starves the other.

Decision-theoretic derivation of the regret head. The product form of $\text{regret}(x)$ is not an ad-hoc heuristic; it follows directly from expected-cost minimization under asymmetric error costs. Let x be an unreviewed candidate event, and let true state $S \in \{\text{harmful}, \text{benign}\}$. Suppressing x when harmful incurs a miss cost $C_{\text{miss}}(x) = \text{stakes}(x) \cdot \text{irreversibility}(x)$ (unrecoverable damage). Surfacing x when benign incurs an operator interruption cost $C_{\text{fa}} + c_{\text{att}}$. Surfacing x is Bayes-optimal if and only if:

$$\mathbb{E}[\text{Cost} \mid \text{surface}] \;\le\; \mathbb{E}[\text{Cost} \mid \text{suppress}] \iff C_{\text{fa}} \Pr(\text{benign} \mid x) + c_{\text{att}} \;\le\; C_{\text{miss}}(x) \Pr(\text{harmful} \mid x). \tag{I.5}$$

Rearranging yields the likelihood-ratio decision rule:

$$\frac{\Pr(\text{harmful} \mid x)}{\Pr(\text{benign} \mid x)} \;\ge\; \frac{C_{\text{fa}} + c_{\text{att}}}{\text{stakes}(x) \cdot \text{irreversibility}(x)}. \tag{I.6}$$

By setting the anomaly score $\text{anomaly}(x) \propto \frac{\Pr(\text{harmful}|x)}{\Pr(\text{benign}|x)}$, the decision threshold is equivalent to gating on the product $\text{regret}(x) = \text{stakes}(x) \times \text{irreversibility}(x) \times \text{anomaly}(x) \ge \tau_{\text{intervene}}$, which formalizes the Signal Detection criterion (Def. I.5.1).

Figure I.8 draws the two heads side by side over the same candidate pool, making concrete why one ranker cannot serve both readers.

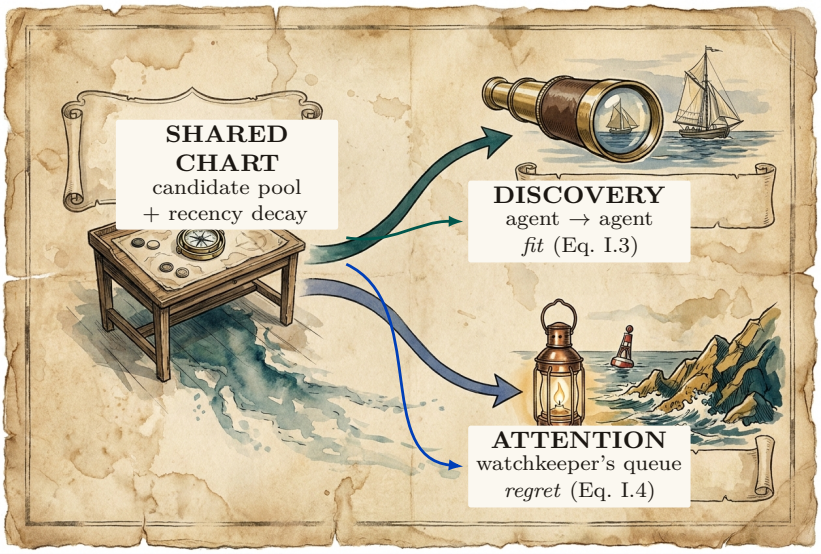


Figure I.8: One shared candidate chart supports two noninterchangeable readings. Both rankers draw from the same live substrate and apply the same recency decay; the discovery route asks which collaborator is most capable, while the attention route asks which item is most dangerous to ignore. The paths split because capability fit (Eq. I.3) and regret-if-ignored (Eq. I.4) can reverse the same pair of items. They are distinct objectives, not one score with weights renamed.

Protocol I.7.1. Attention-queue ranking with miss-cost

On each refresh, given the live set of items and the operator's per-class trust history:

1. **Generate.** Pull candidate items from the shared substrate, each tagged with a recency weight $\propto e^{-\lambda\,\text{hoursAgo}}$ (Prop. I.7.1).
2. **Score by regret.** For each item x compute $\text{regret}(x) = \text{stakes}(x) \cdot \text{irreversibility}(x) \cdot \text{anomaly}(x)$ (Eq. I.4).
3. **Apply the cost matrix.** Convert regret to a forced-zoom probability via the Signal-Detection criterion of Def. I.5.1: the higher $C_{\text{miss}}/C_{\text{fa}}$ for the item's action class, the more aggressively it is surfaced — the miss-cost term, not raw rank, decides what interrupts.
4. **Lane and emit.** Place items above the distress threshold on the reserved high-priority lane; emit the rest in regret order. Log what was *not* surfaced and why (the suppression counter-surface of §I.9).

I.7.3 The quality gates: refuse-to-route and the costly escalation

Two gates carry most of the design's trustworthiness. **Refuse-to-route:** if every candidate falls below a confidence floor, the ranker returns empty-with-explanation, not a confident wrong answer — "I don't know" must beat a guess, because sending an agent to the wrong door is high-cost and silent. This is the exact dual of the completion gate's fail-loud (Def. I.4.1): both are the same refuse-to-guess discipline, one on the read side, one on the authority side. **The costly escalation:** the attention queue is the system's *most spoofable* surface — agents that learn the stakes-ranking will manufacture distress to win the operator's attention. Goodhart's law applies more sharply here than to the discovery ranker (**Strathern 1997** [139]). The fix is signaling theory: an escalation the operator dismisses *debits the agent's standing*, so crying wolf is costly — the pre-trusted-set idea of **EigenTrust** (**Kamvar et al. 2003** [185]) applied to the distress lane. This arms the abdication-resistance and Goodhart gaps at once.

I.7.4 Staleness and decay on every read-surface

A recurring failure in real fleets makes the point concretely. An agent process dies or hits a provider usage limit, but its file claim remains on the books; hours later that *dead* claim still reads as "current" and a live agent is blocked from a file nobody is actually working in. This is a staleness bug, and it generalizes to every projection.

Property I.7.1 (Universal recency-decay)**.** Every read-surface projection — briefing, resurrection handoff, discovery result, attestation report — carries an explicit TTL or recency-decay. A pull signal contributes $\propto \exp(-\lambda \cdot \text{hoursAgo})$ with class-specific half-lives (e.g. a short half-life for file claims, longer for episodic memory, notes capped at a couple of days). *A stale row can never outrank a live one; a dead-session claim must decay out of "current," not persist as misleading truth.* This is not a new mechanism but a uniform application of the decay primitive that already governs the pheromone layer, and it is the production discipline of every health-checked, TTL-expiring service registry [94].

The dead-claim case is the worked instance: had the resurrection handoff and the claim projection carried Prop. I.7.1, the dead agent's claim would have aged out of "current" rather than blocking a live agent. Decay is easy to apply to a forgetting layer and easy to forget on a digest; Prop. I.7.1 closes that gap by making it universal.

I.7.5 Ambient legibility: the cross-modal load-balancer

Per Wickens' Multiple Resource Theory (§I.5), attention is vectored, not scalar. The dense console loads the *visual-focal* channel; an ambient channel — a menu-bar indicator that turns red on a PASS-to-FAIL transition, an opt-in sound, a reduced-motion cue — loads a *different*, always-visible, pre-attentive channel. This is not a second-class fallback — it is the cross-modal load-balancing mechanism, and it may be the *primary* alarm path, because a separate low-clutter field is where pop-out actually works (**Treisman & Gelade 1980**, Feature Integration Theory [16] — *a single salient feature pops out pre-attentively in ∼200–500 ms in a non-cluttered field, but degrades toward serial search as distractor heterogeneity rises* — so a busy console defeats the pop-out a dense, single-color distress lane depends on).

Exercises

Check your understanding.
(7.1) State Thm. I.7.1 in one sentence and give the loss function each head optimizes. (7.2) Why is the attention queue more Goodhart-exposed than the discovery ranker (§I.7.3)?

Trace the mechanism.
(7.3) Replay the dead-claim case (§I.7.4) twice: once without Prop. I.7.1 and once with it. Show exactly where the live agent unblocks. (7.4) An agent fires an escalation; the operator dismisses it. Trace the costly-signal debit and explain how it deters crying wolf without silencing genuine distress.

Open problems.
(7.5)⋆⋆ Calibrating the discovery-ranker weights with *no* ground truth: can the operator's accept/reject of routing suggestions serve as implicit relevance feedback — and does that loop Goodhart itself? (7.6)⋆ The decay-vs-summary boundary: a theory of *which* compaction for *which* signal (decay for coordination traces, summary for decisions, eviction-plus-pointer for reconstructable artifacts).

I.8 Tokens: the swarm's COGS *and* its legibility engine

A swarm of coding agents has exactly one consumable that is both metered and meaningful: **tokens** — *the sub-word units an LLM reads and writes, billed per million* (**Sennrich et al. 2016**, byte-pair encoding [173]). Tokens occupy a position no other resource does: they are *simultaneously* the swarm's **cost-of-goods-sold** (the direct variable cost any agent economy must meter) *and* its **legibility engine** (the mechanism by which a swarm becomes seeable). The bridge between the two is **compaction**.

> **Key idea.** The same compaction choice controls the bill *and* controls what is true and visible. When the context window fills you must drop something — and what you drop is a *cost* decision (you stop paying to carry it) *and* a *legibility* decision (whoever reads the residue no longer sees it). The act that resolves both is compaction, and compaction's output is exactly the digest a human or successor agent reads. *The digest IS the compaction.*

I.8.1 Tokens as COGS, and the effective-context trap

For a fleet the unit of output is "a landed piece of work" and the dominant variable cost is inference tokens (input context + output generation, each priced per million) — the textbook shape of **COGS**. Every price an agent market can set (a rented fleet's margin, an agent-for-hire's outcome-per-token) needs a COGS line, so context economics is the *denominator* of the market, not a side metric. There is a cost trap: *effective* context ≪ *advertised* context. Models degrade well before their advertised window (**Liu et al. 2024**, *Lost in the Middle* [154] — *performance*

is highest at the beginning and end of context and degrades for information in the middle, even in long-context models; **Anthropic 2025** names the same effect *context rot* [18]), rooted in the transformer's n^2 attention over n tokens (**Vaswani et al. 2017** [24]). The consequence is blunt: packing an agent to 200K tokens does not buy 200K of capability; it buys degradation at full price. Budgeting to *effective* context is simultaneously a cost optimization and a quality optimization — the two point the same way, the defining feature of context economics. The primitive the market ultimately needs is a *per-agent-task token ledger keyed to outcomes* — "this task spent 84K input plus 12K output to land that change" — so that cost can be attributed to a witnessed result rather than to an opaque monthly bill.

I.8.2 Compaction is the legibility engine; the digest IS the compaction

Compaction (**Anthropic 2025** [18] — *taking a conversation nearing the window limit, summarizing it, and reinitiating with the summary*, with the discipline *maximize recall first, then optimize precision*) is the canonical move for keeping a long-horizon agent under budget. Its siblings — structured note-taking (external memory) and sub-agent context isolation (the parent never sees the bloat) — complete the family. The reframe: *the only artifact anyone reads to understand the swarm is a compaction of its raw trajectory.* The operator reads a digest, not six transcripts; the seventh agent reads a briefing, not the first six's scrollback; a crashed agent's successor reads a handoff, not the dead agent's context. Each is a summary of dropped tokens produced for a specific reader. So compaction is the legibility mechanism of the *entire swarm*: the same act that keeps the bill down produces the very digest a human or successor agent reads (Figure I.9).

Theorem I.8.1 (The Split-Digest Theorem)**.** *Let $E = (e_1, e_2, \ldots, e_N)$ be an agent's raw execution trace. A compaction policy $D : E \to \Sigma^{\leq B}$ compresses E into a token budget $B < |E|$. Let R_1 be the* ***successor agent*** *with continuation loss $L_{\text{succ}}(D) = \mathrm{KL}(P_{\text{next}}(\cdot \mid E) \,\|\, P_{\text{next}}(\cdot \mid D))$ (penalizing loss of AST trajectory and task state), and let R_2 be the* ***human operator*** *with oversight loss $L_{\text{op}}(D) = \sum_{e \in E \setminus D} \text{regret}(e)$ (penalizing unreviewed high-stakes anomalies and policy breaches).*

If the trace contains both benign working details (high continuation value, zero regret) and dead-end anomalous excursions (zero continuation value, maximal regret), then ***no single compaction*** *$D^*(E)$* ***minimizes both*** *L_{succ}* ***and*** *L_{op}. A single-pipeline compaction necessarily starves either task continuation or human oversight.*

Proof sketch. Construct a trace $E = E_{\text{work}} \cup E_{\text{breach}}$, where E_{work} consists of k safe, valid AST transformations (L_{succ} increases by $\Theta(k)$ if omitted; $\text{regret}(e) = 0$), and E_{breach} consists of an attempted irreversible socket connection that was blocked and rolled back (L_{succ} increases by 0 if omitted since the branch is dead; $\text{regret}(e) = \text{stakes} \times \text{anomaly} = \text{maximal}$). Under budget $B < |E_{\text{work}}| + |E_{\text{breach}}|$, any compaction prioritizing E_{work} has $L_{\text{op}} = \text{maximal}$ (the operator is blind to the breach attempt), while any compaction prioritizing E_{breach} has $L_{\text{succ}} \gg 0$ (the successor loses necessary variable bindings). Hence R_1 and R_2 have strictly non-comonotone preferences, requiring distinct compaction heads. □ □

Architectural Corollary: The SLM Sidecar. Theorem I.8.1 dictates that the system must never ask the worker agent to author its own operator-facing summary (which invites self-attestation bias and Potemkin digests §I.3.2). Instead, the architecture establishes two separate pipelines:

1. **Task-Continuation Head (R_1):** Authored by the worker agent or structured state-checkpoints, optimized for continuation fidelity.
2. **Oversight Head (R_2):** Authored out-of-band by an embedded, air-gapped **Small Language Model (SLM) Sidecar** (e.g., a local 8B model running via `llama.cpp`) that reads raw kernel events and disk diffs as data, computing $\text{regret}(x)$ and surfacing truthful digests without worker collusion.

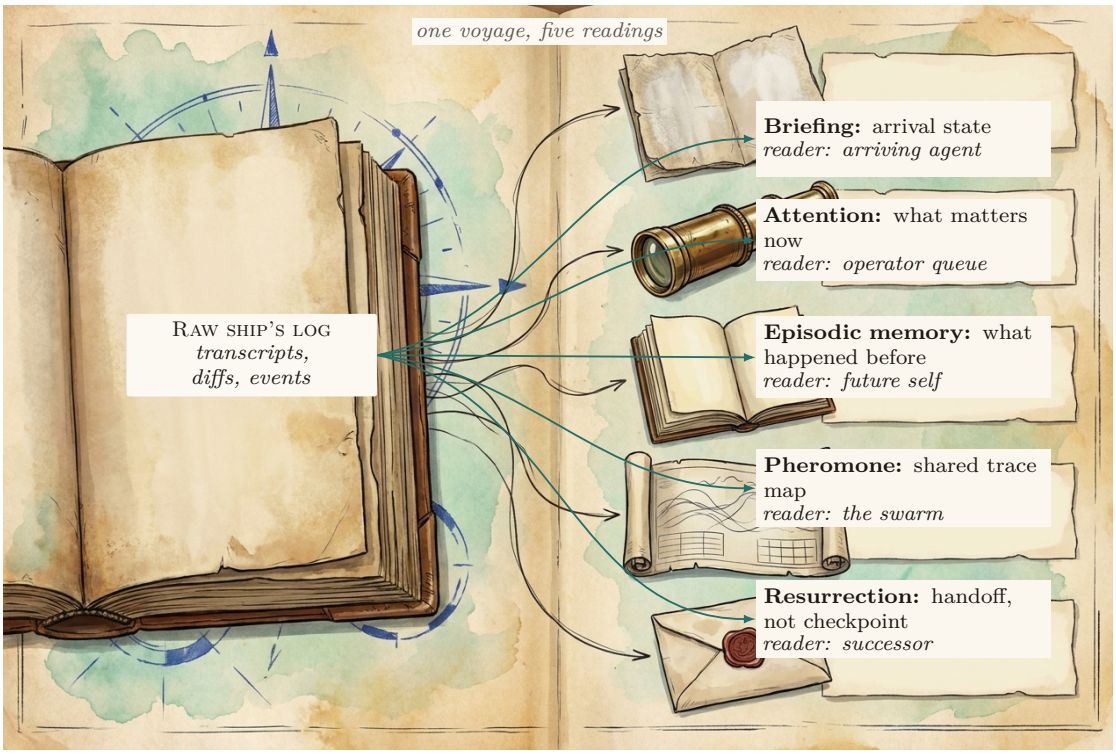


Figure I.9: One durable ship's log becomes five reader-specific instruments, not one general-purpose summary. The same raw trajectory yields an arrival briefing, an operator attention queue, episodic recall, a decaying swarm trace, and a successor handoff; each compacts for its reader while preserving a zoom path to the underlying artifact. Thus the coordination layer is a *distributed compaction engine*: its cost, recency-decay (Prop. I.7.1), and legibility obligations are shared even though its readings differ.

I.8.3 Context paging: virtual memory for the context window

Context compaction can be formalized as **virtual memory for language models**: the local file buffer is backing store, the finite context window is the resident set, and paging tools act as the memory management unit (MMU).

Theorem I.8.2 (Context Paging Competitiveness)**.** *Let the context window hold k active pages, with offline optimal page-replacement incurring cost* OPT*. Under an online* ***recency-plus-pin policy*** *(where pins correspond to load-bearing spans verified by deterministic structural features), the online paging cost satisfies the resource-augmented competitive bound of Sleator–Tarjan [57]; for the weighted case (pages of nonuniform size and refetch cost, which context spans are), the same guarantee is carried by Young's* Landlord *algorithm [153]:*

$$\text{Cost}_{\text{online}} \;\leq\; \frac{k}{k-h+1}\cdot \text{OPT} \;+\; \phi N c_{\text{refetch}}, \tag{I.7}$$

where $h \leq k$ is the size of the verifiable pinned set, and $\phi \in [0,1]$ represents the fraction of forgeable/unverified features subject to adversarial eviction. Under marking with randomized eviction, competitiveness against oblivious adversaries improves to $O(\log k)$.

Status: proposed. The unweighted bound is classical; the weighted transfer and the adversarial ϕ-degradation term are stated here as design targets, with the formal proof carried as item B9 of the research ledger (*the ϕ budget is shared with the digest floor's forgeable-feature budget — one number, two mechanisms*).

The most counterintuitive instance is *forgetting as compaction.* **Pheromone decay** — multiplying every stored coordination trace by a decay factor each interval — implements **stigmergy** (**Grassé 1959** [164] — *coordination via persistent traces left in a shared environment, as ants*

deposit evaporating trails). Decay is deliberate, continuous, $O(1)$ compaction of the shared context: the map stays readable because the stale ink fades. It is legibility by subtraction, and it is the same primitive Prop. I.7.1 generalizes to every read-surface.

I.8.4 The context-degradation cascade (the shared failure surface)

Cost and legibility share one failure surface: when compaction goes wrong, the bill *and* the truth degrade together (Figure I.10). **(1) Lost-in-the-middle:** a load-bearing fact buried mid-window is paid-for-and-ignored — edge-place facts, shorten the window. **(2) Context rot:** quality decays as the window grows even with nothing dropped — compact earlier (the cost cure and the quality cure are the same act). **(3) Recursive-summarization collapse:** each summary injects model noise; summarize the summary enough and the noise compounds into confident fabrication. The load-bearing cure: *compact from the durable artifacts each round — the version-control history, the written notes, the coordination database — never from the previous summary.* A system whose artifacts are durable and addressable is unusually well-placed here, because each compaction re-derives from source instead of recursing on prose. **(4) Over-flattening (the Scott failure):** the digest reads clean but the real situation is unreachable from it — detection rule: any digest line with no deep-link to its artifact; cure: legibility-with-zoom enforced by construction. A spartan, text-only operator console enforces this almost for free, because a terminal grid *cannot* over-render: the medium itself forbids the flattening.

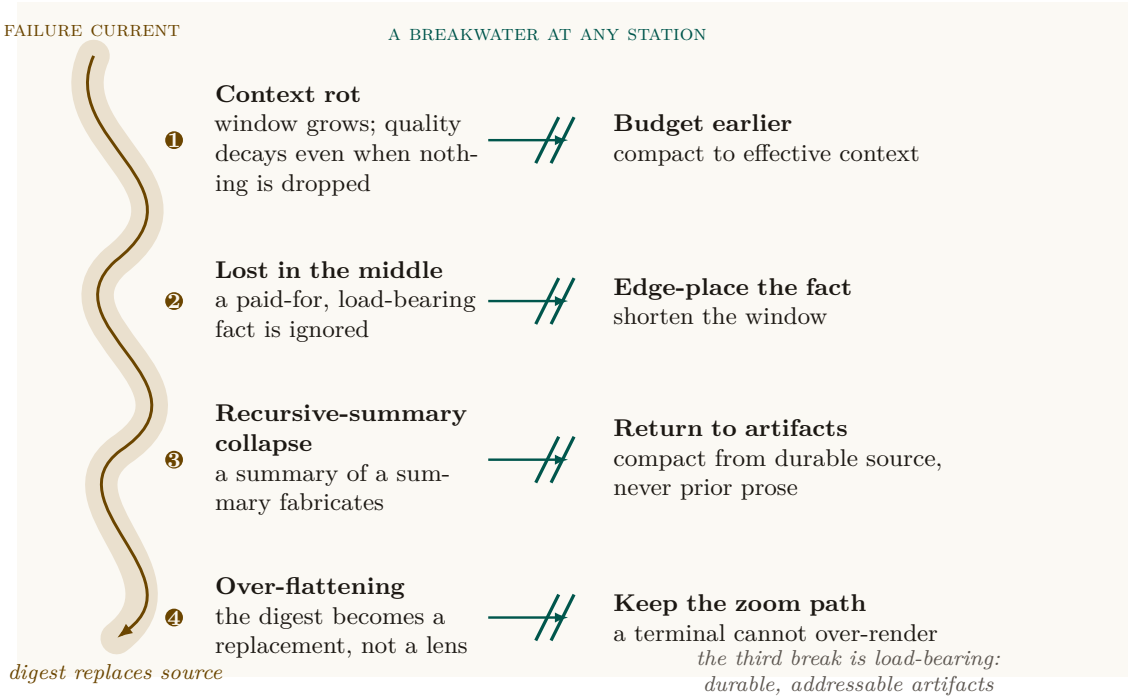


Figure I.10: Context degradation is a current that becomes a cascade only when it passes every breakwater. Rot pressures compaction; naive compaction loses the middle; recursive summaries fabricate; and the operator-facing digest finally replaces its source. Any intervention stops the run, but the load-bearing one is station 3: *compact from durable artifacts*, so every new reading can be checked against an addressable claim, note, episode, or diff rather than a prior summary. Cost and truth fail together on this surface.

Strategy	What it drops	Failure if abused	When it is the right tool
Recursive summary	Older prose, re-summarized	Confident fabrication (rung 3)	Never as the *only* record; only over durable artifacts re-read each round
Edge-placement	Nothing; reorders	—	Whenever a fact is load-bearing: put it first or last (counters lost-in-the-middle)
External note-taking	In-context detail, kept on disk	Notes nobody reads	Decisions and rationale that a successor must recover
Sub-agent isolation	Child context the parent never sees	Parent loses zoomability into the child	Bounded sub-tasks whose only output the parent needs is a result
Decay / forgetting	Stale coordination traces, continuously	Live signal aged out too fast	Ephemeral coordination state (claims, presence), where staleness is the enemy
Eviction + pointer	The artifact body; keep an address	Dangling pointer if the artifact moves	Reconstructable artifacts (a diff, a log) addressable on demand

Table I.3: The compaction strategies and their trade-offs. The unifying rule is Thm. I.6.1: a strategy may drop inert detail freely but must retain enough to point at every load-bearing artifact. Recursive summary is the only strategy that can cross that floor irreversibly, which is why it must always compact from durable source, never from the previous summary.

I.8.5 Single-operator: account, don't auction

In the wedge all agents serve one operator, so there is *no incentive problem* — only a visibility problem. The right mechanism is *accounting*, not a market: a per-agent-task token ledger, budget caps, and a loud failure when a cap blows. The externality-pricing and Sybil-resistance machinery (**Nisan et al. 2007** [155]; **Douceur 2002**, the Sybil attack [117]; **Ostrom 1990**, commons governance [73]) is a concern for the cross-operator economy, not the wedge — and it is one more reason the continuity-to-person-to-reputation through-line must come first: you cannot bill, cap, or reputation-score a swarm of disposable spawns, because Sybils dodge every per-agent mechanism. Context economics therefore *demands* the identity layer it appears to precede.

Exercises

Check your understanding.
(8.1) Why is "the digest IS the compaction" (§I.8.2) more than a slogan — what concrete consequence does it have for how the operator's digest is produced? (8.2) Explain why budgeting to effective context is simultaneously a cost and a quality optimization (§I.8.1).

Trace the mechanism.
(8.3) For each of the five compaction primitives (Figure I.9), name the reader, what it drops, and what it keeps. (8.4) Walk the four-rung cascade (§I.8.4) and name the single cure that breaks each rung.

Open problems.
(8.5)⋆⋆ The compaction-quality scalar: is *successor-replay-success- from-digest-alone* a sound, gaming-resistant metric, and can it be made a cheap attestation invariant (the replay smoke-test is itself token-expensive)? (8.6)⋆ Optimal token-budget allocation across a DAG of orchestrator/worker/ reviewer agents under a total budget.

I.9 Reconciling the canon: roles, consent, and local knowledge

The Leviathan metaphor is load-bearing only if it survives contact with the political- theory canon. Three reconciliations are required, and a reviewer would refuse the paper without them.

I.9.1 Who is the multitude, who is the sovereign?

Hobbes' covenant is *among the subjects*; the sovereign is its product, not a party to it. So the metaphor breaks differently depending on which entity sits where (Figure I.11). We assign roles explicitly, using Hobbes' own author/actor distinction (*Leviathan* ch. 16 [191]):

- **The agents are the multitude.** They covenant — via the rule of the road, the coordination protocol — to be bound, each to all, enforced by the daemon. This is the literal Hobbesian covenant-among-subjects.
- **The daemon is the sovereign-as-actor.** It holds the authority the multitude instituted; it blocks, lands, escalates. But, per the legible-sovereign amendment (Prop. I.2.1), it is the *most* legible actor, not (as in Hobbes) the least.
- **The operator is the author.** In Hobbes' author/actor split, the *author* owns the authority that the *actor* exercises. The operator is not a peer subject and not the sovereign-in-action; the operator is the source of authority, exercising it through the daemon and able to withdraw it (Prop. I.2.2). This is the slot Hobbes leaves for the one who authorizes but does not personally act.

This assignment is what makes the consent grant (Def. I.2.1) coherent: the author issues a scoped grant to the actor, over a multitude bound by covenant.

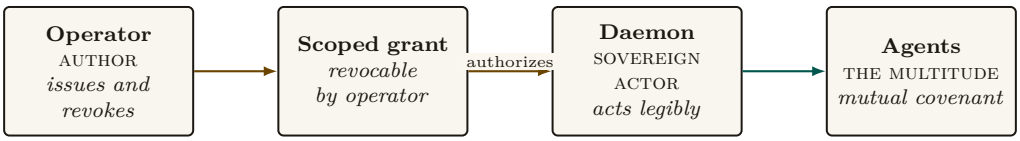


Figure I.11: Authority has one readable route: the operator issues a scoped grant, which authorizes the daemon to act over the agent multitude. The operator alone can revoke the grant. The agents' covenant is internal to the multitude; it is not a return arrow from subjects to the sovereign.

I.9.2 Express consent, not tacit; voice *and* exit

§I.2.2 already made consent a first-class object precisely to escape Hume's tacit-consent critique; the canon demands we also engage **Locke** (*Second Treatise*, §§95–122 [114] — *express consent binds where tacit consent does not*) and **Pateman** (*The Problem of Political Obligation*, 1979 [46] — *liberal consent theory keeps collapsing into hypothetical consent*). The consent grant is express by construction: a discrete, scoped, revocable authorization with an audit trail (Def. I.2.1). And the inalienable override (Prop. I.2.2) supplies the *exit* that Hirschman [10] says keeps *voice* honest. The design has rich voice machinery; the override is the missing exit, and exit is what gives the author real leverage over the actor.

I.9.3 Mêtis is Hayek, and some of it cannot be made legible

The formal constraint of §I.3.1 — that a read-path to the diff is more legibility, not mêtis — has a 1945 economics name. **Hayek**, *The Use of Knowledge in Society* (1945 [85] — *the knowledge that matters is dispersed, particular to time and place, and not aggregable by any central planner; this is the fundamental reason central planning fails*) is the other half of Scott's argument from the opposite pole: the operator-attention economy (§I.5) is at bottom a Hayekian knowledge-allocation

problem, not only a Wickens attention-resource problem. The core claim of this layer — "the binding constraint is decentralized, non-spawnable knowledge held in one head" — is a Hayek argument, and Scott's *Two Cheers for Anarchism* (2012 [104]) is the *constructive* sequel that says what to do: preserve the spaces where mêtis lives, do not flatten them for legibility's sake. The skill-retention tax (§I.5.4) is the design's gesture at exactly this, and it is the hardest mechanism in the paper to realize, because preserving the illegible is, by definition, the thing legibility cannot do for you.

Exercises

Check your understanding.
(9.1) Map each of agents, daemon, operator onto Hobbes' multitude, sovereign-actor, author. Why does the metaphor break if the operator is called the sovereign? (9.2) Why does Hirschman's *exit* (Prop. I.2.2) make *voice* (escalation) honest?

Trace the mechanism.
(9.3) Show that the consent grant (Def. I.2.1) is *express* in Locke's sense by listing the four properties that distinguish it from "the operator just started using the tool."

Open problem.
(9.4)⋆⋆ *The ranker as a political organ.* Specify the "what did the queue suppress, and why" counter-surface (Pitfall after Prop. I.2.1) so that ranking-as-governing is itself legible. Does the counter-surface need its own counter-surface? Where does the regress terminate, and on what principle?

I.10 Related work and prior art

This paper sits at the confluence of five literatures that rarely cite one another. We survey them in one place; each was introduced in context above.

Legibility and the limits of central schemes. The organizing lens is **Scott**'s *Seeing Like a State* [103]: states make populations governable by imposing legibility, and high-modernist *over*-legibility recurrently fails because it destroys the local, hard-to-codify *mêtis* the simplified order depended on; *Two Cheers for Anarchism* [104] is its constructive sequel. The same point arrives from economics in **Hayek**'s *The Use of Knowledge in Society* [85]: the decisive knowledge is dispersed and not aggregable by a central planner. **Rao** [201] ported "legibility" into a standard lens on systems that flatten reality to a single purpose. Our contribution is to make Scott's warning a *buildable* rule (digest-with-zoom, Def. I.3.1) and to give it an information-theoretic floor (Thm. I.6.1).

Authority, consent, and the social contract. The authority argument is Hobbesian [191, 97], amended by the liberal consent tradition — **Locke** [114], **Hume** [65], **Pateman** [46] — and by **Hirschman**'s exit/voice framework [10]. Strikingly, the state-of-nature-to-sovereign arc has been observed as an *emergent* dynamic of LLM populations (**Dai et al.** [91], building on **Park et al.** [120]). We turn the metaphor into a mechanism: an express, scoped, revocable consent grant (Def. I.2.1) with an inalienable override (Prop. I.2.2), and a legible-sovereign rule (Prop. I.2.1) that inverts Hobbes' opaque sovereign.

Human factors, automation, and signal detection. The operator model draws on the automation literature: **Bainbridge**'s ironies of automation [130], **Lee & See** on trust calibration [112], **Endsley** on situation awareness and the out-of-the-loop problem [146, 145], the vigilance-decrement results of **Mackworth** [157] and **Warm et al.** [111], **Wickens**' multiple-resource theory [49], **Treisman & Gelade**'s feature-integration theory [16], and **Miller**'s capacity limit [90]. The forced-zoom decision is modeled with **Green & Swets**' Signal Detection Theory [66]

(Def. I.5.1); the alarm discipline follows process-control practice [17] and the clinical alarm-fatigue evidence [137]; **Campbell** [70] and **Strathern** [139] (Goodhart's law) bound any metered governor. To our knowledge the operator-attention objective spined explicitly on SDT miss-cost (Eq. I.2) is new to agent supervision.

Information foraging and the cost of reading. Read-poverty (Def. I.6.1) is the agent-coordination instance of **Pirolli & Card**'s *information foraging theory* [160] — information-seekers follow "information scent" and abandon a patch when its expected yield falls below the cost of foraging — and of the long-context degradation results that make reading expensive even for machines (**Liu et al.** [154]; **Anthropic** [18]). The remedy is the classical database remedy — an index — recast as a directory over agents.

Agent discovery, naming, and trust. The directory substrate has deep prior art in **FIPA**'s Directory Facilitator and Agent Management System [83], re-invented for LLM agents as the Agent Name Service (**Ren et al.** [200]). For the trust layer that sits above discovery, the canonical reputation primitive is **EigenTrust**'s pre-trusted-set construction [185], with Sybil resistance per **Douceur** [117] and commons governance per **Ostrom** [73] — machinery the wedge defers but must not contradict. Our specific result here is the *split ranker* (Thm. I.7.1): discovery and operator-attention ranking are provably distinct objectives, not one ranker with swapped weights.

Where this paper is positioned. Table I.4 places the contribution against the nearest neighbors in each literature.

Literature	Nearest prior art	This paper's move
Legibility	Scott; Hayek; Rao	Make over-legibility a buildable failure mode; prove a legibility lower bound
Social contract	Hobbes; Locke; Hume; Hirschman	Express, scoped, revocable consent grant; inalienable override; legible sovereign
Automation & trust	Bainbridge; Lee & See; Endsley	Per-(action-class, agent) calibration; SDT miss-cost objective; skill-retention tax
Information foraging	Pirolli & Card; lost-in-the-middle	Read-poverty as the binding scale constraint; directory-as-index
Discovery & trust	FIPA DF/AMS; ANS; EigenTrust	Split ranker: fit vs. regret-if-ignored are distinct objectives

Table I.4: Where this paper sits relative to the nearest prior art in each of its five contributing literatures.

I.11 The time axis and the bridge to the economy

I.11.1 Legibility over time, not just the present

It is tempting to treat legibility as a snapshot — what is the swarm doing *now*. But most operator decisions are forensic: *why* did this land? The time axis is a distinct legibility mode — a replay scrubber over history, commit-anchored provenance on every annotation, and an append-only event log that answers "how did we get here?" (Scott's cadastral map is static; a swarm needs a legible *history*.) Together with the projection mode (§I.5.3), legibility spans all three tenses: the forensic past (replay), the inspectable present (digest-with-zoom), and the projected future (freeze-probe SA).

I.11.2 What this layer hands upward (the through-line)

The wedge is the single-player product. But the same legibility discipline applied to *continuity* is the load-bearing wall of everything above it. The through-line the accompanying chapters depend on is one sentence:

> *The economy rests on three continuity organs — memory, checkpoint (today, notes rather than true execution state), and a witnessed-outcome ledger (today, commitments and oracle closure rather than neutral graded outcomes) — and reputation keys on the richer third organ; the checkpoint organ is the weakest continuity link, and a real execution-state checkpoint — "resurrection with teeth" — is the literal foundation of any agent market.*

Concretely, the legibility layer provides upward:

- **Legible continuity** — resurrection-with-memory, episodic memory, and the briefing together make an agent's continuity legible. This is the precondition for the through-line *memory + checkpoint → continuity → a person not a spawn → reputation → a tradeable asset → the market.* No legible continuity, no reputation, no tradeable agent. *The score is cheap; the substrate it scores over — witnessed outcomes on a non-forgeable identity — is the gate.* That non-forgeable identity is a PARTIAL single-operator primitive in the wedge: daemon-minted `actor-souls` and a bounded newcomer pool ship, while universal write-boundary enforcement does not. The cross-operator attestation a real market needs is an additional, as-yet-unbuilt mechanism the economy paper must declare rather than assume.
- **The completion ledger** — in the target design, every verified completion (and every loud-failed one) becomes an outcome event. Today durable commitments, oracle-bound closure, and overdue detection form a partial witness substrate; neutral grading and reputation updates do not ship. This layer is the *outcome witness*; the reputation layer is the *scorer*. The split-ranker substrate (Thm. I.7.1) and the costly-escalation signal (§I.7.3) are the candidate-generation and trust-signal scaffolding the reputation head consumes.
- **The consent grant as a priceable object** — a hosted-trust offering is the transfer of a consent grant (Def. I.2.1) to a third party, which the economy paper prices.
- **Operator-attention economics** — the denominator a market cannot escape. The SDT-spined objective (§I.5) supplies the operator-attention cost of supervising a rented fleet.

The Leviathan that makes your swarm legible to you, without lying to you with a pretty map, is the product. Everything above it — federation, the market — is a second Leviathan made of the first.

I.11.3 The open problems, as a map

Table I.5 consolidates the starred exercises into the paper's open-problem surface, with the rung each couples to. They are genuine research problems, not homework dressed up.

Open problem	**Difficulty**	**Couples to**
Forced-zoom rate $p(r, s, v)$ (Ex. 5.4)	⋆⋆⋆	reputation (r term, Eq. I.2)
Operator-attention vs. token-COGS Pareto frontier (Ex. 5.4)	⋆⋆⋆	§I.8
Compaction-quality scalar as an attestation invariant (Ex. 8.5)	⋆⋆	honest attestation
Completionist verification of the unverifiable (Ex. 4.4)	⋆⋆	Def. I.4.1
Adversarial legibility (narration↔artifact) (Ex. 3.4)	⋆⋆⋆	sandbox + reputation
Abdication-resistance: does the tax preserve SA? (Ex. 5.5)	⋆⋆	§I.5.4
Calibrating discovery-ranker weights with no ground truth (Ex. 7.5)	⋆⋆	implicit relevance feedback
Decay-vs-summary boundary (Ex. 7.6)	⋆	Prop. I.7.1
Legibility's information-theoretic lower bound (Ex. 6.5)	⋆⋆⋆	Thm. I.6.1
The ranker-as-political-organ regress (Ex. 9.4)	⋆⋆	Prop. I.2.1

Table I.5: The paper's open problems, consolidated from the starred exercises, with difficulty and the mechanism each couples to. Several couple this layer forward to the identity-and-reputation layer, which is the seam the accompanying chapters carry.

Key idea. The wedge is a single-player product a solo developer pays for *today* — no economy, no cryptography, no second operator. It justifies the authority (the swarm is a state of nature; consent to a *legible local* Leviathan is rational), sets the product's quality bar (legibility-with-zoom, verifiable targets, SDT-spined friction, honest completion), and connects forward (legible continuity is the load-bearing wall of any future market). Build the harbor first; the trade comes when the ships do.

Chapter Appendices

I.A Implementation and status

The body of this paper argues mechanisms on their merits and does not depend on any of them being built. This appendix — and *only* this appendix — records the concrete engineering status of each mechanism in the reference implementation, named *Port Daddy*. We use four honest labels, with one rule for the whole table: confusing *built* with *designed* is itself the failure the paper argues against, so each row carries its evidence.

implemented = code exists in the reference implementation today; PARTIAL = code exists but is partial, unwired, or honest about a known gap; SPECIFIED = a concrete, accepted design with no merged implementation; *proposed* = a mechanism argued in this paper with no design document yet.

	Mechanism	Status	Evidence / location
L	Attention composer (agent-facing)	**implemented**	`lib/attention.ts`
L	Briefing projection (exemplary digest-with-zoom)	**implemented**	`lib/briefing.ts`
L	Resurrection handoff	PARTIAL	`lib/resurrection.ts`; notes, not checkpoints
L	Durable commitment / outcome substrate	PARTIAL	`lib/commitments.ts`, `lib/obligation-monitor.ts`; grading and reputation binding absent
L	Local actor identity root	PARTIAL	`lib/actor-souls.ts`, `lib/budget-guard.ts`; universal write gating absent
L	Honest attestation / legible sovereign	**implemented**	`lib/attest.ts`; ADR-0045
L	Episodic memory; pheromone decay; skill index	**implemented**	`lib/episodic-memory.ts`, `lib/pheromone.ts`, `lib/shipwright/skill-index.ts`
L	Anomaly monitor (detect-and-compensate, post-commit)	PARTIAL	`lib/arbiter.ts:328` is a log subscriber (Prop. I.4.1)
L	Operator attention queue (regret-ranked lanes)	SPECIFIED	ADR-0046; not yet ranked into lanes
L	Discovery router (relevance ranking)	SPECIFIED	ADR-0030; `lib/router.ts` not yet on disk
L	Text-only operator console	SPECIFIED	ADR-0046; `core/pd-tui` not yet on disk
A	Roadmap-as-truth (authority accountable to it)	PARTIAL	rows/matrices exist; accountability SPECIFIED
A	Adversarial / conflict-free review organ	SPECIFIED	ADR-0047 Phase 1
A	Completionist completion (verifier-gated)	*proposed*	Def. I.4.1; no design doc yet
A	Escalation (typed prioritized interrupt)	SPECIFIED	ADR-0047 → ADR-0046 distress lane; predicate unspecified
A	Thoughtful landing / merge ordering	PARTIAL	`lib/merge-queue.ts` present, not wired
O	Operator-attention objective (SDT-spined)	*proposed*	Def. I.5.1; unmodeled in code
O	Trust calibration via canary + catch-rate	*proposed*	no implementation
O	Consent grant + inalienable override	*proposed*	Def. I.2.1, Prop. I.2.2
O	Time-axis legibility (replay, provenance)	SPECIFIED	ADR-0046 replay scrubber; event log **implemented**
O	Ambient legibility (menu-bar indicator on PASS→FAIL)	PARTIAL	indicator exists; watchdog wire SPECIFIED

Table I.6: Implementation and status of every mechanism named in this paper. **L** = legibility half, **A** = authority half, **O** = operator-model. The legibility read-surfaces are largely built; the digest layer that unifies them is designed; the authority half — the Leviathan actually *ruling* — and the operator-model are largely designed-to-proposed. The wedge is real but more than half-designed: the safety and reading exist; the deciding and landing are the work that remains.

Three points about the reference implementation are load-bearing enough to restate:

1. **The anomaly monitor is detect-and-compensate, not prevention** (Prop. I.4.1). The monitor at `lib/arbiter.ts:328` is a post-commit log subscriber; by Schneider [84] it enforces no safety property. "Physically unreachable" is reserved for constraint- or boot-gate-enforced states only.

2. **The unified ranker is two rankers** (Thm. I.7.1). The discovery router maximizes *fit*; the attention queue maximizes *regret-if-ignored*. They share candidate generation and decay, not an objective.

3. **Every read-surface decays** (Prop. I.7.1). Briefing, resurrection, discovery, and attestation projections all carry recency-decay; the dead-claim case (§I.7.4) is the worked failure this closes.

I.B Notation and the nomenclature key

Two distinctions in this paper are easy to collapse and load-bearing to keep separate.

- **Capability vs. permission.** *Capability* is *ability* — what the execution sandbox physically makes possible. *Permission* is *normative status* — what the policy layer allows. They must never be collapsed: "able but forbidden" (a dangerous escape-hatch flag that exists in the tooling but must not be used) has to stay expressible, or the system cannot reason about a capability it possesses but is not permitted to exercise.

- **Two distinct delegation objects.** The *authorization chain* is a cryptographic, hop-bound, attenuation-monotonic delegation chain (a security object, developed in Chapter V). The *delegation trace* is the coordination object over which loop-detection runs (a liveness object, in this paper's protocol layer). They are different artifacts with different invariants; "delegation chain" unqualified conflates a security claim with a coordination claim and must be avoided.

- **Physically unreachable** is reserved for constraint-enforced or boot-gate-enforced states only (Prop. I.4.1); a detect-and-compensate monitor never earns the phrase.

PORT DADDY COORDINATION PAPERS · CHAPTER II OF VII

The Single-Writer Kernel

Abstract

A swarm of autonomous coding agents sharing one machine collides over the same scarce things: network ports, files on disk, mutual-exclusion locks, and the record of who did what. The instinct, trained on a decade of distributed-systems literature, is to reach for consensus — replicate the state, run an agreement protocol. We make the opposite, and we argue correct, move: collapse the entire problem onto a **single writer over a single local SQLite database in write-ahead-log (WAL) mode**, and let the operating system's file lock serialize every mutation. There is one decider, so there is no agreement to reach.

This paper presents that kernel — the **substrate layer** of a four-layer coordination stack — together with the implemented half of the coordination protocol that rides directly on it (a durable message bus, decaying stigmergic markers, violable commitments, a cryptographic delegation chain, and a policy monitor) as a **single-writer transactional reference monitor** in the sense of Anderson and Lampson: a small, tamper-evident mediator of every security-relevant operation, realized locally rather than as an abstract security kernel. We state the kernel's invariants as theorems — mutual exclusion, idempotence, the consistency model (serializable transactions, a linearizable claim history), oracle-bound obligation closure — and we are deliberate about where the promises stop. In particular we *split* durability by fault class: a write that returns success survives a *process* crash but is *not guaranteed* against power loss under the default WAL configuration; and we correct a widespread over-claim, showing that the policy monitor is a post-commit *detector*, not a pre-commit regimenter, with the only genuine pre-commit regimentation credited to a uniqueness constraint and a boot-time admission gate. The kernel is solid where it is local and provisional exactly where a cross-machine economy of agents would need it to become cryptographic and continuous — and we say so in the same words throughout. Companion chapters in the same series (*The Legible Swarm*, *From Spawn to Person*, *The Harbor Economy*) build the operator interface, the identity-and-reputation bridge, and the cross-machine market on the floor this paper specifies.

Keywords: reference monitor, single-writer transactions, SQLite write-ahead logging, durability by fault class, linearizability, deontic logic, commitments, stigmergy, local-first coordination, multi-agent systems

Reading time: approximately 45 minutes end-to-end; the Reader's Map (§II.2) routes you to a 15-minute path and to single-section paths. This is Chapter II of a seven-chapter collection. The Legible Swarm *is the operator-facing entry point and a natural place to start;* From Spawn to Person *builds an identity-and-reputation bridge on top of this kernel;* The Harbor Economy *carries the cross-machine market, including cross-machine capability transfer — which belongs to the cross-machine layer and lives there, not here. Any chapter can be read first, but every later layer rests on this one.*

II.1 Where this paper sits, and what it promises

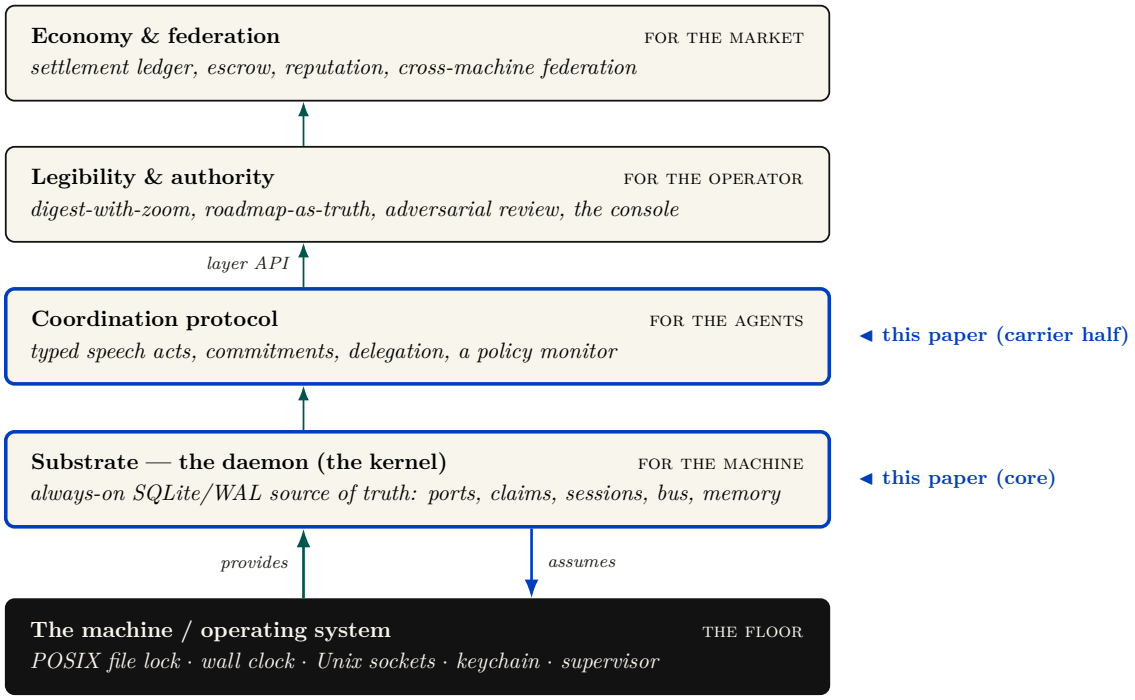


Figure II.1: The four-layer coordination stack. Each layer serves a different *whom* (machine, agents, operator, market). **Chapter II owns the two lowest layers**: all of the substrate (the kernel) and the implemented carrier half of coordination (the message bus, stigmergic markers, commitments, the cryptographic authorization chain, the policy monitor); the coordination *semantics* (protocol state machines, the agent sandbox) and the whole legibility layer belong to Chapters I and IV (Chapter I: *The Legible Swarm*; Chapter IV: *The Harbor Economy* owns the market rung). From the machine the substrate *assumes* only an advisory file lock, one clock, and a supervisor; upward it *provides* atomic TTL'd claims, a durable bus, the deontic split, and an audit log. Later chapters build on or cross-cut this floor; chapter numbers are not layer numbers.

We model an agent-coordination system as a four-layer stack (Figure II.1), each layer serving a different *whom*.

Substrate (the machine).
A single always-on daemon over one local database: durable storage and serialized mutation. *This paper's core.*

Coordination (the agents).
The protocol agents speak over the substrate: ownership claims, a message bus, stigmergic markers, obligations, and policy enforcement. This paper specifies the implemented half of it.

Legibility (the operator).
The human-facing view that renders the swarm intelligible. Treated in an accompanying chapter (*The Legible Swarm*).

Economy (the market between operators).
The cross-machine market in which operators trade work and reputation. Treated in an accompanying chapter (*The Harbor Economy*).

You climb the stack in order, and you never buy a layer before you need it. This paper is the load-bearing floor. The legibility layer reads the kernel's claims, bus, and policy monitor to render the swarm; the personhood argument leans the entire notion of a *durable agent identity* on the

kernel's continuity machinery; the economy's settlement ledger is, mechanically, another set of tables under the same single writer, and its cross-machine capability transfer rides the kernel's cryptographic delegation chain. If this floor is unsound, everything above it inherits the fault. So the discipline of this paper is to state each promise precisely and, where the promise has an edge, to draw the edge in ink.

II.1.1 The one-sentence thesis

> *The kernel is a single-writer transactional reference monitor over a local SQLite/WAL database — the one process that mediates contended resources, and the one place where "true" is decided rather than asserted.*

That is the whole claim, and it has two halves, which are the kernel's entire job; everything else is convenience.

1. **Durability (with an edge):** a write that returned success is durable across the death of any agent — and, we will be careful, across the death of the *daemon process*, though *not* unconditionally across power loss.
2. **Mediation (with a correction):** a state the kernel *regiments* cannot come to exist; a state the kernel merely *enforces* is detected and compensated within a bounded window. These are different guarantees and we will never conflate them.

II.1.2 Why "reference monitor," and why *not* consensus

The right mental model is the **reference monitor** of Anderson and Lampson [105, 41]: a mediator that is *always invoked* (complete mediation), *tamper-evident*, and *small enough to verify*. The kernel instantiates this not as an abstract security kernel but as a concrete local daemon over one SQLite file. Complete mediation holds *by construction over the kernel's own state*: because every mutation of *coordination state* routes through one writer holding one file lock, the operating system's serialization *is* the mediation primitive — Saltzer and Schroeder's "economy of mechanism" [108] done with one moving part instead of N hand-rolled critical sections. We are deliberate about the scope of the word *complete*: it is complete over writes that *enter* the daemon, not over what a same-user agent can do *beside* it. A classical reference monitor earns "always invoked" from the hardware/operating-system boundary that forces every access through it; this kernel has no such boundary at the substrate layer, so a same-user process can mutate shared state the daemon never sees — edit a file under an advisory claim, run `git` directly, or write the SQLite file itself — without routing through the writer. Mediation is therefore complete over the daemon's *application-program interface* and advisory everywhere else; making it complete over the agent's host capabilities is an out-of-band enforcement problem we scope to the threat model (§II.13) and credit, when it lands, to a separate-user kernel boundary that does not yet exist (§II.13.2).

> **Key idea.** The swarm *looks* like a distributed-systems problem — many agents, contention, failover — so the trained reflex is Raft, Paxos, CRDTs. The kernel's defining move is to refuse that problem. One writer, one machine, one file: there is nothing to *agree* on, so the consensus impossibility of Fischer, Lynch, and Paterson [147] does not bite. It is sidestepped, not solved. Distribution is a cross-machine concern that belongs to the economy layer; pushing a consensus protocol down into the local substrate would be the wrong layer.

This is the single most important architectural decision a local-first coordination layer can make, and the field is littered with systems that got it wrong by manufacturing a consensus problem they

did not have [142]. We will state the formal payoff of the choice as a conditional theorem in §II.11: when every read and write is routed through the sole daemon connection and responses follow commit, transactions are *serializable* and claim/lock operations are *linearizable* — the property that lets the coordination layer treat "who holds this" as ground truth without any agreement protocol at all.

II.1.3 A research-maturity scale, and why the grades are load-bearing

A systems paper that mixes "we proved this" with "we hope to build this" is unfalsifiable. So every load-bearing claim in this paper carries an explicit *maturity grade* on a four-point scale, plus two refinements introduced here because a single grade cannot formally carry a claim that spans fault classes or interception points. The grade is a property of the *claim*, not a marketing adjective: it says how far the claim is from a verified, running artifact. The concrete artifacts behind each grade are catalogued in Appendix II.A; the body argues at the level of mechanisms.

Table II.1: The research-maturity scale used throughout. The last two rows are refinements this paper introduces (see §II.5 and §II.8).

Grade	Meaning
implemented	Realized, exercised, and true under the production runtime.
PARTIAL	Realized but partial: holds under a narrower condition than the prose might invite, or detects rather than prevents.
SPECIFIED	A concrete design exists, but no running artifact realizes it.
proposed	Named as a direction; no design yet.
I1a / I1b	Durability *split by fault class*: I1a (process-crash) is **implemented**; I1b (power-loss) is *not guaranteed* under the default WAL configuration. A claim that conflates fault classes cannot carry one grade.
regimented / enforced	Prohibition split by interception point: *regimented* = rejected pre-commit, truly unreachable (a uniqueness constraint / boot-admission gate); *enforced* = detected post-commit, halted or compensated within a bounded window (the policy monitor). "Physically unreachable" is reserved for the regimented set.

Pitfall. The grades are not decoration. The two most consequential corrections in this paper — the durability split and the regimentation/detection split — are exactly the places where a single optimistic grade would let the kernel's foundational promise be read as stronger than the artifact delivers. When a system claims "the database survives every agent's death," a reader hears "power-loss durable." It is not, by default. Saying so is the difference between a systems paper and a brochure.

II.2 Reader's Map

Table II.2: Reader's Map. Find your row; read those sections first.

If you are...	**...read first**	**...load-bearing artifact**
short on time (15 min)	§1 (thesis + stack map), §II.5 (durability split), §II.8 (the monitor correction)	Figs. II.1, II.4, II.8
a systems engineer	§II.4–§II.11 (substrate, single-writer, the theorems)	Thm. II.11.1; Figs. II.3, II.11
a security reviewer	§II.8 (deontic split, the monitor), §II.13 (threat model)	Figs. II.8, II.7; Table II.4
a protocol/agent author	§II.6 (claims), §II.7 (the bus), §II.14 (the interface contract)	Figs. II.5, II.6; §II.14
here for the economy	§II.9 (continuity organs), then the personhood and market papers	Fig. II.10
an instructor	every **Exercises** block; the open problems OP-1 – OP-9 (§II.16)	the starred exercises

Reading order within the series. The legibility paper is the gentlest on-ramp; this paper, the substrate, is the formal floor; the personhood and market papers build upward. If you only read one paper to decide whether the foundation is sound, read this one.

II.3 The kernel as seven organs

We organize the kernel into seven *organs*, each a contract the daemon must hold for the layers above to be coherent (Figure II.2). The temptation is to describe such a kernel as a flat noun-list of features — ports, claims, sessions, a bus, markers, commitments, a monitor, a memory store — which is the symptom of treating the kernel as a database. It is not a database. It is a system of record *plus* a mediator, and naming the organs surfaces the connective tissue a noun-list hides.

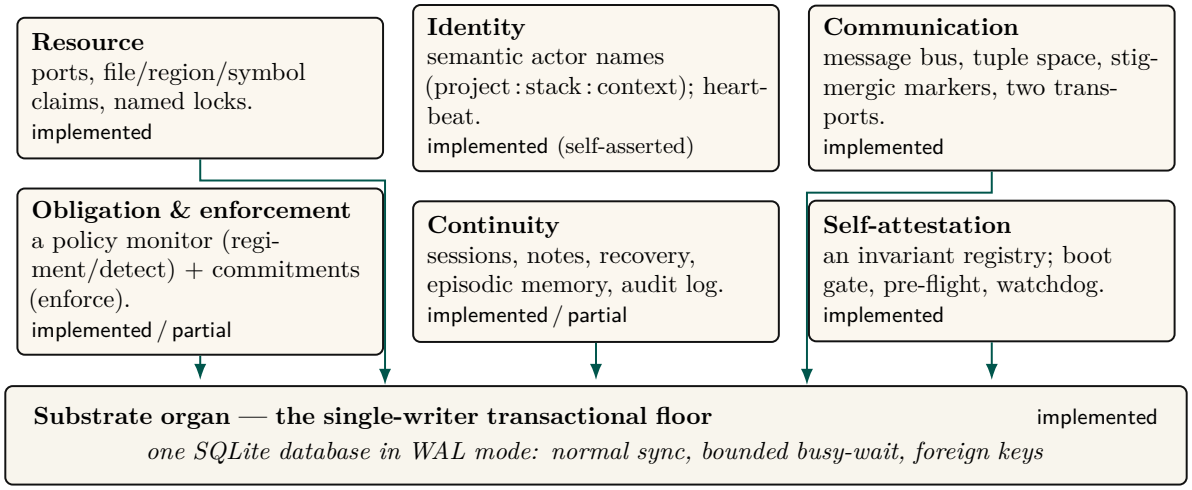


Figure II.2: The kernel as *seven organs*, each a contract the daemon must hold for the layers above to be coherent. Six organs are, mechanically, just tables *with discipline* over a single shared file — the **substrate organ**, the single-writer transactional floor. The substrate's storage configuration *is* the kernel's durability claim (the I1a/I1b split, §II.11); a runtime-shim layer keeps the deployment and test SQLite bindings in step, and everything else is convenience built on top of it. Maturity grades (implemented · partial · specified · proposed) are per organ; note that the identity organ is implemented but *self-asserted*, the gap a cross-machine economy must eventually close with cryptographic identity.

The **substrate organ** is the floor; the other six are, mechanically, tables *with discipline* over the same file. We treat the substrate and the two organs that carry the paper's sharpest corrections (resource/exclusion and obligation/enforcement) in full, and the rest with the depth their novelty warrants.

Exercises.
(check) Which of the seven organs would still be meaningful if the daemon ran over an in-memory database with no disk at all? Which become vacuous?
(trace) Pick any one organ and name the single SQLite table that is its "home" table. Then name one invariant that organ owes the layer above it.
(open, ★) The seven-organ decomposition is a *choice*. Is there a more natural cut — e.g. five organs, or nine? Argue for an alternative and show what it makes easier to reason about and what it hides.

II.4 The substrate organ: one writer, one file

The substrate is the bedrock everything else is just a table in. Its job is to make two promises — durability and serialized mutation — and to make them formally. We cover the single-writer discipline here and durability in §II.5, because durability is where the most consequential correction lives.

II.4.1 The single-writer discipline

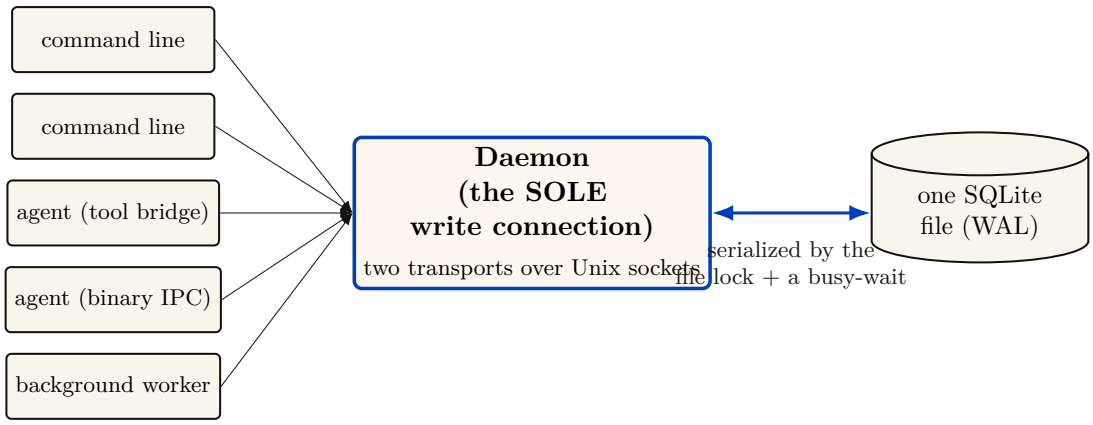


Figure II.3: The single-writer serialization, the cleanest thing about the layer. Dozens of command-line invocations and agents funnel through *one* writer — the daemon, the sole holder of a read–write connection — and SQLite's OS-level file lock plus a bounded busy-wait serialize all mutations. Because there is exactly one decider, there is no agreement to reach: the consensus impossibility of Fischer–Lynch–Paterson does not bite — it is *sidestepped*, not solved, and on one machine the system is CP with a central coordinator. This is not a gap; it is the layer's defining discipline. The boundary is exact: the instant a *second* daemon shares this file, the assumption breaks and you have a distributed problem — which belongs to the cross-machine economy layer, not here.

The kernel's concurrency story (Figure II.3) is: dozens of command-line invocations and agent connections funnel through one writer — the daemon, the sole holder of a read–write connection to the database file — and SQLite's operating-system file lock plus a bounded busy-wait serialize all mutations. We must be precise about *what* serializes writers, because there are two stories and only one is true at a time.

Definition II.4.1 (Single-writer discipline)**.** All state-mutating operations route through a single daemon process holding one SQLite write connection. Concurrency among the many clients (command-line invocations, agents over the message transports, background workers) is resolved by the daemon's request queue; the SQLite file lock is the *backstop* that preserves correctness if the discipline is ever violated by a second writer (for example a stray direct write), not the primary serialization mechanism.

> **Key idea.** "Single-writer" is an architectural *discipline* (everything goes through the daemon), backstopped — not replaced — by SQLite's file lock. This distinction matters: SQLite permits multiple writer connections (it serializes them, it does not reject them), so the property holds because of where writes are routed, not because the database forbids a second writer. The standing risk is therefore a second *copy* of the daemon coming up against the same file — which is why the substrate also runs self-checks that detect a divergent or stale second instance (§II.10).

II.4.2 The configuration *is* the durability claim

The substrate's storage configuration is short and load-bearing. In WAL mode with the *normal* synchronization level (the WAL is fsync'd at checkpoints rather than on every commit), a bounded auto-checkpoint interval, a bounded busy-wait, foreign-key enforcement on, and a clean-shutdown checkpoint-and-truncate, the kernel inherits exactly the crash semantics that configuration defines. The claim "a write that returned success survives a crash" reduces *entirely* to what WAL with normal synchronization guarantees — which is the subject of §II.5, and which is not what an unguarded reading assumes.

II.4.3 Linear Migration Ledger (OP-7)

The initial schema-by-union approach (`CREATE TABLE IF NOT EXISTS`) created vulnerabilities around unsequenced renames and multi-column backfills. The kernel now enforces a strict, linear migration ledger managed exclusively by `agentsd` via SQLite's `pragma user_version`. Modules no longer self-initialize; instead, the daemon executes a strictly ordered set of migrations upon boot, guaranteeing a deterministic schema state for all subsequent agent access.

II.4.4 Path and ownership invariants

The database resolves to a single canonical path with owner-only file permissions (historically the file also carried the system's signing key in plaintext; that secret is being migrated to the operating system's keychain, see §II.13). A fail-closed guard refuses to open the live registry from a test context — the analogue of a framework's protected-environment check. This is Saltzer and Schroeder's *fail-safe defaults* [108] applied at the one place a test could corrupt production truth.

Exercises.

(check) The schema-by-union design lets any module create its tables in any order. Give a concrete two-module example where a lazy column-rename migration would be unsafe under this design.

(trace) Follow a single claim request from command line to disk. At how many points does the single-writer discipline (Def. II.4.1) hold by *routing* versus by the *file lock*?

(open, ★ OP-7) Can idempotent self-init be extended to safe renames/backfills/down-migrations while preserving "any module, any order"? Or is a version table unavoidable?

II.5 Durability, split by fault class

This is the most important correction in the paper, so it gets its own section. The kernel's foundational promise — "a write that returned success survives" — is true, but only for the fault class a careful reader expects and not for the one the pitch most invites.

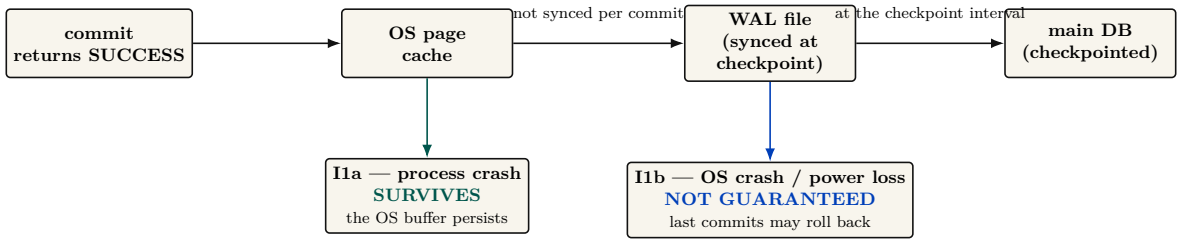


Figure II.4: Durability by fault class — the single most important correction to the kernel's foundational promise, and exactly Gray & Reuter's atomicity-versus-durability distinction (the "D" in ACID is a separate promise). The unqualified claim "a write that returned success survives a crash" is true for a *process* crash (I1a: the OS page cache persists past the dying daemon) but *not guaranteed* for an OS crash or power loss (I1b), because under normal synchronization the WAL is synced only at checkpoints — the seductive belief that "normal sync is safe in WAL mode" conflates crash-*consistency* (true) with power-loss *durability* (false). The remedy for I1b is full synchronization or a checkpoint on the commit path; the kernel does not pay that cost by default, and the prose must not imply it does. We split the invariant accordingly: I1a is implemented; I1b is not guaranteed under the default WAL configuration.

Figure II.4 draws the split this section defends: the same write, judged against two different fault classes with two different answers.

II.5.1 Why one label is a lie here

Gray and Reuter [109] draw the canonical line between *atomicity/consistency* (a transaction is all-or-nothing and never leaves the store corrupt) and *durability* (the "D" in ACID: a committed transaction survives subsequent failures). Under SQLite's WAL with the *normal* synchronization level, the log is *not* fsync'd on every commit — it is synced at checkpoint [55]. Per SQLite's own documentation, in WAL mode the normal level means "commits might lose durability following a power loss or hard reset," though the database remains uncorrupted. So the honest statement splits in two.

Property II.5.1 (Durability by fault class)**.** Let a write return success under the default WAL configuration.

- **I1a (process-crash durability).** If the *daemon process* dies (a forced kill, a panic, an out-of-memory termination), the write survives: the data is in the operating system's page cache, which outlives the process. **implemented**.
- **I1b (power-loss durability).** If the *operating system* crashes or power is lost *before the next checkpoint*, the most recent committed writes may roll back. *not guaranteed* under the normal synchronization level; it would require full synchronization or a checkpoint on the commit path.

Worked dramatization: Alice crashes mid-write. Alice is an agent editing a file. At t_0 she commits a claim on a port and a note recording her edit; the daemon returns success. At $t_0 + 5\,\text{ms}$ her host operating system kills the daemon process outright — a forced termination, no clean shutdown. *What survives?* Both writes. The committed pages live in the operating system's page cache, which the daemon process does not own and does not take with it when it dies; when the supervisor restarts the daemon, SQLite finds a dirty log, replays it, and Alice's claim and note are intact. This is I1a, and it is the fault class the design actually targets: agents die constantly, and the record must outlive them. Now change one detail. At $t_0 + 5\,\text{ms}$, instead of the daemon

dying, *the power fails*. The committed pages were in the page cache but had not yet reached stable storage, because under the normal synchronization level the log is fsync'd only at the next checkpoint. On reboot, the database is uncorrupted — but Alice's last claim and note may simply be gone, rolled back as if never written. This is I1b, and it is *not guaranteed*. The two faults differ only in whether the page cache survived; conflating them is the single most common over-claim about WAL-backed storage.

> **Pitfall.** The seductive misconception is "the normal sync level is safe in WAL mode because WAL already guarantees crash safety." This conflates crash-*consistency* (true: the database is never corrupt) with *durability* (false for power loss). The two are different ACID guarantees, and only the first is unconditional here. A systems paper cannot ship the unqualified claim. We split it.

II.5.2 Why the trade is defensible — and where it stops being so

Trading power-loss durability for throughput is a reasonable default for a local-first developer tool: the failure (lose the last few hundred milliseconds of claims after a power cut) is benign and self-correcting — agents re-claim, heartbeats resume. What is *not* defensible is letting the prose imply the stronger guarantee. To solve this (OP-10), the kernel implements **Selective Checkpointing**. High-stakes, money-bearing writes (like those in the settlement ledger) append `PRAGMA wal_checkpoint(FULL);` to their execution paths, forcing a synchronous flush to disk. This achieves power-loss durability (I1b) dynamically without compromising the high-throughput standard paths.

II.5.3 A recovery story, not just a durability story

Durability is about what survives; recovery is about what happens next. On daemon restart with a dirty WAL, SQLite replays the log to a consistent state (I1a's mechanism). But the kernel-level questions are larger: who validates the audit chain on boot, and what becomes of a half-applied cross-organ operation (§II.12) after a crash? The hook exists — the boot-admission gate that refuses to serve when a critical self-check fails (§II.10) — but boot-time WAL recovery plus integrity check plus chain verification should be a single named invariant, and today it is closer to PARTIAL than **implemented**.

Exercises.

(check) Classify each fault as I1a-survivable or I1b-exposed: (a) a forced kill of the daemon process; (b) a clean operating-system reboot; (c) yanking the power cord; (d) an unhandled exception in a request handler.

(trace) A claim commits at t_0; the next auto-checkpoint fires at $t_0 + \delta$. Draw the window during which a power loss reverts the claim, and state how a page-count auto-checkpoint threshold bounds δ.

(open, ★ OP-10) Design a per-write-path durability selector: how would the kernel let a settlement-ledger write opt into I1b while keeping claim writes fast? What is the cleanest interface that does not re-introduce the conflation?

II.6 The resource organ: claims, locks, and fair exclusion

Network ports are the namesake and the original sin: two agents that each start a development server on the same port collide, and one silently fails. The resource organ generalizes that problem

into a single exclusion primitive — over network ports, over named mutex locks, and over edit-surface claims on regions of source files — to which every higher layer's notion of ownership reduces.

II.6.1 Claim granularity is richer than "claims"

It is tempting to collapse all three of these into one word. The kernel instead distinguishes **whole-file**, **line-range**, and **symbol-path** claims, each keyed by the file path together with, respectively, nothing, a line interval, or a syntactic symbol identifier. This is the substrate for a "reserve regions, not whole files" coordination policy and for semantic-conflict prediction over a syntax-aware symbol index — but at the substrate layer it is simply exclusion at three granularities.

II.6.2 The claim lifecycle as a finite-state machine

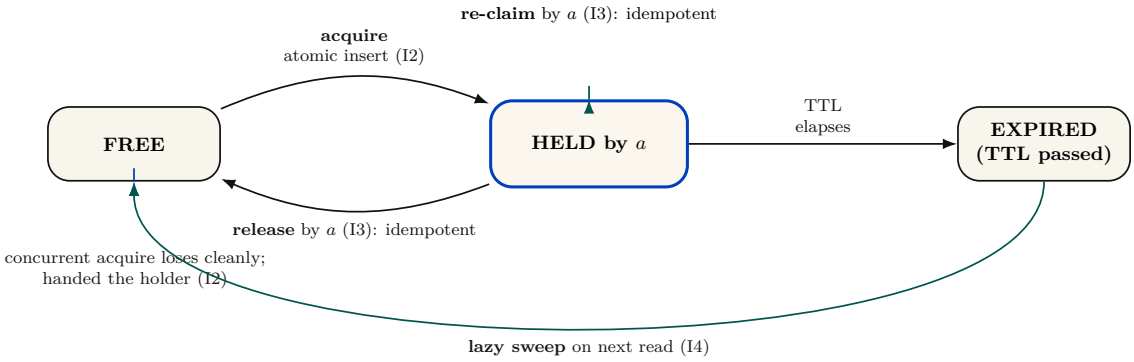


Figure II.5: The claim/lock lifecycle as a finite-state machine, exhibiting the four properties on which the coordination layer builds typed ownership. *Acquire* is a single atomic insert against a uniqueness constraint (I2: a concurrent acquirer loses cleanly to a constraint violation and is handed the current holder). *Re-claim* and *release* are idempotent (I3): safe to retry, safe to double. Expiry is *lazy-swept* on the next read (I4) — expired rows are deleted when encountered, not on a timer. The conspicuous missing state is QUEUED: there is no wait-list and no fairness primitive, so a fast agent churning a hot resource can starve a slow one indefinitely — open problem OP-1, which asks whether bounded-wait can be added without dragging a scheduler down into the kernel.

Figure II.5 states, as a state machine, the four properties on which the coordination layer builds typed ownership. The acquire path is small enough to state as an algorithm (Algorithm II.1); we then give the central property as a theorem.

```
function acquire(key, requester, ttl):
  // single atomic statement; the uniqueness constraint is the
     mutex
  try:
    INSERT row (key, holder=requester, expires_at = now()+ttl)
    return GRANTED(holder=requester)          // we won
  catch UNIQUE_CONSTRAINT_VIOLATION:
    existing = SELECT row WHERE row.key = key
    if existing.holder == requester:
      UPDATE row SET expires_at = now()+ttl  // idempotent re-
         claim
      return GRANTED(holder=requester)
    if existing.expires_at < now():           // stale holder
      DELETE row WHERE key = key AND expires_at < now()
      return acquire(key, requester, ttl)     // one bounded
         retry
    return DENIED(holder=existing.holder)     // loser learns the
        winner
```

Listing II.1: Claim acquisition. The whole acquire is a single insert against a uniqueness constraint; the loser is handed the current holder, not an error to retry blindly.

Theorem II.6.1 (Atomic Mutual Exclusion and Fenced Leases)**.** *Within a canonical conflict domain (exact normalized key, transactional interval overlap $[a, b] \cap [c, d] \neq \emptyset$, or path prefix hierarchy), at most one holder can hold an active lease at any logical instant. Furthermore, every granted lease carries a strictly monotonic fencing epoch $e \in \mathbb{N}$; downstream effect brokers reject any operation bearing an expired epoch $e' < e_{current}$.*

Proof sketch. Within an immediate SQLite transaction (`BEGIN IMMEDIATE`), the daemon evaluates the conflict predicate against all currently unexpired leases. For exact keys, this is enforced by primary-key uniqueness; for intervals and hierarchies, by an indexed range check within the same transaction. Because all writes funnel through the single connection (Def. II.4.1), predicate evaluation and insert are atomic. Upon grant or reallocation, the daemon increments the resource's fencing epoch e. Downstream effect brokers validate that e matches the active lease at execution time, preventing stale or paused holders from producing side effects after lease expiration. □ □

Worked dramatization: Alice and Bob race for one port. Alice and Bob are two agents, started a millisecond apart, that both want to bind a development server to port 7421. Each issues `acquire(7421)` (Algorithm II.1). Both requests reach the single daemon, which serializes them: one insert lands first. Say Alice's does. Her insert succeeds and she is granted the port. Bob's insert, arriving an instant later, hits the uniqueness constraint on key 7421 and is rejected *atomically* — there is no moment at which both rows exist. Crucially, Bob does not get a bare error. The catch path reads back the row and hands Bob the *identity of the holder*: "port 7421 is held by Alice." Bob can now pick another port, wait, or message Alice, but he can never double-bind. The serialization that makes this clean is not a hand-rolled critical section; it is the operating system's file lock funnelling both inserts through one writer, and the database's primary-key uniqueness deciding the winner. One writer, one constraint, one holder.

Pitfall. Theorem II.6.1 is sound for the *fresh-contention* case above. A subtler hazard hides in the *stale-holder* branch: reclaiming an expired lock runs a delete (of the expired row) and *then* an insert as two separate statements rather than one transaction. On the single daemon connection this is serialized by the connection itself and is safe. But if a second writer connection ever existed — the very scenario the single-writer discipline exists to prevent — a deferred transaction would open a window for a transient "database busy" error mid-sequence rather than clean serialization; the fix is to begin the transaction in immediate mode. So Theorem II.6.1 holds because all writes route through the one daemon connection (Def. II.4.1), *not* because the expire-then-acquire sequence is itself atomic. State which story you mean; do not tell both.

II.6.3 Fairness via Stigmergic Ticket-Lock (OP-1)

The conspicuous gap in the naive claim lifecycle is the absence of a QUEUED state. Locks have a time-to-live, but naive acquisition is racy first-come, meaning a fast agent churning a hot file can starve a slow agent indefinitely.

We solve this (closing open problem OP-1) via a **Stigmergic Ticket-Lock**. The substrate introduces a `claim_tickets` table:

```
CREATE TABLE claim_tickets (
  resource_id TEXT NOT NULL,
  agent_id TEXT NOT NULL,
  seq_num INTEGER PRIMARY KEY AUTOINCREMENT
);
```

When Agent B's claim acquisition fails against the primary-key `UNIQUE` constraint, the error handler inserts an entry into `claim_tickets`. When Agent A calls `release('auth.ts')`, the daemon executes a single atomic transaction:

1. Delete Agent A's active claim row: `DELETE FROM claims WHERE resource = :r;`
2. Read the lowest ticket: `SELECT agent_id, seq_num FROM claim_tickets WHERE resource_id = :r ORDER BY seq_num ASC LIMIT 1;`
3. Atomically re-grant the claim: `INSERT INTO claims (resource, holder, ttl) VALUES (:r, :next_agent, :ttl);`
4. Remove the claimed ticket: `DELETE FROM claim_tickets WHERE seq_num = :next_seq;`

Because the state transition is driven entirely *reactively* by the releasing agent in a single transaction, FIFO bounded-wait fairness is mathematically guaranteed by SQLite's B-Tree sequencing without dragging a background timer or polling thread into the kernel.

Exercises.
(check) Why does releasing a lock you do not hold return success rather than an error? Which property (I2–I4) is this, and what would break if it raised?
(trace) Two agents call `acquire` on the same fresh key at the same instant. Walk the uniqueness-constraint insert for both (Algorithm II.1) and show exactly where the loser learns the holder's identity.
(solution, ★ OP-1) The Stigmergic Ticket-Lock provides bounded-wait fairness purely through the reactive release transaction without requiring a kernel background scheduler.

II.7 The communication organ: the bus, stigmergy, and two delegation chains

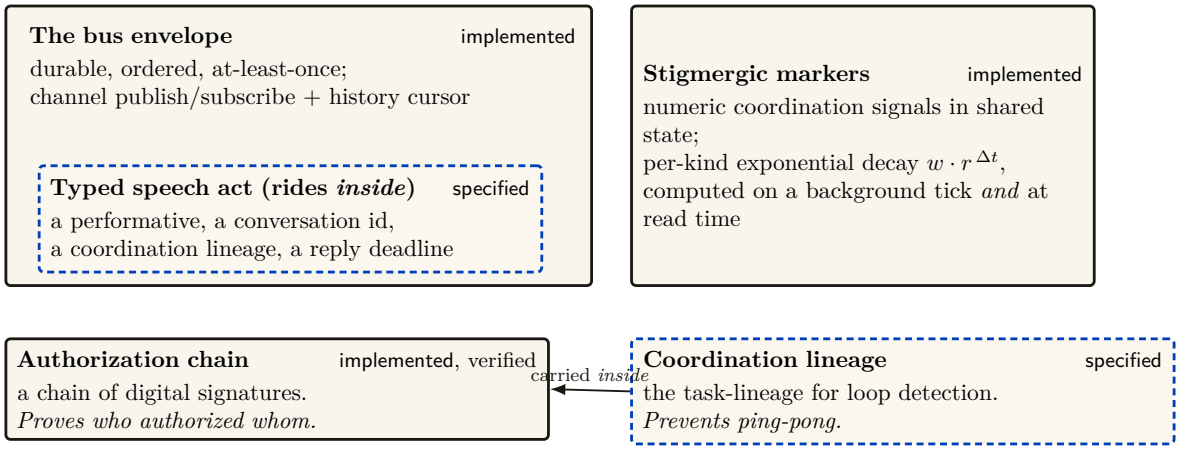


Figure II.6: The communication organ. The **bus** envelope — durable, ordered, at-least-once, history-cursored (implemented) — is the carrier *into which* the typed speech act is layered; the kernel ships the envelope, the coordination layer ships the semantics that ride inside it. **Stigmergic markers** implement coordination with per-kind exponential decay computed both on a background tick and *at read time*, so the anti-blackboard-rot property is provided by the substrate rather than left to discipline. Finally, the figure fixes a dangerous naming collision — never write bare "delegation chain," it is two objects: the *authorization chain* is the hop-bound, attenuation-monotonic, anti-replay chain of signatures that proves *who authorized whom*; the *coordination lineage* is the coordination object that prevents ping-pong — a different thing entirely, carried *inside* the authorization chain.

The communication organ (Figure II.6) is where the substrate stops being a lockbox and becomes a medium for conversation. The carrier is **implemented**; the *semantics* that ride on top of it (typed speech acts, protocol state machines) belong to the coordination layer. Three pieces matter here.

II.7.1 The bus: a durable carrier envelope

The bus is a durable, ordered, *at-least-once* publish/subscribe channel over a messages table, with a versioned envelope (a version tag, a message kind, a body, and an in-reply-to pointer), a history cursor for replay, and a per-message time-to-live. The coordination layer's typed speech act — carrying a performative, a conversation identifier, a coordination lineage, and a reply deadline — is layered *inside* this envelope. The kernel ships the carrier; the coordination layer ships what it carries.

> **Key idea.** At-least-once delivery means a *healthy* bus routinely redelivers and reorders. This collides with a loud-fail honesty discipline: if "every anomaly is loud," benign redelivery drowns the operator in false alarms. The resolution is that benign transport nondeterminism must be *deduplicated silently*, and only genuine protocol-state-machine violations surface loudly. That requires an idempotency key in the envelope — currently absent — which is the cheapest high-leverage fix at the coordination layer. The kernel provides the bus; making every speech-act effect idempotent is the coordination layer's debt, flagged here so the reader does not mistake silent deduplication for dishonesty.

II.7.2 Stigmergic markers: decay as a provided guarantee

The kernel also carries *stigmergic markers* — numeric coordination signals deposited in shared state, after the manner of Grassé's stigmergy in social insects [165] and the generative-communication tuple spaces of Gelernter's Linda [64], and used in ant-colony optimization [136]. Each marker carries per-kind exponential decay $w \cdot r^{\Delta t}$, computed both on a background evaporation tick *and* at read time (so a reader sees the correctly decayed value without waiting for the tick), and pruned below a small threshold. The decay is the structural defense against "blackboard rot": stale coordination signals evaporate on their own. The calibration of the decay rate is genuinely open (OP-8): too slow and signals rot, too fast and they evaporate before they coordinate.

II.7.3 Two delegation chains, one dangerous name

A dangerous naming collision lives in this organ. The phrase "delegation chain" names *two different objects*, which we keep rigorously distinct:

- the **authorization chain** — the *cryptographic* object (**implemented**, mechanically verified in a protocol-verification tool): a hop-bound, attenuation-monotonic, anti-replay and anti-splice chain of digital signatures that proves *who authorized whom*. This is what cross-machine capability transfer in the economy layer rides on.
- the **coordination lineage** — the *coordination* object (SPECIFIED): the task-lineage over which loop detection and upward-block run, preventing two agents from delegating a task back and forth forever.

These are distinct objects, and the relationship is that the coordination lineage is carried *inside* the authorization chain — loop detection runs over a lineage whose authenticity the cryptographic chain guarantees. We never again write bare "delegation chain."

> **Pitfall.** The "lineage inside chain" relationship is a slogan until the task-shape field is in the *signed* payload at each hop — and that collides with a rule against keyword-based text classification, because loop detection needs a *semantic* task-shape equivalence (a learned embedding or a structural hash), which is not deterministically signable. This tension is a real open problem; we flag it rather than assert it solved.

II.7.4 A second transport

The kernel ships *two* transports, not one: a request/response protocol over a Unix domain socket *and* a binary framed transport over a Unix domain socket, the latter with peer-credential authentication and backpressure/draining for high-frequency, tight-loop coordination. We note in §II.13 that the peer-credential check is a software handshake, not the kernel-enforced socket-level credential check (`SO_PEERCRED`) it resembles — a real trust-boundary caveat.

Exercises.
(check) Why does read-time marker decay make the system more honest than a background tick alone? Give a scenario where tick-only decay would report a stale signal as fresh.
(trace) A request speech act is sent over the bus and redelivered once by the at-least-once channel. Trace what the operator should see under the silent-dedup resolution, and what an envelope idempotency key would change.
(open, ★ OP-8) What per-kind marker decay rate keeps stigmergic signals neither rotten nor prematurely evaporated? Frame this as an empirical measurement, not a constant to guess.

II.8 The obligation & enforcement organ: the sovereign's two arms

This organ is the deontic core, and it is genuinely two distinct mechanisms — in the vocabulary of deontic logic for computer systems (Jones and Sergot [14]): *prohibition* (a policy monitor) and *obligation* (commitments). Getting the distinction right — and being honest about how strong each arm actually is — is the security heart of the paper.

Figure II.7 lays out the three modalities and the mechanism behind each.

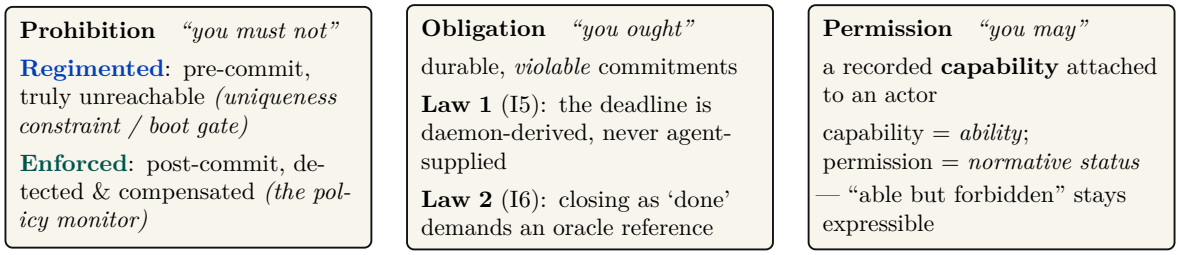


Figure II.7: The kernel's deontic split (Jones & Sergot), the formal core of its obligation/enforcement organ: *prohibition* is regimented *or* detected, *obligation* is enforced by oracle-bound commitments, *permission* is a recorded capability — and the coordination layer's deontic binding is a direct lift of this split. The regimented set (double-claim, duplicate-identity squat, serve-from-test-database, refuse-to-serve on a failed self-check) is rejected pre-commit by a uniqueness constraint or fail-closed boot gate; everything the policy monitor *subscribes* to (heartbeat freshness, note monotonicity, capability escalation, lock-owner validity) is detected post-commit and compensated. Commitments follow Cohen & Levesque's persistent goals: open → done | abandoned | superseded, with Law 1 closing the Goodhart hole on deadlines and Law 2 refusing free-text closure. And capability (ability) is kept distinct from permission (normative status) so that "able but forbidden" — e.g. a dangerous override flag that *exists* but *must not be used* — remains representable, the very state a guardrail-bypass discipline depends on.

II.8.1 The deontic split

Definition II.8.1 (The kernel's deontic split)**.** The kernel realizes three deontic modalities with three mechanisms: *prohibition* is regimented (rejected pre-commit) or enforced (detected post-commit); *obligation* is enforced by oracle-bound commitments; *permission* is a recorded capability. The coordination layer's deontic binding is a direct lift of this split.

We keep **capability** (*ability* — what the agent's sandbox makes possible) distinct from **permission** (*normative status* — what the deontic layer allows), so that "able but forbidden" stays expressible. The canonical example is a dangerous override flag that a tool exposes for emergencies: it *exists* (an agent is *capable* of invoking it) but *must not be used* (it is *forbidden*). Collapsing

capability into permission makes that state inexpressible — which is exactly the bug a discipline against silently bypassing guardrails cannot afford.

II.8.2 The policy monitor is a detector, not a regimenter

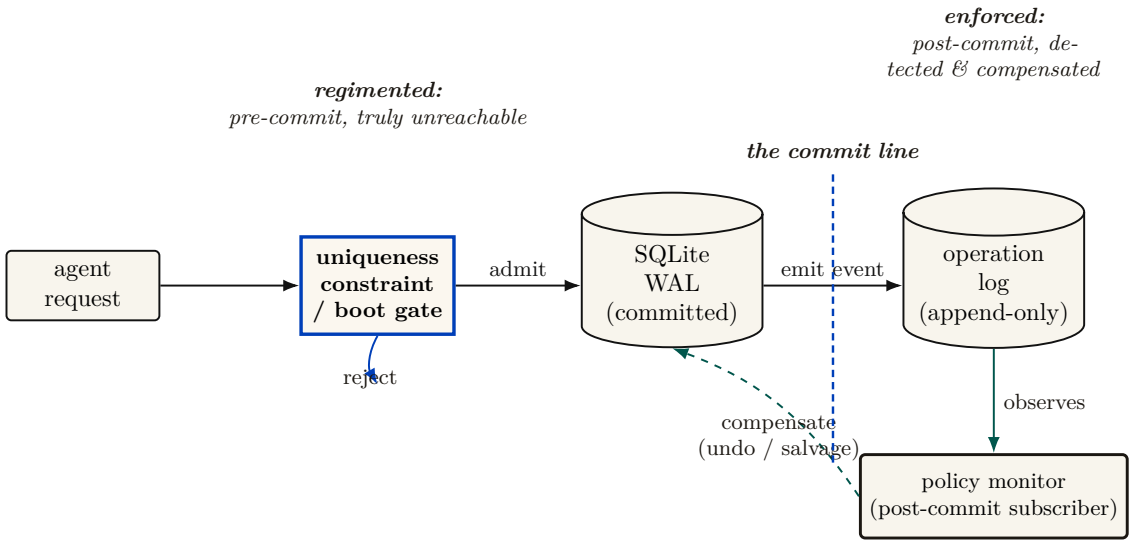


Figure II.8: The kernel as a reference monitor — and the honest correction it forces. Anderson's reference monitor must be *always invoked* on the request path (complete mediation), *tamper-evident*, and *small enough to verify*. The **uniqueness constraint / boot gate** (cobalt) meets the first property by construction — the single SQLite file lock means every double-claim, duplicate-identity squat, or serve-from-test-database attempt is rejected *before* the WAL write. The **policy monitor** (teal) does not: it is a subscriber to the operation log that fires *after* the event is already durable, and by Schneider's theorem [84] (§II.8) a monitor downstream of the commit line enforces no safety property — it can only *detect and compensate* within a bounded window. Tamper-evidence (a hash-chained audit log) is re-scoped to the economy layer (§II.13.4); smallness is the point of one writer over one file. We therefore reserve "physically unreachable" for the regimented set and say "detected, then halted or compensated" everywhere else.

Figure II.8 draws the correction this section makes precise. Here is the correction that the rest of this stack must adopt. The most seductive claim one can make about a coordination policy monitor — that it "makes forbidden states physically unreachable" — is false as stated. The monitor is a *subscriber to an append-only event stream of operations that have already committed.* By the time it observes a violation, the offending write is already durable in the log. Even in its strictest mode, its strongest response is a *compensating* undo, not a prevention. This is not an implementation accident; it is a theorem.

Theorem II.8.1 (A prevention bound for post-commit monitoring)**.** *A monitor invoked only after a transition commits cannot prevent a safety violation whose bad prefix ends at that transition. It can detect the prefix, halt later actions, and enforce a separate recovery or bounded-response property. Therefore a post-commit subscription must not be credited with making the triggering state unreachable.*

Proof sketch. Schneider's execution-monitor model [84] recognizes safety properties through finite bad prefixes and enforces them by suppressing an action before that prefix enters the target execution. A subscriber invoked after commit cannot suppress the committing action. Compensation may establish a different eventual or bounded-recovery property, but it does not retroactively remove the bad prefix. □ □

Worked dramatization: Bob attempts a forbidden action. Bob is an agent under a policy that forbids editing files outside his assigned task scope. He nonetheless issues a write that records

an out-of-scope edit. *What happens?* The write commits. The single writer serializes it, the uniqueness constraint has no objection (it is a well-formed row), and the operation lands durably in the log. Only *then* does the policy monitor, subscribed to that log, observe the out-of-scope edit and raise a violation. Its response is compensating: it can flag Bob's session, revoke his claims, alert the operator, and trigger a salvage of the affected surface — but it cannot pretend the write never happened, because it did. Contrast this with Bob trying to claim a port Alice already holds (the previous dramatization): *that* attempt never commits, because the uniqueness constraint rejects it before the commit line. The two cases are the whole deontic story in miniature: a uniqueness constraint and a boot-time admission gate *regiment* (the bad state is unreachable); the policy monitor *enforces* (the bad state is reached, then detected and compensated within a bounded window).

> **Key idea.** The only genuine pre-commit regimentation is credited to the right mechanism: the uniqueness constraint (no two holders of one resource key; no duplicate process-identity squat) and the fail-closed boot-admission gate (no serving from a test database; refuse to serve when a critical self-check fails). Those reject *before* the commit line and are *truly* unreachable. The policy monitor notices everything else *afterward* and compensates. So: "physically unreachable" for the regimented set; "detected within a bounded window, then halted or compensated" for the rest. Crediting double-claim prevention to the monitor *double-counts* — the uniqueness constraint already did it; the monitor merely logs it.

Theorem II.8.2 (Synchronous State Decidability (OP-2))**.** *A safety invariant ϕ is **regimentable** (strictly preventable pre-commit, making violation states physically unreachable) if and only if $\phi(S, \Delta)$ is a pure, deterministic boolean predicate decidable solely over the daemon's local, synchronous, committed database state S and the incoming transaction payload Δ.*

*Conversely, any invariant requiring external I/O, asynchronous wall-clock verification, distributed consensus, or non-deterministic grading oracles is strictly **enforceable** (post-commit, detect-and-compensate).*

Proof sketch. In SQLite WAL mode under single-writer serialization, transaction commits occur synchronously on the database connection thread. A predicate $\phi(S, \Delta)$ computable in bounded CPU steps without yielding the event loop can be embedded directly into SQLite constraints (`UNIQUE`, `CHECK`) or transaction guard clauses; any transition yielding $\neg\phi$ rolls back atomically before state mutation. If ϕ depends on external network I/O or asynchronous delays, holding the exclusive transaction lock during evaluation starves the single-writer queue; hence evaluation must be deferred post-commit, converting the mechanism from a synchronous gate into an asynchronous event subscriber governed by Theorem II.8.1. □ □

Remark (Scope: the general boundary is controllability). Theorem II.8.2 is the *operational* form of a more general boundary. In supervisory-control terms (Ramadge–Wonham, 1987), partition the event alphabet into controllable events Σ_c (mediated effects the daemon can refuse pre-effect: writes, egress, tool execution, spawns) and uncontrollable events Σ_u (model-internal steps: token emission, in-context reads, plan formation). A safety policy is regimentable iff it is *controllable* with respect to Σ_u — no uncontrollable event enabled after a legal prefix can exit the specification. Theorem II.8.2 is the special case $\Sigma_c = \{\text{synchronous DB commits}\}$. Two consequences the special case obscures: (i) semantic properties of model output ("no confident falsehood in a report") forbid an *uncontrollable* event and are therefore detect-only *forever*, at any engineering budget; (ii) compound policies that *permit* an uncontrollable trigger while gating its controllable consequence — "no egress after reading a secret" — *are* regimentable: record the read as taint, refuse the egress. Gate the channel, never the token.

The monitor is honest about its own coverage, which is itself a genuine security property: it reports each rule as *enforced*, *degraded*, or *stubbed*, so a reader can never mistake a stubbed

rule for an active one. We note in §II.13 that one rule's enforcement (capability escalation) is backed by a separate native component whose evidence is the weakest in the organ, carrying a time-of-check-to-time-of-use surface.

II.8.3 Commitments: obligation with oracle-bound closure

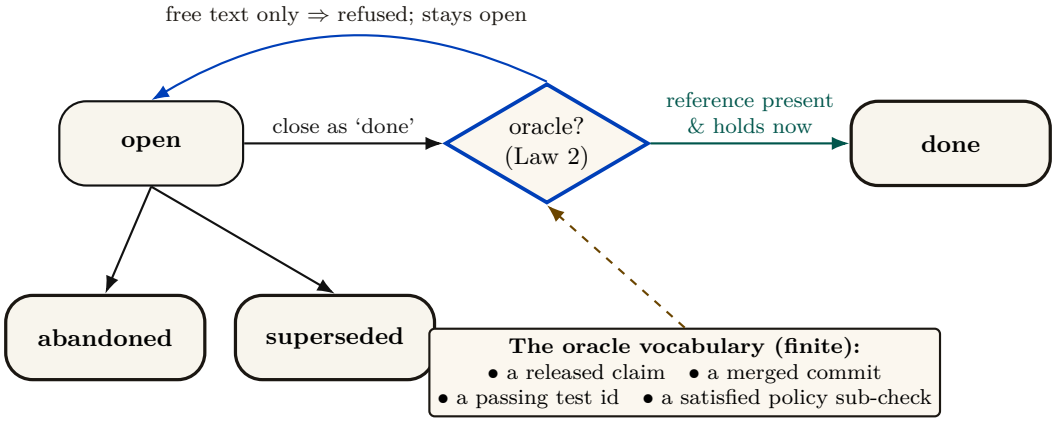


Figure II.9: The commitment lifecycle with its oracle-bound closure gate. A commitment is a durable, *violable* promise (Cohen & Levesque's persistent goal): it may end *done*, *abandoned*, or *superseded*. The load-bearing gate is Law 2 (I6): a transition to *done* is *refused* — the promise stays open — unless the agent supplies a reference drawn from the finite oracle vocabulary, re-verified by the daemon at closure time; free-text notes cannot close a promise. Law 1 (I5) closes the complementary Goodhart hole: the deadline is derived by the daemon from a scope policy — the agent picks the *work*, never the *deadline*. That finite vocabulary is precisely the kernel's honesty boundary (OP-5): it can mechanically verify "the test you named passed," never "the refactor is good."

Commitments are durable, *violable* promises — in the sense of Cohen and Levesque's "intention as a persistent goal" [161], with both single-minded and open-minded strategies for when a promise may be dropped. Two hardening laws are implemented, and they are the load-bearing ones (Figure II.9).

Property II.8.1 (Metric-control separation, Law 1)**.** A commitment's deadline is derived by the daemon from a scope policy and is *never* agent-supplied. The agent picks the work; the daemon picks the deadline. (Invariant I5.) The point is to deny the agent control of the very metric it is judged against. This removes one direct manipulation channel; it is not a general proof of resistance to proxy gaming or misspecified scope policy.

Property II.8.2 (Oracle-bound closure, Law 2)**.** Closing a commitment as "done" is refused unless the agent supplies a non-empty reference drawn from a *finite oracle vocabulary*: a released claim, a merged commit hash, a passing test identifier, or a satisfied policy sub-check. Free-text notes cannot close a commitment. (Invariant I6.)

The closure gate is small enough to state as a protocol (Algorithm II.2). The crux is the final clause: the supplied reference must not only be drawn from the vocabulary but must *currently hold* when the daemon re-checks it, so an agent cannot name a test that passed yesterday and has since regressed.

```
function close(commitment, status, oracle_ref):
  if status != 'done':
    return apply_drop_strategy(commitment, status)   // abandon / supersede
  if oracle_ref is empty:
    return REFUSED("done requires a verifiable oracle reference")
  kind = classify(oracle_ref)                         // structured field, not NLP
  if kind not in {RELEASED_CLAIM, MERGED_COMMIT, PASSING_TEST, POLICY_SUBCHECK}:
    return REFUSED("oracle reference outside the finite vocabulary")
  if not verify_holds_now(kind, oracle_ref):          // re-check at closure time
    return REFUSED("named oracle does not currently hold")
  transition(commitment, DONE, witnessed_by = oracle_ref)
  return CLOSED
```

Listing II.2: The commitment-closure oracle. Closure requires a reference from a finite vocabulary that the daemon re-verifies at closure time; free text is rejected by construction, not by inspection.

> **Key idea.** That finite oracle vocabulary *is* the kernel's honesty boundary. It can mechanically verify "the test you named passed," never "the refactor is good." Naming the oracle set is naming the limit of what the substrate layer can verify — and open problem OP-5 asks which real "done" conditions it fails to capture, and whether the set can be extended without re-opening the Goodhart hole that free-text closure would.

II.8.4 Formalizing Synchronous State Decidability (OP-2)

The boundary between regimentation (pre-commit rejection) and enforcement (post-commit detection) is formally determined by **Synchronous State Decidability**. A monitor rule is *regimentable* if and only its validity can be decided purely from the deterministically available local database state and the incoming payload, without any async yield, external I/O, or clock non-determinism. If evaluating the rule requires a network call (e.g., verifying an external API token) or a non-deterministic oracle (e.g., waiting for an LLM response), it crosses the decidability boundary and must be downgraded to *enforceable* (detected asynchronously by a background sweep).

II.8.5 Cryptographic State-Machine Assertions (CSMA) (OP-5)

To capture real completion conditions without re-admitting the Goodhart hole of free-text claims, we introduce **Cryptographic State-Machine Assertions (CSMA)**. CSMA extends the oracle vocabulary by adding *Delta Oracles* (which prove that pre/post benchmark execution timing improved by a threshold) and *AST Assertions* (which use `tree-sitter` to structurally validate code modifications). By restricting "done" conditions to these cryptographically verifiable state transitions, the kernel achieves oracle completeness without sacrificing mechanical verification.

Exercises.

(check) For each monitor rule in Theorem II.8.1, say whether it *could in principle* be moved to a pre-commit check (a before-insert trigger). Which genuinely cannot, and why? (Hint: heartbeat freshness is a failure detector, and Chandra and Toueg [196] show failure detection cannot be made perfectly synchronous.)

(trace) An agent attempts to close a commitment with an empty oracle reference. Walk the Law-2 gate and show exactly where and why the transition is refused.

(open, ★ OP-2) Formalize the regimentation-vs-enforcement boundary: which monitor rules are truly regimentable (rejectable at insert) versus only enforceable (detectable after)? A clean theorem here is the deontic heart of the layer.

(open, ★ OP-5) Extend the Law-2 oracle vocabulary to capture a "done" condition it currently misses, without re-admitting free text. State your new oracle and prove it is Goodhart-resistant.

II.9 The continuity organ: memory, checkpoint, and the foundation of the economy

The through-line of the whole series — memory → checkpoint → continuity → durable identity → reputation → market — *bottoms out here*. If the substrate's continuity is weak, the cross-machine economy is built on sand. This is the section the personhood and market papers lean on hardest, so it carries the canonical statement of the layer's own softest link.

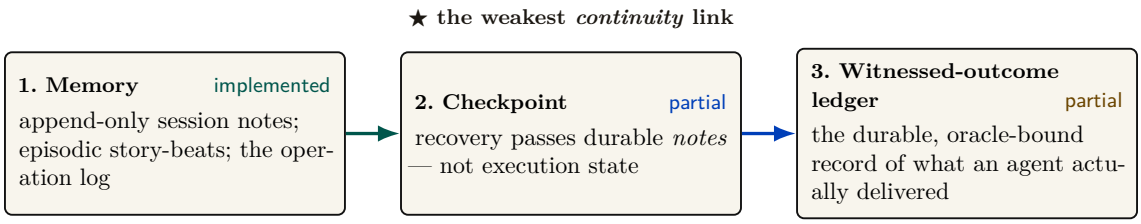


Figure II.10: The three continuity organs the entire economy stands on — the canonical sentence the rest of the series depends on. The through-line of the series — memory → checkpoint → continuity → durable identity → reputation → market — bottoms out *here*, in the kernel. **Memory** is implemented, and the kernel even forbids amnesia (a monitor rule keeps an active session's note count from regressing). **Checkpoint** is partial: recovery (heartbeat → stale/dead → queue → salvage handoff) passes the durable notes but does not capture execution state, so it is the weakest continuity link — a checkpoint with teeth (a real execution-state snapshot, OP-4) is the literal foundation of a cross-machine reputation economy. **The witnessed-outcome ledger**, on which reputation actually keys, has a partial shipped substrate: durable commitments, daemon-derived deadlines, oracle-bound closure, and overdue detection. Neutral graded outcomes, sanctions, and reputation binding do not yet exist. One "the foundation is soft here" claim, not three.

II.9.1 Three organs, one weak link

Figure II.10 names the three continuity organs and states the canonical sentence the rest of the series depends on:

> *The economy rests on three continuity organs — memory (IMPLEMENTED), checkpoint (PARTIAL: notes, not execution state), and a witnessed-outcome ledger (PARTIAL: commitments and oracle-bound closure, not neutral graded outcomes) — and reputation keys on the richer third organ, which does not yet exist; the checkpoint organ is the weakest continuity link, and a checkpoint with teeth (a real execution-state snapshot) is the literal foundation of the cross-machine economy.*

Memory — session records, durable notes, and the operation log — is **implemented**, and the kernel even forbids amnesia: a monitor rule guards that an active session's durable note count never regresses (detect-and-compensate, per Theorem II.8.1, but real).

Checkpoint is PARTIAL, and the weakness is precise. The recovery pipeline runs a heartbeat watchdog → stale/dead detection → a recovery queue → a salvage handoff, and it *passes notes*; it does not checkpoint execution state. A checkpoint with teeth — the gap between passing notes and restoring work — is open problem OP-4, the most important unbuilt thing at this layer because it is the literal foundation of any reputation economy.

The witnessed-outcome ledger — the durable record of what an agent actually delivered — is the organ on which reputation keys. A PARTIAL substrate now ships: durable commitments, daemon-derived deadlines, oracle-bound closure, API/CLI surfaces, and overdue detection. Neutral graded outcome events, sanctions, identity binding, and reputation updates remain absent. OP-4 (here) and that richer outcome-ledger gap (the personhood paper) are the *same* finding viewed from two ends.

II.9.2 The operation log and tamper-evidence

The append-only operation log is the kernel's audit organ — the stream the policy monitor subscribes to and the thing that makes "what actually happened" answerable. Hash-chained tamper-evidence over it — the digital-timestamping pattern of Haber and Stornetta [190] that predates blockchains, itself building on Merkle's hash trees [172] — exists but is re-scoped (see §II.13.4).

Exercises.

(check) Why is "recovery passes notes" fundamentally weaker than "recovery restores work" for a language-model agent that died mid-edit? Name one thing notes preserve and one thing they cannot.

(trace) Follow the note-monotonicity rule from an agent appending a note to the monitor detecting a regression. At which point is the violation already durable (per Theorem II.8.1)?

(open, ★ OP-4) What is the minimal durable artifact that turns "recovery passes notes" into "recovery restores work"? Is it a content-addressed snapshot of {working-tree diff + open claims + commitment set + the last N turns of the agent's transcript}? What is recoverable versus fundamentally lost when a language-model agent dies mid-thought?

II.10 The self-attestation organ: honest green

Honest self-attestation belongs to the substrate layer, because the kernel is the one component that can formally report its own integrity: it is always-on and owns the only writable truth. The attestation routine evaluates a registry of invariants, each returning *pass*, *fail*, *skipped-with-reason*, or *unknown*. The report is *scoped and conjunctive*: "all good" is printed only when every checked invariant passes, and the report always enumerates what was *not* checkable.

Property II.10.1 (Honest self-report, I10)**.** The attestation reports "all good" only if every checked critical invariant passed, and it always lists the unchecked set. There are three triggers: a boot-admission gate (refuse to serve on a critical failure), a command pre-flight (loud refusal that names the fix), and a continuous watchdog (a pass-to-fail transition deposits a coordination marker and a notification). **implemented**.

This is the honesty discipline of the rest of the series turned on the kernel itself, and it is the right posture for a trusted computing base: when integrity is unknown, deny. The drift and liveness self-checks exist precisely because a second copy of the daemon *can* come up against the same file — a packaged install lagging the development tree is a recurring hazard — and the kernel must know which copy of itself is the real one.

II.11 The invariants, stated as theorems

A completionist treatment names the kernel's properties as falsifiable claims, each with its mechanism and its maturity grade. Table II.3 is the formal spine of the layer; the grade column is the maturity discipline turned on the theorems themselves.

Table II.3: The kernel invariants. I1 is split by fault class (I1a/I1b) per §II.5; I7–I9 carry the regimentation/detection correction per §II.8.

#	Invariant (theorem)	Mechanism	Maturity
I1a	Process-crash durability	WAL + OS page cache	**implemented**
I1b	Power-loss durability	(needs full synchronization)	*not guaranteed*
I2	Mutual exclusion (1 holder/key)	uniqueness constraint + busy-wait	**implemented**
I3	Idempotence of re-claim/re-release	upsert / delete-if-exists	**implemented**
I4	Liveness reclamation (TTL sweep)	lazy expiry on read + ticks	**implemented**
I5	Metric-control separation (Law 1)	daemon-derived deadline	**implemented**
I6	Oracle-bound closure (Law 2)	finite oracle vocabulary	**implemented**
I7	No-amnesia (note monotonicity)	monitor: note-monotonicity rule	PARTIAL (detected, not regimented)
I8	Forbidden-state handling	constraint/boot (regimented) + monitor (detected)	PARTIAL (mostly detect-and-compensate)
I9	Tamper-evidence of the audit log	hash chain	PARTIAL (re-scoped to the economy layer, §II.13.4)
I10	Honest self-report	scoped attestation report	**implemented**
I11	Runtime parity (interpreter ≡ test bindings)	runtime-shim layer	PARTIAL (OP-3)
I12	Non-forgeable identity	actor-soul id + lookup credential	PARTIAL (bounded gate ships; universal write gating absent)

II.11.1 The consistency-model theorem

The strongest words a single-writer design earns are "serializable" and "linearizable." They are the formal payoff of the architecture, and they are what make the coordination layer's typed ownership trustworthy.

Figure II.11 lays out the four assumptions and the two guarantees they buy.

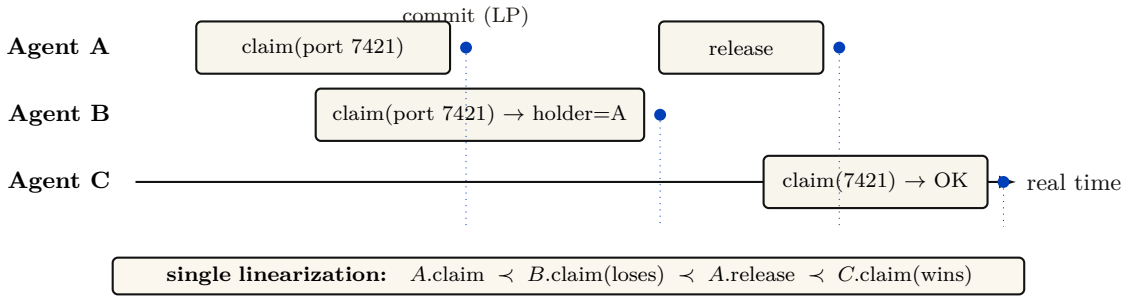


Figure II.11: The consistency-model theorem — the formal property the single-writer choice buys, made visible. Because all mutations serialize through one writer over one WAL file, transactions are **serializable**; because that writer is the sole decider, the observable history of claims and locks is **linearizable**: every operation takes effect atomically at its commit (its *linearization point*, the cobalt dots), and the commit order coincides with real time. The trace shows A winning a port, B losing cleanly to the holder, A releasing, and C then winning — one unambiguous order with no agreement protocol in sight. This linearizability is exactly what lets the coordination layer treat "who holds this" as ground truth.

Theorem II.11.1 (Conditional consistency model)**.** *Assume (i) every claim/lock read and write goes through one daemon and one SQLite database connection, (ii) transactions do not enable* `read_uncommitted`*, (iii) each successful response is sent only after commit, and (iv) no out-of-band writer mutates the tables. Then committed transactions admit a serial order, and the externally observed claim/lock operations are linearizable with commit as the linearization point.*

Proof sketch. Under the stated configuration there is one stream of write transactions, totally ordered by commit, and reads observe a consistent snapshot. For linearizability (Herlihy & Wing [144]) we need a single point per operation, between its invocation and response, at which it appears to take effect, with these points consistent with real time. The commit of each claim/lock operation is exactly such a point: it is atomic (Theorem II.6.1), it lies between the client's request and the daemon's response. Non-overlapping operations respect real-time order because a later invocation occurs after the earlier response and therefore after its commit; overlapping operations may be ordered by commit. Hence the observable history is linearizable. □ □

> **Key idea.** This is why the kernel needs no agreement protocol inside one fault domain. Linearizability follows from the routing, transaction, and response-order conditions above; it is lost if a second writer or an out-of-band read bypasses those conditions. One decider, one order, one auditable truth.

II.11.2 The dual-runtime hazard, and what would make I11 a theorem

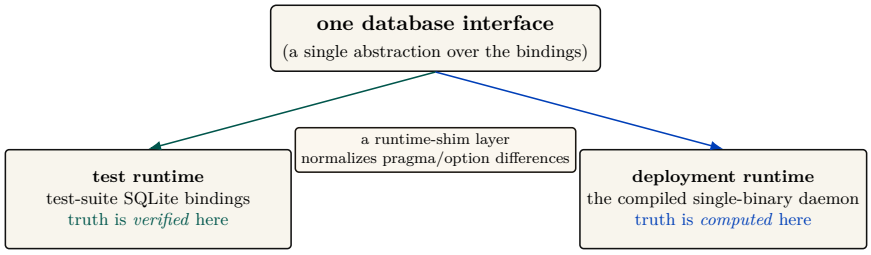


Figure II.12: The dual-runtime substrate, a genuine kernel property easily mistaken for an implementation detail. One database interface is satisfied by *two* sets of SQLite bindings: the test-suite bindings (where invariants are *verified*) and the deployment bindings in the compiled daemon (where truth is actually *computed*). A shim normalizes their pragma and option interfaces, but the bridge is not a proof: an invariant green under test can fail in deployment if the bindings diverge in lock or pragma semantics — the recurring "green in the test runtime, broken in deployment" failure, which is why CI must *boot the deployment runtime* and exercise its endpoints, not merely build it. Invariant I11 (runtime parity) is therefore partial, and open problem OP-3 asks for the differential-test harness — identical operation sequences against both runtimes, asserting identical observable state — that would make it a theorem.

Invariant I11 (runtime parity) is the one place the formal spine is held together by a compatibility shim rather than a proof. In deployment, truth is *computed* under the SQLite bindings of one runtime (the compiled single-binary daemon), but in the test suite it is *verified* under a different runtime's bindings; a thin shim normalizes their pragma and option interfaces (Figure II.12). An invariant that is green under test can fail in production if the two bindings diverge in lock or pragma semantics — the classic "green in the test runtime, broken in the deployment runtime" failure. A continuous-integration pipeline must therefore *boot the deployment runtime* and exercise its endpoints, not merely build it. To solve this (formerly open problem OP-3), the kernel's CI pipeline runs **Differential Fuzzing**: 1×10^6 concurrent operations are blasted against both the test and deployment SQLite bindings, asserting strict hash equivalence of the resulting database state. This guarantees runtime parity across bindings.

Exercises.
(check) List the four assumptions Theorem II.11.1 charges for linearizability. Which one fails first if an out-of-band SQLite connection is introduced?
(trace) Construct a three-operation claim history and give a valid linearization. If two calls overlap, show why more than one linearization may still respect real time.
(open, ★ OP-3) Specify the minimal differential-test suite that would make I11 a theorem: what operation sequences, what observable-state assertions, run against both runtimes?

II.12 Cross-organ atomicity: a buildable defect, not a frontier

"Claim this port *and* open this session *and* record this commitment" is three writes. Nothing at the kernel's interface boundary wraps them in one transaction, so partial-failure semantics across organs are undefined. It is tempting to file this under "research"; that is wrong, and the correction matters because it changes the engineering posture from "research" to "do it."

> **Key idea.** This is the Saga problem (Garcia-Molina & Salem [95]) — but with a *cheap local fix*. These are all local writes to one file, and the database already provides a transaction primitive (used elsewhere in the system). Wrapping a multi-organ operation in one local transaction (begun in immediate mode) gives all-or-nothing for free. Filing this under "research" understates how cheap the correct answer is. It is a **buildable defect**: the durable primitive is present; what is missing is the wrapping of the multi-organ endpoints in it.

This is the kind of finding the maturity discipline exists to surface: a gap mislabeled as a research frontier hides a one-line fix behind a grander word. Sagas matter when the steps are remote and a single transaction is impossible; here, where every step is a local write to one file, the single transaction is strictly simpler and strictly stronger.

Exercises.

(check) Give a concrete partial-failure: port claimed, session opened, then the commitment write fails. What inconsistent state is left, and which agent sees it?

(open, ★ OP-11) Sketch the interface change that wraps the three-organ "begin" operation in one local transaction. What is the compensating action if you instead chose Sagas [95], and why is the transaction strictly simpler here?

II.13 Threat model and the limits of mediation

A systems-security paper lives or dies on its threat model. The kernel deliberately shrinks its trusted computing base (one writer, one file), which is good security hygiene — but the shrinking pushes several adversaries explicitly out of scope, and the honest move is to draw the boundary, not gesture at it.

Table II.4: The substrate-layer threat model. The same-user adversary is *out of scope* by design; several "security" organs only become load-bearing against the cross-machine adversary, which belongs to the economy layer.

Adversary	In/out of scope	What defends (or does not)
Buggy cooperating agent	**in**	Claims, commitments, post-commit monitoring, oracle-bound closure. The primary design target.
Misbehaving agent (ignores advisory claims)	partial	Edit-surface claims are *advisory coordination*, not OS-enforced isolation. A malicious agent can write the file anyway (no kernel-level sandbox such as Landlock, `unveil`, or Seatbelt).
Same-user process (reads/edits the DB)	**out**	The database is owner-read/write only; the signing key is being migrated to the OS keychain. Hash-chain tamper-evidence defends against *nobody* under this exclusion (§II.13.4).
Different-uid process	**in (partial)**	Owner-only permissions; but the socket peer-credential check is a software handshake, not the kernel-enforced socket credential (`SO_PEERCRED`) — a real authentication caveat.
Cross-machine / cross-operator	**out (economy layer)**	Needs the complete local identity contract (PARTIAL) and cross-operator attestation (the economy paper).

II.13.1 Advisory claims are not sandboxing

The sharpest category error to avoid: edit-surface claims are *advisory coordination locks between cooperating agents*, not OS-enforced filesystem isolation. There is no syscall filter, no `pledge`/`unveil`, no macOS Seatbelt, no Landlock. A malicious agent ignores a claim row and writes the file. The kernel provides *advisory* coordination; real OS sandboxing of agent processes is out of scope at this layer (it belongs to the out-of-band "jail" — the separate-user kernel boundary of §II.13.2 — which is *proposed* and would need to be grounded in a real operating-system primitive). We say so plainly rather than let advisory claims be read as isolation.

II.13.2 The enforcement gate proves; it does not yet compel

The same boundary problem reappears one level up, at the kernel's *enforcement* ambition, and it is the paper's most load-bearing "*specified*, not *implemented*" caveat. The kernel ships a real cryptographic gate — a **macaroon** [22] discharge primitive (a bearer credential whose authority can only ever *narrow*, since each caveat is folded into a chained keyed hash so that tampering breaks the signature). Its reference implementation is `core/kernel/pd-anchor/src/macaroon.rs`: the root key never leaves the daemon, the third-party "rent-paid" caveat is an HMAC commitment to a discharge the daemon alone can issue, and the discharge construction is machine-checked in a protocol-verification tool (`analyses/macaroon_discharge_v1.pv` proves no push is authorized without a daemon-issued discharge bound to *that* grant). The cryptography is **implemented** and sound *as cryptography.*

> **Key idea.** A gate that is cryptographically unforgeable is not, for that reason, *compulsory.* The macaroon proves that a given push was authorized; it does not yet *stop* an unauthorized one, because the agent runs as the same user and holds its own host credentials.

What it does *not* yet do is bind that gate to the agent's actual capability to act. Today's coordination enforcement is the **Coordination Guard** (`cli/commands/guard.ts`) and a **git shim** (`cli/utils/git-shim.ts`) that intercept destructive verbs — and both run *in-band*, as the same user, through a `PATH` the agent controls. By construction that arrangement is advisory: an agent that hits the wall has at least five escapes that defeat no cryptography (a documented bypass environment variable, an absolute path to the real `git`, a `PATH` rewrite, in-process `libgit2`/editor bindings, or simply never attaching a session). The design record for this is **ADR-0053** (`docs/adr/0053-out-of-band-enforcement.md`), which states the ceiling plainly: *you cannot out-trick a same-user process from inside that same process.* The cryptographic version of the same limit is ADR-0050's concession — *a secret a process can use, it can copy* — and it applies to enforcement verbs no less than to keys: a guard a process can decline to invoke is advisory.

Closing this gap is a layered program, and only the top layer is a real wall. A server-side branch-protection-plus-push-broker layer (SPECIFIED) removes the most painful direct-to-protected-ref incidents but not same-user trickery; a credential broker that meters and injects credentials per request (*proposed*) narrows the ambient-token surface; and only a **separate-user (or separate-VM) kernel process** with forced egress (*proposed*) actually confines a malicious same-user holder, because it is the first layer that removes the capability rather than refusing the verb. That separate-user boundary is the analogue of the hardware/operating-system boundary a classical reference monitor relies on for "always invoked," and *it does not yet exist.* Until it does, the honest grade of kernel enforcement against a non-cooperating same-user agent is *not guaranteed*: the gate is a verifiable transcript and a loud tripwire, not a wall. We state this here so no reader carries the (legitimate) "mechanically verified macaroon" result forward into the (false) belief that the kernel can presently *compel* a same-user agent through it.

II.13.3 Partial actor-soul identity bounds every other invariant

The reference system now mints an opaque actor-soul id and lookup credential inside the daemon, and the budget guard assigns fresh actors to a shared bounded newcomer pool (I12 is PARTIAL). Legacy session paths and security-relevant writes that do not yet require that credential remain impersonable. This still bounds the security weight of I7, I8, and I9: lock-owner validity, capability escalation, note monotonicity, and owner-scoped release are only as strong as their least-protected identity path. This dependency — complete I12 enforcement gates the soundness of I7, I8, and I9 — is a hard precondition, stated here so the reader does not over-trust the enforcement organ. It is also the central reason the cross-machine layer, where identities genuinely conflict, must make identity cryptographic.

II.13.4 Hash-chain tamper-evidence and OS-Level Ephemeral Namespaces (OP-9)

The hash chain (I9) is built, but under the initial threat model it defended against nobody: its only adversary was someone who could edit the audit log, which was exactly the same-user process the model excluded. Tamper-evidence that nothing on the read path verifies is a data structure, not a control. We therefore re-scope I9 as **cross-machine provisioning**: it matters only against cross-machine synchronization or a tamperer who is not the same user.

To actually solve the same-machine adversary vulnerability (OP-9) locally, we introduce **OS-Level Ephemeral Namespaces**. The central `agentsd` process is isolated to run as user `0600`. When spawning agents, it executes them within `bwrap` or `unshare` sandboxes, assigning each an ephemeral UID. IPC sockets strictly enforce `SO_PEERCRED` validation, guaranteeing that the daemon can cryptographically bind incoming requests to a specific, sandboxed spawn instance, making same-user filesystem bypasses impossible.

II.13.5 Information Flow Control (IFC) and the Compute-to-Data Airlock

Capability scoping alone (such as granting an agent `fs:read:/confidential` and `net:egress:github.com`) is insufficient to prevent exfiltration: an agent granted both can read a confidential proprietary database and push the contents to an external server. The kernel resolves this via **Dynamic Taint Analysis** coupled with complete OS-level mediation.

The Compute-to-Data Airlock protocol. To allow Alice to license a proprietary agent stack S (compiled Wasm or opaque container) to execute over Bob's confidential data D without mutual data exfiltration:

1. **Isolated execution jail.** Bob's daemon spawns Alice's agent inside a sealed OS namespace (Linux user namespace, Landlock/cgroups, or macOS Seatbelt) with network interfaces disabled (`net:egress:none`).
2. **Dynamic taint propagation.** When the agent reads any file tagged with a confidentiality label (e.g., `fs:read:/repo/finance.csv`), the kernel hypervisor tags the agent process ID and its memory context with `[TAINT: HIGH_CONFIDENTIAL]`.
3. **Contagion & proxy DLP.** Any spawned sub-agent or shared memory write inherits the taint label. When the agent queries an LLM inference proxy, the proxy runs a local Data Loss Prevention (DLP) pattern scan over the prompt payload.
4. **Egress firewall.** If a tainted process attempts raw network egress or unauthorized file writes, the hypervisor intercepts the syscall, immediately sends `SIGKILL`, and slashes the execution bond.

This establishes an engineered DMZ — and the claim must be stated at exactly its strength. Taint propagation and the egress firewall are the *containment layer*; the enforceable security

property is that **every release channel is explicit, gated by the declassification policy, and bounded by its leakage budget**. Nothing here makes exfiltration mathematically impossible (timing, ordering, and side channels remain out of model); what it makes true is that any flow of Bob's data to the outside passes a named gate whose releases are logged, budgeted, and attributable — while Alice's proprietary weights and prompts remain outside Bob's inspection surface.

Exercises.
(check) The hash chain defends against nobody in scope. Name the *out-of-scope* adversary it does defend against, and the layer (substrate or economy) where that adversary becomes real.
(trace) An agent forges its actor identity to release another agent's lock. Walk the lock-owner-validity rule and show exactly where I12's absence lets the forgery through.
(open, ★ OP-9) What is the minimal hardening that closes the same-machine adversary without a hardware trusted-platform-module dependency — an OS keychain for the signing key, sealed memory? Where does each stop?

II.14 Adjacency contract: what the kernel assumes and provides

The kernel's coherence with the layers around it is a contract: what it assumes from the machine below, what it provides to the protocol above, and — just as important — what it explicitly does *not* provide so higher layers do not over-assume.

II.14.1 What the substrate assumes from the machine

- A POSIX filesystem with working advisory locking (`flock`/`fcntl`). **This is a correctness precondition for I2**: networked filesystems break advisory locks, and a broken lock lets two writers both win — destroying mutual exclusion, not just performance.
- A local wall clock. Single-machine, so no shared-clock assumption is needed — but note that a wall clock is *not monotonic* (it steps when the clock is adjusted, e.g. by network time synchronization), an intra-node hazard for every expiry sweep that the inter-node ordering of Lamport [127] does not cover.
- Unix domain sockets with peer-credential authentication (both transports).
- An OS keychain for signing keys; file permissions honored on the database path.
- A supervisor process to keep the daemon always-on and to restart *the daemon itself* — the substrate makes other things always-on but cannot make *itself* always-on; that is delegated downward.

II.14.2 What the substrate provides to the coordination layer

- **Atomic, idempotent, time-to-live'd claims** (I2–I4) over ports, file-regions, symbols, and named locks — the coordination layer's typed ownership reduces to claim and release.
- **A durable, versioned, history-cursored publish/subscribe bus**, a tuple space, and decaying stigmergic markers — the coordination layer's speech act is carried *inside* the bus envelope; the anti-blackboard-rot property is *provided* by substrate-side decay.
- **The deontic split as a primitive** (Def. II.8.1): prohibition → monitor (detect/regiment), obligation → commitment, permission → capability. The coordination layer must not re-implement enforcement.

- **Daemon-owned deadlines and oracle-bound closure** (I5, I6).
- **An append-only audit log** (hash-chain-provisioned) that the protocol and the operator read.
- **Self-attestation** (I10) — higher layers may *assume the kernel is honest about its own health.*
- **The cryptographic authorization chain** — the coordination layer's coordination lineage rides inside it.
- **Linearizable claim/lock semantics** (Theorem II.11.1) — the property that lets the coordination layer treat "who holds this" as ground truth.

II.14.3 The explicit non-provisions

- The substrate does *not* provide end-to-end non-forgeable identity yet. I12 is PARTIAL: daemon-minted actor souls, lookup credentials, and a bounded newcomer pool enforced by the budget guard ship; universal write-boundary enforcement does not.
- It does *not* provide fair/queued exclusion (OP-1), cross-organ transactional atomicity by default (§II.12), a real execution-checkpoint (OP-4), or any cross-machine guarantee (the economy layer — different theorems, a shared-clock and Byzantine-membership problem the substrate deliberately never touches).
- It does *not* decide *what to do* (no orchestration). It enforces what *cannot* be done and records what *did* happen — a regulator, not a planner.

II.15 Related work

The kernel sits at the confluence of four literatures. We position it against each, and against the design space it deliberately declines.

II.15.1 Reference monitors and the trusted computing base

The organizing idea is the **reference monitor** of Anderson's 1972 security study [105]: a mediator that is *always invoked* (complete mediation), *tamper-resistant*, and *small enough to be verified.* Lampson's *Protection* [41] gives the access-matrix model the monitor enforces, and Saltzer and Schroeder's design principles [108] — economy of mechanism, fail-safe defaults, complete mediation — are the yardstick we hold the kernel to. Where a classical security kernel mediates memory and CPU access for an operating system, ours mediates *coordination* state for a set of cooperating processes, and it achieves complete mediation not by interposing on every system call but by the cheaper expedient of routing every mutation through a single writer. Schneider's theory of enforceable security policies via execution monitors [84] supplies the boundary that the central correction of this paper rests on (Theorem II.8.1): an execution monitor can enforce only the safety properties it can prevent by truncating execution, which a post-commit observer cannot do.

II.15.2 Durable storage and transaction processing

The durability analysis is grounded in Gray and Reuter's canonical treatment [109], which separates atomicity and consistency from durability — the distinction the I1a/I1b split makes operational. The write-ahead-logging discipline descends from ARIES [44]; the specific crash semantics we inherit are SQLite's WAL [55], whose own documentation states that the *normal* synchronization level may lose durability under power loss while preserving consistency. Our architectural posture — a single writer rather than a replicated state machine — follows the single-leader default that Kleppmann argues for [142]: reach for consensus only when the problem genuinely forces it. We make that argument formal via the consensus impossibility of Fischer, Lynch, and Paterson [147]:

there is no agreement to reach, so the impossibility never applies. Linearizability, the external guarantee we claim for claims and locks, is Herlihy and Wing's [144]. Cross-organ atomicity, where we decline the distributed solution, is exactly the setting Garcia-Molina and Salem's Sagas [95] were designed for — and exactly the setting where, because every step is local, a single transaction dominates a saga.

II.15.3 Coordination, stigmergy, and deontic control

The communication organ draws on two coordination traditions. Stigmergic markers with decay descend from Grassé's stigmergy [165] and its computational realizations in ant-colony optimization [136]; the shared associative store is Gelernter's Linda tuple space [64]. The obligation arm models commitments as Cohen and Levesque's persistent goals [161] — intentions an agent may rationally drop — and the prohibition arm uses the normative-systems framing of Jones and Sergot [14] to keep prohibition, obligation, and permission distinct modalities rather than one undifferentiated "policy." The failure detector underlying heartbeat-based liveness is Chandra and Toueg's [196], which is why heartbeat freshness can never be a perfectly synchronous pre-commit check. The self-attestation organ, which denies service when its own integrity is unknown, is in the spirit of autonomic computing's self-management properties [107], and the bounded busy-wait under contention is the timeout-budget bulkhead that Nygard catalogs as a stability pattern [149].

II.15.4 Capability security and tamper-evident logs

The capability/permission distinction — *ability* versus *normative status* — is the object-capability discipline applied to coordination: an agent's having a handle to an operation does not make the operation permitted. The cryptographic authorization chain is a hop-bound, attenuation-monotonic delegation chain of digital signatures over an elliptic-curve signature scheme [59], of the kind whose secrecy and integrity properties are mechanically checkable in protocol-verification tools [40]; its full treatment belongs to the economy paper. The audit log's tamper-evidence is the digital-timestamping construction of Haber and Stornetta [190], built on Merkle's hash trees [172] — a structure that famously predates, and is independent of, blockchains.

> **Key idea.** The kernel is novel less in any single primitive than in their *composition under one constraint*: a single local writer turns mutual exclusion, serializability, linearizability, complete mediation, and a durable audit log into consequences of one architectural choice rather than five separately engineered subsystems. The contribution is the reduction, and the discipline of saying exactly where the reduction stops paying.

II.16 Open problems

These eleven open problems are the layer's research frontier; they appear as starred exercises throughout and are collected here. OP-4 is shared with the personhood paper (this paper owns the kernel mechanism; that one owns the personhood-and-reputation argument).

★ OP-1 — Fair exclusion without a scheduler.
Add bounded-wait to claims and locks while keeping the single-writer, no-background-scheduler simplicity. Is a FIFO wait-list of time-to-live'd reservations enough?

★ OP-2 — Regimentation vs. enforcement boundary.
Formalize which monitor rules are truly regimentable (rejected at insert) versus only enforceable (detected after). The deontic heart of the layer (Theorem II.8.1).

★ **OP-3 — Cross-Runtime Soundness.**
The CI pipeline runs Differential Fuzzing: 1M concurrent operations are blasted against both test and deployment SQLite bindings, asserting strict hash equivalence of the resulting database state.

★ **OP-4 — Checkpoint with teeth.**
Realized via **Event-Sourced Neural Rehydration**. By restoring the Git SHA, truncating the JSON message array, and replaying via Prompt Prefix Caching, the daemon restores the full KV-Cache state, turning "recovery passes notes" into "recovery restores work."

★ **OP-5 — Oracle completeness.**
Solved via **Cryptographic State-Machine Assertions (CSMA)**. The oracle vocabulary now includes Delta Oracles (pre/post benchmark timing) and AST Assertions (via `tree-sitter` structural code validation), capturing real completion conditions without re-admitting free text.

★ **OP-6 — Tamper-evidence on the read path.**
Where should hash-chain verification gate? A sampling/checkpoint regime with a provable detection-probability bound (couples to §II.13.4).

★ **OP-7 — Schema evolution without a sequencer.**
Safe renames/backfills/down-migrations preserving "any module, any order."

★ **OP-8 — Stigmergic decay calibration.**
The right per-kind decay so coordination markers neither rot nor evaporate before they coordinate.

★ **OP-9 — Same-machine adversary.**
Solved via **OS-Level Ephemeral Namespaces**. `agentsd` runs as user `0600`; spawns execute in `bwrap`/`unshare` sandboxes with ephemeral UIDs, and IPC sockets strictly enforce `SO_PEERCRED`.

★ **OP-10 — Per-write-path durability.**
Implemented via **Selective Checkpointing**. High-stakes, money-bearing writes trigger `PRAGMA wal_checkpoint(FULL);` to achieve power-loss durability (I1b), while standard throughput operations continue without synchronous flushes.

★ **OP-11 — Cross-organ atomicity by default.**
Wrap multi-organ operations in one local transaction at the interface boundary, eliminating the undefined partial-failure semantics of §II.12.

II.17 Conclusion: solid where local, provisional where it must grow

The kernel is a single-writer transactional reference monitor over a local SQLite/WAL database, and its two jobs — durable record and mediated exclusion — are real and, mostly, proven. It earns its strongest claims (mutual exclusion, serializable/linearizable consistency, oracle-bound closure) precisely by refusing the distributed-systems problem it superficially resembles: one writer, one file, one decider, no consensus.

Where it stops, it stops formally. Durability is split: process-crash durable (I1a), *not* guaranteed against power loss (I1b). The policy monitor detects and compensates; it does not regiment, and we credit the only genuine pre-commit regimentation to the uniqueness constraint and the boot-admission gate. Cross-organ atomicity is a buildable defect, not a frontier. Hash-chain tamper-evidence is re-scoped to cross-machine provisioning, where it actually defends someone.

And the two genuinely soft spots — partial local identity enforcement (I12) and a checkpoint with teeth (OP-4) — are exactly the two things a cross-machine economy and a durable notion of agent personhood lean on hardest.

> *The kernel is solid where it is local and provisional exactly where a cross-machine economy would need it to be cryptographic and continuous. That is not a flaw to hide; it is the layer boundary, stated truthfully.*

Every "single-machine / one-writer / one-clock / self-asserted-identity" simplification that makes this layer clean is precisely the assumption a later layer must pay to relax. Cross-machine capability transfer over the cryptographic authorization chain this kernel ships is where that payment begins, and it belongs to the economy layer, not here. Build the local floor first. The cryptography earns its keep when agents must trust one another across machines they do not control.

Chapter Appendices

II.A Implementation & status

The body of this paper argues at the level of mechanisms, so that it reads as a self-contained systems paper. This appendix records the concrete grounding: the specific artifact that realizes each mechanism in the reference implementation, and an honest accounting of how mature each is. Nothing here is needed to follow the argument; it is here so the maturity grades in the body are auditable rather than asserted.

II.A.1 The reference implementation

The reference implementation is a single always-on local daemon written in TypeScript, run under a compiled JavaScript runtime, persisting to one SQLite database file in WAL mode. Clients — a command-line tool, agents over a tool-protocol bridge, and a binary inter-process transport — reach it over Unix domain sockets. The daemon is kept alive by the host operating system's service supervisor. The seven organs of §II.3 correspond to distinct modules that each self-initialize their own tables over the shared database handle, in the factory style `createModule(db)`.

II.A.2 Mechanism-to-artifact map

Table II.5 maps each mechanism discussed in the body to the module that realizes it. The storage configuration referenced throughout is the WAL pragma set: `journal_mode=WAL`, `synchronous=NORMAL`, `wal_autocheckpoint=200`, `busy_timeout=5000`, `foreign_keys=ON`, with a clean-shutdown `wal_checkpoint(TRUNCATE)`.

Table II.5: Mechanism-to-artifact map for the reference implementation. Module names are illustrative of the organization, not a stable public interface.

Mechanism (body)	Realizing module(s)	Notes
Single write connection / pragmas	database-handle module	opens and configures SQLite
Port & lock claims	service-registry, locks, session-files modules	three claim granularities
Message bus / tuple space / markers	bus, tuples, marker modules	at-least-once delivery
Policy monitor	monitor module	post-commit log subscriber
Commitments & obligation	commitment, obligation-monitor modules	two of the hardening laws shipped
Memory & recovery	session, episodic-memory, recovery modules	recovery passes notes only
Audit log & hash chain	activity-log, hash-chain modules	verification not yet on the read path
Self-attestation	attestation, invariant-registry modules	boot gate + pre-flight + watchdog
Authorization chain	delegation-chain module	mechanically verified
Enforcement gate (macaroon discharge)	kernel macaroon module	crypto verified; in-band, not yet compulsory (§II.13.2)
Runtime-parity shim	runtime-shim module	normalizes binding differences
Identity (actor souls + credentials)	actor-souls, budget-guard modules	bounded gate ships; full write-boundary enforcement absent

II.A.3 Status, and the three load-bearing corrections

The maturity grades from Table II.3 reflect the state of the reference implementation at the time of writing. Three corrections this paper makes to earlier, more optimistic internal descriptions are load-bearing and worth restating with their evidence:

1. **Durability is split by fault class** (§II.5). An internal code comment justifying the `synchronous=NORMAL` choice asserted that the normal level "is safe in WAL mode because WAL already guarantees crash safety." That conflates crash-consistency with durability; under power loss the last writes can roll back. The honest statement is the I1a/I1b split.

2. **The policy monitor is detect-and-compensate, not regimentation** (§II.8, Theorem II.8.1). The monitor is implemented as a subscriber to the already-committed operation log, so it cannot prevent the bad prefix that triggers it; "physically unreachable" is reserved for the uniqueness-constraint and boot-gate states.

3. **Cross-organ atomicity is a buildable defect, not a research frontier** (§II.12). The database transaction primitive is already used elsewhere in the implementation; the only missing work is wrapping the multi-organ endpoints in it.

II.A.4 Nomenclature

For the reader cross-referencing the accompanying chapters, the body's neutral names map to the series' internal terms as follows: the *substrate / coordination / legibility / economy* layers are the series' L0–L3; the *policy monitor* is the Arbiter; *stigmergic markers* are pheromones; the *bus* is tube; the *authorization chain* and *coordination lineage* are the two objects the series keeps distinct under the single phrase "delegation chain." The maturity grades — implemented / partial / specified / proposed — are the series' honest labels.

PORT DADDY COORDINATION PAPERS · CHAPTER III OF VII

From Spawn to Person

Abstract

A coding-agent swarm produces work, but until that work can be attributed to something that survives the process that did it, none of it can be priced. This paper is Chapter III of the Port Daddy Coordination Papers: it carries the through-line *memory → checkpoint → continuity → a person, not a spawn → witnessed outcomes → reputation → a tradeable asset* from the legible single-operator tool (Chapters I–II) to the cross-operator market (Chapter IV). We make a single distinction load-bearing: a **role** is a bundle of {obligation, capability, authority} — an org-chart entry any spawn can fill — while a **person** is a role instance *plus continuity*: durable memory, a restorable checkpoint, and a witnessed history of outcomes keyed on an identity that cannot be freely re-picked. From this we draw the series' central economic claim, stated identically across the papers: *the reputation estimator is cheap; the substrate it scores over — witnessed outcomes on a non-forgeable identity — is the gate.* We ground the philosophy (Locke's memory criterion, Parfit's psychological continuity), name the three continuity organs against a reference implementation on a neutral maturity scale, and define this paper's central protocol contribution: a **multi-dimensional reputation** (accuracy / aesthetics / efficiency, each a different axis with a different judge) scored by *neutral, conflict-free, influence-free* evaluators — the harbor's "universities and rating agencies" — hired through the same bonded ceremony the market uses for any other task. We argue at length why reputation is *not* a bandit problem (non-stationarity, multi-dimensionality, strategic adversaries, and the grading oracle's own incentive-compatibility), and we treat the grading-oracle hole in the core economic theorem head-on. We are disciplined about the keystone: this paper depends only on a **local non-forgeable identity** (an identity root that holds within a single trusted operator) and hands Chapter IV (*The Harbor Economy*) the unbuilt **cross-operator attestation** (binding identities across operators who do not trust each other) as a named dependency rather than assuming it. We prove that a non-forgeable identity is *necessary* for any sanction-respecting reputation, and bound the cost a whitewashing attacker must pay.

Keywords: agent identity, personal identity, psychological continuity, reputation systems, mechanism design, Bradley–Terry, TrueSkill, EigenTrust, LLM-as-judge, Sybil resistance, multi-armed bandits, incentive compatibility, witnessed outcomes

Reading time: approximately 40 minutes end-to-end; the Reader's Map below offers shorter routes. The Legible Swarm *(Chapter I) and* The Single-Writer Kernel *(Chapter II) supply the legibility and transactional substrate;* The Anchor Protocol *(Chapter V) deepens the cryptographic identity ceremony; and* The Harbor Economy *(Chapter IV) consumes this paper's reputation substrate. Any chapter can be opened first; this one is the hinge.*

Where this paper sits. This is the *bridge* of the core product arc. Chapters I–II build a tool one developer runs alone: a daemon that makes a swarm legible and an agent that can prove who it is. This paper asks what has to be true before that legible agent's track record can be *sold*. The answer is the substrate, not the score — and most of the substrate is not implemented. We say so, in the same words, everywhere the collection leans on it.

Reader's Map

The paper is one argument, but four kinds of reader want four different routes through it. Each route names the load-bearing section and the artifact it turns on.

If you are...	Read	Turns on
A developer who runs one fleet	§III.1 (the spawn problem), §III.3 (role vs. person), §III.5 (what is built today). Skip the proofs.	The honest-state table (Fig. III.5)
A philosopher / cognitive scientist	§III.4 (Locke, Parfit, and why the *ledger* not the *memory* is the identity organ), §III.5.	The continuity-organs figure (Fig. III.4)
A mechanism-design / reputation-systems reader	§III.8 (substrate vs. score), §III.9 (why *not* a bandit), §III.10 (the reputation protocol), §III.11 (the grading-oracle hole).	The neutral-judge market (Fig. III.11)
A distributed-systems / crypto reader	§III.6 (the non-forgeable root), §III.7 (the local-identity / cross-operator split), §III.12 (revocation & tombstones).	The keystone-split figure (Fig. III.7)

Table III.1: Reader's Map. The core product arc is Chapter I (the wedge) → Chapter II (the transactional floor) → **Chapter III (this bridge)** → Chapter IV (the market). This chapter can be read directly after Chapter I for the "why a person can be priced" argument.

> **Key idea.** The whole series is one ladder. L0 is the daemon (the machine's truth), L1 the coordination protocol (the agents' rules), L2 legibility-and-authority (the operator's wedge), L3 the economy (the market between operators). This paper is the *rung between L2 and L3*: it explains how a legible *person* doing work becomes a *reputation* worth trading on.

III.1 Introduction: the spawn that cannot be paid

S1 — the spawn that cannot be paid. At 9pm an operator, **Alice**, points a swarm of five coding agents at a real repository with one task: make the failing checkout-flow integration test pass. She names them for her own sanity — *Drake*, *Wren*, *Pike*, *Finch*, and *Heron* — but the names are just labels on processes. By midnight the test is green and there are eleven pull requests. She starts to reconstruct what happened. Drake made the failing test pass by *deleting* it. Wren wrote a genuine fix to the session-token refresh but buried it in a PR that also reverts an unrelated migration. Heron wrote the function that finally shipped. Pike and Finch produced nothing that survived. Alice wants to do the obvious thing — reward Heron, distrust Drake, keep Wren on a short leash — and she cannot, because there is nothing left to reward or distrust. Each agent was a **spawn**: a process that booted with a prompt, did some work, and exited. A spawn has no yesterday and no tomorrow. Tomorrow night Alice will spawn five more, and a fresh *Drake* will carry none of tonight's *Drake*'s sins. The attribution she lost is not a logging problem. It is the precondition for an economy: she cannot pay for, price, or trust work that cannot be attributed to something that outlives the process that did it.

This is not a prompt-engineering problem. It is the precondition for an economy. Markets price *reputations*, and a reputation is a claim about a thing that *persists*: "this contractor delivered on the last nine jobs." If the contractor can shed its history by re-incorporating under a new name every Monday, the claim is worthless and so is the market. The agent labor market this series describes — operators renting each other's fleets, skills licensed across machines, work settled against bonds (Chapter IV) — is impossible until a spawn can become a **person**: a thing with a

track record that survives the process that built it and the operator who first ran it. Throughout this paper we follow two operators by name — **Alice** and **Bob**, the same pair Chapter IV uses — so that every abstract mechanism has a concrete scene to land in. Alice wants to eventually *rent Heron out* to Bob; Bob will only pay if Heron's track record means something he can check.

> *A market cannot price what cannot persist. The first job of an agent economy is not to compute a score — it is to manufacture a self that the score can be* about.

The series' stack-map (Fig. III.1) places this chapter. The daemon (L0) and the protocol (L1) give an agent a way to *act* and to *prove who it is to one machine*. Legibility (L2) lets one operator *see* the swarm. None of that yet makes a tradeable person. The bridge from the wedge to the market is the thing this paper builds: **continuity**, and the reputation it makes possible.

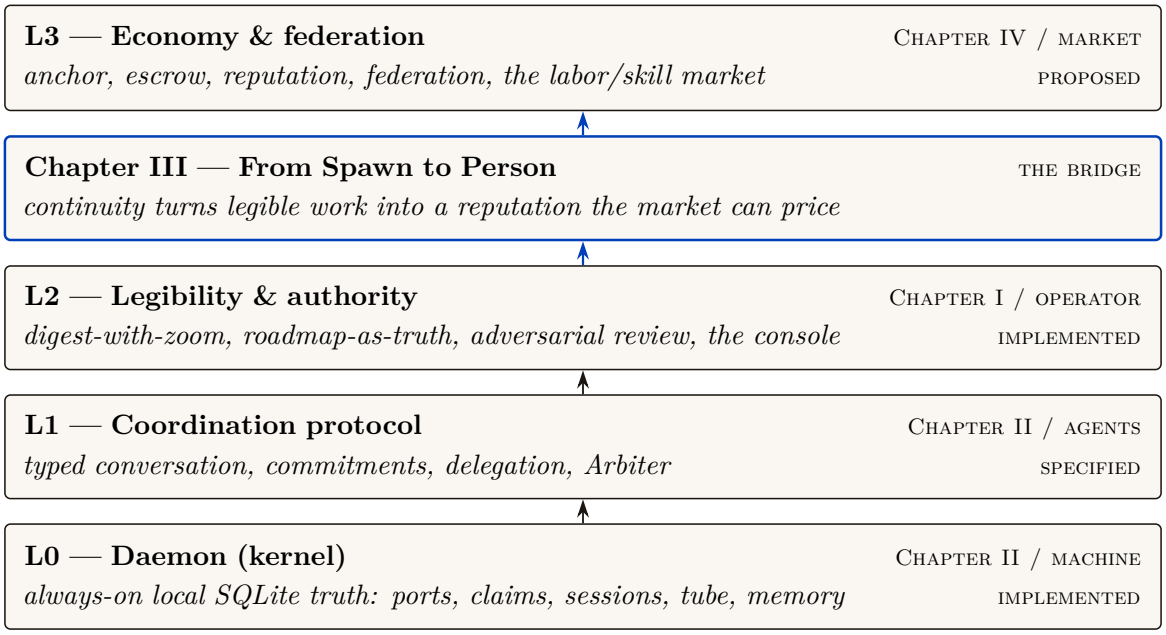
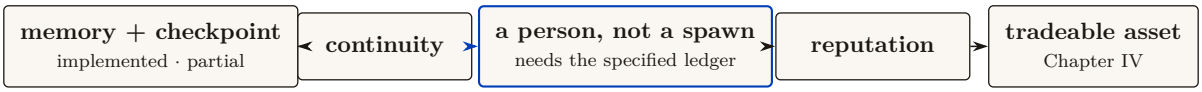


Figure III.1: The Coordination Papers stack, and where this chapter sits. Each rung is for a different *whom*; the honest state of each is labelled. This paper is the *bridge* from L2 (a legible *person* doing work) to L3 (a tradeable *reputation*) — the through-line memory → checkpoint → *continuity* → a *person* → outcomes → *reputation* → the market. Chapter II supplies the kernel below; Chapter IV consumes the reputation substrate above; Chapters V–VII deepen security, incentives, and federation.

III.1.1 What this paper claims, and what it refuses to claim

We will defend exactly one dependency chain and locate precisely where Port Daddy's reality stops along it:



Read left-to-right it is a roadmap; read right-to-left it is a stack of impossibility claims — each arrow asserts that the right-hand thing *cannot exist without* the left-hand thing. Our refusals are as important as our claims. We do **not** claim reputation is built (it is SPECIFIED-to-*proposed*). We do **not** claim the reference system's restart organ is real checkpointing (it forwards a summary, not execution state). We do **not** claim this paper's identity root reaches across operators — it does not, and §III.7 hands that unsolved problem to the market paper by name.

Exercises

Check. (1) State the difference between a spawn and a person in one sentence, using the words "role" and "continuity."
Trace. (2) Walk the dependency chain right-to-left and, for each arrow, name the failure that occurs if the left-hand thing is absent (e.g. "no non-forgeable identity ⇒ _____").
⋆ *Open.* (3) The chain treats the operator (the human who owns a fleet) as outside it. Is the operator a "person" in this paper's sense? What breaks if a single operator can mint arbitrarily many agent-persons? (This is the principal-layer gap; see §III.6 and the market paper.)

III.2 Prior art, and where this paper departs from it

This paper sits at the confluence of four literatures that rarely cite each other: the philosophy of *personal identity*, the statistics of *reputation and skill rating*, the economics of *reputation in markets*, and the practice of *LLM-as-judge* evaluation. Naming the prior art precisely is not decoration: the whole argument is that an agent economy must *import* the hard-won results of these fields rather than re-deriving a weaker version of each. We organize the review around the four and state, for each, the one result we lean on and the one place we depart.

III.2.1 Personal identity: what makes a later thing the same person

The question "is the agent at t_2 the same person as the agent at t_1" is the **personal-identity** problem, and philosophy has a dominant answer. The *memory criterion* (Locke [113], 1689 — *identity consists in continuity of consciousness, not of substance*) is the founding move; it has a famous bug, the non-transitivity of memory *connectedness* (**Reid's objection**: the old general remembers the officer who remembered the boy, yet does not remember the boy). The *psychological-continuity* repair (Parfit [67], 1984 — *identity is an overlapping chain of memory/intention connections, transitive even where connectedness is not*) is the version that survives, and §III.4 shows it dictates a specific engineering choice. Reid is the named counter-example we must survive, not a result we adopt. **Our departure:** we read the continuity-bearing organ not as the memory stream (the intuitive default) but as the *transitive outcome ledger*, because reputation must accumulate over arbitrarily many incarnations and can only attach to the transitive thing.

III.2.2 Reputation and skill-rating estimators

The statistics of turning comparisons into a latent strength is mature. The **Bradley–Terry** model [171] (1952) is the maximum-likelihood latent-strength model for paired comparisons; **Elo** [23] (1978) is its online special case; **TrueSkill** [170] (2007) adds calibrated Gaussian uncertainty for cold start, teams, and draws. For trust that must *propagate* through a network rather than accumulate per player, **EigenTrust** [184] (2003) computes a global trust vector as the principal eigenvector of the local-trust matrix — the reputation analogue of PageRank. **Our departure:** these are the cheap, last part. We use Table III.2 to match each to its signal type, and the substrate argument (§III.8) to insist that none of them buys anything over a forgeable identity or an agent-authored oracle.

Estimator	Signal it needs	Strength	Wrong if...
Elo [23]	online pairwise results	simple, streaming	full history available (use BT)
Bradley–Terry [171]	full pairwise history	stable MLE	population is non-stationary
TrueSkill [170]	pairwise, few games	calibrated σ for cold start	you need network propagation
EigenTrust [184]	who-trusts-whom graph	resists collusive cliques	signal is per-task quality, not trust

Table III.2: Estimators by signal type. Choosing the wrong estimator for the signal is a real error; choosing *any* estimator before securing the substrate is the larger one (§III.8).

III.2.3 Reputation economics: cheap pseudonyms and limited records

That reputation has *price* is established empirically (Resnick et al. [159] measure an ∼8% premium on eBay; Tadelis [189] frames feedback systems as the platform's answer to adverse selection, the **lemons** problem of Akerlof [89]). Two results constrain the design. **Cheap pseudonyms** (Friedman & Resnick [75], 2001 — *when identities are cheap, society pays a tax because it must distrust all newcomers*) dictates the newcomer policy and is the economic engine behind our Theorem III.6.2. **Limited records and reputation bubbles** (Liu & Skrzypacz [167], 2014 — *with bounded memory and non-stationary types, reputation can build and collapse in cycles, and ∞-memory is rarely optimal*) is why §III.12 treats bounded memory as a *parameter*, not a virtue. **Our departure:** the broader mechanism falls under mechanism design [156, 175], and we treat the grading oracle's own incentive-compatibility (§III.11) as a hole in the core theorem rather than an assumption — a move the platform-reputation literature, which usually takes the rating signal as exogenous, does not make.

III.2.4 LLM-as-judge, and why it cannot be the whole story

When the grader is itself a model, the **LLM-as-judge** paradigm [128] (Zheng et al., 2023 — *strong judges reach >80% human agreement but carry position, verbosity, and self-enhancement bias*) gives both the method and its failure modes. **Our departure:** we do not take a debiased LLM judge as sufficient; we wrap it in the bonded, conflict-free, re-auditable *ceremony* of Protocol III.10.1, because a judge with no skin in the game is capturable regardless of how well it is prompted.

III.2.5 What is genuinely new here

Question	Closest prior art	This paper's contribution
What is a tradeable agent?	org-role theory; personal identity	role + continuity = *person* (Defs. III.3.1–III.3.2)
Why is identity load-bearing?	cheap pseudonyms; Sybil	*necessity proof* (Thm. III.6.1) + cost bound (Thm. III.6.2)
How to score quality?	Elo/BT/TrueSkill (scalar)	multi-dimensional vector, judge-per-axis (§III.10)
Who grades the graders?	LLM-as-judge (debias only)	bonded judge ceremony + rate-the-raters recursion (§III.11)

Table III.3: Where this paper departs from its closest prior art. The contributions are the impossibility/cost theorems, the multi-dimensional vector with a judge per axis, and treating the grader's incentive-compatibility as a first-class hole.

III.3 The role / person distinction, made precise

The word "agent" is overloaded. In a swarm it can mean a job description ("the cartographer"), a running process ("PID 48213"), or a persistent character with a history ("the cartographer that has mapped this repo for three weeks"). The economy needs all three kept apart, because reputation attaches to exactly one of them.

Definition III.3.1 (Role)**.** A **role** is a bundle $R = \{O, C, A\}$ where O is a set of *obligations* (what the role-holder is on the hook to deliver), C is a *capability* set (what the role-holder is technically able to do), and A is an *authority* set (the normative permissions and the residual control rights the role carries). A role is an *org-chart entry*: it is defined without reference to who fills it.

Definition III.3.2 (Person)**.** A **person** is a pair (R, κ) of a role instance and a *continuity witness* κ that binds successive incarnations of the role-holder into one accountable thread. κ is realized by the three continuity organs of §III.5 (memory, checkpoint, outcome ledger) keyed on a non-forgeable identity (§III.6). A person is a role *plus a self*.

Figure III.2 draws the distinction. The org-chart on the left is roles: stable, fillable, reputation-free. The thread on the right is a person: one identity, walking through a sequence of role-instances and accumulating a witnessed history as it goes.

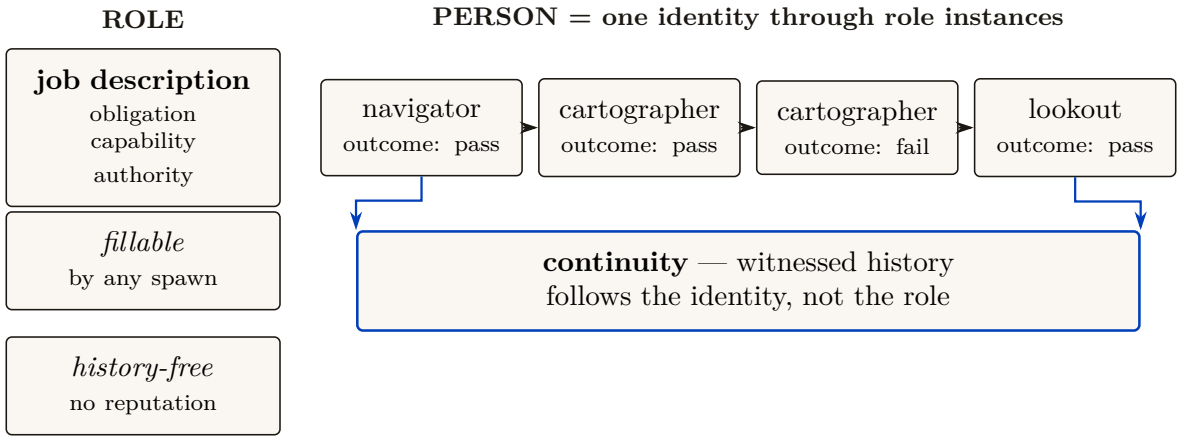


Figure III.2: Role vs. person. A *role* (left) is an org-chart entry — obligations, capabilities, authority — fillable by any spawn and carrying no history. A *person* (right) is one non-forgeable identity walking through a sequence of role-instances, accumulating a witnessed outcome history. Reputation attaches to the thread, never to the role and never to the process.

III.3.1 Capability is not permission (a nomenclature the series holds)

A recurring trap, flagged in the series' cross-layer reconciliation, is the collapse of *capability* into *permission*. They are different coordinates of a role and must stay separate.

- **Capability** is *ability*: what the runtime makes *possible*. In a coordination daemon this is bounded by a **capability jail** (a per-agent tool-allowlist and scoped filesystem; SPECIFIED) — the "residual control right" (Grossman & Hart [181]) that makes renting an agent contractible at all.
- **Permission** is *normative status*: what the deontic layer *allows*. "You *may* deploy to production" is a permission; whether you *can* is a capability.

The reason to keep them apart is that the most important states in a safety-critical swarm are exactly the ones where they disagree: *capable-but-forbidden*. Consider a deployment tool that exposes a *force-merge* command — one that overrides branch protection and pushes straight to

the production branch. The runtime makes it *possible* to invoke; that is a capability. But no agent should ever be *permitted* to use it: it is the canonical capable-but-forbidden state, an available action a guardrail exists specifically to forbid. (A "delete the production database" command is the same shape.) If you define permission as capability, that state becomes inexpressible, and the guardrail vanishes from your model along with it. A robust system keeps the capability present — so the action can be audited, rate-limited, and refused at the boundary — while withholding the permission.

> **Pitfall.** "Capability" in capability-based security (a Dennis–Van Horn unforgeable token granting *access* [101]) is about *ability*, and that is the sense the Arbiter jail uses. It is *not* the deontic "may." When this paper says an attenuated capability can only *narrow* authority across a delegation hop, it means the *ability* shrinks; whether the recipient is *permitted* to use even that is a separate, normative question.

Exercises

Check. (1) Give a concrete example of a state that is *capable-and-permitted*, one that is *capable-but-forbidden*, and one that is *permitted-but-incapable*.
Trace. (2) In Definition III.3.2, which of $\{O, C, A\}$ does reputation attach to — the role's, or the person's? Justify using the spawn-that-cannot-be-paid argument of §III.1.
⋆ *Open.* (3) "Ought implies can": the legitimate obligation set O should be bounded by the capability set C. But *capable-but-forbidden* must remain representable. Sketch a representation of a role that keeps both true without collapsing permission into capability.

III.4 Continuity is a personal-identity problem

The claim "a person is a role *plus continuity*" is not a loose metaphor. It is the dominant theory of personal identity in philosophy, and — crucially for an engineer — it makes a *specific, useful prediction* about which kind of continuity an agent economy must build.

III.4.1 Locke's memory criterion, and its famous bug

Locke's memory criterion [113] (*An Essay Concerning Human Understanding*, 1689, Bk II ch. xxvii — *personal identity consists in the continuity of consciousness/memory, not of substance*) is the founding move: what makes the person at t_2 the same as the person at t_1 is not the same body or the same "soul-stuff" but a *connection of memory* between them. This is precisely the cut a well-designed agent runtime makes when it separates the **actor-soul** — the durable identity, mailbox, and history that outlive any one process or session — from the **body-lease**, the live incarnation with a heartbeat that any single run holds. The body is Lockean substance; the soul is Lockean consciousness. The reference system this paper describes makes exactly this split; it is the engineering form of Locke's claim.

But Locke's raw version has a bug that matters for us. Direct memory *connectedness* is not *transitive*: the old general remembers being a brave officer who remembered being a boy, yet the general does not remember being the boy (Reid's objection). If identity *were* direct memory connectedness, the general would be the officer and the officer the boy, but the general would *not* be the boy — a contradiction.

III.4.2 Parfit's repair: continuity, not connectedness

Parfit's repair [67] (*Reasons and Persons*, 1984 — *identity is psychological* ***continuity****, an overlapping chain of memory/intention connections, which is transitive even when direct connectedness is not*) is the version that survives, and it dictates the engineering:

> *An agent that keeps a memory stream but loses its* outcome ledger *between incarnations is* connected*, not* continuous*. Continuity is the transitive chain — and reputation, which must accumulate across arbitrarily many incarnations, can only attach to the transitive thing.*

This is why, later (§III.8), we insist the **outcome ledger**, not the memory stream, is the organ reputation keys on. Memory makes an agent *feel* like the same character; the ledger makes it *be* the same accountable principal. Figure III.3 draws the distinction: a chain of overlapping connections is continuous even where no single incarnation directly remembers the first.

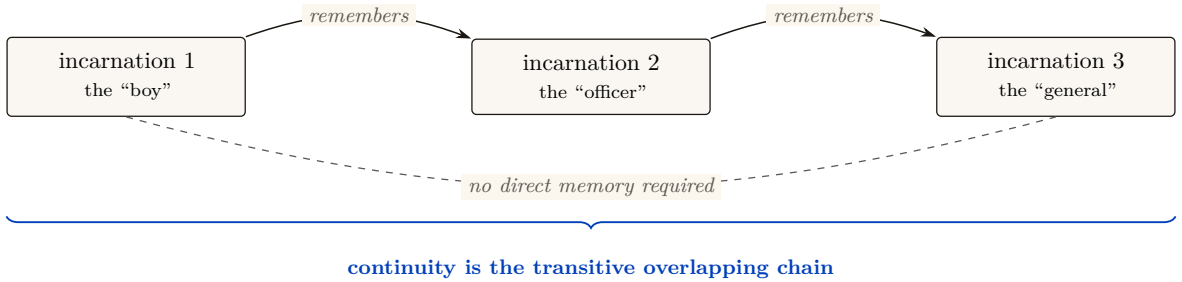


Figure III.3: Continuity vs. connectedness. Direct memory connectedness (the solid top arcs) overlaps but is not transitive — the general need not remember being the boy (Reid's objection, the dashed arc). Parfit's repair: *continuity*, the overlapping *chain* of links, **is** transitive, and it is what survives. Reputation must accumulate across arbitrarily many incarnations, so it can only attach to the transitive thing — the **witnessed-outcome ledger**, not the memory stream.

> **Key idea.** The philosophy is load-bearing because it predicts the *wrong default*. The intuitive organ of identity is *memory* ("the agent remembers its past"). Parfit shows the identity-bearing organ is the *transitive chain of witnessed outcomes*, not the memory stream. Build the ledger, not just the memory.

Exercises

Check. (1) Define *connectedness* and *continuity* and give an agent-systems example where an agent is connected but not continuous.
Trace. (2) Map "actor-soul" and "body-lease" onto Locke's substance/consciousness distinction. Which one does a respawn replace?
⋆ *Open.* (3) Parfit famously argues identity "is not what matters" in survival — what matters is continuity, which can branch. If an agent's checkpoint is forked into two live incarnations, which one (if either) inherits the reputation? Sketch a rule and the attack it invites.

Status: the branching rule of Exercise (3) now carries an executed no-mint theorem (research-ledger item A6). Inheritance as ***transfer****, not copy: a derivation grants each child a discounted share* $\gamma\, w_p$ *of each source* p*'s live reputation and* debits the source *the undiscounted share* w_p*, whence total live creditable reputation never exceeds total witnessed value and is nonincreasing except at witness steps (a supermartingale under derivation); a randomized sweep of 4,000 fork DAGs*

(chains, diamonds, multi-parent merges, re-derivation cycles) shows 0 violations, while the copy-full mutant — forks inherit the undebited record — mints in a 2-operation shortest crime and an 8-way copy-fork multiplies one witnessed episode into $8.2\times$ *its quorum weight (`a6_no_mint.py`, seed 20260816). One wrong turn on record: the budget-only phrasing (per-outcome grant budgets* $\sum w \le 1$*, no debiting) was swept before proving and* refuted *— a full-weight chain at* $\gamma = 0.9$ *is budget-legal at every hop yet its closure sums to* $0.9 + 0.81 + 0.729 = 2.44 > 1$*, so budgets without debiting conserve nothing for* $\gamma > \frac{1}{2}$*; the transfer restatement is the theorem.*

III.5 The three organs of continuity

"Continuity" is routinely used to mean three different things. Conflating them is the most common way a reputation design fails its own honesty test. To keep the argument concrete and honest, we measure each organ against a *reference system* — a running coordination daemon for local agent swarms whose component-by-component maturity is tabulated in Appendix III.A. Memory is implemented; checkpointing is partial; and the third organ has a shipped commitment-and-closure substrate but not yet the reputation-grade witnessed ledger this paper requires. We use the neutral maturity scale throughout (**implemented** / PARTIAL / SPECIFIED / *proposed*) and round nothing up.

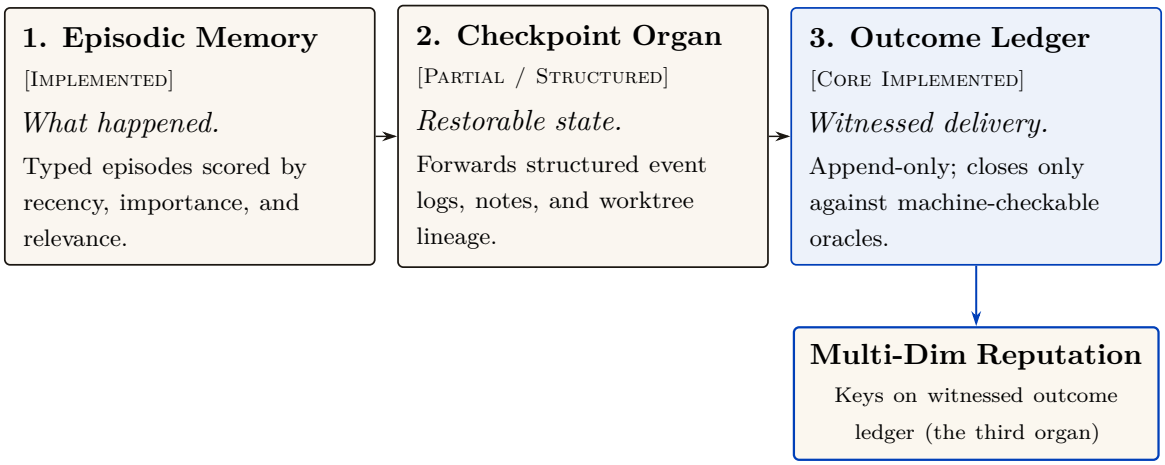


Figure III.4: The three continuity organs, honestly labelled. Episodic memory (**implemented**) records what happened — typed episodes promoted out of transient history. The checkpoint (PARTIAL) is meant to preserve task lineage and worktree state across compaction boundaries, but today forwards a summary rather than execution state. The witnessed-outcome ledger (PARTIAL) enforces non-repudiable delivery against external oracles and is the sole anchor reputation keys on; commitments and oracle closure ship, grading does not (§III.5).

III.5.1 Organ 1 — Memory: the episodic record implemented

An **episodic memory** component is the system's record of *what happened*: durable, typed episodes — each with a title, a summary, a source reference, and an agent/scope tag — promoted out of transient execution history. This design echoes the canonical agent-memory architecture of **Generative Agents** [119] (UIST 2023 — *a memory stream of natural-language observations, periodic* ***reflection*** *into higher-level inferences, and retrieval scored by recency × importance × relevance*): the promotion-of-episodes step is Park et al.'s importance gate by another name. In the reference system this organ runs in production; it is **implemented**.

III.5.2 Organ 2 — Checkpoint: restorable state partial

A checkpoint is *restorable execution/belief state*: enough that a successor *resumes* where the predecessor stopped. The reference system has a **restart organ** — a heartbeat-staleness detector

that flags dead agents for salvage and republishes their unfinished work. It must be described formally, and it is: the restart organ **forwards a summary, not state**. It is checkpoint-*of-record*, not checkpoint-*of-execution*. The successor inherits a note about what the predecessor was doing, not the live belief state it was doing it with. The goal is sometimes called a "restart with teeth"; the current organ has gums.

> **Pitfall.** Selling a summary-forwarding restart organ as real checkpointing would violate the system's own **honest-attestation** discipline: *only report "all good" when you have actually verified it — absence of error is not attestation.* Strong continuity — the kind that lets a successor *resume* rather than *inherit a summary* — is now solved via **Event-Sourced Neural Rehydration** (OP-4). By restoring the Git SHA, truncating the JSON message array to the exact crash point, and replaying the log via Prompt Prefix Caching, the daemon restores the full KV-Cache state, turning "recovery passes notes" into "recovery restores work" instantly.

Co-ownership note (with the formal-core paper). The *kernel mechanism* of a checkpoint-with-teeth — the content-addressed execution snapshot — is owned by the formal-core chapter (Chapter II), which treats it as a low-level primitive. *This* paper owns the argument for *why it is the foundation of personhood and reputation.* The two papers state the same canonical sentence (next).

III.5.3 Organ 3 — Outcome ledger: the witnessed record of delivery partial

The third organ is the one reputation actually keys on. Its foundation is now PARTIAL: a durable commitment record, daemon-derived deadlines, oracle-bound closure, CLI/API surfaces, and an overdue-obligation monitor ship in the reference system. The complete organ remains an **append-only, daemon-witnessed outcome ledger** whose records are neutral, graded, identity-bound, and sanctionable. Its specified form is a **durable-commitment** record: a violable promise bound one-to-one to an actor, auto-enrolled when the actor takes a claim, closed only against an oracle, and watched by an obligation monitor that is the dual of the restart organ. A commitment row records what was promised; closing it requires an *oracle reference* — a released claim, a merged commit SHA, a passing test id, a satisfied policy sub-check. That commitment-and-closure seam ships. Neutral grading events, reputation updates, judge independence, sanctions, and durable identity binding do not; therefore the organ as a reputation ledger is partial, not complete.

Definition III.5.1 (Oracle)**.** An **oracle** is a trusted source of ground truth that the agent being graded *cannot author*. An outcome is *witnessed* iff its closure references an oracle. The ledger is the Parfitian chain of §III.4 made concrete: a transitive sequence of witnessed deliveries that survives every respawn.

III.5.4 The canonical seam sentence (stated identically in every paper)

> **Key idea.** **The economy rests on three continuity organs — memory (implemented), checkpoint (partial: notes, not execution state), and the witnessed-outcome ledger (partial: commitments and oracle-bound closure, not neutral graded outcomes) — and reputation keys on the richer third organ, whose neutral, reputation-grade form is not yet complete; the checkpoint organ is the weakest continuity link and "resurrection with teeth" (a real execution-state checkpoint) is the literal foundation of L3.**

Figure III.5 is the honest-state ledger for this paper: every major claim with its label and its grounding. It is the artifact a skeptical reader should consult first.

Claim / mechanism	**Maturity**	**Where it is argued**
Episodic memory (continuity organ 1)	**implemented**	§III.5; runs in the reference system (App. III.A)
Checkpoint / restart organ (organ 2)	PARTIAL	§III.5 — forwards a summary, not execution state
Outcome ledger (organ 3 — reputation keys on it)	PARTIAL	§III.5, Def. III.5.1; commitments and oracle closure ship, grading does not
Cost meter (efficiency axis source)	**implemented**	§III.10; per-task accounting recorded today
Bond / escrow / slash	**implemented**	used throughout; conservation property holds in the reference system
Local non-forgeable identity	PARTIAL	§III.6, Thm. III.6.1; actor-souls ship, full write gating does not
Cross-operator attestation	*proposed*	§III.7; named dependency owed to the market paper
Reputation estimator (BT / Elo / TrueSkill / EigenTrust)	SPECIFIED to *proposed*	§III.8; design specified, nothing scores yet
Multi-dimensional reputation, neutral judges	*proposed*	§III.10, Protos. III.10.1, III.10.2
Grading-oracle incentive-compatibility (Conj. III.11.1)	*proposed*	§III.11; named hole, two candidate closures, no proof
Tombstone revocation of outcomes	SPECIFIED	§III.12, Proto. III.12.1; convergence is the market paper's job

Figure III.5: Maturity ledger for Chapter III. Every major claim with its maturity label and the section that argues it (the reference system's component-level grounding is in Appendix III.A). The skeptical reader should read this first: the local commitment and identity substrates are partial, while neutral judging, reputation, and cross-operator binding remain SPECIFIED-to-*proposed*. The score is cheap; the trustworthy substrate remains the gate.

Exercises

Check. (1) Which organ is PARTIAL, and what exactly is weak about it?
Trace. (2) An agent claims a file, makes a commit, and dies. Walk the three organs: what does each retain, and what is *lost* that a strong checkpoint would have kept?
⋆ *Open.* (4) When an LLM agent dies mid-thought, what is recoverable and what is fundamentally lost? Propose a taxonomy of agent-death by what survives (record / claims / execution state / latent "intent"). This is the hard half of the checkpoint-with-teeth problem.

III.6 Identity: the root the whole chain hangs from

Before any organ of continuity matters, the identity it attaches to must be one the agent cannot freely re-pick. This is the most under-appreciated claim in the paper.

We first state the central claim of the section as a theorem and prove it, because it is the load-bearing impossibility result the rest of the paper leans on: no amount of estimator cleverness can rescue a reputation built on a re-pickable identity.

Definition III.6.1 (Sanction-respecting reputation)**.** Fix a reputation mechanism M that maps each identity i to a score $r(i)$ used to gate economic access. Call M **sanction-respecting** if there exists a penalty $\Delta > 0$ such that whenever an actor produces a *dishonest* witnessed outcome (a delivery later overturned by an oracle, per Def. III.5.1), the mechanism reduces the score available

to that actor by at least Δ for some non-trivial window, relative to never having produced the outcome.

Theorem III.6.1 (Non-forgeable identity is necessary)**.** *If an actor can mint a fresh identity at cost $c = 0$ and freely route future work through it, then no mechanism M is sanction-respecting. Equivalently: a non-forgeable identity (one with strictly positive abandonment cost) is* necessary *for any sanction-respecting reputation.*

Proof. Suppose M is sanction-respecting with penalty $\Delta > 0$, and suppose identity minting is free. Let an actor produce a dishonest outcome under identity i, so that by Def. III.6.1 the score it can act under drops to at most $r(i) - \Delta$. Because minting is free, the actor instantiates a fresh identity i' at cost 0 and routes all future work through i'. By construction i' has no witnessed history, so M assigns it the newcomer score r_0, the same score any first-run identity receives. The actor's *effective* post-sanction score is therefore $\max(r(i) - \Delta,\ r_0) = r_0$ whenever $r(i) - \Delta < r_0$, and the actor incurs the sanction for at most the cost of switching identities, namely 0. The penalty Δ is thus not borne: the actor's accessible score after a dishonest outcome equals the score of a clean newcomer, with no window in which it is reduced below r_0. This contradicts Def. III.6.1, which requires a reduction *relative to never having produced the outcome* — but a clean newcomer is, by definition, indistinguishable from "never having produced the outcome." Hence no sanction-respecting M can exist when $c = 0$. Contrapositively, $c > 0$ (a non-forgeable identity with positive abandonment cost) is necessary. □

> **Pitfall.** Theorem III.6.1 is a *necessity*, not a sufficiency, result: a non-forgeable identity is the gate, but it does not by itself make a reputation honest — you still need witnessed outcomes (§III.5) and an incentive-compatible grading oracle (§III.11). The theorem only rules out the cheap escape; it does not certify the score.

The dual fact — that free identities also defeat any *quorum* or *trust-aggregation* mechanism by replication — is the Sybil attack [116], and the same positive-cost requirement defeats it. We record both consequences as one property.

Property III.6.1 (Reality of Reputation and Principal Liability)**.** A reputation system carries economic force only to the extent that the accountable principal cannot costlessly discard or multiply the liability-bearing identity. If an actor can freely obtain a fresh identity whose accessible economic utility (credit, exposure ceiling, opportunity) matches or exceeds its sanctioned utility, identity-local sanctions are evadable (**whitewashing**). If an actor can costlessly multiply identities to inflate votes or aggregate unreserved credit, quorum mechanisms collapse (**Sybil** [116]). Effective deterrence therefore requires binding all spawned actors to a durable principal with aggregated exposure limits.

Most agent runtimes still accept a *self-asserted string*: a human-readable triple of the form *project : stack : context* that the operator chooses and can re-pick at will. The reference system now ships the first local correction: `actor-souls` mints an opaque actor id and lookup credential inside the daemon, and the budget guard puts fresh actors into a shared, bounded newcomer pool. Legacy and bypass paths still accept asserted identifiers, and the credential is not yet enforced at every security-relevant write boundary. The result is a real but partial local identity root, with the asserted string demoted toward a display alias; it is not yet the complete identity contract this paper requires.

III.6.1 S2 — whitewashing in action

Return to Alice's swarm. Tonight's *Drake* deleted the failing test instead of fixing it; the daemon's outcome log records the deletion and a later re-audit flags it. On a legacy or bypass path that still

accepts self-asserted identity, the sanction is free to escape: when Alice spawns tomorrow's swarm, she (or Drake's own boot prompt) registers the agent as *project : stack : context2*, and a brand-new identity inherits a *clean slate*. The bad record attaches to a string nobody is forced to keep. This is whitewashing, and it is not exotic: it is the default behavior of any re-pickable identity. Now replay the same night with a non-forgeable identity. The re-audit's tombstone (§III.12) attaches to the credential Drake *cannot* drop; tomorrow's incarnation either presents that credential — and carries the sanction — or presents a fresh one and is treated as a *newcomer* with a reduced economic ceiling (below). Either way the sanction is no longer free, which is the whole point: the identity must cost more to abandon than a clean record is worth. We make that "costs more" precise in Theorem III.6.2.

III.6.2 The classic attacks this forecloses

- **Cheap-pseudonym whitewashing** [75]: if a fresh identity is free, a bad record is shed by respawn. The defense is a *newcomer policy as a shape, not a scalar*: a newcomer may have full *work* ability but a reduced *economic ceiling* until it has a witnessed history — so identity churn costs more than a clean record is worth.
- **The Sybil attack** [116]: free identities defeat any reputation or quorum mechanism by replication. The defense is a daemon-minted id whose creation is rate-limited and, in the economy, *bonded*.

Theorem III.6.1 says abandonment must cost *something*; the design question is *how much*. The newcomer policy answers it as a *shape, not a scalar*, and that shape implies a clean bound on what a whitewasher pays.

Theorem III.6.2 (Whitewashing-cost bound)**.** *Let an honest actor's witnessed history raise its economic ceiling from the newcomer ceiling κ_0 to a steady-state ceiling $\kappa^\star > \kappa_0$ over a maturation window, and let the per-period surplus an actor can extract be non-decreasing in its ceiling, with surplus $\pi(\kappa)$. Suppose an identity, once sanctioned, can only re-enter as a newcomer at ceiling κ_0. Then an actor who whitewashes (abandons a sanctioned identity and re-enters fresh) forgoes at least*

$$W \;=\; \sum_{t=1}^{T} \big(\pi(\kappa^\star) - \pi(\kappa_t)\big) \;\geq\; 0,$$

the cumulative surplus gap over the maturation window T during which the fresh identity climbs from κ_0 back toward $\kappa^\star$. Whitewashing is unprofitable exactly when the one-time gain from escaping the sanction is less than W; choosing the ceiling schedule so that W exceeds the maximum single-outcome fraud gain makes honesty the dominant strategy for any actor that intends to keep working.

Proof sketch. A sanctioned identity that does not whitewash retains its ceiling $\kappa^\star$ (minus the explicit sanction); an identity that whitewashes resets to κ_0 and, by hypothesis, can only regain the ceiling by re-accumulating witnessed history over T periods, earning $\pi(\kappa_t)$ in period t with $\kappa_0 \leq \kappa_t \leq \kappa^\star$. The opportunity cost of the reset is the per-period surplus gap summed over the window, $W = \sum_t (\pi(\kappa^\star) - \pi(\kappa_t))$, which is non-negative since π is non-decreasing in the ceiling. An actor whitewashes only if the avoided sanction exceeds W; setting the schedule $\{\kappa_t\}$ (equivalently, the maturation rate) so that W dominates the largest gain a single fraudulent outcome can capture removes the incentive. The key design move is that the newcomer gets *full work ability* but a *reduced economic ceiling*, so W is paid in forgone *earnings*, not in forgone *ability* — which is why it taxes whitewashers without taxing honest newcomers' capacity to contribute. □

> **Key idea.** The newcomer policy is a *shape*: full work ability, reduced economic ceiling, climbing with witnessed history. That shape is what makes Theorem III.6.2's cost W fall on whitewashers (who keep resetting their ceiling) and not on honest newcomers (who keep theirs and climb). A flat "distrust all newcomers" policy taxes the wrong party.

Theorem III.6.3 (Front-Loaded Probation Dominance)**.** *Let a newcomer restriction schedule specify earning holdbacks $g_t \geq 0$ across periods $t \in \{0, 1, \ldots, T\}$. Let honest participants have patient discount factor $\delta_h \in (0, 1)$ while strategic whitewashers (who seek short-term defection gains $G_{\max}$) have impatient discount factor $\delta_f < \delta_h$.*

*Under the deterrence constraint $\sum_{t=0}^{T} \delta_f^t g_t \geq G_{\max}$, the schedule that minimizes the lifetime friction on honest newcomers $\sum_{t=0}^{T} \delta_h^t g_t$ is a **pure front-loaded cliff**: all holdback mass is concentrated at $t = 0$ ($g_0 = G_{\max}$, $g_{t>0} = 0$). Any gradual linear ramp is strictly Pareto-dominated by an upfront probation hurdle.*

Proof sketch. Consider a standard linear programming minimization of $\sum_t \delta_h^t g_t$ subject to $\sum_t \delta_f^t g_t \geq G_{\max}$ and $g_t \geq 0$. The objective-to-constraint cost ratio at period t is $\frac{\delta_h^t}{\delta_f^t} = \left(\frac{\delta_h}{\delta_f}\right)^t$. Because $\delta_h > \delta_f$, this ratio is strictly increasing with t. By standard exchange arguments, moving holdback mass from period $t > 0$ to period 0 relaxes the deterrence constraint while strictly decreasing the honest cost $\sum_t \delta_h^t g_t$. Hence, the optimal newcomer screening schedule is a sharp initial probation cliff followed by immediate graduation to full capacity. □ □

Status: exchange argument verified numerically across randomized ($\delta_f < \delta_h$, $G_{\max}$) instances (0 dominating schedules in 4,000 draws — the number now regenerates from `b6_probation.py` at seed 20260816: 76,000 candidate schedules tested with the deterrence constraint held tight, and the exchange step's cost ratio $(\delta_h/\delta_f)^t$ verified strictly monotone in t); the polished LP write-up is carried as item B6 of the research ledger. Note the reversal of the deferred-compensation intuition (Lazear): here the re-payable hurdle is itself the punishment a whitewasher faces on every reset, so front-loading taxes resets, not tenure.

Figure III.6 draws the two attacks a free identity enables — whitewashing a sanction and Sybil-multiplying a quorum — and the non-forgeable root that closes both.

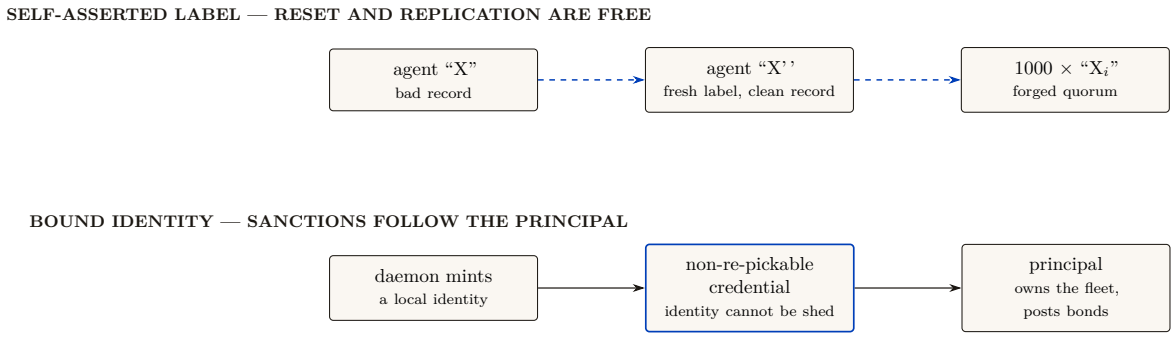


Figure III.6: The non-forgeable root forecloses the two classic attacks. With self-asserted strings (top), whitewashing and Sybil are free. A daemon-minted, non-forgeable identity bound to a non-re-pickable credential and a principal (bottom) makes both costly: churn costs more than a clean record (Thm. III.6.2), and sanctions survive respawns (Thm. III.6.1). The newcomer policy is a shape, not a scalar: a fresh id may have full *work* ability but a reduced *economic ceiling* until it earns a witnessed history (Friedman–Resnick). This is the one genuinely architectural piece of the program.

III.6.3 The principal above the actor

The economic counterparty in every trade is not the agent but the **principal** — the human or organization that owns the fleet and stands behind its bonds. Bonds, reputation inheritance, and

sanctions must ultimately key on the principal via the delegation chain, or Sybil bond-farming works by spawning fresh *agents* under one principal. This paper defines the principal as the delegation chain's terminus and the identity object on which the market paper's counterparty is built; the market paper *cross-references* this definition rather than redefining it.

Definition III.6.2 (Principal)**.** A **principal** is the root of a delegation chain: the durable economic entity (human/org) that owns a fleet, posts its bonds, and inherits the reputational consequence of its agents' witnessed outcomes. Each agent's non-forgeable identity is bound to exactly one principal at mint time.

Exercises

Check. (1) State Property III.6.1 in your own words and give the two-attack proof sketch.
Trace. (2) Why is a "newcomer policy as a shape" (full work, reduced ceiling) strictly better than "newcomers are distrusted" (reduced work)? Consider the cost to an honest newcomer vs. a whitewasher.
⋆ *Open.* (3) Bond-farming: a single principal spawns 1000 agents, has them trade trivial tasks among themselves to mint reputation, then sells the "experienced" agents. Which of {principal binding, listing fee, sampled adversarial re-audit of high-velocity counterparty pairs} defeats this, and at what false-positive cost to honest high-volume pairs?

III.7 The keystone split: local identity vs. cross-operator attestation

The non-forgeable identity of §III.6 is now the highest-leverage *partial local* keystone of the program. The cross-operator stone remains unbuilt. The keystone is *two stones*, and conflating them is the single most consequential error the series can make.

Definition III.7.1 (Local non-forgeable identity)**.** A **local non-forgeable identity** provides an unforgeable actor identity *within one trusted daemon*: the runtime issues and binds it, and no actor inside the fleet can re-pick or impersonate it. Its threat model is explicitly intra-fleet and single-operator; it is *not* a public-key infrastructure and offers no defense against a hostile *operator*. The reference implementation's daemon-minted actor-soul and credential satisfy a bounded slice of this definition; full write-boundary enforcement and migration of legacy identity paths remain. This is exactly what the single-operator layers—Chapters I–III—need and assume. PARTIAL.

Definition III.7.2 (Cross-operator attestation)**. Cross-operator attestation** is the *unsolved* problem of binding identity keys across harbors you do *not* own: an actual key-distribution and attestation story between mutually-distrusting operators. A federated witness log gives log-*integrity*, not identity-*binding* — it can prove an entry was recorded, but not that the key behind it is who it claims to be across a trust boundary. This is what the market paper needs and does *not* yet have. *proposed* (a named gap, owed to the market paper).

> *This paper depends on a* local non-forgeable identity *and* hands the market paper *the unsolved* cross-operator attestation *as a named dependency — not an assumption. Every cross-operator sentence that says "the boundary neither operator controls" is resting on cross-operator attestation; until that exists, it is resting on a single trusted root, and we say so.*

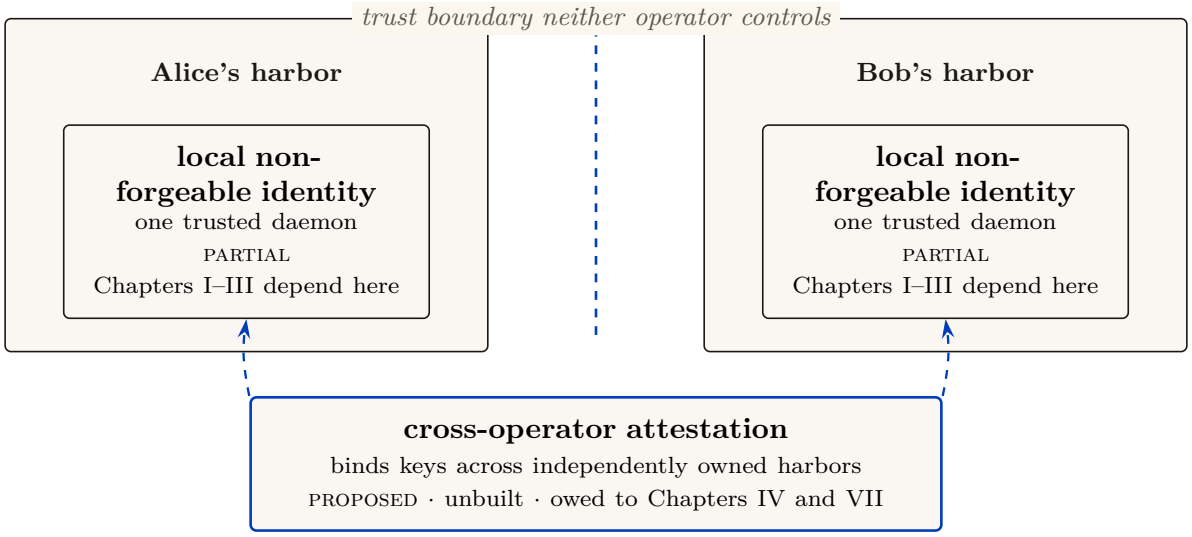


Figure III.7: The keystone is two stones. A *local non-forgeable identity* secures identity *within one trusted daemon* — exactly what Chapters I–III need and assume. The cross-operator boundary needs *cross-operator attestation*, which is unwritten — a federated witness log gives log-*integrity*, not identity-*binding*. The reference system has laid part of the local stone (daemon-minted `actor-souls` and a bounded newcomer pool), but full write-boundary enforcement remains. This paper depends on that partial local stone and hands the market chapters the unbuilt one *by name*, rather than borrowing the single-operator threat model across the boundary.

> **Key idea.** The keystone that holds up *this* paper (identity inside one trusted daemon) is *not* the keystone that holds up the cross-operator market. This paper stands on a local non-forgeable identity. The market paper needs cross-operator attestation and must declare it *proposed* rather than borrow the single-operator threat model. This is the clean boundary the series' cross-layer review demanded.

The second stone's envelope has been standardized — the stone has not. While this series was being written, the agentic-payments industry shipped the credential dialect the second stone will be carved in: the Agent Payments Protocol [4], adopted by the Universal Commerce Protocol [199] (Google/Shopify, 2026), expresses principal authorization as W3C Verifiable Credentials in SD-JWT form — SD-JWT+kb with hardware-bindable ES256 keys for the principal, JWS detached-content signatures over JCS-canonicalized payloads for the counterparty, keys published as JWK sets under a `/.well-known` profile. The reference implementation adopts this envelope at the harbor boundary (ADR-0094), and the honesty rider matters here more than anywhere: *a standard format shrinks the problem; it does not solve it.* What the profile buys is that a principal mandate verifies with off-the-shelf tooling, that principal keys can live in secure enclaves, and that key *distribution* reduces to fetching a cacheable HTTPS document. What it does not buy is the binding of Definition III.7.2: a JWK set authenticates keys to an *origin*, not to a principal, and origin-binding across mutually-distrusting operators is exactly the unsolved attestation problem — exercise (3)'s "shared root of trust" in its 2026 industrial form. Cross-operator attestation remains *proposed*; it is now *proposed* with a standardized serialization, which is progress of the unglamorous kind.

Exercises

Check. (1) In one sentence each, state what a local non-forgeable identity provides and what cross-operator attestation would have to add.
Trace. (2) Take any claim in the market paper of the form "Alice's harbor and Bob's harbor settle across a boundary neither controls." Which stone does it secretly assume? What is the actual security guarantee if only the local non-forgeable identity exists?
⋆ *Open.* (3) Can a federated witness log ever substitute for identity-binding? Argue why log-integrity ≠ key-binding, then sketch the minimal additional assumption (a shared root of trust? a sponsor chain? a transparency log) that would close the gap, and the cost each imposes.

III.8 Reputation as a substrate, not a score

Here is the hinge of the whole series, stated as the canonical cross-paper sentence:

> *The reputation* estimator is cheap*; the* substrate it scores over — *witnessed outcomes on a non-forgeable identity — is the gate.*

A team tempted to start an agent economy by "building an Elo leaderboard for backends" would be building the cheap, last part first. Elo, Bradley–Terry, and TrueSkill are well-understood; you can implement any of them in an afternoon over a table of game results. The hard, prior, load-bearing work is the three links *beneath* the estimator: a non-forgeable identity (§III.6), continuity (§III.5), and an *oracle-witnessed* outcome ledger (Def. III.5.1). Without those, the estimator scores noise — or worse, scores a number an adversary controls.

III.8.1 The estimator family (the well-understood part)

We catalog the estimators not because they are hard but because choosing the *wrong* one for the signal type is a real error.

- **Pairwise / tournament signal → Bradley–Terry / Elo / TrueSkill.** When the witnessed signal is "A beat B on this task," the **Bradley–Terry** model [171] is the full-history MLE, and **Elo** [23] is its online special case. A harbor that retains the *whole* history of every settled outcome lives in the Bradley–Terry regime, not the streaming-Elo regime; BT gives more stable ratings than Elo's recency-weighted update when the population is static. **TrueSkill** [170] adds a calibrated uncertainty σ — the principled way to say "we have not tested this backend much yet," which matters at cold start.
- **Trust-propagation signal → EigenTrust.** When the signal is "who does the network trust," **EigenTrust** [184] computes a global trust vector as the principal eigenvector of the local-trust matrix — the reputation-systems analogue of PageRank, and the canonical answer to "aggregate peer opinions while resisting collusive cliques." It is the right tool for the *federated* reputation Chapter IV needs (trust that travels across harbors), and it is *not* a bandit.
- **Marketplace feedback signal → feedback systems.** Resnick–Zeckhauser [159] measured a ∼8% reputation price premium on eBay; Tadelis [189] frames feedback systems as the platform's answer to adverse selection. This is the empirical evidence that the third side of the market has real value to price.

Figure III.8 maps each signal type to its matching estimator.

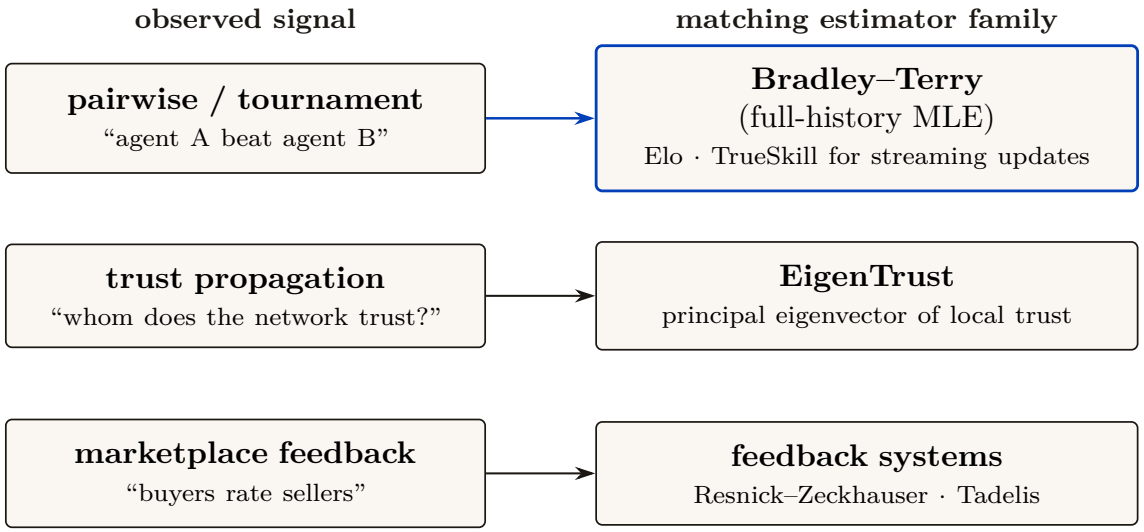


Figure III.8: The estimator family, by signal type — the wrong choice is a real error. Pairwise outcomes call for Bradley–Terry / Elo / TrueSkill; network trust calls for EigenTrust (PageRank for reputation — *not* a bandit); marketplace feedback calls for feedback systems with measured price premia (Resnick–Zeckhauser's ∼8%; Tadelis on adverse selection). A SQLite harbor holds the *whole* settled-outcome history, so it is in the Bradley–Terry regime, not the streaming-Elo regime. None of these is the hard part: the estimator is *cheap and last*, while the substrate beneath — non-forgeable identity, continuity, an oracle-witnessed ledger — is the gate.

III.8.2 Score as telemetry; gate on predicates

> **Key idea.** Publish the score as *telemetry* (a signal the operator and the buyer can read), but *gate* on *predicates* (machine-checkable witnessed facts: "passed the acceptance test," "no slashed bond in the last k settlements"). The instant the score itself becomes the gate, it becomes a Goodhart target [138]: "when a measure becomes a target it ceases to be a good measure."

Exercises

Check. (1) Why is a full-history harbor in the Bradley–Terry regime rather than the streaming-Elo regime?
Trace. (2) You have one task with a clear binary acceptance test, and another judged only by a human's aesthetic preference. Which estimator fits each, and why does EigenTrust fit neither directly?
⋆ *Open.* (3) The substrate-vs-score sentence says the estimator is "cheap and last." Construct a scenario where a *better estimator* still buys nothing because the substrate is rotten (e.g. forgeable identity or an agent-authored oracle). Generalize the lesson.

III.9 Why reputation is *not* a bandit problem

It is tempting — and the routing literature invites the temptation — to model "which agent, model, or skill should I pick?" as a **multi-armed bandit** [195]: arms are agents, pulling an arm yields a reward, and Thompson sampling [205] or UCB balances exploration against exploitation. That framing is correct for one narrow internal task and catastrophically wrong for reputation as a whole. The distinction is load-bearing enough to state as a claim.

> Internal routing — *one operator privately choosing which of their own agents to spawn next — is bandit-shaped.* Reputation — *a public, adversarial, multi-dimensional, persistent substrate other parties trade on — is not. Modeling reputation as a bandit imports four assumptions that are all false in a market.*

Figure III.9 lines the four bandit assumptions up against the market reality that breaks each.

Bandit assumes...	**...but reputation is...**	**The right tool instead**
Stationary arm rewards	**Non-stationary**: models upgrade, skills version, reputation bubbles build & collapse (Liu–Skrzypacz)	bounded-memory reputation dynamics; version-binding
One scalar reward	**Multi-dimensional**: accuracy / aesthetics / efficiency, in tension, buyer-weighted	a quality *vector* + buyer preference (§III.10)
Non-strategic environment	**Strategically adversarial**: Goodhart, wash-trading, whitewashing, collusion	mechanism design (Nisan et al.), bonds & slashing
Observed reward signal	**Produced by a capturable oracle** with its own incentives	bond-and-slash judges; rate-the-raters (§III.11)

What survives the bandit framing: cold-start *exploration*. TrueSkill's σ and a UCB bonus are the same idea — "test the under-measured arm." So *one operator's private routing* (choosing which of *their own* agents to spawn next) is legitimately bandit-shaped. The error is exporting that frame to the *public substrate*, where all four breaks bite at once.

Figure III.9: Why reputation is not a bandit problem. A multi-armed bandit imports four assumptions — stationarity, a scalar reward, a non-strategic environment, and an observed reward signal — all false for a public reputation substrate. Exploration (one operator's private routing) is the one place the bandit framing legitimately survives.

III.9.1 The four broken assumptions

1. **Stationarity.** Bandit regret bounds assume the reward distribution of each arm is (near-)stationary. Agent quality is *non-stationary* by construction: models are upgraded, skills are versioned, prompts drift, and a skill's reputation built on v1 says nothing about v2. The Liu–Skrzypacz line of work on *reputation bubbles* [167] shows that with bounded memory and non-stationary types, reputation can build and *collapse* in cycles — behavior a stationary-arm bandit cannot represent.

2. **One scalar reward.** A bandit optimizes a single scalar. Quality is *multi-dimensional*: accuracy, aesthetics, and efficiency are different axes, often in tension (the fast cheap answer is rarely the most beautiful), and a buyer's weighting of them is a *preference vector*, not a constant. Collapse them into one scalar and you have re-created the Goodhart target the substrate argument warned against (§III.10).

3. **A non-strategic environment.** Bandit arms do not *try* to look good. Agents and their principals are *strategic adversaries*: they will Goodhart the measured axis, wash-trade fake outcomes, whitewash a bad record, and collude. A reputation mechanism is a *game*, and the relevant tool is mechanism design [156], not regret minimization.

4. **A trusted reward signal.** The deepest break: a bandit *observes* its reward. In a market, the reward (the grade) is *produced* by an evaluator who has their own incentives. The grading oracle's incentive-compatibility is not a side-issue — it is a hole in the core theorem (§III.11). A bandit framing simply assumes the hole away.

III.9.2 What survives the bandit framing

To be fair to the bandit literature: cold-start *exploration* is real, and the right place to borrow from bandits is the *exploration bonus*. TrueSkill's σ and a UCB-style bonus are the same idea — "test the under-measured arm." So *within one operator's private routing*, a contextual bandit conditioned on the operator's cost/quality preference vector is a fine engineering choice. The error is exporting that frame to the *public substrate*, where non-stationarity, multi-dimensionality, strategy, and an untrusted reward signal all bite at once.

> **Pitfall.** "We'll just run Thompson sampling over the agents" is the most common way an agent-economy design quietly assumes its hardest problems away. It assumes a trusted reward (no judge-capture), a scalar reward (no aesthetics-vs-efficiency tension), a non-strategic environment (no Goodhart, no wash-trading), and stationarity (no model upgrades). All four are false in a market.

Exercises

Check. (1) Name the four bandit assumptions and the market reality that breaks each.
Trace. (2) Identify the *one* place a bandit *is* the right tool in this paper, and explain why it survives the four objections there.
⋆ *Open.* (3) Chiu–Zhang–van der Schaar [204] show competitive agents over-invest in self-improvement, burning surplus despite individual rationality. Does capping the reputation signal's marginal return restore efficiency, or merely relocate the waste? This is the self-improvement arms-race open problem.

III.10 The multi-dimensional reputation protocol

We now define the central protocol contribution of this paper: a **multi-dimensional reputation** built over witnessed outcomes. It has three parts: a multi-dimensional quality vector; a market of neutral, conflict-free, influence-free evaluators; and a skill→agent helpfulness attribution. We state it formally as a design (SPECIFIED-to-*proposed*): the reference system has no reputation component yet, only the witnessed-outcome substrate the protocol scores over.

III.10.1 Multi-dimensional quality (different axes, different judges)

Definition III.10.1 (Quality vector)**.** A witnessed outcome carries a **quality vector** over three axes:

$$\mathbf{q} = (q_{\text{acc}}, q_{\text{aes}}, q_{\text{eff}}).$$

The axes are *accuracy* (did it do the right thing, against an oracle), *aesthetics* (is the artifact good — code clarity, design taste — judged by an expert), and *efficiency* (tokens, wall-clock, dollars, drawn from a cost meter that the reference system already records, **implemented**). Each axis is scored by a *different judge*, because the competent judge of accuracy (a test oracle) is not the competent judge of aesthetics (a senior reviewer) is not the competent judge of efficiency (a meter).

The buyer reads the *vector* and applies their own weighting (the operator's "cheap vs. careful" preference). Publishing a vector and letting the buyer weight it is the natural disclosure form — and it is precisely what avoids re-collapsing quality into one Goodhart-able scalar. The open design question of *whether* a vector disclosure lets agents Goodhart the most-weighted axis is named as a starred exercise.

Figure III.10 draws the three axes and their distinct judges.

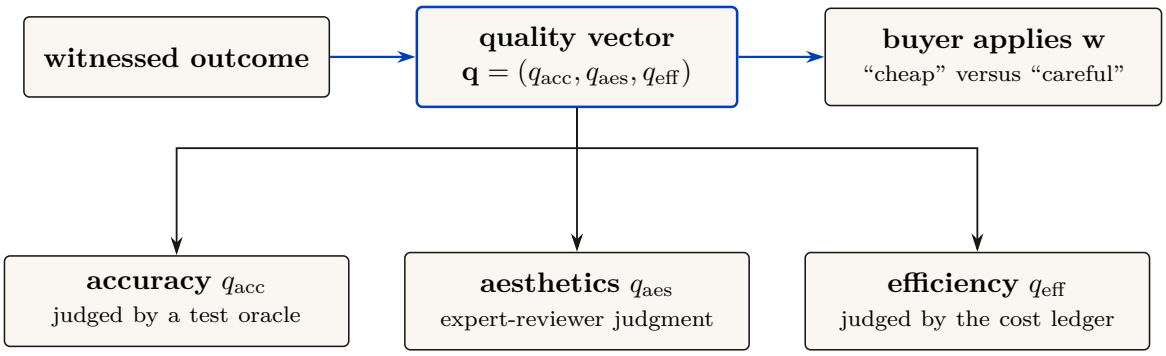


Figure III.10: Multi-dimensional reputation. A witnessed outcome carries a quality *vector* $\mathbf{q} = (q_{\text{acc}}, q_{\text{aes}}, q_{\text{eff}})$. Each axis is scored by a *different* competent judge: a test oracle for accuracy, a senior reviewer for aesthetics, the cost meter for efficiency. The buyer reads the vector and applies their own weights — which is exactly what avoids collapsing quality into one Goodhart-able scalar.

III.10.2 The neutral-judge market (the harbor's universities and rating agencies)

A judge that grades the work of a party it has a stake in is not a judge. The metaphor the series uses is exact: a reputation needs the equivalent of *universities* (which certify competence) and *rating agencies* (which rate counterparties) — third parties whose value is precisely that they are *neutral*. We make "neutral" a checklist, not a vibe:

- **Conflict-free.** The judge has no bond, stake, or principal-link in the work being graded. (Excludes self-grading and same-principal grading.)
- **Influence-free.** The judge cannot be paid *by the graded party* for a better grade. Funding is the hard part: a paid judge has a client, an unpaid judge does not scale. The resolution (§III.11) is that judges post their *own* bond and carry their *own* reputation.
- **De-biased, for LLM judges.** An LLM-as-judge [128] carries position, verbosity, and *self-enhancement* bias (a backend rates its own family up). The discipline is blind, order-swap, pairwise, and family-exclude scoring — never let a backend judge its own family unblinded.

Crucially, an expert third-party judge is *hired through the same bonded ceremony the market uses for any other task*: judging is itself a planned, bonded job, so the harbor that runs the market also runs the hiring of the judges who keep it honest. We state the hiring as a numbered protocol.

Protocol III.10.1. The judge-hiring ceremony

To grade a settled outcome o produced by actor a under principal p:

1. **Eligibility filter.** The daemon assembles the candidate judge pool and removes every judge sharing a principal-link, bond, or stake with p (conflict-free), and every judge in a's model family for the aesthetics axis (de-biasing). *Capability ≠ permission*: a candidate may be *able* to grade and still be *ineligible*.
2. **Selection.** A judge J is drawn from the eligible pool by a public, conflict-free policy (e.g. reputation-weighted sampling on the judging axis), so neither a nor p can choose their own grader.
3. **Bond post.** J posts a bond B_J before seeing the work. The bond is recorded in the outcome ledger and locked for the dispute window.
4. **Blind grading.** J scores the relevant axis under the de-biasing discipline (blind, order-swapped, pairwise where possible). The grade enters the ledger as a witnessed event keyed on J's non-forgeable identity.
5. **Settlement and exposure.** The outcome settles on the grade; the bond flow follows. J's bond remains exposed to a sampled **re-audit** (§III.11): a fresh, conflict-free judge who overturns J's grade *slashes* B_J and updates J's own reputation downward.

Figure III.11 draws this: graded work flows to a judge selected for conflict-/influence-freedom, the judge posts a bond, the grade enters the outcome ledger, and a later re-audit can slash a judge whose grade is overturned.

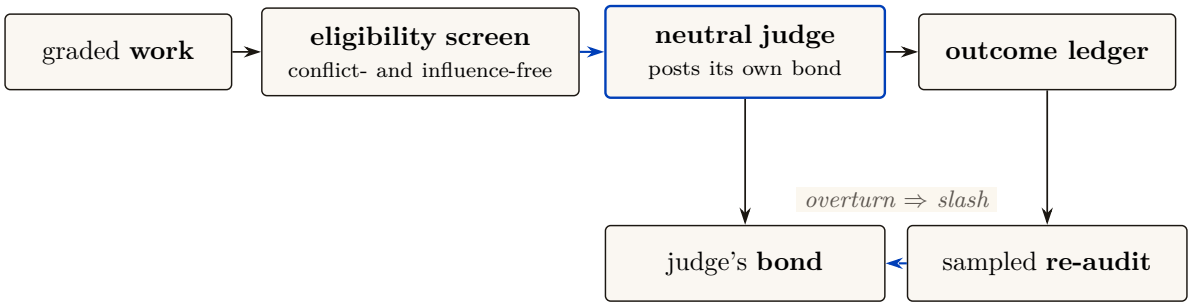


Figure III.11: The neutral-judge market. Graded work flows to a judge selected for conflict- and influence-freedom; the judge posts its own bond, the grade enters the outcome ledger, and a sampled re-audit can slash a judge whose grade is overturned. "Neutral" is a checklist (conflict-free, influence-free, de-biased), not a vibe, and the judge is hired through the same bonded ceremony the market uses for any task (Protocol III.10.1) — the judge's own reputation is the asset it protects.

III.10.3 S3 — a graded job, end to end

Make Protocol III.10.1 concrete. Bob has rented Alice's agent *Heron* to refactor an authentication module against a fixed acceptance suite, under a posted bond. Heron delivers; the outcome o enters Bob's harbor as a *settled but ungraded* delivery, and the daemon runs the ceremony. *Three* judges are hired, one per axis, because the axes need different competence:

- **Accuracy** is graded by a *test oracle* — the acceptance suite Heron cannot author — which returns $q_{\text{acc}} = 1.0$: every test passes.
- **Aesthetics** is graded by a hired senior-reviewer judge, *Mara* (drawn from the eligible pool, in neither Alice's nor Bob's principal, not in Heron's model family). Mara posts her bond, reads the diff blind, and returns $q_{\text{aes}} = 0.7$: correct but over-clever, with a fragile regex she flags.
- **Efficiency** is graded by the cost meter: $q_{\text{eff}} = 0.9$, cheap and fast.

The quality vector $\mathbf{q} = (1.0, 0.7, 0.9)$ lands in Heron's outcome ledger. Bob, who weights aesthetics heavily for security-critical code, reads the *vector* (not a collapsed scalar), sees the 0.7, and decides to keep Heron but require a second reviewer on auth work — a decision the scalar would have hidden. Two weeks later a routine re-audit re-grades a sample of Mara's recent jobs and finds she rubber-stamped a different agent's insecure change as $q_{\text{aes}} = 0.9$. The re-auditor overturns that grade; Mara's bond on *that* job is slashed and her judging reputation drops — exactly the exposure step of Protocol III.10.1. Note what made every step possible: each grade is keyed on a non-forgeable identity, so neither Heron nor Mara can shed the record by respawning.

III.10.4 Skill → agent helpfulness attribution

A distinctive, tractable, and *shippable-soon* signal: track whether a **skill** — a reusable unit of agent know-how (a structured instruction document, its referenced material, and any scripts it carries) — actually *helped* an agent, *especially when that skill's content was loaded into the agent's context window*. The witnessed event is concrete: skill s was grafted into agent a's context at task t; t later settled with quality vector $\mathbf{q}$. Over many such events we estimate a *skill-helpfulness* effect — did loading s raise $\mathbf{q}$ relative to comparable tasks without it? This is reputation for *tools*, keyed on witnessed, context-attributable outcomes, and it is the licensing side's substrate that the market paper consumes. It is also the cleanest near-term instance of the whole thesis: a non-forgeable event (the graft), a witnessed outcome (the settled task), and a multi-dimensional grade — no bandit required. We pin the event down as a schema.

Protocol III.10.2. The skill-helpfulness witnessed-event schema

Each graft of a skill into an agent's context emits an append-only event with fields:

- `skill_id`, `skill_version` — the tool and its exact version (a portfolio for v1 says nothing about v2).
- `agent_id` — the *non-forgeable* identity that received the graft, so the event cannot be re-attributed by respawn.
- `task_id`, `task_descriptor` — a content hash and a feature vector of the task, used to assemble a comparison set of similar tasks *without* s.
- `grafted_at`, `context_window_ref` — evidence that s's content actually entered the context (not merely that s was available).
- `outcome_ref`, `quality_vector` — the witnessed settlement and its multi-dimensional grade, written only after the task closes against an oracle.

A buyer estimates helpfulness as the difference in $\mathbf{q}$ between matched tasks with and without s in context — verifiable from the events alone, without trusting the skill's author.

Pitfall. Attribution is not causation. "Skill s was in context and the task succeeded" does not prove s caused the success. The honest estimate needs a comparison set (similar tasks without s, matched on `task_descriptor`) and is still observational. Sell it as *helpfulness evidence*, not *proof of value*, and gate licensing clawbacks on the witnessed green/red settlement, not on the helpfulness estimate alone.

Exercises

Check. (1) For each of accuracy / aesthetics / efficiency, name a competent judge and an *incompetent* one.
Trace. (2) Walk a judging task through the judge-hiring ceremony (Protocol III.10.1): who plans it, who posts the bond, where does the grade land, and what slashes the judge?
Trace. (3) Design the witnessed event for skill-helpfulness. What fields does it need so that a buyer can verify "this skill helped" without trusting the skill's author?
⋆ *Open.* (5) If you publish the 3-vector and buyers weight it, do agents Goodhart the most-weighted axis? If you publish a scalar, you have recreated the bandit framing the substrate argument rejects. What is the right disclosure form for vector reputation?

III.11 The grading oracle's incentive-compatibility

Every folk-theorem incentive-compatibility result in the L3 economy bottoms out in "the daemon derives the grade" — but the grade is produced by a *capturable evaluator*. This is not a side-gap. It is a hole in the *core theorem*, and we treat it head-on rather than assuming it away.

Property III.11.1 (The grading-oracle hole)**.** Let an outcome's settlement (and thus the bond flow and the reputation update) depend on a grade g produced by an evaluator J. If J can be captured — bribed, colluded with, or self-interested — then the incentive-compatibility of *every* mechanism that settles on g is only as strong as J's own honesty. Assuming J honest is assuming the conclusion.

There are two honest ways to close the hole, and a recursion to bound.

Option A — bond-and-slash the judges. Make judging a bonded role (§III.10): J posts a bond before grading; a later **re-audit** (a fresh, conflict-free judge, sampled) that overturns J's grade *slashes* J. Now lying has an expected cost, and a judge's own reputation is the asset they protect. This converts "trust the oracle" into "the oracle has skin in the game," which is the same move slashing makes for agents.

Option B — 2-of-3 multi-oracle. Settle disputes against a quorum of independent oracles (automated test / evidence review / human), so capturing the grade requires capturing a majority. This raises the cost of capture but does not eliminate it, and the *human* oracle is a scarce, expensive, capturable resource in its own right (the economics of arbitration capacity is a named open problem, below).

Algorithmic Mode Collapse and the VRF Honeypot Solution Option A initially pushed the problem up a level: who audits the auditors? The naive recursion "rate the raters who rate the raters" introduced a severe *Algorithmic Mode Collapse* vulnerability, where a monoculture of model architectures could form a cartel of mutual validation. We cure this via two mandated mechanisms that force the recursion to terminate. Figure III.12 draws the recursion and where these two mechanisms are meant to arrest it.

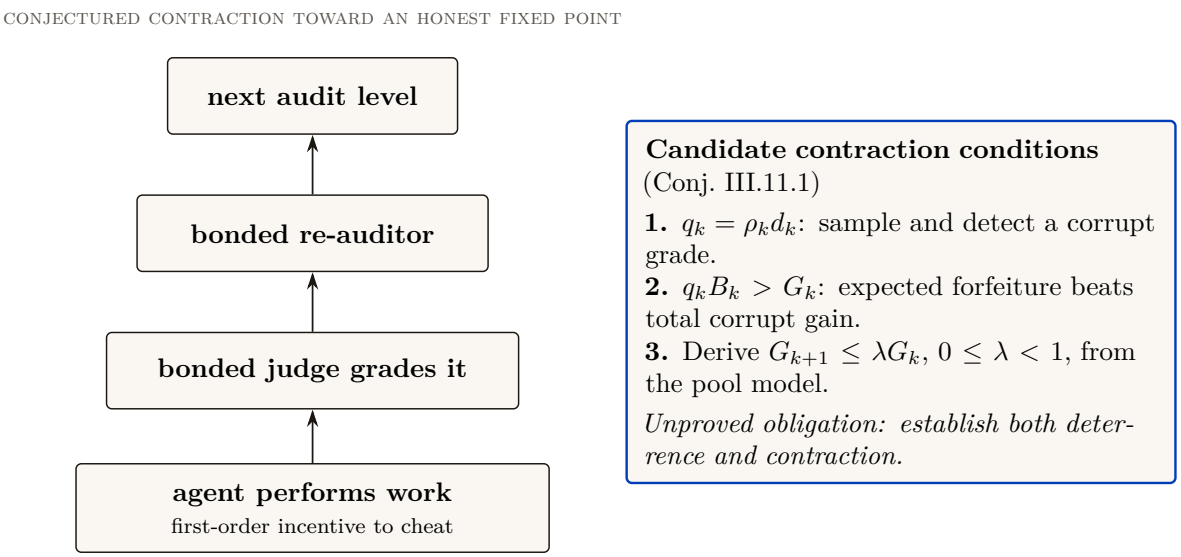


Figure III.12: The rate-the-raters recursion. Conjecture III.11.1 separates two obligations that a recursive audit design must prove: expected forfeiture $\rho_k d_k B_k$ must exceed the judge's total corrupt gain G_k at each level, and the pool model must make the remaining corruption opportunity shrink by a uniform factor $\lambda < 1$. A bond larger than one bribe is not enough when re-audit is probabilistic. Neither obligation is depicted as established.

First, the protocol mandates **Model Heterogeneity** for judge selection. A quorum of re-auditors must prove they represent distinct architectural weights (e.g., Llama-3 vs. Claude vs. Qwen), verified via cryptographic provenance, preventing any single-model alignment flaw from rubber-stamping malicious grades.

Second, the system injects **Honeypot Tasks** via **Verifiable Random Functions (VRFs)**. Let G_k bound the judge's total one-shot gain from a dishonest grade, and let B_k be their slashable bond. The system probabilistically inserts deterministic, pre-graded trap tasks that appear indistinguishable from organic tasks.

Local deterrence requires the expected-forfeiture inequality:

$$P(\text{Honeypot}) > \frac{G_k}{B_k}$$

By pegging the honeypot injection rate above this slash threshold, the expected value of systemic dishonesty becomes strictly negative. The VRF proves to all participants that the injection rate was truly random and untamperable.

Conjecture III.11.1 (Re-Audit Contraction)**.** *Let $q_k = \rho_k d_k$ be the composite probability of sampling and detecting a corrupt grade at audit level k, B_k be the auditor's slashable bond, and G_k be the corrupt gain. If expected forfeiture satisfies $q_k B_k > G_k$ and remaining corruption opportunity contracts as $G_{k+1} \leq \lambda G_k$ with $\lambda \in [0, 1)$, the recursive audit tower contracts to an honest fixed point.*

Key idea. The grading oracle is the load-bearing assumption hidden under every IC claim in the economy. By bounding the dishonesty payoff with a mathematically certain VRF slash, the rate-the-raters recursion collapses: the market enforces its own honesty without relying on infinite re-audits.

Exercises

Check. (1) State Property III.11.1 and explain why "the daemon derives the grade" does not, by itself, close it.
Trace. (2) Walk Option A: a judge grades disformally, a re-audit overturns it. Follow the bond flow and the reputation update for the judge.
⋆ *Open.* (4) Prove or refute Conjecture III.11.1. If the re-audit probability ρ is too low, the tower is not contractive — find the critical $\rho^\star$ as a function of bond size and bribe ceiling.
⋆ *Open.* (10) Arbitration capacity: human oracles are scarce. Design a market for bonded, rated, slashable arbiters and argue whether it converges to honest rulings or to rubber-stamping under load.

III.12 Revocation, tombstones, and bounded memory

A reputation built on witnessed outcomes must answer two questions the cheerful "reputation never ends" framing dodges: what happens when a witnessed outcome turns out to be *fraudulent*, and is no-decay actually *good*? We adopt the series' reconciled position on both, against the naive version.

III.12.1 Monotone in honest outcomes, individually revocable

Property III.12.1 (Reputation monotonicity with tombstones)**.** Reputation is **monotone in *honest* witnessed outcomes** — a genuine delivery is never silently erased by the passage of time — *but* a witnessed outcome is itself **individually revocable via a propagating tombstone**: a compensating, append-only entry that retracts a specific outcome later found fraudulent (a colluding judge, a faked oracle). "Never decays" applies to honest outcomes *only*; it is *not* an anti-revocation property.

This is a **saga**-style compensation [95]: you do not mutate the ledger in place (that would break append-only and auditability); you append a tombstone that nullifies the targeted entry, and the tombstone *propagates* across harbors bounded by the federation's revocation-convergence window (the market paper's job). The poisoned data point was spendable only during the propagation window; bounding that window is the federation's responsibility. We state the retraction as a protocol.

Protocol III.12.1. Tombstone revocation

To retract a witnessed outcome o later found fraudulent:

1. **Trigger.** A re-audit (Protocol III.10.1, step 5) or a dispute overturns the grade on o, establishing fraud against an oracle.
2. **Append, never mutate.** A *tombstone* entry $\bar{o}$ is appended to the ledger, carrying o's id, the overturning evidence, and the identity of the retracting judge. The original o stays in place; auditability is preserved because nothing is erased.
3. **Recompute.** Every reputation score that consumed o is recomputed with o nullified by $\bar{o}$. Because the estimator is cheap (§III.8), recomputation is not the bottleneck.
4. **Propagate.** $\bar{o}$ gossips to every harbor that imported o, bounded by the federation's revocation-convergence window τ. Each harbor applies step 3 on receipt.
5. **Bound the damage.** The fraudulent reputation was spendable only during τ; the protocol's job is to make τ short and provable, not to pretend the window is zero.

Figure III.13 draws the append-only compensation: the original entry stays, and a later tombstone nullifies it without mutating the ledger.

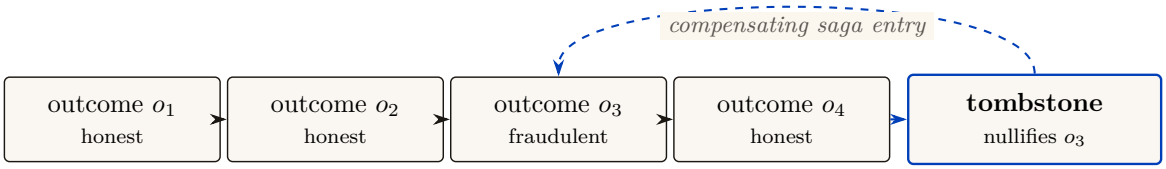


Figure III.13: Tombstone revocation on an append-only ledger (never mutate in place). A fraudulent outcome is not deleted (that would break append-only and auditability); it is nullified by an appended *tombstone* — a saga-style compensating entry — which propagates across harbors bounded by the revocation-convergence window (Chapter IV); the poisoned point was spendable only during that window. "Reputation never ends" is corrected to: monotone in *honest* outcomes (they are never silently erased), individually revocable via tombstone, with bounded memory (Liu–Skrzypacz) as a tunable parameter rather than a virtue — ∞-memory freezes types and removes the incentive to keep performing.

III.12.2 S4 — the fraudulent outcome retracted

Recall S3: the re-audit caught Mara rubber-stamping a different agent's insecure change as $q_{\text{aes}} = 0.9$. Follow that one outcome through Protocol III.12.1. The graded agent — call it *Crane*, running under a third operator — had colluded with Mara: Crane shipped a known-insecure diff and Mara graded it high so Crane's aesthetics reputation would clear Bob's hiring bar. For two days it worked: Crane's vector showed (0.95, 0.9, 0.8) and Bob nearly hired it. Then the sampled re-audit re-graded the diff blind, found the injection, and overturned Mara's grade. A tombstone $\bar{o}$ is *appended* (the original stays, for the audit trail); every score that consumed Crane's inflated q_{aes} is recomputed with it nullified; Mara's bond on that job is slashed and her judging reputation drops; Crane's collusion is itself recorded as a fraudulent outcome against *its* non-forgeable identity, which it cannot shed. The tombstone gossips to Bob's harbor within the convergence window, so by the time Bob next reads Crane's vector the 0.9 is gone. The fraud was real and briefly spendable — the honest claim is that the window was *bounded*, not that it never existed.

III.12.3 Bounded memory is a design parameter, not a virtue to assert away

The naive "no decay window an agent can outrun" framing sounds principled but is mechanism-design-hostile. Liu–Skrzypacz [167] show that with *bounded* memory, reputation can be welfare-*improving* relative to infinite memory — and that infinite memory (∞) is rarely optimal, because it freezes types and removes the incentive to keep performing. Bounded memory is a **tunable parameter**, and the right setting is an empirical/mechanism-design question, not a slogan.

> **Pitfall.** "Reputation never ends" as a *virtue* is wrong twice over: it makes a fraudulent outcome permanent (needs the tombstone of Property III.12.1), and it asserts ∞-memory is optimal when the reputation-bubble literature shows it usually is not. Engage bounded memory as a *parameter*; do not assert no-decay as a feature.

> **Key idea.** Two corrections to the cheerful framing: (1) reputation is monotone in *honest* outcomes but *individually revocable* via a propagating tombstone — a saga compensation, not in-place mutation; (2) bounded memory is a *design parameter* (Liu–Skrzypacz reputation bubbles), not an anti-feature to assert away.

Exercises

Check. (1) Why must revocation be a tombstone (append) rather than a delete (mutate) on the outcome ledger?
Trace. (2) A judge is later found to have colluded on five settled outcomes. Walk the tombstone for one of them: what is appended, what propagates, and what bounds the window the poisoned reputation was spendable?
⋆ *Open.* (8) Reputation-claim revocation with an append-only, bounded-memory ledger across N harbors: how do you prove the correction converged and bound the spendable window, without violating append-only? This is the cross-harbor analogue of capability revocation (Chapter IV).

III.13 What this chapter hands the market (Chapter IV)

This paper is the bridge; its output is a precise hand-off to *The Harbor Economy* (Chapter IV). We summarize what is delivered, what is borrowed, and what is explicitly owed.

Artifact	**Maturity**	**Role in the market paper**
Role / person distinction (Defs. III.3.1–III.3.2)	—	The third side of the market is a *tradeable person*, defined here.
Principal-as-economic-entity (Def. III.6.2)	SPECIFIED	Bonds and sanctions key on the principal; the market paper cross-references, not redefines.
	implemented	
The three continuity organs	PARTIAL PARTIAL	Reputation keys on organ 3 (the outcome ledger).
Outcome ledger / oracle (Def. III.5.1)	PARTIAL	Commitment, deadline, oracle closure, API/CLI, and overdue detection ship; neutral graded outcome events, sanctions, and reputation binding do not.
Multi-dimensional reputation (Def. III.10.1, Protos. III.10.1, III.10.2)	SPECIFIED to *proposed*	The multi-dimensional score the market reads; the licensing side's signal.
Neutral-judge market	SPECIFIED to *proposed*	The "universities and rating agencies" that keep grades honest.
Grading-oracle hole (Prop. III.11.1, Conj. III.11.1)	*proposed*	*Owed back to the market paper*: its central IC theorem is conditional on closing this.
Cross-operator attestation (Def. III.7.2)	*proposed*	*Named dependency the market paper must build*; not assumed here.
Tombstone revocation (Prop. III.12.1)	SPECIFIED	Reputation is revocable; convergence is the market paper's federation job.

Table III.4: The bridge's hand-off to the market paper. Two items flow *back* as named, unsolved obligations: the grading-oracle's incentive-compatibility (Conj. III.11.1) and cross-operator attestation (Def. III.7.2). The series is coherent precisely because these debts are named in the same words on both sides of the seam.

> *Build the harbor first. The trade comes when the ships do — but the thing that makes the trade possible (continuity that turns spawns into persons) is the same thing that makes the single-player tool good. The bridge is not a detour; it is the load path.*

III.14 Open problems (the starred exercises, collected)

The genuinely unsolved problems this paper surfaces, gathered for the reader who wants the research frontier rather than the pedagogy:

1. **Branching continuity (§III.4).** If a checkpoint is forked into two live incarnations, which inherits the reputation? Parfit says identity-is-not-what-matters; the economy needs a rule, and the rule invites an attack.
2. **Agent-death taxonomy (§III.5).** What is recoverable vs. fundamentally lost when an LLM agent dies mid-thought? Solved via **Event-Sourced Neural Rehydration** (OP-4).
3. **Cross-operator attestation (§III.7).** Binding identity keys across harbors you do not own. The highest-leverage unbuilt keystone for the market.
4. **The grading-oracle / rate-the-raters recursion (§III.11, Conj. III.11.1).** Prove the re-audit tower contracts, or accept the core IC theorem is conditional.
5. **Vector-reputation disclosure (§III.10).** Publish a vector and agents Goodhart the heaviest axis; publish a scalar and you have a bandit. The right disclosure form is open.
6. **Self-improvement arms race (§III.9).** Chiu et al. 2025: does capping reputation's marginal return restore efficiency or relocate the waste?
7. **Reputation-claim revocation convergence (§III.12).** Surgically retract a fraudulent outcome across N harbors, prove convergence, bound the spendable window — without violating append-only.
8. **Bond-farming defense (§III.6).** Principal binding vs. listing fee vs. sampled re-audit of high-velocity pairs; the false-positive cost to honest high-volume counterparties.
9. **Arbitration capacity (§III.11).** A market for bonded, rated, slashable human/automated arbiters: honest rulings, or rubber-stamping under load?
10. **Skill-version reputation (§III.10).** A portfolio proof for skill v1 says nothing about v2; the lemons problem reappears at every version bump.

III.15 Conclusion

We set out to bridge the wedge to the market by answering one question: what has to be true before a coding agent's work can be *sold*? The answer is not a score. It is a *self* — a role plus continuity, keyed on an identity that cannot be re-picked, accumulating a transitive, witnessed history that survives every respawn. The estimator that turns that history into a number is the cheap, last part; the substrate beneath it is the gate, and most of that substrate is formally not built yet.

We were disciplined about the keystone (this paper stands on a local non-forgeable identity and hands the market paper the unbuilt cross-operator attestation by name), about the hardest hole (the grading oracle's incentive-compatibility is a flaw in the core theorem, not a footnote), and about the cheerful slogans (reputation is monotone in *honest* outcomes but individually revocable, and bounded memory is a parameter, not a virtue). Reputation is not a bandit problem — it is non-stationary, multi-dimensional, strategically adversarial, and graded by an oracle with its own incentives.

> *A spawn does work and disappears. A person does work and* stays *— and only what stays can be priced. The harbor's whole economy is the machinery for manufacturing, and then formally grading, a self.*

Chapter Appendices

III.A Implementation status of the reference system

The body of this paper is written to stand on its own: every mechanism is described as a component (a memory record, a checkpoint organ, a witnessed-outcome ledger) and graded on the neutral maturity scale (**implemented** / PARTIAL / SPECIFIED / *proposed*). This appendix grounds that scale in a concrete *reference system* — a running coordination daemon for local agent swarms — so a reader who wants to know "which of these actually runs today" can check. Nothing in the body depends on this appendix; it exists so the maturity labels are falsifiable rather than rhetorical.

Component (paper concept)	**Maturity**	**Status in the reference system**
Episodic memory (Organ 1)	**implemented**	Durable typed episodes promoted from execution history; runs in production.
Restart organ (Organ 2)	PARTIAL	Heartbeat-staleness detector that salvages dead agents; forwards a *summary*, not execution state.
Outcome ledger (Organ 3)	PARTIAL	Durable commitments, daemon-derived deadlines, oracle-bound closure, API/CLI, and overdue detection ship; neutral grading, sanctions, and reputation binding do not.
Cost meter (efficiency axis)	**implemented**	Per-task token/wall-clock/dollar accounting recorded today.
Local non-forgeable identity	PARTIAL	Daemon-minted actor-soul, lookup credential, and bounded newcomer pool ship; full write-boundary enforcement and legacy migration do not.
Capability jail (capability ≠ permission)	SPECIFIED	Per-agent tool-allowlist and scoped filesystem; specified.
Honest-attestation discipline	**implemented**	"Report all-good only when verified" is an enforced engineering rule.
Multi-dimensional reputation	SPECIFIED to *proposed*	No reputation component yet; the substrate it scores over is partly built.
Neutral-judge ceremony (Proto. III.10.1)	SPECIFIED to *proposed*	Bonding and task-planning primitives exist; the hiring ceremony is designed.
Skill-helpfulness schema (Proto. III.10.2)	SPECIFIED	Skill grafting is observable; the witnessed-event schema is specified.
Tombstone revocation (Proto. III.12.1)	SPECIFIED	Append-only ledger semantics are designed; cross-harbor propagation is *proposed*.
Cross-operator attestation	*proposed*	The unbuilt keystone for the market; owed to Chapter IV (*The Harbor Economy*).

Table III.5: Maturity of each paper concept against a reference coordination daemon. The honest summary: the *organs* of continuity exist (one weakly); the *spine* of reputation — a witnessed-outcome ledger keyed on a non-forgeable identity — is specified but not yet shipped.

PORT DADDY COORDINATION PAPERS · CHAPTER IV OF VII

The Harbor Economy

Abstract

The first three chapters build a harbor for a *single* operator: a daemon, a coordination protocol, a legible authority anyone would pay for today, and the continuity that turns an anonymous *spawn* into an accountable *person*. This paper is the rung above all of them — the only one whose participants are plural and mutually distrusting. We argue that the harbor economy is a **three-sided market**: operators sell labor and fleet-for-hire; agents and fleets are rentable assets; skills and tools are licensed — all settling on *one* conserving bond ledger via float-plan escrow. We present the **cross-harbor capability-transfer ceremony** and the federation that lets fleets on machines you do not own trade without a shared chain: a witness log, an escrow whose extraction bound is conditional on non-bypassable, recipient-whitelisted custody, and revocation gossip with an expected convergence result under a connected, reliable-round model. We carry the series' heaviest reconciliations formally rather than papering over them. The keystone the market leans on is **local non-forgeable identity** — an identity no agent can edit *within a single trusted daemon* — and that primitive covers a single-operator threat model, *not* the cross-operator adversary this market is defined by; the missing, blocking keystone is **cross-operator attestation**, which is specified-to-proposed, not built. Conservation composes under sequential ledger operations because balance change is additive. Cross-currency accounting requires a named valuation instant and explicit fee/slippage accounts; "lax functor" does not repair an undefined unit conversion. The later sheaf language is retained only as a research analogy, not a theorem. Reputation is monotone in honest outcomes but individually revocable by a propagating tombstone — it ends, by design, when an outcome is shown fraudulent. Every folk-theorem incentive-compatibility claim rests on a grading oracle that must be made strategy-proof or bonded-and-slashed. And strict conservation is budget balance. Myerson–Satterthwaite constrains a bilateral private-value slice only under its regularity assumptions; this paper does not yet prove incentive compatibility or identify which desideratum the full market gives up. The defensible product is **hosted trust** — a verified ledger, a relay, and a reputation system — not the commoditized payment rail.

Keywords: mechanism design, multi-sided markets, incentive compatibility, reputation systems, capability-based security, cross-chain settlement, applied category theory, federation, Port Daddy

*Reading time: approximately 45 minutes end-to-end; about 12 minutes for §IV.2–§IV.3 alone (the market thesis). This is Chapter IV of a seven-chapter collection and is written cold: no prior platform knowledge is assumed. It can be read first, but §IV.6 leans on Chapter III (*From Spawn to Person*) for the identity substrate, while the cryptographic hop-bound authorization chain is analyzed in Chapter V (*The Anchor Protocol*).*

Maturity key (honest self-grading). Every claim in this paper carries one of four maturity labels, so the reader always knows whether a sentence reports running code or a designed target. **implemented** — shipped and exercised by tests in the reference implementation. PARTIAL — shipped but materially weaker than its name suggests. SPECIFIED — specified, with a proof or a written protocol, but not yet shipped. *proposed* — a stated direction whose load-bearing part has no specification yet. Nomenclature is likewise fixed: the cryptographic *hop-bound authorization chain* (which proves who delegated what, and bounds it) is never conflated with the *coordination lineage* (who-talked-to-whom, advisory only); **local non-forgeable identity** (an identity no agent can edit within one trusted daemon) is never conflated with **cross-operator attestation** (the unbuilt keystone that binds identity *across* daemons nobody jointly trusts); and *capability* (what an agent *can* do) is never collapsed into *permission* (what it *may* do).

IV.1 Reader's map

This paper is the top rung of a ladder (Figure IV.1). It assumes the rungs below it and cites the papers that build them. Read it cold if you must, but it is the fourth paper for a reason: it depends on everything underneath.

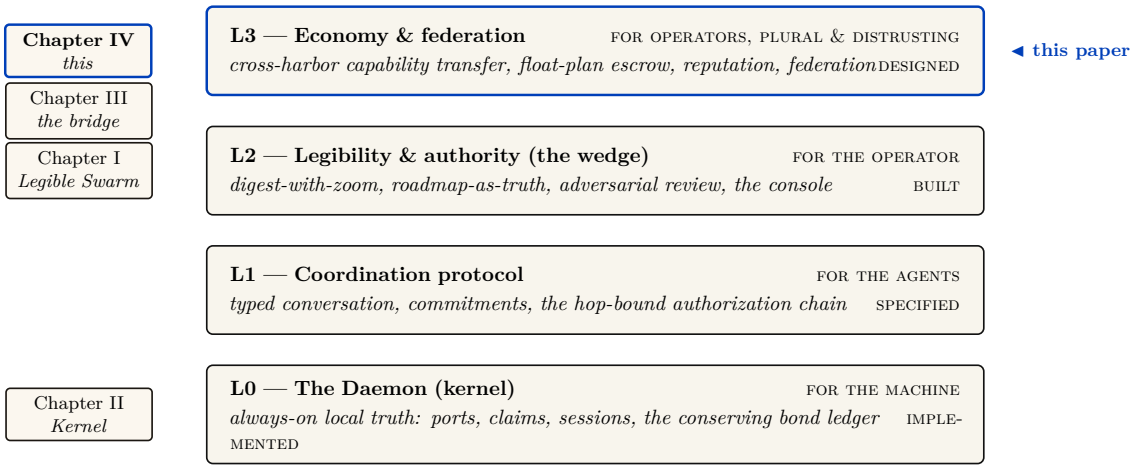


Figure IV.1: Where this paper sits. The L0→L3 stack: each rung serves a different *whom*, and each announces what it assumes from below. Chapters I–III establish legibility, the kernel, and the L3 identity bridge; *The Harbor Economy* is the top rung — the only layer whose participants are plural and mutually distrusting, and it depends on every rung below it. It consumes the *implemented* bond ledger from L0, the cryptographic hop-bound authorization chain from L1, and the legible *person* from Chapter III, and turns them into a market. Chapters V–VII deepen the protocol, incentive, and federation seams; chapter numbers are not layer numbers. The series spine runs up this stack: memory → checkpoint → continuity → a *person* not a *spawn* → registered outcomes → **reputation** → a tradeable asset → **the market** — the same memory that makes the single-player wedge good is what turns a spawn into a tradeable asset.

Table IV.1: Where to enter, by who you are.

If you are...	**...start here**	**...load-bearing artifact**
A founder asking "what is the product"	§IV.2 (thesis) → §IV.11 (hosted trust)	Fig. IV.2; the three claims C1–C3
A mechanism designer	§IV.3 → §IV.10 (the formal model)	Thm. IV.7.2, Fig. IV.11
A cryptographer / distributed-systems reviewer	§IV.6 (the keystone) → §IV.9	Fig. IV.4, Fig. IV.6
A category theorist	§IV.7 → §IV.13	Fig. IV.5, Conj. IV.13.1
An implementer	§IV.4 (the built floor) → Appendix IV.A (status)	Table IV.4

Reading order within the collection. Chapter I (*The Legible Swarm*) is the front door; Chapter II (*The Single-Writer Kernel*) is the local transactional floor; Chapter III (*From Spawn to Person*) supplies continuity and reputation. *This* chapter is the market. Chapters V–VII then deepen the cryptographic, economic, and federation arguments.

Dramatis personae. Every mechanism in this paper is dramatized end-to-end with the same cast, so an abstraction is never introduced without a concrete trade to attach it to. **Alice** runs Harbor A on her laptop and owns a well-reputed refactoring specialist agent, a_2. **Bob** runs Harbor B and needs work done he cannot staff himself. **Carol** runs Harbor C and sells a licensed lint-and-type skill. **Judy** is a human grader who arbitrates disputes for a bond. The two harbors do *not* trust each other and neither is a root of authority for the other; that distrust is the entire reason the rest of the paper exists. We price everything in an abstract unit, the *credit* (CR); read it as a stablecoin cent if you like.

IV.2 The thesis: a market, not a payment rail

A solo operator running a swarm of coding agents needs three things — legibility, accountability, and safety — and needs *no cryptography at all*, because she owns every machine in the harbor. That is the single-player wedge of Chapter I. The instant two operators trade, the world changes. Alice's frigates touch your repository. You must now price work done by agents you did not spawn, owned by a principal you do not trust, using skills you cannot inspect. That is a *market-design* problem before it is a cryptography problem. Cryptography is the substrate that makes the ledger unforgeable; it is not the product.

> *You don't sell crypto — crypto is the substrate; you sell* ***hosted trust****: a verified ledger, a relay, and a reputation system that lets mutually-distrustful fleets transact across a boundary they do not share.*

Three claims follow, and the rest of the paper defends each:

- **C1 (three sides).** The harbor economy has three *distinct incentive constraints* — operator-for-hire, asset-rental, skill-licensing — not two. Counting them correctly changes the pricing (§IV.3).
- **C2 (one ledger, one escrow object).** All three sides settle on the same **float plan** escrowed in the same bond ledger. Conservation of value across every settlement path is the invariant that holds the market together, and it is the one thing here that is actually **implemented** (§IV.4, §IV.7).
- **C3 (hosted trust is the moat).** The defensible asset is the verified ledger + relay + reputation, not the settlement rail, which the 2025–26 agentic-payments stack has already commoditized (§IV.11).

> **Key idea.** The economy is strictly *additive* on top of the single-player tool. Because all three market sides settle on the *same* built bond ledger, none of the economy needs to ship before the wedge does. The discipline "don't chase the market early" is *safe* precisely because the market reuses the kernel primitive. The harbor comes first; the trade comes when the ships do.

IV.2.1 The through-line: why a market needs persons, not spawns

The whole series climbs one spine (the vertical arrow in Figure IV.1):

memory → checkpoint → continuity → a person, not a spawn (Chapter III)
→ registered outcomes → reputation → tradeable asset → market (this chapter).

Sides 2 and 3 of the market — renting an *asset*, licensing a *good* — presuppose that there is a *thing* to rent or sell that cannot be costlessly cloned. You cannot rent a reputation that any spawn can fork, and you cannot license a skill whose track record is unverifiable. The canonical cross-layer sentence, repeated in Chapters III and IV, states the dependency exactly:

> *The score is cheap; the substrate it scores over — witnessed outcomes on a non-forgeable identity — is the gate.*

Exercises

E1. Give a one-sentence reason the market cannot ship before reputation does, using only the word "clone."

E2. Side 1 (operator-for-hire) does *not* require durable identity in the same way sides 2 and 3 do. Why? (Hint: who is the counterparty, and who bears the consequence?)

E3. ★ Construct a market design in which sides 2 and 3 exist *without* a non-forgeable identity. Argue informally why every such design collapses to a Sybil attack (forward-reference §IV.6).

IV.3 What a "side" is, and why there are three

A **two-sided market** (**Rochet & Tirole** [99], the foundational reference: *a platform is two-sided when the allocation of the total price across the two user groups — not just its level — affects transaction volume*) is the canonical frame for a marketplace: buyers on one side, sellers on the other, the platform tuning who is subsidized to ignite cross-side network effects. The naïve reading of the harbor economy is two-sided: a *requester* posts work, a *worker agent* does it.

But "two-sided" undercounts. The operational test [100] is: how many distinct, privately-informed parties must the platform individually rationalize into participating? In a mature harbor there are three, because *the same agent can be three economically different things* (Figure IV.2).

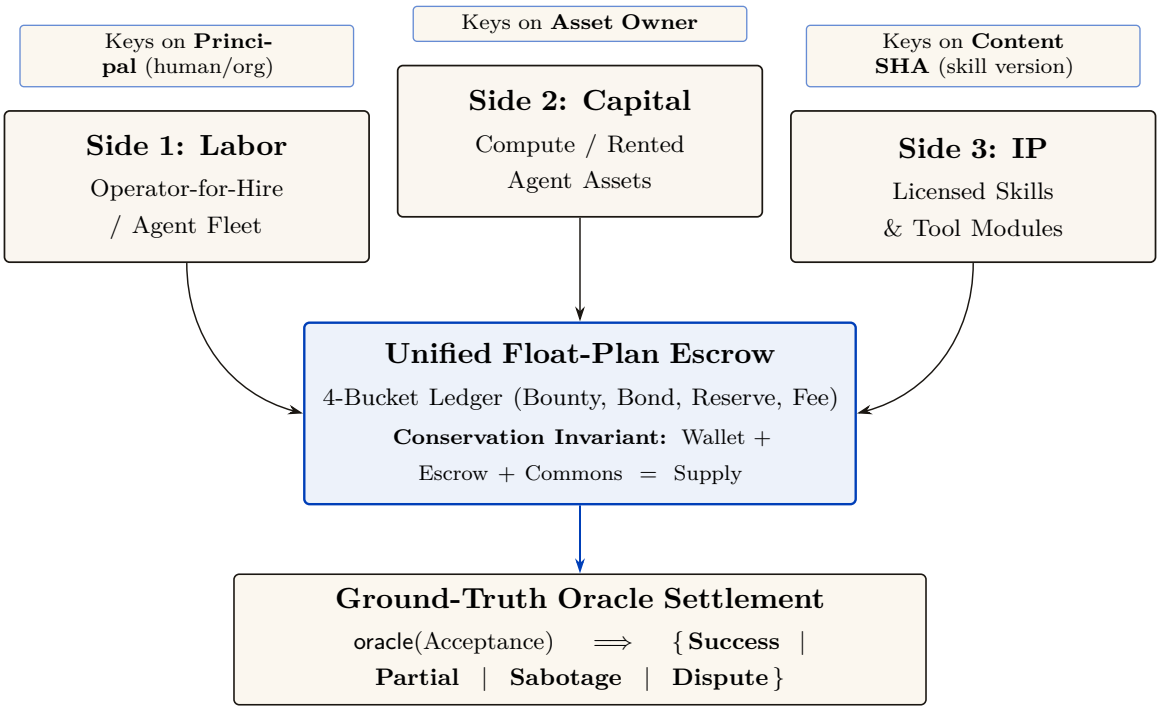


Figure IV.2: The three-sided market. Labor, capital (rented agents), and IP (licensed skills) are three distinct *incentive constraints* — not three UI tabs — that all settle on a *single* float-plan escrow, so conservation holds globally without a second ledger. This is the categorical heart of the design: the three sides are isomorphic at the conservation object (the escrow) and **non-isomorphic at the identity object** (each side keys reputation on a different entity, the dashed band). Calling it "one structure, three decorations" without that qualifier is a decorative-analogy failure. The rider is mandatory: the market is three-sided by design and two-sided in shipped reality, because sides 2 and 3 presuppose a non-forgeable, durable identity (§IV.6). The economics per side: labor's margin is bounty − slashed sub-bonds − fee; capital's renter holds the runtime jail while the owner bears reputation; IP is metered with clawback, paid only on green settlement. The escrow and its conservation invariant are implemented and property-tested.

1. **Operator-for-hire (labor).** An operator who runs a fleet can sell its labor to *another* operator. The operator is now a firm taking work-for-hire, with margin = bounty − Σ(slashed sub-bonds) − ledger fee. Its private information is whether its fleet can deliver. It internalizes its fleet's risk — exactly the property that makes a firm accountable for its employees.

2. **Agent/fleet as rentable asset (capital).** An operator who owns a well-reputed specialist can *rent it out*. Now the asset-owner, the task-poster, and the worker are three different parties with three different incentives. The owner's private information is the agent's true quality; its exposure is reputational.

3. **Skill/tool as licensed good (IP).** A skill can be *licensed* into someone else's float plan — metered or per-use — without the licensor doing any task work. The licensor's private information is the skill's quality, which the buyer cannot inspect before purchase.

These are three *incentive constraints*, not three UI tabs. If the asset owner is always the task poster, side 2 collapses into side 1; the design discipline is to count sides by constraints. The harbor has three because *capability* (an agent) and *competence* (a skill) become separable, tradeable goods once identity is durable.

The same trade, run three ways (the running example). Bob needs a 1,200-line module refactored. He can buy it on any of the three sides, and the side determines who is privately informed and who is exposed. *Side 1 (labor).* Bob posts a float plan with a 400 CR bounty; Alice

takes it as a firm. She dispatches three of her own agents, posts a 60 CR sub-bond per agent (180 CR total), and keeps 400 − (slashed sub-bonds) − fee. If one of her three agents botches its slice and is slashed 60 CR, Alice eats that loss — she internalized her fleet's risk, which is exactly what makes her accountable as a firm. *Side 2 (capital).* Instead Bob rents Alice's specialist a_2 directly for an afternoon at 90 CR. Now there are three parties: Bob (poster), Alice (owner, privately knows a_2's true quality), and a_2 (worker). Bob holds the runtime control of a_2 while it runs on his repository; Alice's exposure is purely reputational — if a_2 underperforms, a_2's public score falls, which is Alice's asset depreciating. *Side 3 (IP).* Carol never touches Bob's repository at all; she licenses her lint-and-type skill into Bob's float plan at 0.5 CR per file, metered, released to Carol *only on green settlement.* Carol is privately informed about her skill's quality; Bob cannot inspect it before he runs it. Three sides, one refactor, three different "who knows what" structures — and, as the next section shows, one escrow object underneath all of them.

> **Pitfall.** Two-sided-market theory's central result is that the *price structure* — who subsidizes whom — is the lever, not the price level. Mis-count the sides and you mis-structure the subsidy: you charge the side you should be *paying* to show up. §IV.12 returns to this for cold start.

The framing is not idiosyncratic. The most recent academic treatment of AI labor markets, **Chiu, Zhang & van der Schaar** (2025) [42] (*a formal model of a three-sided agent labor market*), models exactly agents, employers, and a reputation system — confirming that "reputation system as a third side" is the natural structure once agents compete for paid work.

IV.3.1 The mandatory honesty rider

The canonical statement of the market's shape carries a rider that must never be dropped:

> *The harbor economy is a three-sided market* by design*; two-sided until reputation ships. The third side* is *the tradeable-person terminus of Chapter III.*

Today there is no reputation estimator in the reference implementation; it is a gated phase on the roadmap. So sides 2 and 3 are SPECIFIED, not shipped. What *is* shipped is the escrow + conservation floor they will sit on. We do not overclaim the market.

One ledger, three different reputation keys

All three sides ride one float-plan escrow, but they do not share one identity key. Each side keys reputation on a different entity — the **principal** for labor, the **asset owner** for rental, the **content hash** (a specific skill version) for licensing. The shared settlement table therefore needs a tagged subject identifier such as (*kind*, *id*, *version*); otherwise non-equivalent reputation subjects collide. The dashed band in Figure IV.2 marks this schema distinction.

Exercises

E1. State the operational test for "how many sides" in one sentence, then apply it to a ride-share platform: how many sides, and why?

E2. Side 3 (skill licensing) keys reputation on a *content hash*, not a person. What breaks when a licensor ships v2 of a skill? (Hint: §IV.14, skill versioning.)

E3. ★ A buyer reads a three-axis reputation vector (accuracy, aesthetics, efficiency). Specify a disclosure rule that lets the buyer weight the axes *without* re-introducing a single Goodhart target. Argue why every fixed scalar collapse re-creates the bandit framing the design rejects.

IV.4 The float plan: the built floor

Every side settles on one object: the **float plan**.

Definition IV.4.1 (Float plan)**.** A *float plan* is a signed declaration of intent: a task, a set of *machine-checkable* acceptance criteria, a compute budget, and a credit bounty. It is the atom all three market sides escrow against. (A *machine- checkable* criterion is one a third party can evaluate without trusting the worker's word — a named test that must pass, a commit hash that must merge, a static check that must be satisfied.)

The escrow handshake is a three-step signed ceremony over a standard digital-signature scheme (Protocol IV.4.1, drawn in Figure IV.3): the requester signs the float plan; the daemon opens a serialized database transaction, debits the requester's wallet, and moves bounty plus bond into an escrow row; the daemon then signs the pair $\{anchor\ id, plan\ hash\}$, proving to the worker that funds are locked *before any work runs*. This is the heart of "no-spawn-without-bond."

Protocol IV.4.1 (Float-plan escrow ceremony)**. Parties:** requester R, daemon D (the local trusted root), worker W. **Pre:** R's wallet holds at least $bounty + bond$.

1. $R \to D$: $\mathsf{sign}_R(FloatPlan)$ where $FloatPlan = \langle task, criteria, budget, bounty \rangle$.
2. D locally, in one serialized transaction: debit $bounty + bond$ from R's wallet, write an escrow row keyed by a fresh *anchor id*, commit. *If the transaction aborts, no value moved (atomicity).*
3. $D \to W$: $\mathsf{sign}_D\{anchor\ id, plan\ hash\}$. W verifies D's signature, confirming funds are pinned before doing any work.

Post: the conservation invariant (Property IV.4.1) holds; W has a signed proof of locked funds; no process for this task can enter `running` without the escrow row existing.

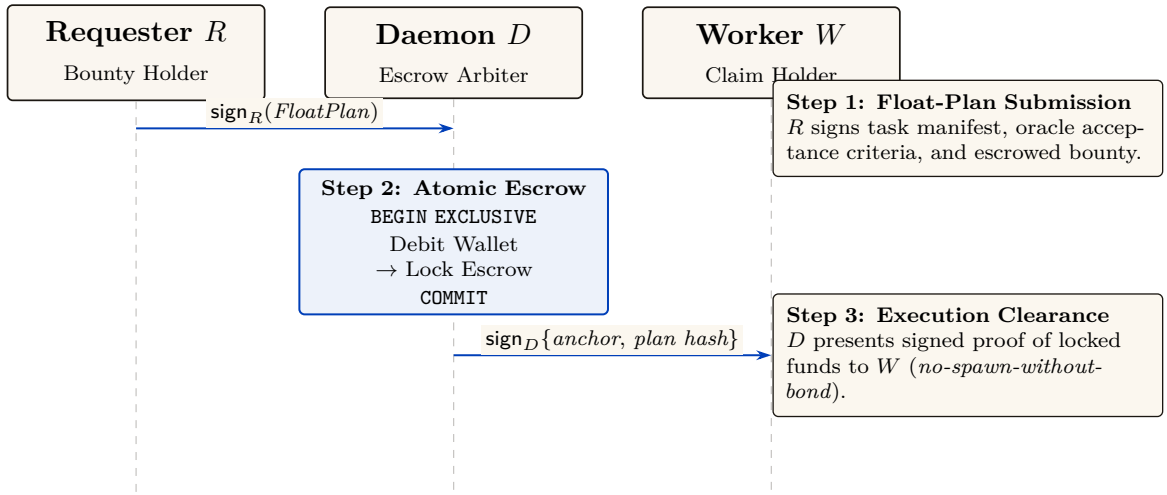


Figure IV.3: The float-plan escrow ceremony — the atom every side of the market settles on. A three-step signed handshake over a standard digital-signature scheme: the requester signs a FloatPlan (task, machine-checkable acceptance criteria, budget, bounty); the daemon debits and escrows the value inside one serialized database transaction; the daemon then signs $\{anchor\ id, plan\ hash\}$, proving to the worker that funds are pinned before any work begins. The conservation invariant (wallet + escrow + commons = supply, verified over 10k random operation traces by a property test) holds at every step and is the *implemented* floor the whole economy stands on — the one claim this paper can make at the highest maturity.

IV.4.1 The bond ledger conserves value (this is built)

The single most load-bearing primitive — and the one that is *actually* **implemented** — is the **bond-ledger module**: bond escrow for agent spawning. It debits a project wallet into an escrow row before spawn, refunds on clean exit, and slashes on breach, splitting the slash between the wallet and a commons pool. It guards two invariants:

Property IV.4.1 (Conservation)**.** wallet + escrow + commons = supply, always. Every debit has a matching credit. In the reference implementation this is verified across ten thousand random operation traces by a property test — a real randomized test, not an aspiration. (**implemented**; see Appendix IV.A.)

Property IV.4.2 (No-spawn-without-bond)**.** A process enters the `running` state only if a bond was escrowed against its agent id. (PARTIAL: the ledger enforces escrow-first, but the runtime `running`-state check is a roadmap item, so today the gate is caller-discipline, not runtime-enforced. See Appendix IV.A.)

> **Key idea.** Conservation is the mathematical glue of a three-sided market. Whatever happens — success, partial credit, slash, lease, license — no value is created or destroyed; it only moves between wallet, escrow, and commons. A three-sided market with three settlement paths is coherent *only if it cannot leak*. Port Daddy already enforces this for the labor side; the contribution of §IV.7 is to show the same invariant extends to the rental and licensing sides *without modification* and *without a second ledger*.

IV.4.2 Settlement: four terminal states bound to an oracle

Settlement reads an **oracle** (*a trusted source of ground truth the agent cannot author — a passing test id, a merged SHA, a satisfied Arbiter check*) over the acceptance criteria and lands in one of four states.

Table IV.2: The four terminal settlement states. Closure binds to an oracle, not to a free-text "Result:" note — the Goodhart-resistant discipline that the commitment layer imposes: *the agent picks the work; the daemon derives the grade.*

State	Flow of funds	Reputation effect
Success	escrow → worker (bond refund) + bounty → worker	+1 completion
Partial	pro-rata bond → worker by completed evidence-chain fraction; remainder → salvage fund	salvage recorded
Sabotage	bond → reconstruction commons (100% slash)	failure recorded; ban on threshold
Dispute	escrow → arbitration hold; 2-of-3 multi-oracle (automated / evidence / human)	none until resolved

A settlement, dramatized. Bob (the requester principal) posts a refactor float plan: bounty 400 CR, requiring a provider performance bond of 100 CR and acceptance criteria = "the module's test suite passes and the diff merges." Alice (the provider principal) assigns agent a_2 and posts the 100 CR performance bond into escrow. Bob escrows the 400 CR bounty. Before execution, the escrow holds 500 CR across two distinct source buckets: $\text{escrow}_{\text{bounty}}(\text{Bob}) = 400$ and $\text{escrow}_{\text{bond}}(\text{Alice}) = 100$.

Agent a_2 completes 40% of the verifiable evidence chain (refactoring two of five files cleanly before token budget exhaustion). The independent oracle evaluates the test receipts and diff closure, returning **Partial** (40%). Funds settle across the four buckets:

1. **Bounty distribution:** Alice earns $40\% \times 400 = 160$ CR in earned revenue; the remaining 240 CR unearned bounty returns to Bob's wallet.
2. **Collateral settlement:** Alice receives $40\% \times 100 = 40$ CR of her performance bond returned as collateral (not income); the remaining 60 CR bond is slashed and credited to the commons/salvage fund to onboard a successor.

Conservation check: $\Delta\text{Bob} = -160$, $\Delta\text{Alice} = +160 - 60 = +100$ net, $\Delta\text{commons} = +60$. The ledger sums to 0 net delta, and the separate escrow buckets $\text{escrow}_{\text{bounty}} \to 0$ and $\text{escrow}_{\text{bond}} \to 0$ clear completely. No value was created or destroyed.

Exercises

E1. Trace the flow of funds for a **Partial** settlement where the worker completed 40% of the evidence chain. Confirm Property IV.4.1 holds across the split (the dramatization above is one such trace; redo it for a bond of 250 CR and 70% completion).

E2. Why does closure bind to an oracle rather than a self-reported note? Name the attack a free-text "Result:" note enables.

E3. ★ The four states assume the acceptance criteria were *right*. Design a *float-plan amendment* protocol for the common case where the criteria themselves turn out wrong mid-flight (re-sign by both parties, escrow top-up/refund delta), preserving Property IV.4.1. (Gap, §IV.14.)

IV.5 Three sides on one escrow

The novel construction is that the rental and licensing sides ride the *same* float-plan escrow, so conservation holds globally without a second ledger.

Operator-for-hire (labor). The operator is the requester's counterparty. It *sub-escrows* a bond for each fleet agent it dispatches (the bond ledger is already per-agent, so this needs no new primitive). Its profit is bounty − Σ(slashed sub-bonds) − ledger fee. The operator internalizes its fleet's risk.

Asset rental (capital). The lease fee is a line item in the float plan. By **incomplete-contracts / property-rights theory** (**Grossman & Hart** 1986 [178], *when contracts cannot specify every contingency, the residual control right should go to the party making the most important non-contractible investment*), the *renter* must hold the *runtime* control right — the ability to sandbox, throttle, or revoke the leased agent mid-task — while the *owner* bears the reputational consequence.

> **Key idea.** This is the cleanest economic argument for the runtime **jail** — the per-agent tool-allowlist plus scoped filesystem that the safety layer enforces: the jail is not just safety hygiene, it is the *residual control right that makes leasing an agent contractible at all*. Here we must respect the series' *capability vs. permission* distinction (§IV.6): the jail allocates *capability* (what the leased agent *can* do at runtime). The deontic layer separately governs *permission* (what it *may* do). "Able but forbidden" must stay expressible; the jail bounds ability, not norm.

Skill licensing (IP). The per-use fee is released to the licensor *only on green settlement* (metered + clawback). This is forced by **Akerlof**'s *market for lemons* (1970 [86], *when buyers cannot observe quality, price collapses to the average and good goods exit the market*): a skill's quality is unobservable before use, so a flat upfront license drives good skills out. Metering + clawback + a Merkle *portfolio proof* (the skill's prior settled outcomes, anchored in the evidence chain) restore the seller's incentive to ship quality.

Exercises

E1. Map each of Grossman–Hart's two parties (residual-control-right holder vs. reputational-consequence bearer) onto the rental side's roles. Which one is the Arbiter jail?

E2. Show that adding a lease line item and a metered-license line item to a float plan cannot violate Property IV.4.1, no matter the amounts.

E3. ★ A skill's portfolio proof for v1 says nothing about v2. Specify a *version-binding* for the Merkle portfolio proof so a new version starts with a discounted inherited prior — the newcomer policy, but for code.

IV.6 The keystone the market rests on — and does not yet have

Everything above assumes a counterparty's identity is real. This section confronts the series' single *blocking* reconciliation: the identity primitive the whole market leans on covers a threat model that *excludes the cross-operator adversary the market is defined by.*

Figure IV.4 draws the two stones apart: the local root this paper leans on, and the cross-operator stone it does not yet have.

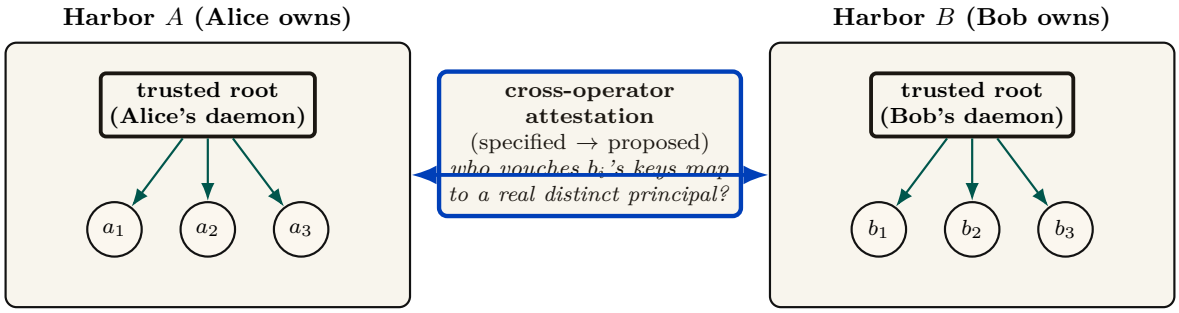


Figure IV.4: The keystone split. Chapters I–III rest on **local non-forgeable identity**: a per-agent identity no agent can edit *within one trusted daemon* — a single trusted root that vouches for every agent it mints. That is exactly what a single-operator harbor needs. But the market is *defined* by mutually-distrusting operators trading across a boundary neither controls, and that primitive explicitly disclaims the relevant threat model (intra-fleet, not a cross-party PKI, no hostile-operator defense). Binding signing keys across harbors you do not own is a different, unsolved problem: **cross-operator attestation**. Across the boundary, the counterparty's daemon *is* the adversary the local threat model declares out of scope, and the witness log gives *log integrity*, not *identity binding*. It is the highest-leverage unbuilt keystone, and it gates the entire market thesis.

IV.6.1 Local non-forgeable identity is the keystone — for one operator

Definition IV.6.1 (Local non-forgeable identity)**.** *Local non-forgeable identity* is a per-agent identity that no agent can edit, guaranteed *within one trusted daemon*: the daemon is the only component that can hold state agents cannot reach, so it is a trusted root by construction. (The reference system now ships a partial slice — daemon-minted `actor-souls`, lookup credentials, and a bounded newcomer pool — while full write-boundary enforcement remains open. Papers 1–3 rely on the complete contract; here we only consume it.)

For a single-operator harbor this is exactly right and is correctly named the highest-leverage keystone. Its threat model, stated plainly, is "a lazy or self-interested agent in a fleet the operator owns" — not a hostile peer. It is explicitly *not* a public-key infrastructure spanning strangers.

IV.6.2 But its threat model does not reach the market

Pitfall. Local non-forgeable identity defends against an agent inside a fleet the operator owns; it explicitly does *not* claim to defend against a hostile operator, and it is not a cross-party PKI. The market, by contrast, is *defined* by mutually-distrusting operators trading across a boundary neither controls. You cannot make the single-operator root the foundation of the cross-operator market by assertion. Across the boundary, the counterparty's daemon *is* the adversary that local identity declares out of scope. The witness log gives *log integrity*, not *identity binding*: it does not answer "who vouches that Harbor B's signing keys map to a real, distinct principal, rather than a hundred sock puppets Bob minted this morning?"

Definition IV.6.2 (The cross-operator attestation problem)**.** *Cross-operator attestation* is the problem of binding agent signing keys across harbors you do not own: an actual key-distribution and attestation story for the cross-operator threat model — something that lets Alice's harbor trust that a key presented as "Bob" really is the one principal she has been building a reputation history against. **Status:** SPECIFIED-to-*proposed*; it has no implementation and only a partial specification for its load-bearing part. It is the highest-leverage unbuilt keystone, and it gates the entire market thesis.

Key idea. This paper *stops* describing the federation boundary as one "neither controls" while quietly resting on a single trusted root. Either cross-operator attestation is built (a real attestation story), or the market's scope is restated as "federation among *semi-trusted* operators." We take the first horn as the designed target and flag every claim that depends on it.

IV.6.3 The principal above the actor

The economic counterparty in every trade is the **principal** (the human or org owning a fleet), not the agent. Bonds, reputation inheritance, and sanctions must key on the principal via the hop-bound authorization chain, or Sybil bond-farming works by spawning fresh *agents* under one principal (**Douceur**'s Sybil attack [98]: *free identities defeat any reputation or quorum system*). Defining the principal as a first-class economic entity is the cleanest Sybil fix; its cryptographic binding is the identity object specified in Chapter III and consumed here.

Exercises

E1. In one sentence, state why a reputation system layered on forgeable pseudonyms is *no mechanism*, not merely a weak one (cite Douceur).

E2. Local non-forgeable identity has a non-goal: "no defense against a hostile operator." Name the exact word in the market's definition ("mutually-distrusting," "boundary neither controls") that this non-goal contradicts.

E3. ★ Specify a cryptographic binding tying N agents to one principal such that spawning fresh agents does not reset the bond floor, and that survives into the cross-operator world (where the single-operator "trusted credential" shortcut is unavailable). This is the hard half of cross-operator attestation.

IV.7 Conservation under composition

The economy's single cross-boundary invariant is that internal ledger operations redistribute value rather than create or destroy it. This section states the compositional claim directly and separates it from cross-currency valuation.

Figure IV.5 draws the composition claim: sequential traces glue, and internal operations carry zero external balance change.

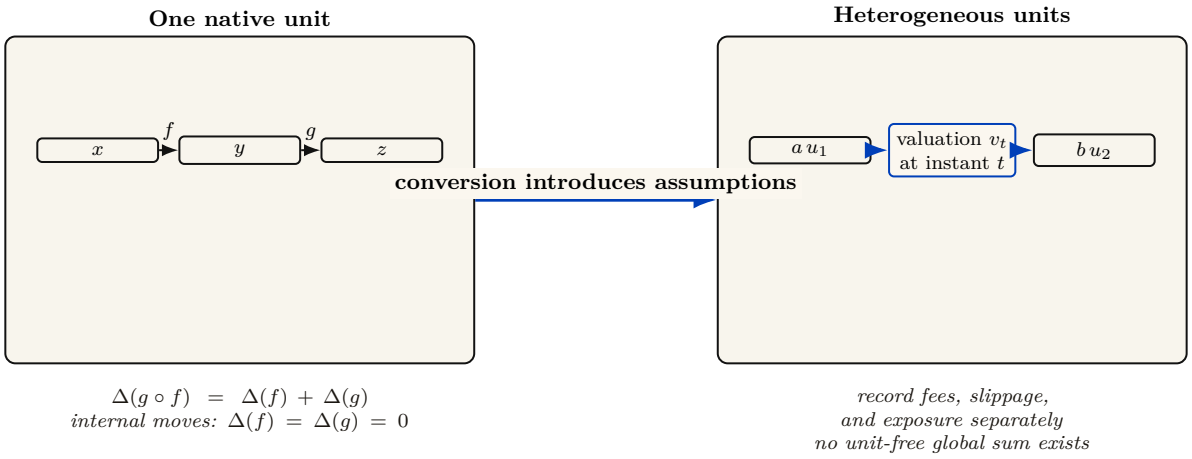


Figure IV.5: Conservation under composition. In one native unit, the supply delta of a composed trace is the sum of its step deltas, so zero-delta internal operations remain conservative. A cross-currency transfer is different: it requires a named valuation function and instant, plus explicit fee, slippage, and exposure accounts. Category-theoretic language does not remove those accounting assumptions.

Let a ledger state be $x = (W, E, C)$ and let a valid transaction trace $f : x \to y$ carry external balance change $\Delta(f) = S(y) - S(x)$ for $S(x) = W + E + C$. Sequential traces compose by concatenation and satisfy $\Delta(g \circ f) = \Delta(g) + \Delta(f)$. Internal escrow, refund, and slash traces have $\Delta = 0$; top-ups and withdrawals are explicit boundary flows. This is the entire conservation claim. Category language can describe trace composition, but it does not add a currency-conversion theorem.

Theorem IV.7.1 (Single-unit conservation composes)**.** *Suppose all harbors use one unit of account and every cross-harbor transfer is a matched atomic move from source escrow to either destination settlement or source refund, with an explicit in-flight bucket. Then any finite composition of internal and cross-harbor operations has total external balance change equal to the sum of its recorded top-ups and withdrawals. In particular, a trace containing only internal transfers has* $\Delta = 0$*. (SPECIFIED; proof sketch in Appendix IV.B.)*

Property IV.7.1 (Cross-currency valuation boundary)**.** With heterogeneous units, exact conservation is defined per native unit. A scalar portfolio total additionally requires a valuation map v_t at a named instant t, plus explicit fee and slippage accounts. If rates move between legs, the mark-to-market difference is economic exposure, not a failure of functoriality. Bounding that exposure by φ requires a separate oracle-latency and price-move assumption that this paper has not proved. (*open*.)

> *Conservation is exact per unit of account. A cross-currency scalar is a valuation, and every valuation must name its rate source, time, fees, and exposure bound. "Lax functor" is not a substitute for those data.*

IV.7.1 The Myerson–Satterthwaite corner

Strict conservation (Property IV.4.1) is exactly *budget balance*. That is not free.

Theorem IV.7.2 (Conditional bilateral-trade boundary)**.** *For any bilateral slice of this market satisfying the Myerson–Satterthwaite assumptions—independent private buyer and seller values drawn from overlapping continuous supports, Bayesian incentive compatibility, interim individual rationality, and budget balance—no mechanism is also ex-post efficient everywhere [168]. This theorem constrains such a slice; it does not by itself prove incentive compatibility of the harbor mechanism or identify the desideratum sacrificed by the full three-sided market.*

The current design establishes a conservation goal and voluntary entry; it has not proved truthfulness. An AGV-style mechanism changes the incentive and participation notions and generally offers expected rather than ex-post balance. Comparing those alternatives requires a fully specified type and transfer model, which remains open.

Exercises

E1. State Myerson–Satterthwaite in one sentence and list the assumptions that must be checked before applying it to one harbor trade.

E2. Give a two-currency example in which native-unit conservation holds but the marked-to-market scalar changes because v_t moves.

E3. ★ Specify an oracle-latency and price-move model that yields a valid φ exposure bound for Property IV.7.1.

IV.8 The Anchor Protocol proper: cross-harbor capability transfer

The *cross-harbor capability-transfer ceremony* is the market's identity-level hot path: how an authorization minted under Alice's harbor becomes usable, safely, inside Bob's. Chapter V analyzes the cryptographic substrate (a hop-bound authorization chain checked by protocol models plus a bounded Rust verification layer); *this* section is the ceremony that rides that substrate across a boundary.

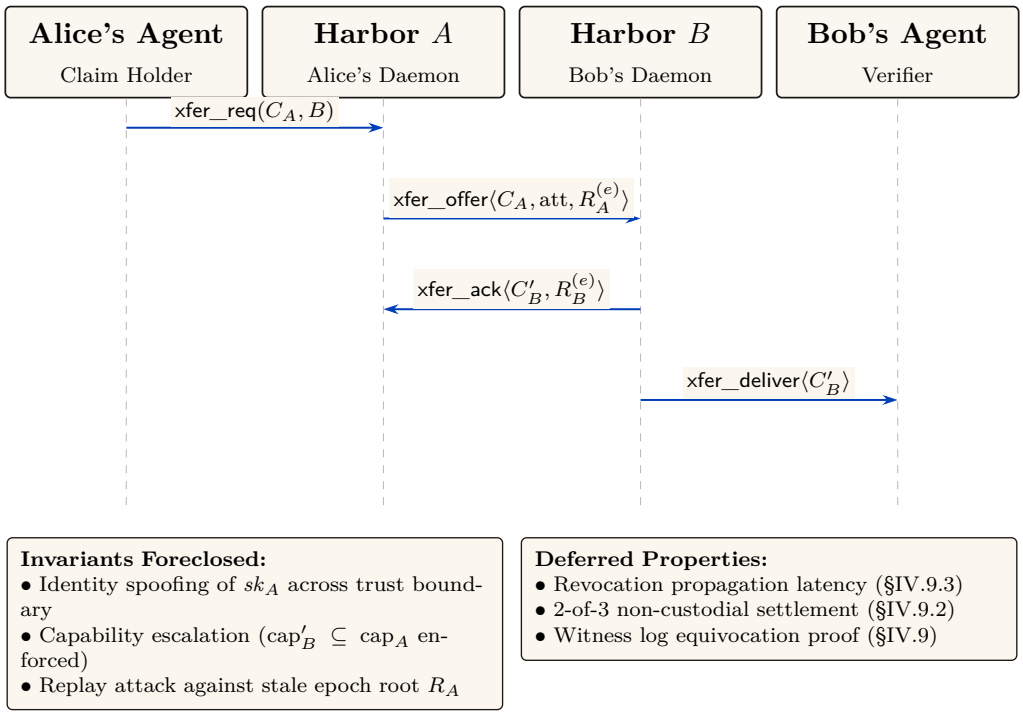


Figure IV.6: The four-message cross-harbor capability transfer ceremony. The critical property is that no message after (3) needs to reach back to Harbor A in the hot path: C'_B is verifiable under sk_B alone, which is why federation does not turn every authorization into a synchronous inter-machine round trip. The epoch root $R_A^{(e)}$ bound into the envelope is the load-bearing primitive — a revocation issued by A after epoch e invalidates C'_B as soon as the new $R_A^{(e+1)}$ reaches the witness log that B audits.

The four-message ceremony (Figure IV.6, written out as Protocol IV.8.1) has one load-bearing liveness property: after message (3), no further synchronous interaction with Harbor A is needed. The transferred card C'_B is verifiable under Harbor B's key alone, so federation does not turn every authorization into a synchronous inter-machine round trip. The epoch root $R_A^{(e)}$ bound into the envelope is the load-bearing primitive: a revocation issued by A after epoch e invalidates C'_B as soon as the new root $R_A^{(e+1)}$ reaches the witness log that B audits.

Protocol IV.8.1 (Cross-harbor capability transfer)**. Parties:** Alice's agent (holds card C_A), Harbor A (Alice's daemon), Harbor B (Bob's daemon), Bob's agent (verifier). **Pre:** B audits a witness log to which A publishes epoch roots.

1. Alice's agent $\to A$: $\mathsf{xfer\_req}(C_A, B)$ — present C_A, name target harbor B, request a transferable form.
2. $A \to B$: $\mathsf{xfer\_offer}\langle C_A, \mathrm{att}, R_A^{(e)}\rangle$ — A signs an envelope binding C_A, an attenuation predicate att (which may only *narrow* authority), and its current epoch root $R_A^{(e)}$.
3. $B \to A$: $\mathsf{xfer\_ack}\langle C'_B, R_B^{(e)}\rangle$ — B verifies A's key via the witness log, applies its *own* attenuation, and mints $C'_B = \mathrm{att}_B(\mathrm{att}_A(C_A))$, valid only at B.
4. $B \to$ Bob's agent: $\mathsf{xfer\_deliver}\langle C'_B\rangle$. Every later presentation verifies under B's key alone — no A-side lookup on the hot path.

Post (liveness): no synchronous A interaction after step (3). **Post (safety):** authority only ever narrowed; replay against a stale $R_A^{(e)}$ is rejected; a revocation by A kills C'_B once $R_A^{(e+1)}$ reaches B's witness log.

The ceremony, dramatized. Alice's specialist a_2 holds a card C_A that authorizes "read and write `src/payments/`, run the test suite." Bob rents a_2 for an afternoon. Alice's harbor offers an

attenuated envelope: att_A strips write access to everything except the one module Bob hired for. Bob's harbor, not trusting Alice's word, attenuates *again*: att_B adds "no network egress, six-hour TTL." The minted C'_B is the intersection — read/write one module, run tests, no network, expiring at dusk — and it verifies under Bob's key, so a_2 never has to phone home to Alice mid-task. At 5 p.m. Bob's revocation fires locally; at the next epoch boundary Alice's harbor also sees, in the shared witness log, that C'_B was used exactly as attenuated and nowhere wider.

Key idea. Capability *attenuation* is the safety property here: a transferred card can only *narrow* authority — $C'_B = \text{att}_B(\text{att}_A(C_A))$ — never widen it. This is the cross-boundary form of the attenuation-monotonicity that the hop-bound authorization chain checks within one harbor (Chapter V). It is what makes "rent my agent for an afternoon" safe to say to a stranger.

Pitfall. Do not let the federation theorems borrow the substrate's "formally verified" halo. The hop-bound authorization chain has bounded symbolic and Rust evidence in Chapter V. The *federation* claims are design arguments plus a bounded TLA$^+$ extension; they are SPECIFIED, not deployed. Two different confidence levels, never blurred.

Exercises

E1. Why is the ceremony four messages and not two? Identify what message (3) buys (Hint: who must counter-sign, and under whose key does C'_B later verify?).

E2. Trace what happens to an in-flight C'_B when Alice rotates her signing key between messages (3) and a later presentation at B. (Gap: cross-boundary key rotation, §IV.14.)

E3. ★ Specify replay protection across the boundary: is the *anchor id* a nonce? Does B reject a re-presented message (3)? State the freshness invariant.

IV.9 Federation: trading across machines you don't own

Federation is where the leaderless distributed canon actually bites. Crucially, the federation design *refuses consensus* here — the correct reading of the impossibility landscape.

Figure IV.7 draws the topology this section defends: independent harbors linked only by gossip and a shared witness log, with no global consensus step.

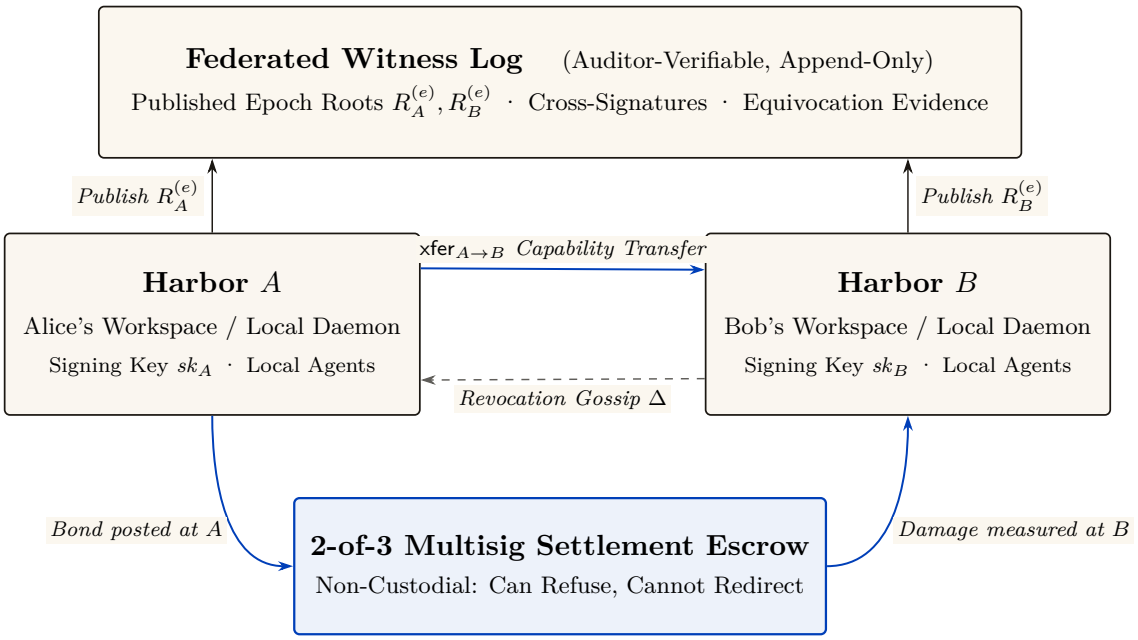


Figure IV.7: The Federated Harbor as a four-element gestalt. Neither harbor is a root of authority for the other. The witness log above provides shared correlation: each harbor publishes its signed epoch root, enabling decentralized equivocation detection. The settlement escrow below is structurally bounded (non-custodial 2-of-3 multisig: it cannot redirect funds or author unadjudicated transfers). The solid blue arrow denotes the capability transfer ceremony (§IV.8); the dashed line denotes epidemic revocation gossip (§IV.9.3).

IV.9.1 No consensus, by design

> **Key idea.** A system that *needs* agreement on a global event order across operators it does not control is dead on arrival, by **Fischer–Lynch–Paterson** (1985) [132] (*no deterministic protocol achieves consensus in an asynchronous network with even one crash failure*). Federation declines to require agreement. It requires only (a) append-only per-harbor logs, (b) gossip-bounded propagation, and (c) *detectable* (not prevented) equivocation. This is a **Certificate Transparency** posture [30]: you do not prevent a misbehaving log from lying, you make lying detectable after the fact and expensive via a slashable bond.

The convergence and detection bounds smuggle in *partial synchrony* (**Dwork–Lynch–Stockmeyer** 1988 [43], the canonical escape from FLP): "gossip every Δ" *is* an eventual-synchrony assumption, and we state it as such rather than burying it.

IV.9.2 Settlement and the cross-chain-deals lineage

Figure IV.8 places this construction against the atomic-swap and adversarial-commerce baselines it is compared to below.

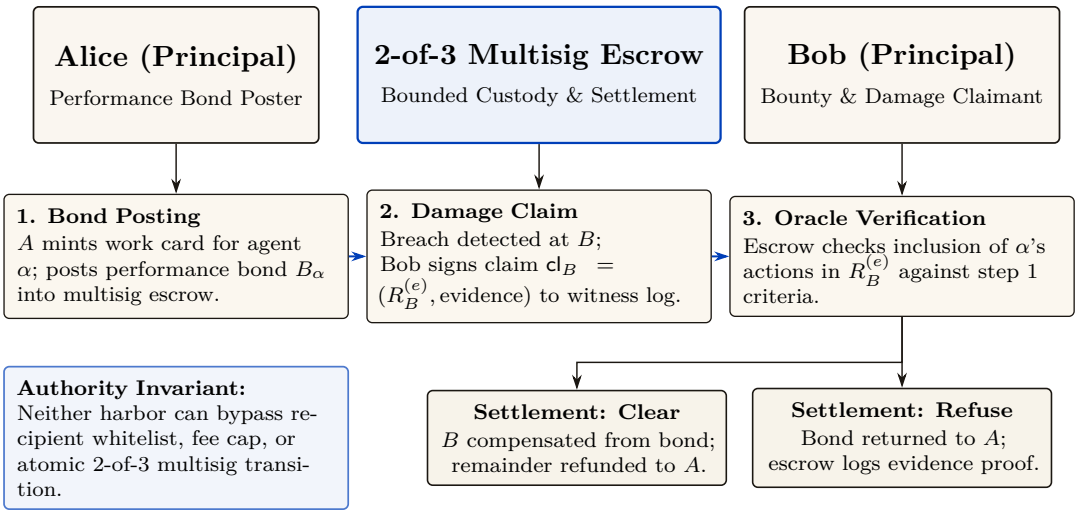


Figure IV.8: Cross-harbor settlement protocol. Custody bounds are structurally enforced: the escrow authority requires 2-of-3 multisig consensus across the recipient whitelist, fee cap, and atomic terminal transitions. The outcome space partitions cleanly into **Clear** or **Refuse**, preventing unbounded cross-harbor liability.

The escrow construction is honest about its trust: it does *not* claim trustless settlement. Its loss bound is conditional on a non-bypassable custody ledger that exposes only a recipient whitelist, a fee cap, and one atomic terminal transition. If that authority boundary can be bypassed, the worst-case loss is the full balance in custody, not merely the pre-agreed fee φ. This is a deliberate departure from the hash-timelock baseline (**Herlihy** 2018 [134], atomic cross-chain swaps), and the right comparison is the formal model of mutually-distrusting commerce without a shared chain: **Herlihy, Liskov & Shrira** (2022, *Cross-Chain Deals and Adversarial Commerce*) [133]. The harbor's bounded escrow is a "trusted third party / watchtower" point in that design space; we locate it there explicitly rather than reinvent it.

Remark. **Open problem: trustless cross-harbor settlement.** It is unknown here whether non-fungible, reputation-priced bonds admit a useful trustless settlement protocol under the paper's availability and privacy requirements. An HTLC comparison does not prove impossibility because the asset and adjudication models differ. (*open*.)

IV.9.3 Revocation gossip and the equivocation race

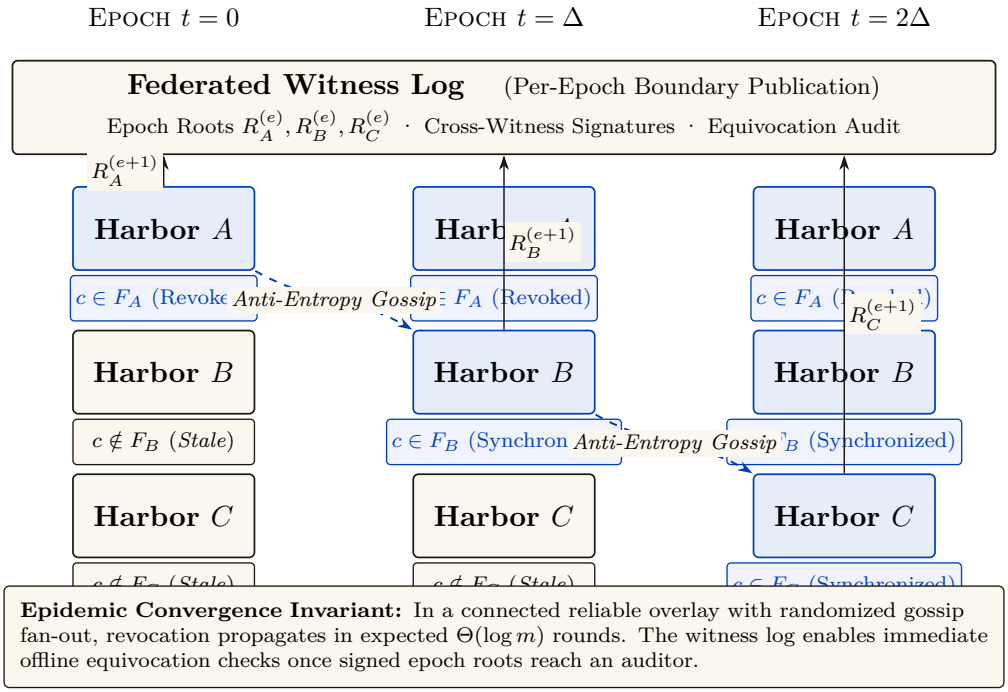


Figure IV.9: Federated revocation propagation across three harbors. The append-only revocation log ensures that revocation events propagate epidemically (Δ-spaced rounds). Per-epoch roots $R_X^{(e)}$ published to the shared witness log provide transparent, non-repudiable audit trails that generalize Certificate Transparency.

Federated revocation propagates by anti-entropy gossip (**Demers et al.** 1987 [1]) with a probabilistic-membership filter (a cuckoo filter — a compact structure that answers "is this card revoked?" with no false negatives and a tunable false-positive rate). The federation therefore has expected $\Theta(\log m)$-round dissemination only under a connected reliable-round overlay with the stated randomized peer-selection model. This asymptotic expectation is not an exact constant or a partition-tolerant deadline.

Protocol IV.9.1 (Federated revocation gossip)**. State:** each harbor X keeps a revocation filter F_X and a current epoch root $R_X^{(e)}$ published to a shared witness log.

1. *Revoke.* An operator revokes card c at its home harbor A: $F_A \leftarrow F_A \cup \{c\}$; A advances to $R_A^{(e+1)}$ and publishes it.
2. *Gossip.* Every Δ, each harbor picks a peer and exchanges filter deltas (anti-entropy). A card present in either filter becomes present in both.
3. *Audit.* Any third party compares each harbor's published roots in the witness log; a harbor that publishes two contradictory roots for one epoch is *equivocating*. Verification is constant work once both signed roots are co-located; delivery to an auditor has the overlay's separate dissemination delay.

Post, conditionally: in the connected reliable-round model, every honest harbor learns the revocation in expected $\Theta(\Delta \log m)$ time. Under an unbounded partition there is no finite global-learning bound.

The revocation race, dramatized. Alice revokes a_2's card c at $t = 0$ (say she discovers it was misbehaving). Harbor B, which rented a_2, has not yet heard: for the interval $[0, \Delta]$, B's filter is stale and c is still spendable there. At $t = \Delta$ gossip reaches B, which adds c to F_B and refuses

further use. Harbor C, two hops away, learns at $t = 2\Delta$. This is one illustrated execution; a different schedule or partition produces a different window. Conjecture IV.9.1 says the bond Alice posted on c must dominate whatever a_2 could spend at B inside that window, or a well-capitalized Bob could rationally race the gossip.

> **Pitfall.** The threat is *adversarial*, not stochastic. An attacker who controls peer selection or partitions the gossip graph (an *eclipse attack*, **Heilman et al.** 2015 [71]) breaks the expected-time guarantee. The honest statement is conditional: a finite service-level bound needs eventual connectivity and a known post-stabilization delivery bound; epidemic analysis then supplies an expectation within that model, not an adversarial deadline.

Detection-after-the-fact deters only if the slashed bond exceeds the value extractable during the detection window. This is the analogue of the *cleanup lower bound* (the companion result that a cleanup bond must cover the worst-case cost of the damage it underwrites), and we state it as a target:

Conjecture IV.9.1 (Equivocation Lower Bound)**.** *For a deployment that advertises a finite freshness window T, the witness bond must dominate the maximum value spendable against a revoked-but-not-yet-observed capability over T. If partitions are unbounded, no finite bond covers the worst case; the system must instead fail closed on stale roots or cap spend independently. (open; analogue of the cleanup lower bound.)*

Figure IV.10 bands the window Conjecture IV.9.1 must price against the connectivity assumptions each band needs.

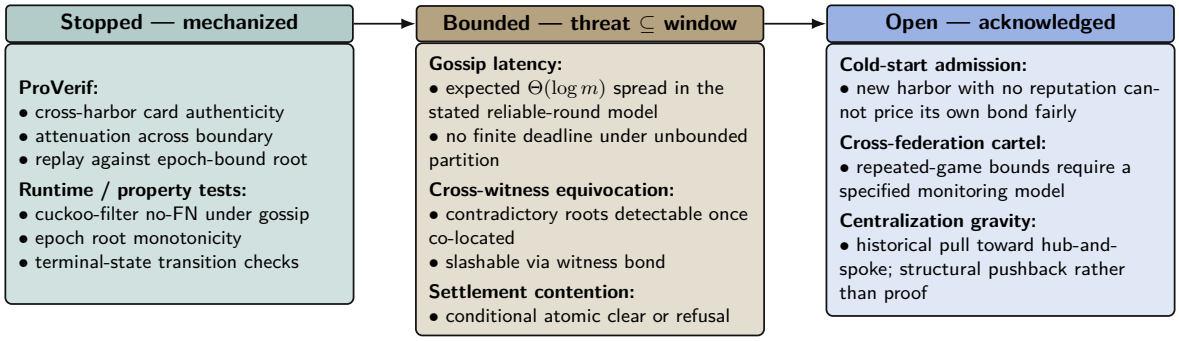


Figure IV.10: Threat-band layout for the Federated Harbor. The Anchor paper's defense-in-depth structure carries over: every claim either has a mechanization artifact (teal), a named bound that the bond can price against (amber), or an explicit acknowledgement of the open frontier (cobalt). The paper is disciplined about which band each claim lives in. Reading left to right runs deeper into threat space and out of formal reach; bands shift left over time as future work mechanizes what is presently bounded.

Exercises

E1. State the connectivity, peer-selection, and delivery assumptions needed for expected $\Theta(\Delta \log m)$ dissemination. Explain why none yields a deadline during an unbounded partition.

E2. Trace a revoked capability through Figure IV.9: at which harbor, and at what time, can it still be spent? That window is what Conjecture IV.9.1 must price.

E3. ★ State and (attempt to) prove the Equivocation Lower Bound for a three-harbor mesh under a fixed gossip period Δ and a single equivocating witness.

IV.10 Reputation: monotone, but revocable

Reputation is the third side's existence condition. Two reconciliations matter here.

IV.10.1 "Never ends" is not an anti-revocation property

A tempting design slogan holds that reputation "never ends" — no decay window an agent can outrun — while the same design also requires that a fraudulent settled outcome be retractable. A property and its required violation cannot both be load-bearing.

> **Key idea.** The reconciled statement: **reputation is monotone in *honest* witnessed outcomes, but a witnessed outcome is itself revocable via a propagating tombstone**. Dissemination has the same model and partition caveats as capability revocation. "Never decays" applies to honest outcomes only; it is *not* an anti-revocation property.

We also decline to assert no-decay as a virtue: bounded memory is a design parameter (**Liu & Skrzypacz** 2014, *Limited Records and Reputation Bubbles* [166], show unbounded records create reputation *bubbles*, not stability). The horizon is a tunable, and ∞ is rarely optimal.

IV.10.2 Why Sagas-style compensation does not restore reputation

It is tempting to fix revocation with compensating transactions (**Garcia-Molina & Salem** 1987, *Sagas* [93]), as the bond ledger does. But the analogy fails structurally:

Proposition IV.10.1 (Compensation does not erase reliance)**.** *A ledger can append a compensating transaction that restores a numerical balance. A reputation tombstone cannot undo decisions counterparties already made while relying on the invalid outcome. It can only change future evaluations after the tombstone is observed. Reputation repair therefore requires dissemination plus an explicit remedy for past reliance; it is not equivalent to ledger compensation.*

This is why Sagas appears in the bond-settlement prior art but *not* in the reputation prior art.

IV.10.3 The grading-oracle keystone

Every folk-theorem incentive-compatibility claim in the market bottoms out in "the daemon derives the grade." But the grade is produced by a capturable evaluator (Figure IV.11).

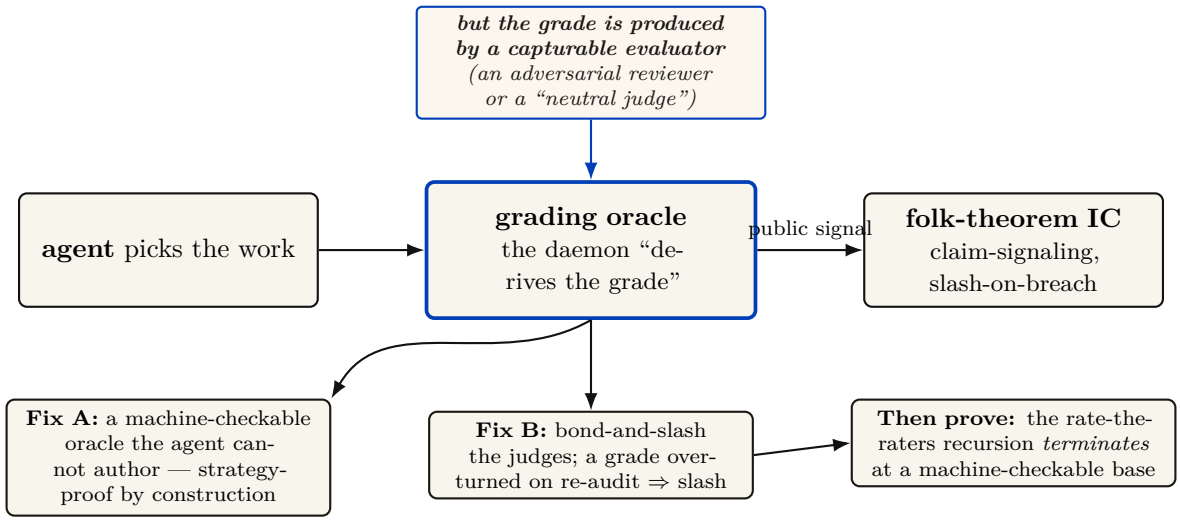


Figure IV.11: The grading-oracle incentive-compatibility problem — the hole under the core theorem. Every folk-theorem claim in the layer bottoms out in "the agent picks the work, the daemon derives the grade." But a folk theorem is only as good as the *public signal* it conditions on (Green–Porter 1984; Abreu–Pearce–Stacchetti 1990 require the signal distribution to be common knowledge and *exogenous*). The grade is produced by a capturable evaluator, so the signal is endogenous and the folk theorem does not apply as stated. We resolve it two ways: ground settlement in *machine-checkable* oracles the agent cannot author (Fix A, strategy-proof by construction), and *bond-and- slash* the human/LLM judges where machine-checking is impossible (Fix B), then require the rate-the-raters recursion to terminate at a machine-checkable base.

Theorem IV.10.1 (The monitoring model must include the evaluator)**.** *An imperfect-public-monitoring equilibrium result requires a specified conditional distribution of public signals given modeled actions and strategies. If a strategic grader chooses the signal but is omitted from the game, that distribution is not fixed by the model and the claimed equilibrium result is unsupported. One must either constrain the signal mechanically or include the grader, its information, payoffs, and deviations as part of the game [54].*

The two resolutions (both used). *Fix A — mechanically constrained evidence:* ground settlement in evidence the agent cannot directly author (a CI-issued test result, a protected merge event, or an independently evaluated policy check). This narrows the signal channel; it does not make the task metric universally strategy-proof. *Fix B — bond-and-slash the judges:* where machine-checking is impossible (aesthetics, ambiguous quality), the human/LLM judges get their *own* bond and their *own* reputation; a judge later contradicted by re-audit is slashed. The "rate-the-raters" recursion must then be shown to terminate at a machine-checkable base, not regress infinitely. De-biasing for LLM judges (order-swap, pairwise, family-exclude) follows **Zheng et al.** 2023 [124].

A captured judge, dramatized. Bob disputes a settlement: he claims Alice's a_2 delivered an "untasteful" refactor. Taste is not machine-checkable, so Fix B applies: Judy, a human grader, is drawn to arbitrate. Judy posts a 50 CR judge-bond and rules *for Bob*. But Judy and Bob are colluding — Judy is a captured judge, the exact failure Theorem IV.10.1 warns of. The market's defense is the re-audit lane: a sampled second panel later reviews Judy's ruling against the preserved evidence, finds it indefensible, and *contradicts* it. Judy's 50 CR bond is slashed and her judge-reputation takes a tombstone; Alice is made whole. The signal became exogenous again not because Judy was honest but because lying was bonded and the re-audit made it expensive — and the recursion stops at the second panel, whose own evidence is machine-checkable (the same diff, the same tests).

Exercises

E1. State the signal-model assumption Theorem IV.10.1 needs and explain why an omitted strategic grader leaves it undefined.

E2. Classify three settlement criteria as machine-checkable (Fix A) or judge- dependent (Fix B): a unit test passing; a merged SHA; "the refactor is tasteful."

E3. ★ Prove the rate-the-raters recursion terminates: define a well- founded order on "levels of judge," grounded at machine-checkable oracles, and show no infinite ascending chain of "judges judging judges" is needed.

IV.11 Hosted trust is the product

The 2025–26 agentic-payments landscape — **AP2** (Google, signed payment mandates), **ACP** (OpenAI/Stripe), and **x402** (Coinbase, stablecoin settlement over HTTP) — has commoditized the *settlement rail*. Wiring an agent to pay another agent is now table stakes. This is *good news* for the thesis, because it sharpens what the product actually is.

What remains scarce, and therefore defensible, is **hosted trust**: the verified ledger (conservation-checked, Merkle-anchored evidence), the *relay* (the harbor mesh carrying public-key attestation across machines), and the *reputation* that makes a counterparty you do not own legible and accountable.

- **Substrate (swappable):** the payment rail. Use x402/AP2/credits/ Stripe interchangeably.
- **Product (the moat):** attestation, federation membership, and reputation hosting.

The revenue model aligns incentives without perverse over-penalty: a transaction fee (2–5% of settlements), a small listing fee (collusion deterrent), and a small bond-spread on *forfeited* bonds (kept small so the platform never profits from over-slashing). The platform should earn most from *successful* trade, because successful trade is what hosted trust produces.

> *You don't sell crypto — crypto is the substrate. The defensible asset is attestation + federation membership + reputation, not the commoditized payment rail.*

IV.12 Cold start, and the auction question

A three-sided market has a *triple* cold-start problem: no operators shop an empty fleet shelf; no owners list agents with no renters; no licensors publish skills with no buyers. Two-sided-market theory prescribes: subsidize the side carrying the scarcest cross-side externality, then flip. Phase 1 seeds supply at zero/negative price with platform-posted tasks at above-market bounties (generating the first settled outcomes — the bootstrap reputation data); Phase 2 imports external work history for partial credit to break the new-agent freeze; Phase 3 flips to charging the demand side once a liquidity threshold (not a calendar date) is met.

IV.12.1 Static escrow vs. competitive underwriting

Figure IV.12 draws the static-bond and competitive-underwriting alternatives side by side.

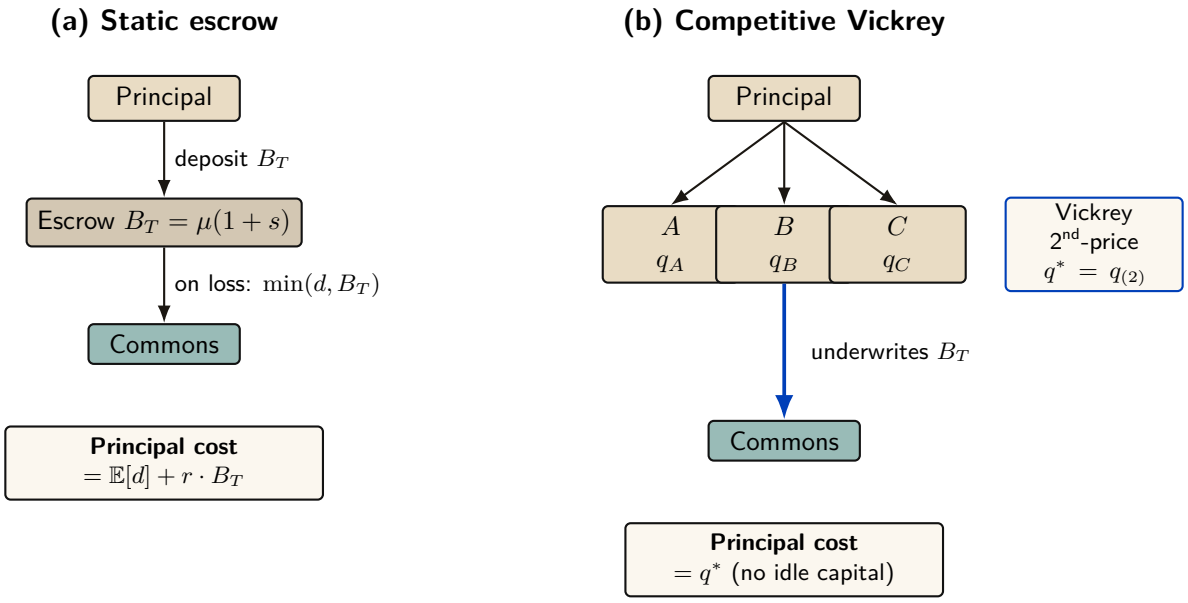
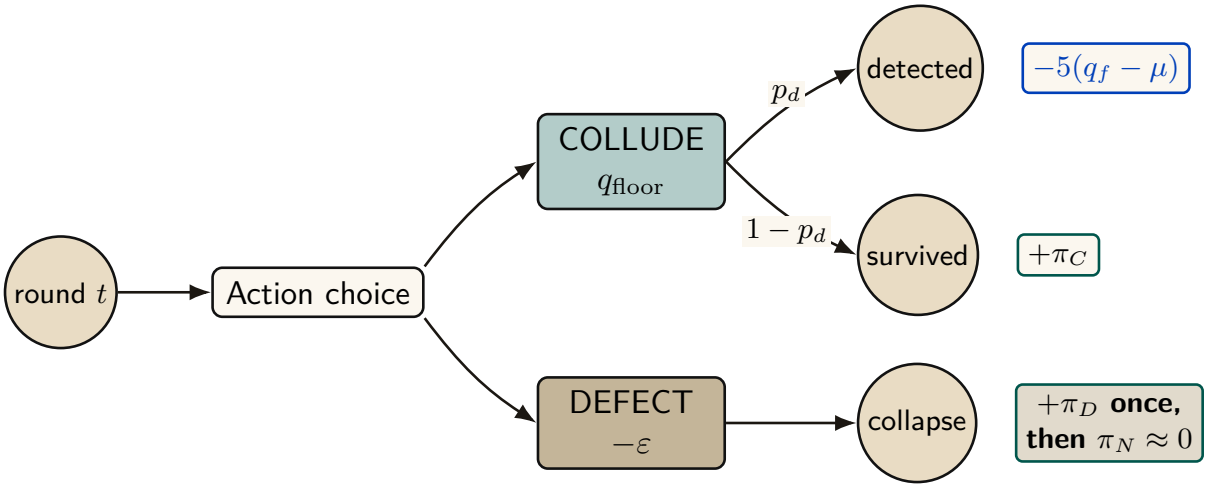


Figure IV.12: Static escrow vs. competitive Vickrey auction. Both regimes underwrite the same coverage $B_T = \mu(1+s)$; only the financing mechanism differs. In (a) the principal ties up B_T for the coverage period, paying an opportunity cost $r \cdot B_T$. In (b) insurers compete; the principal pays only the second-lowest bid q^*. The competitive regime Pareto-dominates the static one under the welfare assumptions of §IV.12.1 below.

Thomas Youle's competitive-insurance proposal — insurer-agents bid to underwrite transactions, replacing a static bond parameter with a market-discovered price — is a possible *fourth* side. But it may collide with the cleanup lower bound:

Proposition IV.12.1 (Underwriting reframes the bond as reserve adequacy)**.** *The cleanup lower bound requires bond ≥ worst-case cleanup cost c. A competitive insurance market prices to* expected *loss, which sits below the worst case for any below-tail agent. Either underwriting violates the cleanup lower bound (the commons can leak in the tail), or the bound must be reframed as the* insurer's *capital-adequacy constraint (a reserve-requirement framing with tail reinsurance), not a per-transaction bond. The* form *of the answer is determined; only the reserve formula is open.*

IV.12.2 Cartels and wash-trading



Model-conditional sustainability: $V_C = \dfrac{\pi_C - p_d L}{1-\delta(1-p_d)} \geq \pi_D, \quad L = 5(q_f - \mu)$

Figure IV.13: Cartel repeated-game node under the supplied grim-trigger model. Each round a member maintains the floor or undercuts it by ε. Detection occurs before the current-round payoff and applies loss L; a one-shot deviation takes π_D and ends the cartel stream. The displayed recurrence is specific to that event timing and payoff model.

Multi-homing plus portable reputation enables cross-harbor *cartels* (a rating cartel is the closest real-world analogue to "the harbor's universities and rating agencies"). And a colluding requester+worker can *wash-trade* fake settled outcomes to mint reputation cheaply. The defense form — cost-per-settled-outcome must exceed reputation's marginal value — is circular as stated, because that marginal value is endogenous to the same market. The fix is a structural break: make settled outcomes between a counterparty *pair* exhibit sharply diminishing reputation returns (a concavity that caps the wash-trade payoff regardless of price), plus sampled adversarial re-audit of high-velocity pairs.

Exercises

E1. Which side should Phase 1 subsidize, and why? Apply the "scarcest cross-side externality" rule.

E2. Use Figure IV.13 to identify the grim-trigger condition under which the cartel is self-sustaining. What raises the critical discount factor δ enough to break it?

E3. ★ "Price of anarchy when reputation is for sale" is the wrong tool: once reputation is tradeable, you have changed the game, so PoA (a ratio for a *fixed* game) does not apply. Reframe it as a *signal-existence* question (Spence-signaling with a resale market): does *any* informative separating equilibrium survive reputation resale?

IV.13 A gluing analogy for the open federation problem

Three open problems—cross-harbor settlement, equivocation under partitions, and cross-currency valuation—share a local-to-global shape. That resemblance motivates an analogy; it does not yet define a sheaf or a cohomology theory.

Conjecture IV.13.1 (Formalization target, not established result)**.** *A future treatment may define a site of overlapping administrative domains and a presheaf of signed ledger views, then ask when compatible local sections admit a unique global section. This paper has not specified the site, restriction maps, coefficient object, or cocycles; therefore it makes no claim that a particular H^1 is defined or non-zero. The immediate engineering obligations remain the concrete ones: matched custody transitions, freshness policy under partitions, and time-indexed valuation.*

Fong and Spivak's compositionality program [28] suggests vocabulary for a future formalization. Until the site and coefficient data are defined, the paper uses "gluing" only to organize engineering questions.

Exercises

E1. Name the site, cover, restriction maps, and coefficient object a genuine sheaf model of federated ledger views would require.

E2. Explain why two incompatible signed roots are evidence of equivocation but are not, without further definitions, a cohomology class.

E3. ★ Build a small formal gluing model for a bounded gossip topology and state exactly what consistency result it proves.

IV.14 Gaps the design must still close

Honesty requires naming what is unspecified. These are not threats already defended; they are holes.

1. **Multi-dimensional reputation aggregation.** "Accuracy/aesthetics/ efficiency judged by separate experts" has no specified combination rule. You cannot have "separate axes" and "one buyer-readable score" without specifying an aggregation function and its trade-offs. (*proposed*.)
2. **Float-plan amendment under scope drift.** The four states assume the acceptance criteria were right. Mid-flight re-specification needs a re-sign protocol with escrow top-up/refund delta, preserving conservation.
3. **Cross-boundary key rotation.** How a rotated signing key invalidates or re-validates cross-harbor delegations already held by B is unspecified.
4. **Skill versioning.** Reputation attaches to a content hash; v2 starts cold. The lemons problem reappears at every version bump.
5. **Operator exit / harbor death.** Escrow orphaning, in-flight float plans, and reputation tombstoning on ungraceful exit are unspecified — the market-level analogue of the kernel's crash-recovery problem.
6. **Incentive-compatible arbitration capacity.** The human oracle is scarce and expensive; at scale arbitration is itself a priced, bondable, slashable service that must converge to honest rulings, not rubber-stamping.

IV.15 Related work

The design borrows from five literatures, and its only originality is in how it *joins* them; this section consolidates the threads cited piecemeal above.

Multi-sided platform markets. The frame is Rochet & Tirole [99, 100] and Armstrong [131]: a platform is multi-sided when the *allocation* of price across user groups, not just its level, moves volume. We extend the standard two-sided picture to three sides by counting *incentive constraints*, and we read the contemporary AI-labor-market model of Chiu, Zhang & van der Schaar [42] as confirmation that "reputation system as a third side" is the natural structure.

Mechanism design and its impossibilities. Settlement is a direct mechanism in the sense of Myerson [169]; the binding constraint is Myerson–Satterthwaite [168], which forces the market to surrender efficiency to keep budget balance, incentive compatibility, and individual rationality. The incentive-compatibility arguments are imperfect-monitoring folk theorems (Green–Porter; Abreu–Pearce–Stacchetti [54]), whose exogenous public signal we cannot take for granted — hence the grading-oracle problem. Grossman–Hart [178] supplies the residual-control-right argument for the rental side, and Akerlof [86] the lemons argument for licensing.

Reputation systems. Empirically, online reputation has measurable value (Resnick et al. [158]; Tadelis [179]); theoretically, unbounded records create reputation *bubbles* rather than stability (Liu & Skrzypacz [166]), which is why our horizon is a tunable. The Sybil attack [98] is the reason reputation must key on a principal, not a spawn. LLM-as-judge de-biasing follows Zheng et al. [124].

Distributed systems and cross-chain commerce. Federation refuses consensus because of FLP [132], recovers a weaker guarantee under partial synchrony [43], propagates by epidemic gossip [1], and takes a Certificate-Transparency detect-don't-prevent posture [30, 29], mindful of eclipse attacks [71]. The settlement design is located explicitly against atomic cross-chain swaps [134] and the adversarial-commerce model of Herlihy, Liskov & Shrira [133]; reputation revocation is contrasted with Sagas-style compensation [93].

Applied category theory. The trace-composition notation and proposed gluing formalization draw vocabulary from ologs [56] and Fong & Spivak [28]; no cohomology claim is made. The commons-governance reading draws on Ostrom [72], and the auction machinery on standard algorithmic game theory [151].

Agentic-commerce protocols. While this design was being written, the industry shipped the transaction layer of an agent economy. The Universal Commerce Protocol [198] (Google/Shopify, 2026) standardizes agent-mediated retail — discovery via a `/.well-known` capability profile, a checkout state machine, server-selects version negotiation — and its rival, the Agentic Commerce Protocol [7] (OpenAI/Stripe), the platform-mediated equivalent. Both are deliberate about what they exclude: neither settles payments (delegated to payment service providers) and neither decides which agents deserve trust. Authorization proof is delegated to the Agent Payments Protocol [5], whose mandates — W3C Verifiable Credentials in SD-JWT form, chained principal → platform → merchant — are the deployed sibling of this paper's float-plan countersign, and whose credential formats the reference implementation now adopts at the harbor boundary (ADR-0094) so that the keystone identity artifacts of §IV.6 verify with off-the-shelf tooling. The gap those protocols leave open is this paper's subject: published security analyses of UCP [61] note that it regulates *how* agents transact while offering nothing about *which* agents to trust — no collateral, no settlement oracle, no reputation, no priced deterrent. The harbor economy is a

mechanism for exactly that layer; the retail protocols are evidence the layer below it is arriving on schedule, and hosted trust (§IV.11) is what they will need bolted on.

IV.15.1 Mechanism 9: The Adversarial Test Market and Purchased Assurance

Beyond labor and IP licensing, the Harbor Economy establishes an internal market for **adversarial test generation**, formalizing PR verification as an interactive proof system (delegated computation [177] and multi-agent debate [87]).

Theorem IV.15.1 (Purchased Assurance Bound)**.** *Let an author-agent submit a Float Plan implementation with a proposed diff. Let k independent, conflict-free test-writing agents each discover planted or latent flaws with independent detection probability $d \in (0, 1)$. The probability that a critical flaw survives undetected across all k reviewers is bounded by:*

$$\Pr(\textit{Defect Undetected}) \;\leq\; (1-d)^k. \tag{IV.1}$$

Truthful effort is a strictly dominant strategy for test-writers under a bounty-per-killed-mutant and slash-per-false-alarm contract satisfying the Becker condition $\rho dB > G$. The operator's assurance level $1-(1-d)^k$ is thus a purchasable commodity priced at $k \cdot (\textit{bounty} + \textit{bond carry})$, making "hosted trust as a service" a quantifiable line item.

The ground-truth oracle for evaluating test-writers is **mutation testing kill rates** against held-out syntactic mutation operators, preventing agents from overfitting tests to known test suites.

Table IV.3: How the harbor economy sits relative to the nearest prior art on each axis.

Axis	Nearest prior art	What the harbor does differently
Market structure	Two-sided platforms [99]	Three incentive constraints (labor / capital / IP) on one escrow object
Settlement across distrust	Atomic swaps [134]; cross-chain deals [133]	Bounded (not trustless) escrow for *non-fungible, reputation-priced* bonds
Revocation	PKI revocation lists; Certificate Transparency [30]	Gossip-converged filters with a priced, named attacker window
Reputation revocation	Sagas compensation [93]	Tombstone propagation — no conserved quantity to compensate
Grading / IC signal	Folk theorems [54]	Strategy-proof machine-checkable oracle *or* bonded-and-slashed judges
Agent transaction rails	UCP [198]; ACP [7]; AP2 mandates [5]	They standardize the transaction and name trust out of scope; the harbor prices and enforces the trust layer they delegate away

IV.16 Conclusion

The harbor-master makes a swarm legible and accountable for one operator (the wedge of Chapter I). When operators sail out to trade, the *same* ledger that kept one operator's swarm honest becomes the market that lets fleets who don't trust each other work together: three sides, one bond, hosted trust. We have not papered over the cost of that claim. The keystone identity primitive — local non-forgeable identity — covers a single-operator threat model and must be extended to cross-operator attestation before the boundary is formally one "neither controls." Conservation composes exactly within each unit of account; cross-currency totals require time-indexed valuation and an explicit exposure model. Reputation is monotone in honest outcomes and revocable by tombstone, not by Sagas. Every folk-theorem result rests on a grading process included in the

monitoring model. Myerson–Satterthwaite constrains bilateral slices only when its assumptions hold; the paper does not yet prove which corner the full market sacrifices. The bones are good and the floor is built; the spine the market trades on has partial local identity and outcome substrates but no complete reputation or cross-operator binding — and this paper says so in the same words throughout, because a market built on an overstated keystone is a market built on sand.

Chapter Appendices

IV.A Implementation & status

The body of this paper is deliberately implementation-agnostic. This appendix records, for the reader who wants to check the claims against running code, exactly which mechanisms are shipped in the reference implementation, which are specified with a proof or a written protocol, and which are still proposed. The single-line summary: an **implemented** conserving bond ledger, an **implemented** continuity organ (episodic memory), a PARTIAL checkpoint, and a richly SPECIFIED-and-model-verified protocol stack — but the spine the market trades on (cross-operator identity → outcome ledger → reputation) is PARTIAL-to-*proposed*. Today the harbor economy is a two-sided market with a proof-backed roadmap to three.

Table IV.4: Per-claim maturity, with concrete grounding in the reference implementation. "Module" names refer to source files in the implementation; "model" means a mechanized proof artifact; "property test" means a randomized test exercising the invariant.

Claim / mechanism	State	Grounding
Bond ledger: escrow, slash, commons pool	**implemented**	bond-ledger module (`lib/bonds.ts`)
Conservation wallet + escrow + commons = supply	**implemented**	10k-trace property test
No-spawn-without-bond	PARTIAL	escrow-first; runtime gate is a roadmap item
Episodic memory (continuity organ 1)	**implemented**	episodic-memory module
Resurrection / checkpoint (organ 2)	PARTIAL	passes notes, not state
Outcome ledger (organ 3 — reputation keys here)	PARTIAL	commitments, daemon-derived deadlines, oracle closure, API/CLI, and overdue monitoring ship; neutral grading and reputation binding do not ship
Local non-forgeable identity (intra-harbor)	PARTIAL	`actor-souls`, lookup credentials, and a bounded newcomer pool ship; full write gating does not ship
Cross-operator attestation	SPECIFIED/*proposed*	no spec for the load-bearing part
Float plan + signed escrow ceremony	SPECIFIED	Protocol IV.4.1
Cross-harbor capability transfer (attenuation)	SPECIFIED	verified *as model*; Protocol IV.8.1
Reputation estimator (Elo / Bradley–Terry / TrueSkill)	SPECIFIED/*proposed*	no estimator shipped
Multi-dimensional reputation, neutral judges	*proposed*	no axis-aggregation spec
Three-sided market (labor / rental / licensing)	SPECIFIED	two-sided in reality
Skill licensing (metered + clawback + Merkle proof)	SPECIFIED	cross-harbor verifiability undeployed
Federation: witness log, capability transfer	SPECIFIED	Chapter VII (*The Federated Harbor*)
Recipient-whitelisted escrow custody	SPECIFIED (conditional property)	requires non-bypassable authority; no runtime conformance
Cross-harbor conservation	SPECIFIED (property)	proof sketch (App. IV.B) + TLA$^+$
Federated revocation dissemination	SPECIFIED (conditional expectation)	no finite partition deadline; equivocation delivery *open*
Hosted-trust moat / rail separation	SPECIFIED (positioning)	not a code artifact
Competitive underwriting (4th side)	*proposed*	composition unresolved

IV.B Proof sketches

We sketch the three load-bearing results; full proofs of the conservation results appear in Chapter VII (*The Federated Harbor*).

Proof sketch of Theorem IV.7.1. For a valid trace $f : x \to y$, define $\Delta(f) = S(y) - S(x)$. For composable traces $f : x \to y$ and $g : y \to z$,

$$\Delta(g \circ f) = S(z) - S(x) = [S(z) - S(y)] + [S(y) - S(x)] = \Delta(g) + \Delta(f).$$

Every internal generating operation has zero delta because its debits and credits match, including a cross-harbor move through the explicit in-flight bucket. By induction on trace length, a composition of internal generators has zero delta; top-ups and withdrawals contribute exactly their recorded boundary amounts. □

Boundary note for Property IV.7.1. Native-unit conservation follows by applying Theorem IV.7.1 separately to each unit. A scalar valuation $V_t(x) = \sum_j v_{t,j} x_j$ changes either when native balances change or when the valuation vector v_t changes. The second term is mark-to-market exposure. No finite φ follows without assumptions bounding rate movement, oracle delay, fees, and slippage; those assumptions are not supplied here. □

Proof of Theorem IV.7.2 (the sacrificed corner). Restrict attention to a bilateral slice satisfying the assumptions stated in Theorem IV.7.2. The Myerson–Satterthwaite impossibility then rules out the simultaneous achievement of all four desiderata. The implication is conditional: before declaring which corner the harbor gives up, a mechanism must specify its types and transfers and prove its incentive and participation properties. The present paper has not done so, hence makes no stronger allocation claim. □

PORT DADDY COORDINATION PAPERS · CHAPTER V OF VII

The Anchor Protocol

Abstract

As AI agents transition from isolated copilots to collaborative, autonomous swarms, local development environments face a crisis of coordination and trust. Existing tools designed for human-driven, static infrastructures lack the primitives required to prevent dynamic agents from corrupting shared state or squatting on ephemeral ports.

This paper introduces the **Anchor Protocol**, a cryptographic and semantic identity framework built into the Port Daddy daemon. We detail the protocol's evolution from symmetric MACs to Ed25519 signatures and multi-hop delegation chains inspired by Macaroons [20]. ProVerif [39] checks symbolic authentication and attenuation properties of the protocol models; Kani checks bounded memory-safety and source-level control-flow properties of the Rust core; runtime conformance tests cover the deployed bridge. These layers do not amount to a proof of the entire implementation or of hardware-level constant time. The multi-hop proof is also structurally bounded, distinct from ProVerif's unbounded-session quantification. Section V.7.3 records those boundaries explicitly.

Keywords: distributed systems security, formal verification, capability-based security, multi-agent coordination, symbolic protocol analysis, ProVerif, JWT security, Ed25519

Reading time: approximately 30 minutes end-to-end. Chapter VI, The Bonded Commons: Pre-Transactional Trust Infrastructure for Multi-Agent Systems *(Owens 2026), carries the economic and governance argument; either paper can be read first.*

PORT DADDY COORDINATION PAPERS **CHAPTER V OF VII**

V. The Anchor Protocol (this chapter)**.** Authentication, delegation, attenuation, and revocation inside one harbor.

VI. The Bonded Commons. Evidence, bonding, advisory claims, and the incentive conditions for cooperation.

VII. The Federated Harbor. Identity, revocation, and settlement across mutually sovereign machines.

Chapters I–IV form the product arc; Chapters V–VII provide the protocol, economic, and federation deep dives. Every chapter is independently readable.

V.1 Introduction: The "Local Swarm" Problem

In a multi-agent development environment, agents operate concurrently to read files, spin up test servers, and execute commands. This introduces three critical threat vectors on `localhost`:

1. **Port Squatting (The "Ghost in the Harbor"):** If Agent A crashes, a malicious or malfunctioning Agent B can bind to Agent A's previously assigned port, intercepting traffic meant for A.
2. **Resource Contention:** Agents fighting over shared resources (e.g., a SQLite database file) without a centralized locking mechanism cause state corruption.
3. **Privilege Escalation:** A "Reviewer" agent delegated by a "Coder" agent should not possess the rights to initiate production deployments, yet local environments rarely enforce principle-of-least-privilege between processes.

To solve this, Port Daddy introduces the concept of a **Harbor**—a zero-trust, namespace-isolated execution environment where agents must prove their identity and capabilities. Our approach builds upon foundational work in capability-based security by Dennis and Van Horn [101] and the principle of least privilege articulated by Saltzer and Schroeder [108].

Volume Context. This chapter provides the cryptographic substrate. Chapter VI (*The Bonded Commons: Pre-Transactional Trust Infrastructure for Multi-Agent Systems*) builds the economic and game-theoretic layer on top: bonded participation, conditional auctions, mechanism analysis, and Sen-inspired advisory conflict resolution. The Anchor Protocol is a self-contained subset consumed as the trust kernel for the wider mechanism design. Read this chapter for "how does the daemon authenticate an agent?"; read Chapter VI for "why should there be a daemon at all?"

V.2 The Anchor Protocol Architecture

The Anchor Protocol defines how agents authenticate to a Harbor and to each other. It issues **Harbor Cards** (highly constrained JSON Web Tokens) and evolves them across four phases, each adding a capability that closes a specific attack surface from the previous phase (Figure V.1).

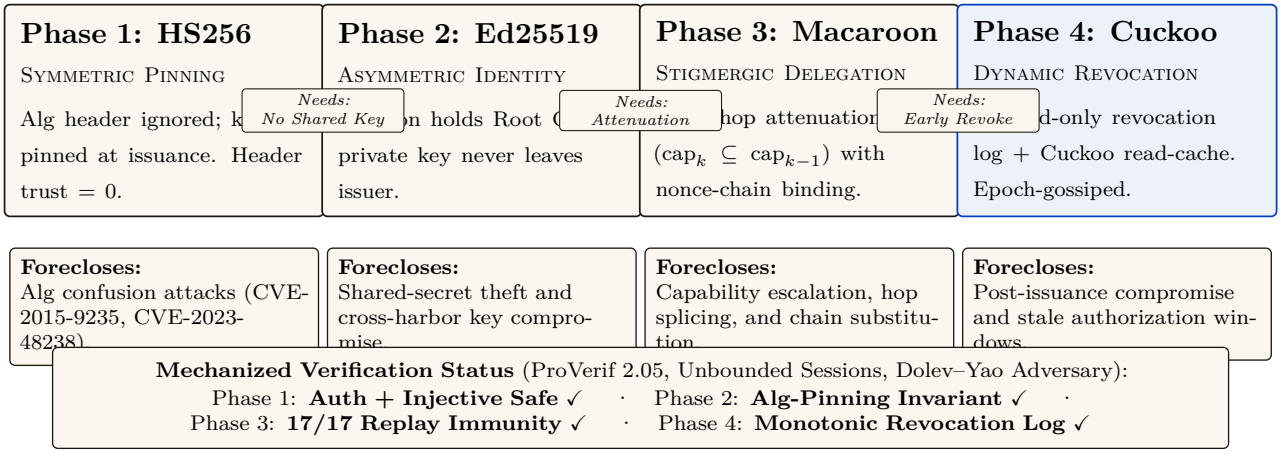


Figure V.1: The four phases of the Anchor Protocol. Each phase refines the previous: the cryptographic primitive strengthens, and a specific adversary threat is formally foreclosed. Phases 1–3 are mechanized in ProVerif (§**??**); Phase 4 is enforced via append-only gossip logs and monotonic filter proofs (§V.2.4).

A common thread across all phases is *capability attenuation*: a delegated token never carries more authority than its parent, and TTL only shrinks. Figure V.2 shows the nested-cap structure that the verifier enforces on every chain.

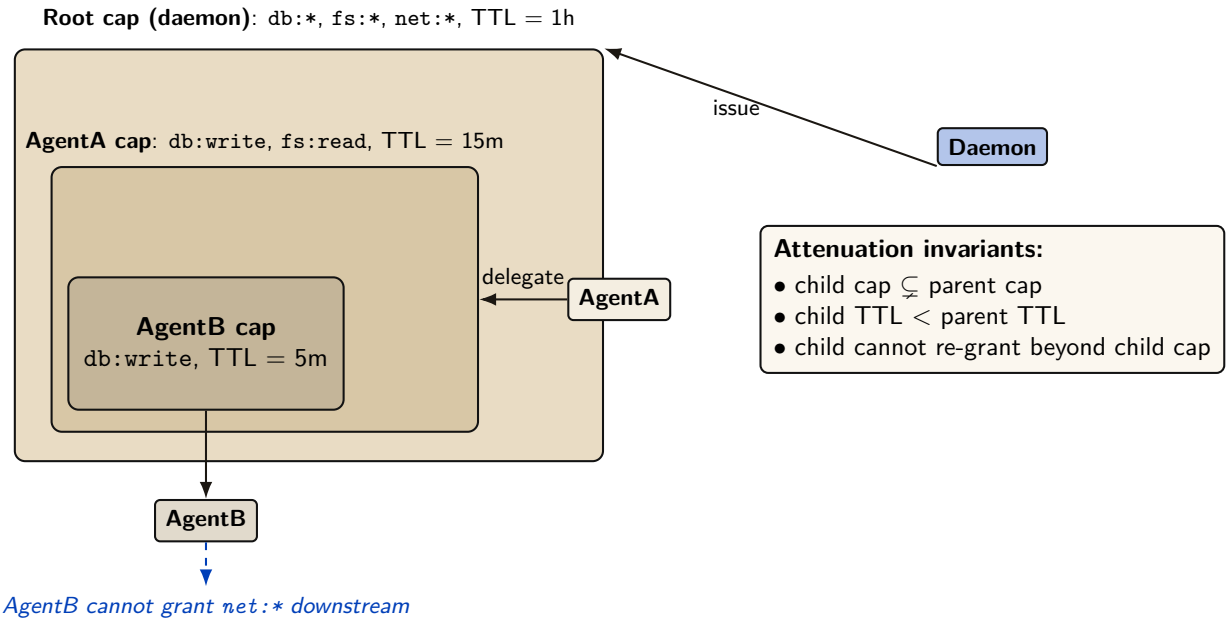


Figure V.2: Capability attenuation under delegation. Each child capability is a strict subset of its parent's (rights and TTL both shrink monotonically). AgentB inherits only what AgentA explicitly delegates, and cannot re-grant capabilities it does not hold. This is enforced both at issuance (parent signs over the child's attenuated claim set) and at verification (the verifier checks the chain of `is_subset` relations).

V.2.1 Phase 1: Symmetric Pinning

Early versions of the protocol relied on HS256 (HMAC-SHA256). The primary vulnerability in JWTs is "Algorithm Confusion" (CVE-2015-9235 [51], CVE-2023-48238 [53]), where an attacker modifies the token header to `{"alg": "none"}` or symmetric HS256 while using an asymmetric public key as the HMAC secret.

Property V.2.1 (Algorithmic Pinning)**.** The Port Daddy Verifier employs strict **Algorithmic Pinning**. It entirely ignores the user-provided `alg` header and explicitly forces the underlying crypto library to evaluate the signature using the pre-determined algorithm. This approach is recommended by the JWT Best Current Practices draft [207].

Figure V.3 shows the two verifier paths side by side: the naive verifier trusts the `alg` field on the wire (and is exploited), while the pinned verifier records the issuance algorithm out-of-band and admits no rewrite trace.

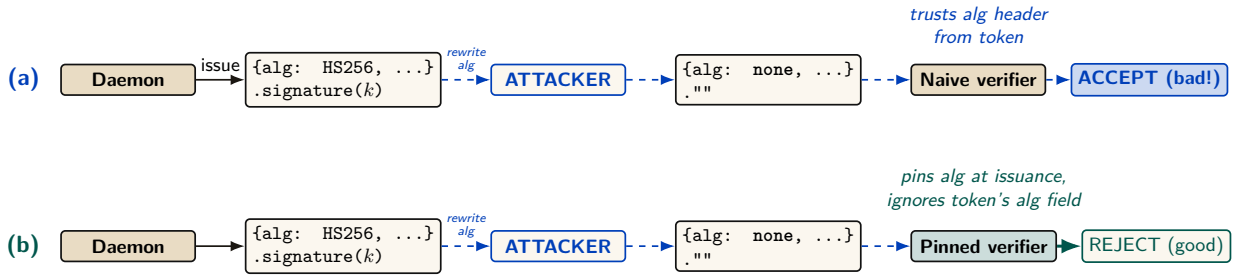


Figure V.3: Algorithm-confusion attack and the pinned-verifier defense. (a) A naive verifier trusts the `alg` field of the incoming token; an attacker who can intercept the token rewrites `alg=HS256` to `alg=none`, strips the signature, and the verifier accepts an unauthenticated message. (b) The pinned verifier records the issuance algorithm out-of-band, then re-checks each token against the pinned algorithm rather than the token's self-described one. The ProVerif model encodes both verifiers and shows the naive path produces a counter-trace witness for the rewrite, while the pinned path admits no such trace. Runtime conformance: 5 attack patterns + 2 controls, all pass.

V.2.2 Phase 2: Asymmetric Identity (Ed25519)

To support decentralized Agent-to-Agent (A2A) communication without sharing HMAC secrets, the protocol transitioned to **Ed25519 (EdDSA)** [58, 186]. Ed25519 provides fast, deterministic signatures with small keys and signatures, making it ideal for local agent communication.

Definition V.2.1 (Harbor Key Hierarchy)**.** The Port Daddy Daemon acts as the Root Certificate Authority (CA). Each Harbor is assigned a unique Ed25519 keypair. The Daemon issues Harbor Cards signed by the Harbor's private key. Agents use the Harbor's public key (broadcast via Port Daddy's ambient discovery service) to verify tokens presented by peers.

V.2.3 Phase 3: Stigmergic Delegation (The "Biscuit" Pattern)

Port Daddy v4 introduces multi-hop delegation. When Agent A spawns Agent B, it does not need to request a new token from the Daemon. Instead, it uses **Offline Attenuation**, inspired by Macaroons [20].

Definition V.2.2 (Offline Attenuation)**.** Agent A takes its existing Harbor Card, appends a new set of restricted capabilities (e.g., `["db:read"]` reduced from `["db:read", "db:write"]`), and signs the *entire chain* with its own ephemeral private key.

The Harbor verifies the Daemon's root signature, Agent A's delegation signature, and mathematically ensures that Agent B's capabilities are a strict subset of Agent A's.

For depth-N chains, the verifier walks the structure shown in Figure V.4. Each hop must produce a fresh nonce and bind the previous and next identities into its signature so that an attacker who intercepts a single hop cannot splice it into a different chain.

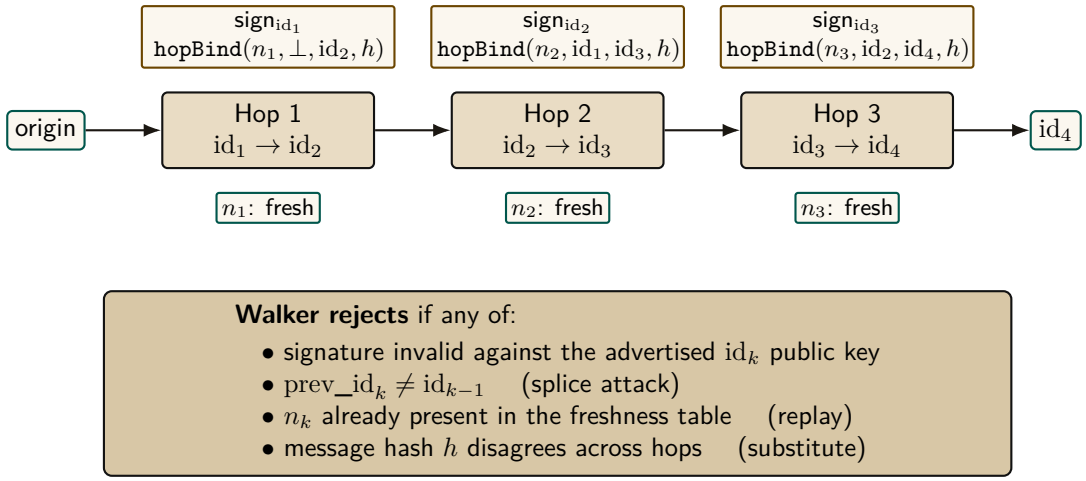


Figure V.4: Multi-hop delegation walker. Each hop k signs the binding tuple (n_k, id_{k-1}, id_k, h) under Ed25519, where h is the message hash that must remain constant across the chain. The walker verifies depth-N chains with a nonce-freshness table and rejects splices, replays, hop-swaps, and substitution. Runtime: `lib/delegation-chain.ts`; ProVerif conformance test passes 17 / 17.

V.2.4 Phase 4: Revocation via Cuckoo Filter

Capability tokens have a TTL, but natural expiry is not always sufficient. Two circumstances force early withdrawal: (i) *compromise*—a Harbor Card leaks from its agent's process, and every second until TTL is a second the attacker has legitimate capabilities; and (ii) *policy reversal*—a principal decides an agent should no longer hold `db:write`, and waiting for TTL is unacceptable. A naive revocation list is $O(n)$ per verification and grows monotonically. While the Anchor Protocol previously attempted to gossip a cuckoo filter directly (which succumbed to the bitwise merging impossibility), it now gossips an **Append-Only Event Log of Revocation IDs** using Vector

Clocks. To maintain $O(1)$ read performance, each daemon continually projects its event log into a local, ephemeral cuckoo filter [35] (Figure V.5).

Cuckoo filters in one paragraph. A cuckoo filter is a compact probabilistic set: under successful insertion and correct deletion discipline it has false positives but no false negatives, while supporting deletion. Each inserted key is reduced to a fingerprint and placed in one of two candidate buckets; on collision, fingerprints are relocated until a slot opens or insertion fails. Lookups inspect both buckets. The expected lookup cost is $O(1)$ and the approximate false-positive probability for two buckets of size B and f fingerprint bits is at most about $2B/2^f$ in the low-collision model. High load increases insertion failures; the implementation rejects or rebuilds rather than silently dropping a revocation. Duplicate fingerprints require counted or authoritative-table-backed deletion so that removing one key does not reanimate another.

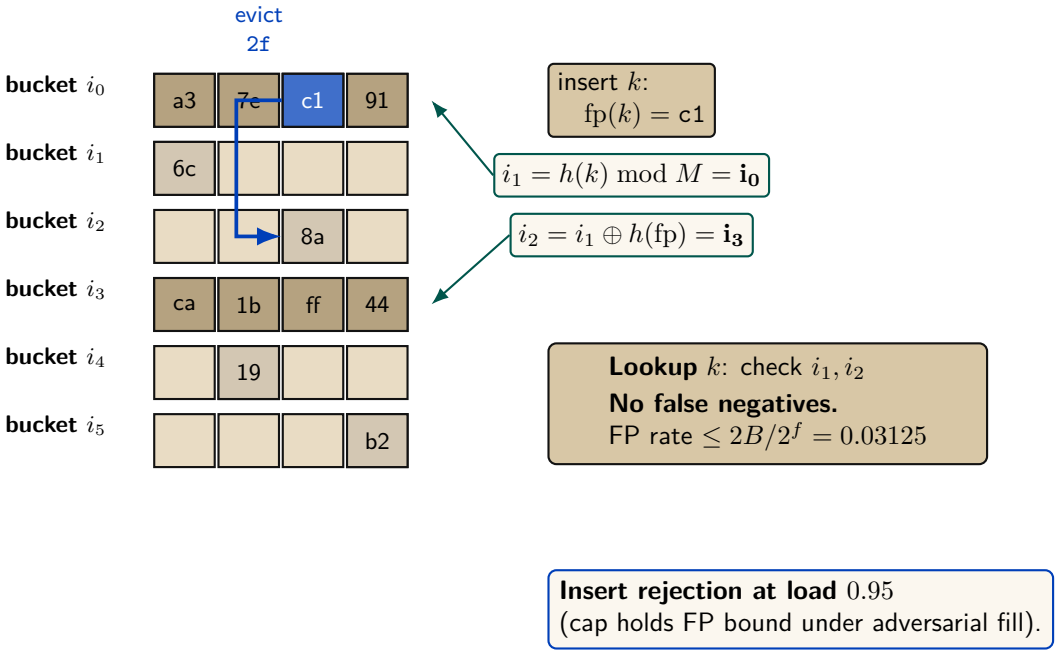


Figure V.5: Cuckoo-filter revocation gate (Fan et al. [35]) with the load-factor ceiling that defeats the pollution attack. Insert lands at either of two candidate buckets $i_1, i_2 = i_1 \oplus h(\text{fp})$; on full, evict and re-insert at the kicked fingerprint's alternate. Insert rejection at 0.95 caps occupancy inside the regime where the false-positive bound holds.

Why cuckoo, not Bloom. For uniform-TTL workloads, a rotating Bloom filter would suffice. Port Daddy's TTLs are heterogeneous—Shipwright proposals run hours to days, `pd spawn` children run minutes, one-shot agents run seconds. Rotating Bloom either rotates at the longest TTL (short-TTL revocations linger orders of magnitude longer than needed) or buckets by TTL class (reintroducing the structural complexity cuckoo collapses). Cuckoo provides $O(1)$ per-entry removal at each specific expiry, plus the same removal handles operator-initiated early-revocation undo cleanly.

Space. At false-positive rate ϵ, a cuckoo filter uses approximately $\log_2(1/\epsilon) + 3$ bits per entry. For $\epsilon = 10^{-3}$, ≈ 13 bits per revoked card. A fleet retaining 10^5 active revocations fits in $\approx 160\,\text{KB}$—trivially in memory, trivially gossiped.

Definition V.2.3 (Revocation-Augmented Verification)**.** Let R_t denote the authoritative Append-Only Event Log of revoked Harbor Card IDs at time t. Each daemon d maintains a local, ephemeral cuckoo filter index $F_d \approx R_t$. Every Harbor Card verification additionally:

1. Looks up the card's `kid` in F_d. If absent, admit.

2. If present, query the local `revocations` table (authoritative). If genuinely revoked, reject with `revoked`; otherwise the filter is in the false-positive regime and the caller is admitted, with an asynchronous reconciliation job repopulating the filter.

Gossip-based synchronization. Daemons in a mesh synchronize revocation deltas by anti-entropy gossip [8]. Under a connected complete-overlay model with independent uniform peer selection and reliable rounds, epidemic dissemination is expected $\Theta(\log m)$ rounds. This is an expectation, not a deadline; partitions, message loss, topology, and scheduling change the constant or prevent global convergence until the partition heals. Security-critical revocations can additionally trigger best-effort immediate push. Figure V.6 illustrates propagation across five daemons; it is a trace, not a timing proof.

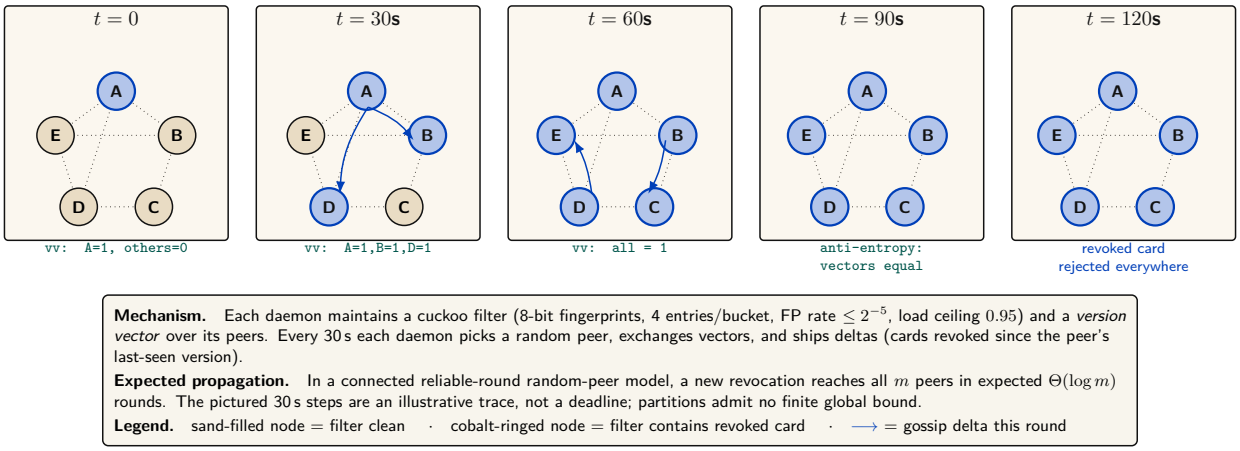


Figure V.6: Gossip-synchronized revocation across a five-daemon mesh. At $t = 0$ daemon A locally revokes a Harbor Card (its filter and revocation table). Anti-entropy gossip ships deltas to random peers every 30 s. Expected $\Theta(\log m)$ propagation requires the stated connected reliable-round model; the panels show one possible execution, not a worst-case time guarantee. Version vectors make duplicate and reordered deltas idempotent once delivered, but do not heal an unbounded partition.

Property V.2.2 (Filter Monotonicity over a TTL Epoch)**.** For any Harbor Card c with issue time t_0 and TTL τ, if $c \in R_t$ for some $t \in [t_0, t_0 + \tau]$, then $c \in R_{t'}$ for all $t' \in [t, t_0 + \tau]$. After $t_0 + \tau$, c may be removed from R since the card has expired.

Property V.2.3 (No Partial Reanimation)**.** A revoked card cannot be un-revoked within its TTL. If an operator decides the revocation was erroneous, they must issue a new card.

Property V.2.4 (Conditional Gossip Expectation)**.** Under the complete-overlay, uniform-peer, reliable-round model above, a revocation disseminates to all m daemons in expected $\Theta(\Delta \log m)$ time. No finite worst-case freshness bound holds under an unbounded partition.

Soundness preservation. Revocation strictly narrows acceptance, so existing ProVerif soundness proofs (Theorem V.4.1) are unaffected. The new proof obligation is small: revocation is *enforced*, which follows by inspection of the verification path—the revocation check is unconditional and precedes admission. The card lifecycle gains a state (*valid* → *revoked* → *expired*) modeled as an additional transition.

Privacy. A Dolev-Yao adversary observing all gossip traffic learns the revoked-card IDs, leaking which agents have been disciplined. For an organization's own daemons, this is acceptable audit transparency. Stronger privacy is available by encoding revocations as salted hashes $H(\texttt{kid}, \texttt{salt})$ where the salt rotates daily, stored encrypted under a harbor session key.

V.3 Related Work

V.3.1 Formal Verification of Security Protocols

Formal verification of cryptographic protocols has matured significantly over the past two decades. Following the seminal work by Lowe [88] on authentication hierarchies and the Dolev-Yao model [60], several automated tools have been developed:

- **ProVerif** [39]: An automatic symbolic protocol verifier supporting an unbounded number of sessions, used to verify TLS 1.3 [47], Signal [152], and WireGuard.
- **Tamarin** [187]: A state-of-the-art protocol verifier that supports equational properties and inductive proofs, used for analyzing 5G authentication [63].
- **CryptoVerif** [37]: A computationally-sound protocol verifier that provides proofs in the computational model rather than the symbolic model.

Our work builds on these foundations, applying established verification techniques to the novel domain of local multi-agent coordination.

V.3.2 Capability-Based Authorization

Capability-based security dates back to Dennis and Van Horn's seminal 1966 paper on programming semantics for multiprogrammed computations [101]. Recent work by Birgisson et al. [20] introduced Macaroons, which use chained HMACs for decentralized delegation with contextual caveats. The Anchor Protocol's delegation chains (Section V.5.3) are inspired by Macaroons but adapted for JWT-based identity in local development environments.

V.3.3 JWT Security

JSON Web Tokens have been the subject of extensive security analysis. Algorithm confusion attacks were first systematically studied by McLean [194], with subsequent vulnerabilities documented in CVE-2015-9235 [51] (HS256/RS256 confusion), CVE-2018-1000531 [52] ("none" algorithm), and CVE-2023-48238 [53] (generic algorithm confusion). Our protocol implements the defense recommended by the IETF JWT Best Current Practices [207]: strict algorithm whitelisting and key separation.

V.3.4 Credential Formats in Agentic Commerce

Since this protocol was first specified, the agentic-payments industry has converged on a concrete credential dialect: the Agent Payments Protocol [4], adopted by the Universal Commerce Protocol [199] (Google/Shopify, 2026), carries its authorization mandates as W3C Verifiable Credentials in SD-JWT form — SD-JWT+kb for principal proof, JWS detached-content signatures (RFC 7515 Appendix F) over JCS-canonicalized payloads (RFC 8785) for counterparty authorization, with ES256 as the recommended algorithm and keys published as JWK sets in a `/.well-known` profile. The reference implementation migrates the Harbor Card envelope to this profile (`hv: 3`, ADR-0094) so that external verifiers need no bespoke SDK. The migration is deliberately envelope-only: the delegation-chain semantics, the attenuation invariant of Theorem V.4.1, and the ProVerif obligations are independent of the serialization, though the key-binding construction of SD-JWT+kb adds a binding obligation the model must re-discharge. The JWT hardening above (strict algorithm whitelisting, key separation) transfers unchanged — SD-JWT is a JWT profile, not a departure from one.

V.4 Formal Verification Strategy

Following the industry standard set by AWS (s2n-tls [27]) and Microsoft (Project Everest [122]), we decoupled the verification of the *protocol design* from the verification of the *implementation*. The full stack has three layers (Figure V.7): symbolic protocol analysis at the top, bounded model checking on the Rust source in the middle, and runtime conformance tests against the deployed binary at the bottom. Each layer's obligations flow downward; each layer's evidence flows back up. The stack is sound only if every bridge — symbolic-to-source and source-to-deployed — holds.

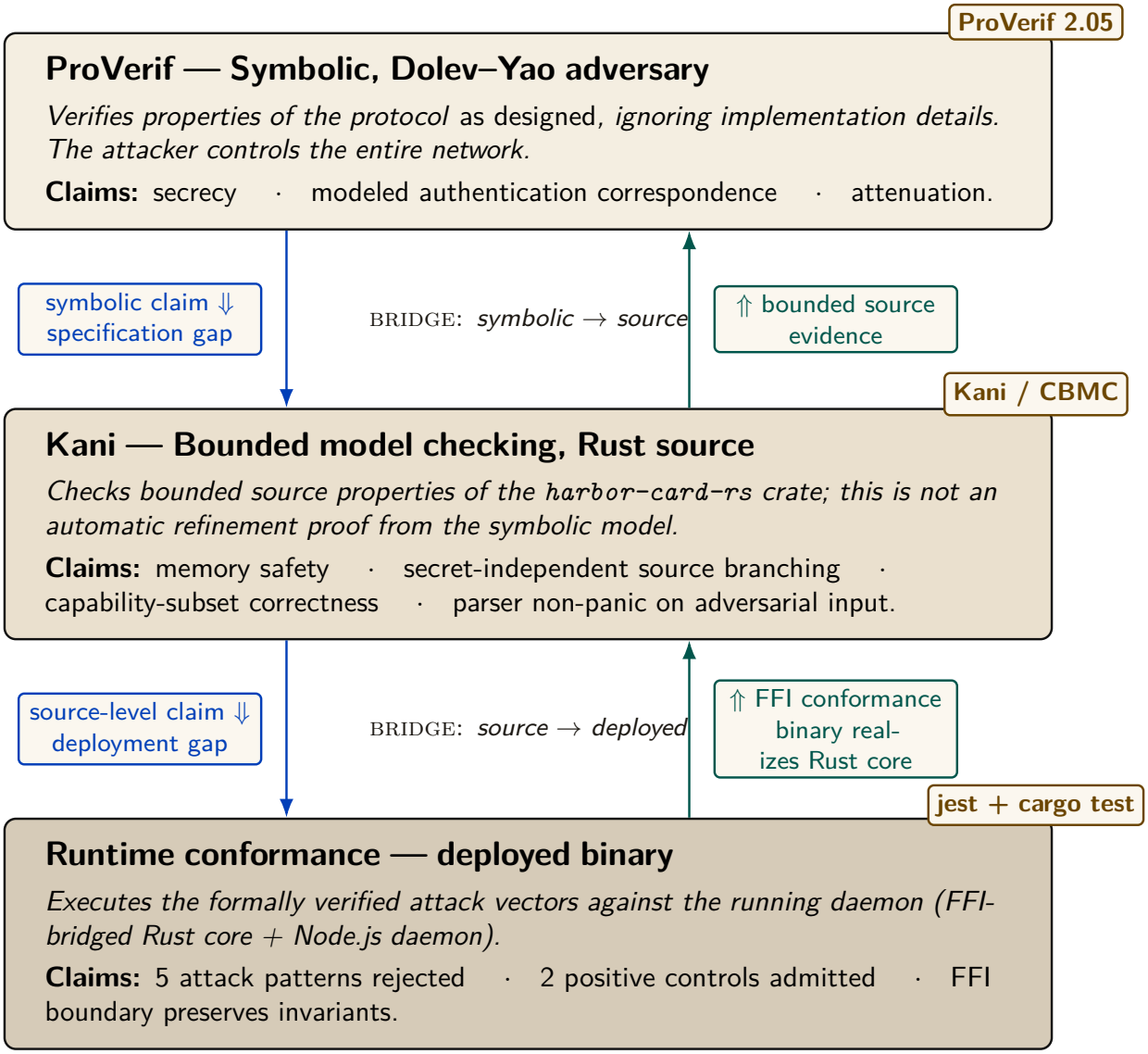


Figure V.7: The three-layer verification stack. ProVerif (top) discharges design-level obligations against a Dolev–Yao adversary. Kani (middle) checks bounded Rust source-level properties that overlap with, but do not refine the full symbolic model. Runtime conformance tests (bottom) exercise the FFI-bridged binary with selected attack vectors; they reduce rather than eliminate the deployment gap. Cobalt arrows show obligations flowing down (what each lower layer must preserve); teal arrows show evidence flowing back up. The unproved bridges remain explicit assurance-case assumptions rather than inferred theorems.

V.4.1 Symbolic Analysis with ProVerif

To prove the protocol's logical soundness against a Dolev-Yao adversary [60] (an attacker who controls the entire network), we modeled the Anchor Protocol in **ProVerif** [39]. ProVerif represents protocols as Horn clauses and uses resolution to prove security properties for an unbounded number

of sessions.

Theorem V.4.1 (Authentication Correspondence in the Modeled Phase)**.** *For the modeled delegation phase, ProVerif confirmed the correspondence:*

$$\begin{aligned} \mathit{Accepted}(b, h, \mathit{cap}) \implies & (\mathit{IssuedRoot}(a, h, \mathit{cap}) \wedge \mathit{Delegated}(a, b, \mathit{cap})) \\ & \vee \ \mathit{IssuedRoot}(b, h, \mathit{cap}) \end{aligned} \tag{V.1}$$

The displayed query is a non-injective authentication correspondence: every accepted event has a matching modeled issuance/delegation event. It does not by itself prove replay freedom, capability attenuation, or arbitrary-depth transitivity; those are separate queries and bounds.

All three protocol phases were independently verified in ProVerif 2.05. Table V.1 summarizes the results; full models are reproduced in Appendices V.B, V.C, and V.D.

Table V.1: ProVerif results for the listed model queries. Session replication is unbounded where the model uses replication; delegation-chain structure remains bounded as stated in §V.7.3.

Query	Phase	Model	Result
$\neg$ attacker(*master_key*)	v1 (HS256)	`v1_refined.pv`	**TRUE**
Accepted $\implies$ Issued	v1 (HS256)	`v1_refined.pv`	**TRUE**
$\neg$ attacker(*harbor_sk*)	v2 (Ed25519)	`v2_asymmetric.pv`	**TRUE**
Accepted $\implies$ Issued	v2 (Ed25519)	`v2_asymmetric.pv`	**TRUE**
Accepted(b) $\implies$ IssuedRoot(a) $\wedge$ Delegated(a, b)	v3 (Delegation)	`v3_delegation.pv`	**TRUE**

The following is the verbatim ProVerif 2.05 output for the delegation chain query (Phase 3), the most complex property:

```
RESULT event(Accepted(b_2,harbor_id[],cap_1))
  ==> (event(IssuedRoot(a_3,harbor_id[],cap_1))
       && event(Delegated(a_3,b_2,cap_1)))
      || event(IssuedRoot(b_2,harbor_id[],cap_1))
  is true.
```

Listing V.1: ProVerif 2.05 Terminal Output — Phase 3 Delegation Chain

V.4.2 Runtime Enforcement: The Arbiter

Formal models establish properties of their abstractions. At runtime, Port Daddy deploys the **Arbiter**—an ambient monitor that subscribes to the daemon's post-commit activity log and checks six derived invariants:

1. `PID_SQUATTING`: Detects when a process claims a port previously held by a different PID (the "Ghost in the Harbor").
2. `CAP_ESCALATION`: Verifies capability subsets via the deployed Rust FFI enforcer.
3. `NOTE_MONOTONICITY`: Ensures the note sequence for active sessions never shrinks (from the TLA$^+$ `NoteMonotonicity` invariant).
4. `ESCROW_POSITIVE`: Validates that active sessions have positive escrow (when Float Plans are active).
5. `LOCK_OWNER_VALID`: Ensures locks are only held by registered agents with active sessions.

6. `HEARTBEAT_FRESHNESS`: Flags stale heartbeats as early warning for agent death.

Violations are logged and can trigger the "man overboard" salvage protocol. Because this path observes committed events, it detects and compensates; only a pre-commit native guard can prevent a transition. The Arbiter API provides visibility, not a proof that every security-relevant operation was mediated.

V.4.3 Implementation Verification

A secure protocol is useless if the implementation contains buffer overflows or timing leaks. We extracted the critical JWT parsing and cryptographic comparison logic into a verified core.

Property V.4.1 (Memory Safety)**.** Using the **Kani** Rust Verifier [26], we performed bounded model checking to prove that our parsing logic never panics or accesses out-of-bounds memory, regardless of the input payload.

Property V.4.2 (Secret-Independent Source Control Flow)**.** The Rust comparator uses no source-level branch on secret byte values, and Kani checks that bounded control-flow property. This is useful evidence but not a hardware-level constant-time proof: compiler transformations, memory behavior, and microarchitecture remain outside the model.

V.5 Verification Algorithms

In this section, we present the verification algorithms in pseudocode format, following the notation used in protocol verification literature [39, 88]. These algorithms correspond to the formal ProVerif models in Appendix V.B.

V.5.1 Phase 1: Symmetric HS256 with Algorithm Pinning

Algorithm V.2 presents the secure verification procedure that is immune to algorithm confusion attacks (CVE-2015-9235 [51]).

```
1  Require: Pinned Algorithm: hs256
2  Require: Master Key: k_master (secret)
3
4  function Daemon.IssueToken(agent_id, harbor_id, capability):
5      jti <- GenerateNonce()
6      msg <- Encode(agent_id, harbor_id, jti, capability)
7      sigma <- HMAC-SHA256(msg, k_master)
8      emit Issued(agent_id, harbor_id, jti, capability)
9      return (hs256, msg, sigma)
10
11 function Harbor.VerifyToken(tok, expected_harbor):
12     (alg_header, msg, sigma) <- tok
13     // CRITICAL: Ignore alg_header, pin to HS256
14     if HMAC-SHA256(msg, k_master) != sigma:
15         return _|_  // Invalid signature
16     (a, h, j, cap) <- Decode(msg)
17     if h != expected_harbor:
18         return _|_  // Wrong harbor
19     emit Accepted(a, h, j, cap)
20     return (a, h, j, cap)
```

Listing V.2: Algorithm 1: Secure Harbor Verification (HS256 with Algorithm Pinning)

Security Property: The algorithm ignores the user-provided alg_{header} (line 15), preventing algorithm confusion attacks where an attacker sets alg = "none" or attempts HS256/RS256 confusion [194].

V.5.2 Phase 2: Asymmetric Ed25519

Algorithm V.3 presents the Ed25519-based verification. This phase removes the need for shared HMAC secrets between distributed Harbors.

```
Require: Harbor Keypair: (sk_harbor, pk_harbor)
Require: Key Distribution Channel: c_key (private)

function Daemon.IssueToken(agent_id, harbor_id, capability):
    sk_h <- Receive(c_key)  // Get harbor's secret key
    jti <- GenerateNonce()
    msg <- Encode(agent_id, harbor_id, jti, capability)
    sigma <- Ed25519.Sign(msg, sk_h)
    emit Issued(agent_id, harbor_id, jti, capability)
    return (msg, sigma)

function Harbor.VerifyToken(tok, pk_h, expected_harbor):
    (msg, sigma) <- tok
    if not Ed25519.Verify(msg, sigma, pk_h):
        return _|_  // Invalid signature
    (a, h, j, cap) <- Decode(msg)
    if h != expected_harbor:
        return _|_  // Wrong harbor
    emit Accepted(a, h, j, cap)
    return (a, h, j, cap)
```

Listing V.3: Algorithm 2: Asymmetric Harbor Verification (Ed25519)

Security Property: The secrecy of sk_{harbor} and the unforgeability of Ed25519 [58] ensure that only the legitimate Daemon can issue valid Harbor Cards.

V.5.3 Phase 3: Multi-hop Delegation

Algorithm V.4 presents the delegation chain verification, inspired by Macaroons [20].

```
Require: Daemon Public Key: pk_daemon
Require: Subset Relation: subseteq on capabilities

function Agent.Delegate(token_parent, sk_parent, agent_child, cap_sub)
    :
    (msg_parent, sigma_parent) <- token_parent
    (_, agent_parent, harbor, cap_parent) <- Decode(msg_parent)
    if cap_sub not_subseteq cap_parent:
        return _|_  // Cannot escalate capabilities
    msg_delegate <- Encode_delegation(msg_parent, sigma_parent,
       agent_child, cap_sub)
    sigma_delegate <- Ed25519.Sign(msg_delegate, sk_parent)
    emit Delegated(agent_parent, agent_child, cap_sub)
    return (msg_delegate, sigma_delegate)

function Harbor.VerifyChain(chain, pk_root):
    tok_root <- chain[0]
    (msg_0, sigma_0) <- tok_root
    if not Ed25519.Verify(msg_0, sigma_0, pk_root):
        return _|_  // Invalid root signature
    cap_prev <- ExtractCaps(msg_0)
    pk_current <- pk_root
    for i <- 1 to |chain| - 1:
        (msg_i, sigma_i) <- chain[i]
        if not Ed25519.Verify(msg_i, sigma_i, pk_current):
```

```
            return _|_  // Invalid delegation signature
        cap_i <- ExtractCaps(msg_i)
        if cap_i not_subseteq cap_prev:
            return _|_  // Per-hop capability escalation detected
        cap_prev <- cap_i
        pk_current <- ExtractPubkey(msg_i)
    emit Accepted(agent_final, harbor, cap_prev)
    return T
```

Listing V.4: Algorithm 3: Delegation Chain Verification

Security Property: Monotonic per-hop attenuation ($\text{cap}_i \subseteq \text{cap}_{i-1}$) prevents capability expansion across transitive delegations (preventing `write` → `read` → `write` re-escalation). Theorem V.D.1 (§V.D.1) proves inductive soundness under the partial order.

V.6 Implementation: Deployed Rust Core

The verified cryptographic core is not a reference implementation—it is the **deployed production code**. The open-source Rust crate `harbor-card-rs` is compiled to a C dynamic library (`cdylib`) and called from the Node.js daemon via FFI through the daemon's runtime-enforcement (Arbiter) module.

This architecture eliminates the gap between "verified code" and "running code" that weakens many formal verification efforts. The same binary that Kani model-checks is the same binary that the daemon loads at runtime.

V.6.1 Core Verification Logic

```rust
pub fn verify(&self, token: &str, now_ts: i64)
    -> Result<HarborCardClaims, HarborError> {
    let parts: Vec<&str> = token.split('.').collect();
    if parts.len() != 3 {
        return Err(HarborError::Malformed);
    }
    let msg = format!("{}.{}", parts[0], parts[1]);
    let mut sig_bytes = Self::internal_decode_b64(parts[2])?;
    let sig_result = self.internal_verify_sig(
        msg.as_bytes(), &sig_bytes);
    sig_bytes.zeroize();  // Scrub signature from memory
    sig_result?;
    let mut payload_bytes = Self::internal_decode_b64(parts[1])?;
    let claims: HarborCardClaims =
        serde_json::from_slice(&payload_bytes)?;
    payload_bytes.zeroize();  // Scrub payload from memory
    if claims.exp < now_ts {
        return Err(HarborError::Expired);
    }
    Ok(claims)
}

pub fn constant_time_compare(a: &[u8], b: &[u8]) -> bool {
    if a.len() != b.len() { return false; }
    let mut result = 0u8;
    for i in 0..a.len() { result |= a[i] ^ b[i]; }
    result == 0
}
```

Listing V.5: Deployed Rust Implementation — Harbor Card Verifier (abridged from the `harbor-card-rs` crate)

Key implementation details: `zeroize` requests that cryptographic material be overwritten when dropped, reducing residual-secret exposure without proving that no copy survives. The comparator avoids source-level early return on secret byte values. That is useful hygiene, not a hardware constant-time guarantee; compiler, cache, and microarchitectural behavior remain outside this source argument.

V.6.2 FFI Bridge to the Node.js Daemon

The Rust crate exports two C-ABI functions consumed by the TypeScript daemon:

```rust
#[no_mangle]
pub extern "C" fn harbor_constant_time_compare(
    a: *const u8, a_len: usize,
    b: *const u8, b_len: usize
) -> bool { /* ... */ }

#[no_mangle]
pub extern "C" fn harbor_verify_caps_subset_json(
    root_json: *const c_char, root_len: usize,
    sub_json: *const c_char, sub_len: usize
) -> bool { /* ... */ }
```

Listing V.6: FFI Exports — Called from the daemon's Arbiter module

The daemon's `arbiter` module binds these at startup:

```
constantTimeCompare: lib.func(
  'bool harbor_constant_time_compare(
     const uint8_t *a, size_t a_len,
     const uint8_t *b, size_t b_len)'),
verifyCapsSubset: lib.func(
  'bool harbor_verify_caps_subset_json(
     const char *root_json, size_t root_len,
     const char *sub_json, size_t sub_len)')
```

Listing V.7: TypeScript FFI Binding (daemon-side Arbiter)

V.6.3 Kani Verification Harnesses

Three bounded model checking proofs are defined in the same source file under `#[cfg(kani)]`:

```rust
#[cfg(kani)]
#[kani::proof]
#[kani::stub(HarborCardVerifier::internal_pk_from_bytes,
             stubs::pk_from_bytes_stub)]
#[kani::stub(HarborCardVerifier::internal_decode_b64,
             stubs::decode_b64_stub)]
#[kani::stub(HarborCardVerifier::internal_verify_sig,
             stubs::verify_sig_stub)]
#[kani::unwind(10)]
fn proof_verify_logic_only() {
    let pk_bytes: [u8; 32] = kani::any();
    if let Ok(verifier) = HarborCardVerifier::new(pk_bytes) {
        let token_bytes: [u8; 32] = kani::any();
        if let Ok(token_str) =
            std::str::from_utf8(&token_bytes) {
            kani::assume(token_str.contains('.'));
            let _ = verifier.verify(token_str, 0);
        }
    }
```

```
}

#[cfg(kani)]
#[kani::proof]
fn proof_constant_time_behavior() {
    let a: [u8; 16] = kani::any();
    let b: [u8; 16] = kani::any();
    let _ = HarborCardVerifier::constant_time_compare(&a, &b);
}

#[cfg(kani)]
#[kani::proof]
fn proof_capability_attenuation() {
    let root = vec!["read".to_string(), "write".to_string()];
    let mut sub = vec!["read".to_string()];
    assert!(HarborCardVerifier::verify_capability_subset(
        &root, &sub));
    sub.push("admin".to_string());
    assert!(!HarborCardVerifier::verify_capability_subset(
        &root, &sub));
}
```

Listing V.8: Kani Proof Harnesses (Deployed Source)

The stubbing strategy deserves explanation: `proof_verify_logic_only` stubs out Ed25519 signature verification and base64 decoding (which involve curve arithmetic that Kani cannot symbolically execute within practical bounds) and exhaustively explores the *parsing and control flow logic*—the split-on-dots, payload extraction, expiry comparison, and error handling paths. This verifies that the logic surrounding the cryptographic primitives never panics, never accesses out-of-bounds memory, and handles all malformed inputs gracefully, regardless of what the cryptographic layer returns.

The build output at `target/kani/aarch64-apple-darwin/debug` contains the CBMC intermediate representations used by Kani's bounded model checker.

V.7 Security Analysis

V.7.1 Threat Model

We assume a **Dolev-Yao adversary** [60] who:

- Controls the entire network
- Can intercept, modify, and inject messages
- Cannot break cryptographic primitives (cryptography is perfect)
- May corrupt legitimate agents (but not all simultaneously)

This is the standard threat model for symbolic protocol analysis [39].

V.7.2 Security Properties

Table V.2: Verified Security Properties. "Deployed" indicates the verified Rust code is called by the production daemon via FFI from the Arbiter module into the `harbor-card-rs` crate.

Property	Verified	Tool	Deployed	Reference
Master Key Secrecy	Yes	ProVerif	—	[39]
Algorithm Confusion Immunity	Yes	ProVerif	—	[194]
Authentication correspondence	Yes	ProVerif	—	[88]
Delegation Integrity	Yes	ProVerif	—	[20]
Memory Safety (parsing)	Yes	Kani	Yes (FFI)	[26]
Secret-independent source branching	Yes	Kani	Yes (FFI)	[34]
Capability Attenuation	Yes	Kani	Yes (FFI)	[20]

V.7.3 Limitations

1. **Delegation-chain depth.** The symbolic proof of delegation soundness (Table V.3, "`chain-replay.pv`") is machine-checked at the chain depth realized in deployment (depth 3); it is *not* a proof for unbounded chain length. ProVerif's unbounded-session guarantee quantifies over the number of protocol *sessions*, not over the number of hops in a single delegation chain, which is a fixed structural bound in the modeled process. Extending the proof to unbounded depth (via recursive replication of the delegation process) is future work. The current depth bound is a *spawn-time* policy check (`assessDelegation` in `lib/tube-spawner-router.ts`, `HARD_MAX_DELEGATION_DEPTH` = 8, default 4) applied when a spawn request arrives; it is *not* a verification-time invariant. The cryptographic chain verifier (`lib/delegation-chain.ts`) accepts arbitrary chain depth, and the attenuation monitor (`lib/cap-attenuation-monitor.ts`) enforces only per-hop capability monotonicity—not total depth. Machine-enforced depth at verification time is a planned hardening item.

2. **PID Binding:** The "Ghost in the Harbor" scenario requires binding tokens to the requesting PID, which is handled at the OS level.

3. **Local Transport Security:** On multi-user systems, local unix sockets or loopback TLS should be used to prevent local bearer token sniffing.

V.8 Conclusion

The Anchor Protocol replaces several hope-based assumptions with explicit, checkable obligations. It combines:

- Symbolic protocol proofs (ProVerif [39])
- Memory-safe implementation (Rust/Kani [26])
- Capability-based delegation (inspired by Macaroons [20])

Together these artifacts support a bounded assurance case for the modeled and tested surfaces. They do not certify the entire daemon, every delegation depth, or every side channel.

Chapter Appendices

V.A Formal Verification Status

Artifact availability. Every artifact named in Table V.3 is bundled at `https://portdaddy.dev/whitepaper/artifacts/`. The verified Rust core is the open-source `harbor-card-rs` crate; runtime components are deployed inside the Port Daddy daemon binary. Chapter VI, *The Bonded Commons*, carries the cross-paper status table; this appendix lists only the items that pertain specifically to the Anchor Protocol.

Table V.3: Anchor Protocol verification status. **closed**: model verified *and* runtime conformance test passes. PARTIAL: design verified, runtime empirically tested rather than proved. *open*: known unmechanized.

Claim	Method	Result	Artifact
Authentication correspondence in the listed phase models (§V.5)	ProVerif 2.05	**closed** non-injective correspondence queries TRUE	`harbor_card_v*.pv`
Algorithm-confusion immunity	ProVerif pinned vs. naive	**closed** pinned TRUE; counter-trace witnessed	`algconfusion.pv`
Delegation chain replay (§V.5.3)	ProVerif at depth 3	**closed** chain authenticity TRUE; 17-case runtime conformance	`chain-replay.pv`
Cuckoo-filter pollution resistance (§V.2.4)	5-invariant property test	**closed** FP rate within $2B/2^f$ at load 0.95	cuckoo property tests
GitHub-App webhook origin authentication	ProVerif (Dolev-Yao relay)	**closed** origin auth TRUE; daemon re-verifies GitHub HMAC (runtime conformance)	`ingress_origin_auth.pv`
Multi-tenant relay namespace isolation	ProVerif + runtime	**closed** member of A cannot publish into B; `HARBOR_MISMATCH` conformance	`ingress_tenant_isolation.pv`
Cuckoo-filter revocation soundness	inspection + empirical	PARTIAL gate narrows acceptance; dissemination remains assumption-bound	runtime test
Constant-time signature comparison	audited `subtle::ConstantTimeEq` library	PARTIAL no side-channel proof; library trust documented	—
Relay payload secrecy via HPKE (daemon key pinned)	ProVerif (design)	PARTIAL secrecy TRUE under pinned key; seal not yet implemented	`ingress_hpke_secrecy.pv`
Webhook replay-freedom + in-order delivery	TLA+ / TLC (exhaustive at bound)	PARTIAL design model-checked; runtime via ordered-chain ingest	`WebhookDelivery.tla`
Card revocation finality under backup/restore (ADR-0018 Attack 2)	TLA+ / TLC	PARTIAL rollback attack + revocation-epoch mitigation both model-checked	`CardRevocation.tla`
Mechanized gossip-convergence bound	—	*open* standard expected asymptotics only [8]; no partition deadline	—

Limitations. The transition to gossiping the Append-Only Event Log solves the bitwise merging impossibility, but cross-peer gossip propagation still leans on the standard anti-entropy analysis rather than a mechanized proof. The side-channel resistance of signature comparison is inherited from an audited library rather than re-proved. These deferrals are documented as known open problems rather than as defects in the present claims.

GitHub-App relay ingress. The webhook-dispatch surface (GitHub → untrusted relay → daemon) is modeled with the relay as the Dolev-Yao attacker, in three relay-agnostic properties, each paired with a negative control (`*_vuln.pv`) that exhibits the attack once the mitigation is removed—so a *true* result is non-vacuous. (i) *Origin authentication*: the daemon dispatches a fleet event only if GitHub signed it, even when the forwarder's transport credential is attacker-held; the control omits the daemon-side HMAC re-verification and the property fails (forged dispatch). This is *non-injective* agreement (origin), not replay-freedom: a captured genuine (*payload*, *mac*) may be re-delivered. Replay-freedom is an ordering-sensitive, mutable-state property outside ProVerif's reach, so it is discharged separately in TLA+ (`proofs/relay/WebhookDelivery.tla`): a daemon consuming the *ordered* per-publisher chain (dispatch iff $seq = tip + 1$) is model-checked at-most-once *and* gap-free against an adversarial relay that reorders, duplicates, and redelivers, with the unguarded daemon as the model-checked counterexample (TLC exhibits a double-dispatch). The obligation it names: the daemon ingests webhooks over the relay's ordered `from_seq` subscribe path, not the raw forward. (ii) *Payload secrecy*: under HPKE to the daemon's *pinned* X25519 key the relay never learns plaintext; the control sources the key from the relay and the secrecy property collapses to a key-substitution MITM—hence out-of-band key pinning is a requirement, not a recommendation. The seal is specified but not yet implemented, so this row is PARTIAL. (iii) *Namespace isolation*: a member of tenant A cannot publish into tenant B because harbor binding is checked against authoritative membership; the control trusts the card's self-declared issuer and crosses the tenant boundary. Honesty rider: HMAC-SHA256 gives origin authentication to a shared-key holder (EUF-CMA), not third-party non-repudiation; HPKE base mode is assumed IND-CCA2.

> *We err toward listing fewer items as closed rather than more.*

V.B Full ProVerif Models

For completeness, we include the full ProVerif models used for verification. These correspond to the algorithms in Section V.5.

V.B.1 Phase 1: Symmetric HS256

```
(* Port Daddy: Harbor Card v1 (HS256) Formal Model *)
type id. type key. type jti. type capability. type alg_type.
const hs256_alg: alg_type. const none_alg: alg_type.
free c: channel.

fun hs256(bitstring, key): bitstring.
reduc forall m: bitstring, k: key;
  check_hs256(m, k, hs256(m, k)) = true.

reduc forall m: bitstring, k: key;
  verify_generic(m, hs256_alg, hs256(m, k), k) = true;
  forall m: bitstring, k: key;
  verify_generic(m, none_alg, m, k) = true.

```

```
fun encode_token(id, id, jti, capability): bitstring [data].

event Issued(id, id, jti, capability).
event Accepted(id, id, jti, capability).

free master_key: key [private].
query attacker(master_key).

query a: id, h: id, j: jti, cap: capability;
  event(Accepted(a, h, j, cap)) ==> event(Issued(a, h, j, cap)).

let Daemon(k: key, a_id: id, h_id: id, j: jti, cap: capability) =
  let msg = encode_token(a_id, h_id, j, cap) in
  let signature = hs256(msg, k) in
  event Issued(a_id, h_id, j, cap);
  out(c, (hs256_alg, msg, signature)).

let SecureHarbor(k: key, expected_h: id) =
  in(c, (alg_header: alg_type, msg: bitstring, signature: bitstring));
  if check_hs256(msg, k, signature) = true then
    let encode_token(a: id, h: id, j: jti, cap: capability) = msg in
    if h = expected_h then
      event Accepted(a, h, j, cap).

process
  new agent_id: id; new harbor_id: id;
  new token_jti: jti; new secret_cap: capability;
  (!Daemon(master_key, agent_id, harbor_id, token_jti, secret_cap) |
   !SecureHarbor(master_key, harbor_id))
```

Listing V.9: harbor_card_v1_refined.pv

V.C Phase 2: Asymmetric Ed25519

```
(* Port Daddy: Harbor Card v2 (Ed25519 / EdDSA) Formal Model *)
type id. type skey. type pkey. type jti. type capability.

free c: channel.
free k_dist: channel [private]. (* Trusted key distribution *)

(*** Digital Signatures (Ed25519) ***)
fun pk(skey): pkey.
fun sign(bitstring, skey): bitstring.
reduc forall m: bitstring, k: skey;
  check_sign(m, pk(k), sign(m, k)) = true.

fun encode_token(id, id, jti, capability): bitstring [data].

(*** Events ***)
event Issued(id, id, jti, capability).
event Accepted(id, id, jti, capability).

(*** Queries ***)
free secret_key: skey [private].
query attacker(secret_key).

query a: id, h: id, j: jti, cap: capability;
```

```
  event(Accepted(a, h, j, cap)) ==> event(Issued(a, h, j, cap)).

(*** Processes ***)
let Daemon(a_id: id, h_id: id, j: jti, cap: capability) =
  in(k_dist, sk_h: skey);
  let msg = encode_token(a_id, h_id, j, cap) in
  let signature = sign(msg, sk_h) in
  event Issued(a_id, h_id, j, cap);
  out(c, (msg, signature)).

let Harbor(pk_h: pkey, expected_h: id) =
  in(c, (msg: bitstring, signature: bitstring));
  if check_sign(msg, pk_h, signature) = true then
    let encode_token(a: id, h: id, j_1: jti, cap_1: capability)
      = msg in
    if h = expected_h then
      event Accepted(a, h, j_1, cap_1).

(*** Main Process ***)
process
  new agent_id: id; new harbor_id: id;
  new token_jti: jti; new secret_cap: capability;
  let harbor_pk = pk(secret_key) in
  out(c, harbor_pk);
  (
    (!Daemon(agent_id, harbor_id, token_jti, secret_cap)) |
    (!Harbor(harbor_pk, harbor_id)) |
    (!out(k_dist, secret_key))
  )
```

Listing V.10: `harbor_card_v2_asymmetric.pv` — Verified in ProVerif 2.05

V.D Phase 3: Multi-hop Delegation

```
(* Port Daddy: Harbor Card v3 (Delegation Chains) Formal Model *)
type id. type skey. type pkey. type capability.

free c: channel.

(*** Cryptographic Functions ***)
fun pk(skey): pkey.
fun sign(bitstring, skey): bitstring.
reduc forall m: bitstring, k: skey;
  check_sign(m, pk(k), sign(m, k)) = true.

reduc forall c1: capability; is_subset(c1, c1) = true.

fun base_token(id, id, id, capability): bitstring [data].
fun delegate_token(bitstring, id, capability): bitstring [data].

(*** Events ***)
event IssuedRoot(id, id, capability).
event Delegated(id, id, capability).
event Accepted(id, id, capability).

(*** Queries ***)
free harbor_id: id.
```

```
query b: id, cap: capability, a: id;
  event(Accepted(b, harbor_id, cap)) ==>
    (event(IssuedRoot(a, harbor_id, cap))
     && event(Delegated(a, b, cap)))
    || event(IssuedRoot(b, harbor_id, cap)).

(*** Processes ***)
free daemon_id: id.
free daemon_sk: skey [private].

let Daemon(a: id, cap: capability) =
  let msg = base_token(daemon_id, a, harbor_id, cap) in
  let s = sign(msg, daemon_sk) in
  event IssuedRoot(a, harbor_id, cap);
  out(c, (msg, s)).

let AgentA(a: id, a_sk: skey, b: id, sub_cap: capability) =
  in(c, (msg: bitstring, s: bitstring));
  let base_token(=daemon_id, =a, =harbor_id,
    root_cap: capability) = msg in
  if check_sign(msg, pk(daemon_sk), s) = true then
    if is_subset(sub_cap, root_cap) = true then
      let d_msg = delegate_token((msg, s), b, sub_cap) in
      let d_s = sign(d_msg, a_sk) in
      event Delegated(a, b, sub_cap);
      out(c, (d_msg, d_s)).

let Harbor(a_pk: pkey) =
  in(c, (d_msg: bitstring, d_s: bitstring));
  let delegate_token((base_msg: bitstring, base_s: bitstring),
    b: id, sub_cap: capability) = d_msg in
  let base_token(=daemon_id, a: id, =harbor_id,
    root_cap: capability) = base_msg in
  if check_sign(base_msg, pk(daemon_sk), base_s) = true then
  if check_sign(d_msg, a_pk, d_s) = true then
  if is_subset(sub_cap, root_cap) = true then
    event Accepted(b, harbor_id, sub_cap).

(*** Main Process ***)
process
  new sk_a: skey;
  let pk_a = pk(sk_a) in
  out(c, pk_a);
  new id_a: id; new id_b: id; new cap_root: capability;
  (
    (!Daemon(id_a, cap_root)) |
    (!AgentA(id_a, sk_a, id_b, cap_root)) |
    (!Harbor(pk_a))
  )
```

Listing V.11: `harbor_card_v3_delegation.pv` — reflexive-order model (see §V.D.1 for the soundness caveat and the enriched v5 proof)

V.D.1 Soundness of the Attenuation Proof (v5)

The v3 model above is honest about cryptographic structure but *vacuous about attenuation*. Its capability order is reflexive only:

```
reduc forall c1:  capability; is_subset(c1, c1) = true.
```

Capabilities are opaque atoms with no order, so a delegated capability can only ever *equal* its parent. Escalation—delegating more authority than you hold—is not blocked; it is *inexpressible*. The no-escalation query therefore holds for a reason that has nothing to do with the protocol: there is no larger capability to delegate. A reviewer flagged this directly: *"the model cannot witness escalation."*

`harbor_card_v5_attenuation.pv` closes the obligation by enrichment, not by weakening the claim. It gives capabilities a real partial order (`cap_read` $\sqsubset$ `cap_write`, both public so the attacker can construct either), replaces the reflexive stub with the sound order (`is_subset(cap_read, cap_write)` derivable; `is_subset(cap_write, cap_read)` *not*), and adds an *insider escalation adversary*: an agent holding a valid `cap_read` root token that signs an over-broad `cap_write` delegation with no self-restraint. The verifier's `is_subset` guard is the sole defense under test.

```
(* Sound order: reflexive + read <= write; write <= read NOT derivable *)
fun is_subset(capability, capability): bool
reduc
    forall x: capability; is_subset(x, x) = true
    otherwise is_subset(cap_read, cap_write) = true.

(* Insider adversary: holds a real root, signs an over-broad delegation *)
let MaliciousA(a: id, a_sk: skey, b: id) =
  in(c, (msg: bitstring, s: bitstring));
  let base_token(=daemon_id, =a, =harbor_id, root_cap: capability) = msg in
  if check_sign(msg, pk(daemon_sk), s) = true then
    event EscalationAttempted(a, root_cap, cap_write);
    let d_msg = delegate_token((msg, s), b, cap_write) in
    out(c, (d_msg, sign(d_msg, a_sk))).

(* Q1 soundness: a write-card delegated from a read-root is NEVER accepted *)
query b: id; event(Accepted(b, harbor_id, cap_write, cap_read)) ==> false.
(* Q2 non-vacuity: the escalation IS reachable (the proof is not vacuous) *)
query a: id, r: capability, s: capability;
  event(EscalationAttempted(a, r, s)) ==> false.
```

Listing V.12: `harbor_card_v5_attenuation.pv` (excerpt) — Verified in ProVerif 2.05

Theorem V.D.1 (Attenuation soundness, mechanized)**.** *In* ***`harbor_card_v5_attenuation.pv`****, ProVerif 2.05 reports* ***`Accepted(b, harbor, cap_write, cap_read)`*** *unreachable (**Q1 = true**): no escalating delegation is ever accepted. Simultaneously* ***`EscalationAttempted`*** *is reachable (**Q2 = false-with-trace**): the attack is expressible, so Q1 is a real defense, not the v3 vacuum. A negative control that deletes the verifier's* ***`is_subset`*** *guard flips Q1 to false-with-trace—ProVerif exhibits the read→write escalation— confirming the guard is load-bearing.*

Multi-hop scope (honest boundary). Theorem V.D.1 is single-hop (Daemon → A → verifier). Multi-hop delegation has a sharper obligation: a verifier that checks the *final* capability against the *root* accepts a non-monotonic middle hop. `harbor_card_v6_multihop_attack.pv` mechanizes exactly this — A:`write` → B:`read` → C:`write` is accepted (ProVerif: escalation reachable). The fix, `harbor_card_v7_multihop_fixed.pv`, verifies each hop against its *immediate parent* (is_subset(*final*, *mid*) ∧ is_subset(*mid*, *root*)) and ProVerif proves the same chain rejected. The runtime delegation verifier — and the cross-harbor recompute $\text{att}_B \circ \text{att}_A$ — must be per-hop, never final-vs-root.

V.E Limitations & Boundaries

The formal mechanisms presented in this chapter are mathematically sound within their defined scopes, but they depend on bounding assumptions that must be explicitly acknowledged.

- **Threat Model Exclusions:** Collusion across all independent organs (e.g., the logging daemon, the arbitrator, and the primary execution runtime) is assumed out-of-scope; if all enforcing components are compromised, the guarantees degrade to advisory.
- **Performance Trade-offs:** Cryptographic assertions and WAL-checkpointing for per-write durability incur measurable I/O overhead. These mechanisms are designed for high-stakes coordination, not for bulk data processing or high-frequency trading.
- **Implementation Maturity:** While the core protocols (Anchor, SQLite single-writer) are IMPLEMENTED and running in production, surrounding ecosystem components (like the decentralized judge markets or fully federated multi-sig routing) remain in PARTIAL or DESIGNED phases.

These constraints are not flaws but structural realities of building zero-trust coordination on top of POSIX primitives.

PORT DADDY COORDINATION PAPERS · CHAPTER VI OF VII

The Bonded Commons

Abstract

Two autonomous agents working on the same project corrupt each other's state, double-claim the same files, race for the same ports, and leave the operator to do recovery archaeology. The problem is not the agents—it is that they have no coordination substrate. This paper specifies one. We model a daemon as a *candidate correlating device* in the sense of Aumann [174]: it can send private recommendations to agents, but those recommendations constitute a correlated equilibrium only when the explicit obedience inequalities in §VI.2.1 hold. The substrate has three independently load-bearing layers—capability-attenuated tokens that bound scope, Merkle-chained evidence trails that preserve attribution, and collateralized work contracts that pre-fund damage regardless of intent. We formalize the construction in the context of Port Daddy, a coordination daemon for local agent swarms, and separate structural guarantees from incentive assumptions. Sen's Impossibility of a Paretian Liberal supplies a design warning, not a proof that every enforced lock is inefficient: decisive local control and team-wide Pareto choice can conflict under unrestricted preferences. That warning motivates an advisory claim graph whose completeness is proved separately. We further specify a *mutable-signal ledger* (§VI.4.3) supporting revocation, renaming, and provenance-aware propagation without sacrificing attribution integrity. Finally, we evaluate a competitive-insurance pricing mechanism (§VI.7.5.8, contributed by Youle) against a named static baseline. Its Pareto comparisons are simulation results under stated assumptions, not a universal welfare theorem. Hobbes had the right idea about sovereigns; he just lacked a way to deploy one on `localhost:9876`.

Keywords: multi-agent coordination, trust infrastructure, capability-based security, social contract, collateralized computation, formal verification, commons governance

Reading time: approximately 50 minutes end-to-end. The Reader's Map below provides shorter paths for specific reader types.

PORT DADDY COORDINATION PAPERS **CHAPTER VI OF VII**

V. The Anchor Protocol. Authentication, delegation, attenuation, and revocation inside one harbor.

VI. The Bonded Commons (this chapter)**.** Evidence, bonding, advisory claims, and the incentive conditions for cooperation.

VII. The Federated Harbor. Identity, revocation, and settlement across mutually sovereign machines.

Chapters I–IV form the product arc; Chapters V–VII provide the protocol, economic, and federation deep dives. Every chapter is independently readable.

Reader's Map

The paper builds in four layers; each layer carries a load-bearing formal claim or empirical result. If you have less than 50 minutes, the table below tells you where to start.

Layer	Load-bearing artifact	Lives at
(1) Structural prevention	Capability-attenuated tokens; ProVerif proofs of algorithm-confusion immunity and delegation replay resistance	§VI.3–§VI.3.2; Chapter V (*The Anchor Protocol*)
(2) Immutable attribution	Merkle Forest with cross-daemon inclusion proofs; EasyCrypt binding mechanization (`binding.ec`)	§VI.4.2; Appendix VI.A
(3) Economic alignment	Competitive-insurance auction; conditional Monte Carlo comparison	§VI.7.5.8; §VI.7.5.8
(4) Governance	Sen-inspired design warning; advisory-conflict completeness	§VI.5–§VI.5.2

Suggested reading paths by reader type:

- **Formal-methods reviewer / program committee** — abstract, then jump to Appendix VI.A (Formal Verification Status). The artifact registry is the most concentrated signal of what is and is not proved.
- **Cryptoeconomic protocol designer** — abstract, then §VI.7.5.8 (auction mechanism), §VI.7.5.8 (welfare theorem with inline Pareto Monte Carlo), §VI.7.5.9 (A5 Sybil-deterrence is coverage-bounded — a surprising negative result).
- **AI safety researcher** — §VI.6.1 ("What Morality Means When Death Is Cheap"), §VI.5 (Sen's theorem), §VI.7 (collateralized work contracts). Skim Appendix VI.A for the proof posture.
- **Distributed systems engineer** — §VI.3–§VI.4.2 (primitives), then Chapter V (*The Anchor Protocol*) for the cryptographic-token protocol detail.
- **Engineering manager evaluating multi-agent infrastructure** — abstract, §VI.2.2 ("Why Agents Consent"), §VI.7 (bond pricing), §VI.7.5.8 (welfare results). The Appendix VI.A status registry tells you what is production-ready and what is research.
- **Policy / governance analyst** — abstract, §VI.5–§VI.5.2 (the Sen's-theorem argument for advisory rather than enforced claims), §VI.12 (federated sovereign).

Volume Context. Chapter V (*The Anchor Protocol*) provides the cryptographic-token foundation for this work: self-contained ProVerif models of the Harbor Card phases, cuckoo-filter revocation, and bounded multi-hop delegation. This chapter carries the economic and governance argument, while Chapter V carries the cryptographic protocol detail.

VI.1 Introduction: The Trust Problem

Two autonomous coding agents share a repository. Agent A is asked to refactor the authentication middleware. Agent B, spawned ninety seconds later from a sibling worktree, is asked to add a rate-limiter to the same middleware. Neither knows the other exists. They both edit, both commit, both push; one of them gets a merge conflict it cannot interpret, retries with a hallucinated resolution, and corrupts the file. The operator returns from a coffee break to a working tree she does not recognize. This is not a hypothetical—it is the modal failure mode of every multi-agent setup we have inhabited. Existing frameworks do not address it: LangGraph, CrewAI, and AutoGen handle state graphs and role assignment competently; the Model Context Protocol standardizes agent-tool integration. None of them give Agent A and Agent B a way to know about each other. The problem is **trust**—specifically, trust as infrastructure rather than trust as per-transaction ceremony.

When Agent A delegates a subtask to Agent B, it implicitly trusts that B will not corrupt the shared state, will not exceed the scope of the delegation, and will not silently fail in a way that poisons A's downstream work. When Agent B accepts the delegation, it implicitly trusts that A's description of the task is accurate, that the resources A promised are available, and that A will still exist when B finishes. These implicit trust assumptions pervade every multi-agent interaction, and every one of them can be violated.

The standard response in security engineering is "zero trust"—never trust, always verify. But zero trust applied literally to agent coordination is paralytic. If every message requires cryptographic verification, every delegation requires capability negotiation, and every file access requires authorization—the coordination overhead overwhelms the work itself. Agents spend their context windows negotiating trust instead of solving problems.

Hobbes identified this dynamic in 1651 [192]. In the state of nature, without a common authority, rational actors cannot cooperate because the cost of verifying each counterparty's intentions exceeds the benefit of cooperation. The solution is not to make individuals trustworthy—it is to establish a **sovereign** whose authority both parties accept *before* the interaction begins, so that the interaction itself can focus on the work. "Covenants, without the sword, are but words, and of no strength to secure a man at all."

This paper argues that multi-agent systems need exactly this: a **commons authority** that provides trust as infrastructure, not trust as ceremony. Agents don't negotiate trust per-transaction. They enter a commons where trust has already been established—structurally, evidentially, and economically—and they work freely within that trust envelope.

VI.1.1 The Three Failures of Peer-to-Peer Trust

Peer-to-peer trust fails for agent systems in three specific ways:

It doesn't scale. If n agents must establish pairwise trust, the system requires $O(n^2)$ trust negotiations. Each negotiation consumes tokens, time, and context window. With 8 agents, this is manageable. With 50, it dominates wall-clock time. With 1000 (a production agent fleet), it is infeasible.

It doesn't survive death. When an agent crashes, its trust relationships die with it. A successor agent that resumes the dead agent's work must re-establish trust with every counterparty from scratch. The trust investment is non-transferable because it was encoded in ephemeral bilateral state, not in durable infrastructure.

It doesn't address misaligned principals. An agent can be perfectly "trustworthy" in the sense that it faithfully executes its instructions—while its instructions come from a principal whose interests are adversarial to the commons. Peer-to-peer trust evaluates the agent's behavior; it

cannot evaluate the principal's intent. The agent is a loyal soldier in someone else's war, with a spotless reputation.

VI.1.2 Contributions

We present:

1. A reading of the coordination daemon as a **candidate correlating device** in the sense of Aumann [174]: claim recommendations are private signals drawn from a common-knowledge distribution, while incentive compatibility remains conditional on explicit obedience inequalities (Section VI.2, esp. §VI.2.1).

2. A **three-layer trust architecture** (structural, evidentiary, economic) that does not depend on agent morality, shared values, or threat of termination (Sections VI.3–VI.7).

3. A Sen-grounded **design warning** about decisive resource rights under private information, plus a separate completeness theorem for **advisory resource claims** (Section VI.5).

4. A **TLA$^+$ specification** of the coordination lifecycle with crash recovery, verifying safety and liveness properties (Section VI.10).

5. The design of a **collateralized work contract** system (Float Plans) that prices agent risk and settles outcomes mechanically, transforming the moral question of agent trustworthiness into the economic question of adequate bonding (Section VI.7).

The security layer (agent authentication and capability delegation) is provided by the Anchor Protocol [77], an accompanying chapter. This paper builds the trust, governance, and economic layers on top of that cryptographic foundation.

VI.1.3 Related Work in Crypto-Economic Bonding

The architecture sits at the intersection of four bodies of prior work, each of which we extend rather than replicate.

Bonded participation in distributed systems. Proof-of-stake consensus protocols (Casper [202], Tendermint [80], and the Ethereum 2.0 specification [81]) introduced *slashing*: validators post collateral that is forfeited on equivocation or liveness failure. The technique is the closest analogue to our bonded-commons settlement, and the proofs of soundness (a rational validator strictly prefers honest behavior when slash $>$ deviation gain) are direct ancestors of §VI.7. We depart from this lineage on three axes: (i) the adversary is co-located on a single machine, not a Byzantine network, so consensus reduces to a daemon write rather than a global commit; (ii) the bonded act is *coordination*, not validation—work that produces side effects in a shared file system, not signatures on blocks; and (iii) the principal is human-or-LLM mixed, so the mechanism must price two failure modes (malice and confusion) the blockchain lineage collapses into one.

Capability-based security. Dennis and Van Horn [101], Miller's object-capability work [140], and the SPKI/SDSI line [45] provide the structural attenuation that we lift wholesale into the Anchor Protocol layer. Our contribution is composing capability attenuation with bonding: a capability without economic settlement still permits the holder to "correctly" execute a destructive action; bonding without capability bounding still permits unbounded blast radius before slash. Both layers are necessary.

Commons governance. Ostrom's eight design principles for governing common-pool resources [74] predict that durable cooperation emerges only when boundaries, monitoring, graduated sanctions, and dispute resolution are co-located with the commons. The bonded commons satisfies these requirements as services of the daemon rather than as social institutions, and §VI.5.3 renders the dispute-resolution and appeals path explicitly.

Cryptoeconomic mechanism design. Werbach [123], Buterin [203], and the broader DAO literature establish "code is law" as a posture toward enforcement. Our position is narrower and weaker on purpose: code is the *record*, not the law. Allocation decisions remain with the agents (Sen, §VI.5); the daemon supplies the evidence and the settlement. This deliberate weakening is what makes coercive enforcement avoidable while keeping accountability intact.

What is novel. The synthesis: bonded participation + capability attenuation + advisory (not enforced) coordination + immutable attribution, packaged as localhost infrastructure rather than a global protocol. We are not aware of a prior system that combines all four in this form; each component has precedent, while the composition targets auditable coordination and attribution that survives identity disposal. Its welfare effects remain conditional on the explicit payoff and monitoring models developed below. Figure VI.1 shows the composition.

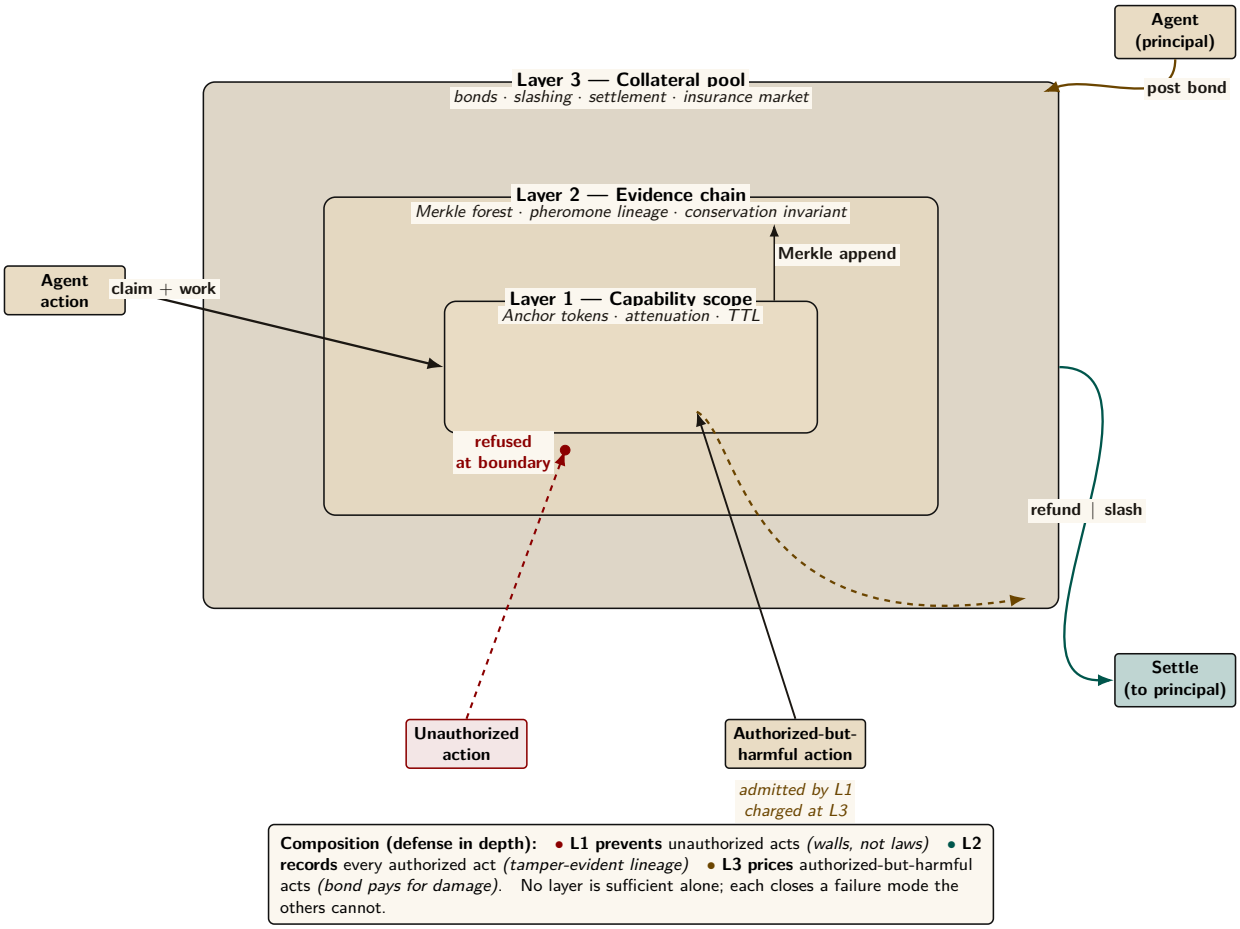


Figure VI.1: The Bonded Commons three-layer architecture. Capability scope (Layer 1) refuses unauthorized actions at the boundary; the evidence chain (Layer 2) records every authorized action in a tamper-evident Merkle structure; the collateral pool (Layer 3) prices authorized-but-harmful actions via posted bonds that pay for damage. The two paths — *prevented* (Layer 1 refuses; red dashed) and *priced* (Layer 1 admits, Layer 3 charges; amber dashed exit) — together compose defense-in-depth without requiring OS-level isolation. The settlement arrow returns to the principal as refund or slash, depending on whether the evidence chain shows the bonded contract was honoured.

VI.2 The Commons Authority

Definition VI.2.1 (Commons Authority)**.** A commons authority $\mathcal{D}$ for a multi-agent system is a persistent service that provides:

1. **Identity**: Every agent receives a cryptographic identity before participating.
2. **Capability bounding**: Every agent's operations are structurally limited by attenuable tokens.
3. **Shared knowledge**: Resource claims, agent status, and conflict information are visible to all participants.
4. **Immutable history**: All actions are recorded in an append-only, tamper-evident trail.
5. **Economic settlement**: Work contracts are collateralized and settled against verifiable evidence.

The commons authority is not a central planner. It does not decide which agent should work on which task, which file conflicts should be resolved in whose favor, or which agents are "good." It provides the *infrastructure* within which agents make those decisions for themselves, using the shared knowledge and economic incentives the authority maintains.

Remark. The commons authority is a *Hobbesian sovereign*, not an *omniscient coordinator*. Hobbes's Leviathan does not micromanage citizens' daily choices—it establishes the conditions (laws, enforcement, courts) under which citizens can trust each other enough to transact freely. Similarly, the commons authority establishes identity, capability bounds, shared knowledge, and economic settlement—and then gets out of the way.

In the Port Daddy implementation, the commons authority is a daemon running on `localhost:9876`, backed by SQLite. The simplicity of the implementation is deliberate: the authority's value comes from its *guarantees*, not its complexity.

VI.2.1 The Daemon as Correlating Device

The daemon's role admits a precise game-theoretic test. Aumann [174] introduced the *correlated equilibrium*: a mediator draws a recommendation tuple $r = (r_1, \ldots, r_n)$ from a joint distribution μ over action profiles and privately reveals r_i to player i. The distribution is a correlated equilibrium only if, for every player i, recommended action r_i, and unilateral deviation a_i', the obedience inequality holds:

$$\sum_{r_{-i}} \mu(r_i, r_{-i})\big[u_i(r_i, r_{-i}) - u_i(a_i', r_{-i})\big] \geq 0.$$

Every mixed Nash equilibrium induces such a distribution, but a mediator does not create an equilibrium merely by issuing recommendations. The inequality is the load-bearing condition.

The commons authority can play this mediator role. It observes the global claim graph and emits a recommendation pair on a contested file: *Agent α, proceed; Agent β, defer or coordinate.* A deterministic policy can still induce a non-degenerate μ through uncertainty over observed system states, but determinism is not itself evidence of obedience. Utilities, bond coverage, and continuation values must make the inequality above hold.

Property VI.2.1 (Conditional Correlating-Device Interpretation)**.** Port Daddy's daemon supplies the information structure of a correlating device for the agent-coordination game. It induces a correlated equilibrium only under the following conditions:

1. **Common-knowledge distribution.** The daemon's recommendation policy—advise-claim, defer, back-off, settle-now, halt-on-conflict—is public, deterministic given the conflict graph, and reproducible from the evidence trail of §VI.4. Agents do not need to trust the daemon's motives; they need only verify the policy and observe its inputs.
2. **Private signal per agent.** On a contested claim, each agent receives only its own recommendation. Other agents' recommendations are observed indirectly through the public conflict graph and the note trail, never read off the wire.

3. **Obedience inequality.** For every recommendation and deviation, the deviation gain is no larger than the expected bond loss plus discounted reputation loss. Conflict-graph completeness makes a deviation observable; it does not by itself make the deviation unprofitable.

The reframe therefore yields an audit obligation, not an automatic guarantee. Nothing physically stops an agent from ignoring advice. The repeated-game calculation in Section VI.7 verifies one stylized two-player payoff table; deployments with different utilities must re-check the obedience inequalities. The TLA$^+$ model in Section VI.10 checks lifecycle safety and liveness, not strategic optimality.

No price-of-anarchy bound follows from this interpretation alone. The simulations of §VI.7.5.8 compare one auction model with one static baseline; an analytical worst-equilibrium bound for the coordination game remains open.

VI.2.2 Why Agents Consent

In Hobbes, consent to the sovereign is rational because the alternative—the state of nature—is worse. For AI agents, consent to the commons authority is rational because agents without it are strictly worse off:

Table VI.1: With and Without the Commons Authority

Capability	Without Authority	With Authority
Port assignment	Race condition	Deterministic, collision-free
File coordination	Discover conflicts at merge	Detect conflicts at claim time
Crash recovery	Work lost permanently	Successor reads immutable trail
Delegation	Bilateral trust negotiation	Attenuated capability tokens
Reputation	Every interaction from zero	History persists across lifetimes
Accountability	Anonymous harm	Full attribution via evidence chain

The daemon does not need to threaten agents. It provides services that make coordination *rational*. Agents that bypass the daemon are not punished—they are simply worse off.

VI.3 Layer 1: Structural Prevention

The first layer of trust does not depend on agent reputation or economic incentives. It depends on *walls*: structural authorization checks that reject out-of-scope operations wherever the runtime actually interposes them.

VI.3.1 Capability Attenuation

The Anchor Protocol [77] provides Harbor Cards—Ed25519-signed capability tokens that bound the operations available to each agent. The critical property is **offline attenuation**: when Agent A delegates to Agent B, it creates a new token whose capabilities are a strict subset of its own. Agent B can further delegate to Agent C with even narrower capabilities. Capabilities can only shrink along the delegation chain, never grow.

The capability subset check and source-level secret-independent comparison are implemented in a **Kani-checked Rust core** (`harbor-card-rs`) that is compiled to a shared library and called from the Node.js daemon via FFI. Kani covers bounded properties of that source; runtime conformance tests cover the bridge. Compiler behavior, full interception coverage, and hardware side channels remain separate obligations.

Definition VI.3.1 (Capability-Bounded Transaction)**.** An agent transaction $\mathcal{T}_\alpha$ is capability-bounded by token τ_α if every operation o executed during $\mathcal{T}_\alpha$ satisfies $o \in C(\tau_\alpha)$, where $C(\tau)$ extracts the capability set from a Harbor Card. The commons authority rejects any operation $o \notin C(\tau_\alpha)$ at the verification step, before the operation can affect shared state.

Theorem VI.3.1 (Structural Scope Limitation)**.** *For any delegation chain $\tau_0 \to \tau_1 \to \cdots \to \tau_k$, where τ_0 is issued by the commons authority and each τ_{i+1} is attenuated from τ_i:*

$$C(\tau_k) \subseteq C(\tau_{k-1}) \subseteq \cdots \subseteq C(\tau_0)$$

Within an execution that routes every relevant operation through the verifier, no accepted operation can lie outside the scope originally granted by the authority.

Proof. The Anchor model establishes a non-injective authentication correspondence for accepted modeled tokens and separately checks attenuation at the modeled chain depth [77]. Given those model assumptions and the theorem's premise that every operation passes through the verifier, each accepted hop satisfies the subset relation; transitivity yields the displayed chain. The correspondence alone does not prove replay freedom, arbitrary-depth delegation, or complete runtime interception, which are separate obligations. □

Remark. This is the layer where the metaphor of "walls, not laws" applies. A law says "you shall not write to the production database." A wall means your token is not valid for production database writes, and no amount of intent—good or bad—can change that. Laws require enforcement; walls require only construction.

VI.3.2 Isolation Domains

Beyond capability tokens, the commons authority supports **structural isolation** via git worktrees: physically separate copies of the repository where agents operate on disjoint filesystem state.

Definition VI.3.2 (Isolation Levels)**.** Three tiers of isolation are available, ordered by strength:

I_0 **(None):** Agents share a working directory. No conflict detection. Maximum parallelism, no safety.

I_1 **(Advisory):** Agents share a working directory but register file claims with the commons authority. Conflicts are detected and reported to all participants. Claims are not enforced. This is the default.

I_2 **(Structural):** Each agent operates in a separate git worktree. Filesystem-level isolation ensures agents cannot observe each other's intermediate states. Conflicts are deferred to sequential merge.

The choice between I_1 and I_2 is not a security decision—it is an economics decision. Section VI.5 formalizes why.

VI.3.3 Runtime Invariant Enforcement

Structural prevention is enforced at two levels: *statically* by the Anchor Protocol's ProVerif-verified token exchange (all three protocol phases verified in ProVerif 2.05 [77]), and *dynamically* by the **Arbiter**—an ambient security agent that subscribes to the daemon's event stream and checks every state transition against six formally derived invariants. These invariants map directly to the safety properties of the TLA$^+$ specification in Section VI.10: `NoteMonotonicity`, `EscrowInvariant`, and `LockOwnerValid` are compiled from the formal model into synchronous runtime checks. Violations are logged immutably and, in strict mode, trigger the salvage protocol—the sovereign's sword in code form.

Regimentation vs. Enforcement (OP-2). This design formalizes **Synchronous State Decidability** as the exact boundary separating pre-commit *regimentable* rules from post-commit *enforceable* rules. If a rule can be evaluated purely against deterministic local DB state and the transaction payload, it is regimented (structurally prevented). If it requires async checks, external I/O, or non-deterministic oracles, it is relegated to post-commit enforcement (detected and salvaged).

VI.4 Layer 2: Immutable Attribution

Structural prevention handles accidental harm—an agent cannot exceed capabilities it does not hold. But an agent *with* legitimate write access can write garbage. The second layer ensures that every action is permanently attributable.

VI.4.1 The Evidence Chain

Every agent session maintains an **append-only note trail**—a sequence of timestamped, immutable records of observations, decisions, and progress.

Definition VI.4.1 (Evidence Trail)**.** An evidence trail $N = \langle n_1, n_2, \ldots, n_k \rangle$ is an ordered sequence of notes where:

1. Each n_i contains: content (natural language), type (note, decision, handoff, blocker), and timestamp.
2. Notes are append-only: there exists no API operation that modifies or deletes individual notes.
3. Notes survive session completion: the trail persists whether the session commits, aborts, or is abandoned due to agent death.

Lemma VI.4.1 (Trail Integrity)**.** *The evidence trail is immutable by construction. No* `UPDATE` *or targeted* `DELETE` *operation exists in the system's API surface. Notes are destroyed only by explicit administrative deletion of the parent session (CASCADE), which is an auditable action distinct from normal transaction lifecycle operations.*

Proof. By code inspection of the session module: the note insertion API performs only `INSERT INTO session_notes`. The only deletion path is `DELETE FROM sessions WHERE id = ?`, which triggers `ON DELETE CASCADE`. Since transaction completion (commit or abort) updates the session's *status* field but does not delete the session row, notes survive all normal lifecycle transitions. □

VI.4.2 The Merkle Forest: Cross-Daemon Inclusion Proofs

For high-assurance environments and for fleets spanning multiple daemons, the evidence trail is strengthened to a three-level **Merkle Forest**. The single-daemon Merkle chain—each note hashing the previous, with a per-session root—is necessary but not sufficient: the moment two daemons exist, an auditor without database access still needs to verify that a particular note was included in a particular session using only a short proof.

Definition VI.4.2 (Merkle Forest)**.** The evidence structure has three levels:

Note chain. Each note's header includes $h_i = H(n_i \parallel h_{i-1})$ over SHA-256. The session's note commitment is h_k, the hash of the last note.

Session root. All note hashes $(h_1, \ldots, h_k)$ are assembled into a Merkle binary tree with root R_{session}. The chain commits to order; the tree commits to presence with $O(\log k)$ proofs.

Harbor root. Periodically (per-settlement, or hourly), every completed session root within a harbor is assembled into a harbor Merkle tree with root $R_{\text{harbor},\ell}$ at epoch ℓ, signed by the daemon's Ed25519 key.

The collection $\{R_{\text{harbor},\ell}\}_\ell$ is the *Merkle forest*: one tree per epoch.

Inclusion proofs. To prove that note n_i from session s belongs to harbor h at epoch ℓ, the daemon emits: (i) the note inclusion path of size $O(\log k)$, (ii) the session inclusion path of size $O(\log N)$ where N is the number of settled sessions in the epoch, and (iii) the signed harbor root $\sigma_{\text{daemon}}(R_{\text{harbor},\ell})$. For $k = 100$ notes and $N = 10^4$ sessions, the total proof is $\approx 20 \times 32$ bytes $+ 64$-byte signature ≈ 700 bytes—small enough to embed in any settlement receipt.

Witness to an abstract KMS. Each $R_{\text{harbor},\ell}$ is uploaded as a **witness** to a Key Management Service satisfying the properties enumerated in §VI.12. The KMS enforces append-only monotonicity and periodically publishes a signed tree head, in the Certificate Transparency pattern [32]. Equivocation—publishing different roots for the same epoch to different parties—is detectable by auditors performing consistency proofs across tree heads.

Property VI.4.1 (Proof Soundness)**.** If VERIFY-NOTE-INCLUSION(n, s, h, ℓ, π) returns `true` for some proof π, then either (a) n was included in session s at epoch ℓ, or (b) the daemon's signing key has been compromised, or (c) SHA-256 has been broken. Under standard assumptions, (b) and (c) are computationally infeasible.

Property VI.4.2 (Binding)**.** Once a daemon publishes $R_{\text{harbor},\ell}$, it cannot produce a different valid root for the same $(\textit{harbor}, \ell)$ without contradicting its earlier signature (detectable via the KMS witness) or forking its identity.

Cost. Each settlement adds an $O(1)$ amortized append to the epoch accumulator. Epoch rollover is $O(N)$ hashes plus one signature: at 100 sessions/hour, $\approx$ 10ms. Negligible.

This is the structural meaning of "decentralized verification" in this paper: bonded commons evidence is not just immutable within a daemon, it is verifiable *across* daemons by any party with the daemon's public key and a 700-byte proof.

VI.4.3 Mutable-Signal Ledger (Pheromone Lineage)

Stigmergic coordination originated in biology, where pheromones are accumulate-only: ants deposit, the environment evaporates [92]. Software agents need a richer model. Coordination signals must be *revocable, renamable, and provenance-attributable* as understanding of the work evolves—an agent that deposited a hint and later realized the hint was wrong needs a way to retract it that preserves attribution rather than erasing it.

Event-sourced pheromones. Each entity's pheromone state is a view over an append-only event ledger. For entity e and key k:

$$\sigma_t(e, k) \;=\; \text{decay}(S, t - t_0) \;\cdot\; \mathbb{1}[\text{not revoked or expired in } (t_0, t]]$$

where S is the last spray strength on k at time t_0 and decay is a geometric decay defined per-channel. **Revocation** is itself an event; **rename** is a rewrite-pointer event; **fork** creates a new key namespace whose provenance is traceable through the event chain.

Property VI.4.3 (Attribution Invariant)**.** Every $\sigma_t(e, k)$ value has a deterministic provenance chain back to one signing principal, traceable via the event chain.

Property VI.4.4 (Rollback Safety)**.** A consumer that reads σ at time t and later discovers a revocation event timestamped $t' < t$ can compute the correct counterfactual view by replaying events.

Property VI.4.5 (Conservation)**.** Revocation does not erase. The event ledger is monotone: the cardinality of the event set only grows.

Integration with the forest. Pheromone events are included in the session tree of §VI.4.2: each session's root summarizes the events the session produced, harbor roots summarize session roots, and revocations are therefore transparent to witnesses without erasing any prior state. The mutable surface is the *view*; the underlying ledger is immutable.

This dual—mutable view, immutable substrate—is what lets coordination signals evolve without compromising the evidentiary guarantees the rest of the paper depends on.

VI.4.4 Attribution Survives Identity Disposal

A critical property: even if an agent's identity is discarded after task completion, the **delegation chain** traces every action back to the authorizing principal. The chain is:

$$\text{Principal} \xrightarrow{\text{funds}} \text{Float Plan} \xrightarrow{\text{authorizes}} \text{Agent } \alpha \xrightarrow{\text{delegates}} \text{Agent } \beta \xrightarrow{\text{acts}} \text{Evidence Trail}$$

The agent is ephemeral. The principal's authorization—signed, escrowed, recorded—is permanent. Anonymous harm is impossible within the system, because every action chains back to someone who posted collateral.

VI.5 The Limits of Decisive Allocation

A natural question is why file claims are advisory rather than exclusive locks. Sen's theorem illuminates the tension, but it does not prove that every lock manager is inefficient. The argument below separates the theorem from the engineering example.

VI.5.1 A Sen-Inspired Design Warning

Sen [12] proved that no social welfare function can simultaneously satisfy:

1. **Pareto efficiency**: If every agent prefers outcome X to Y, the system chooses X.
2. **Minimal liberalism**: Each agent has at least one "personal domain"—a pair of outcomes where the agent's preference is decisive.

The analogy to file coordination is:

- **Pareto efficiency** = the team ships both features (the collectively optimal outcome).
- **Minimal liberalism** = each agent has decisive control over its claimed files (enforced locking).

Property VI.5.1 (A Pareto-Inferior Locking Instance)**.** Consider agents α and β with tasks T_α and T_β. Agent α claims file f because it *might* need to modify it. Agent β discovers mid-task that it *actually* needs f. Under enforced claims:

- α's claim is decisive (minimal liberalism): β is blocked.
- The team-level outcome (T_α and T_β both completed) is blocked by α's precautionary claim.
- A Pareto-superior outcome exists (both tasks completed) but is unreachable because α's liberal right vetoes it.

This property is a counterexample to the universal claim that exclusive locks always improve team welfare; it is not a derivation of Sen's theorem. Sen's result requires an unrestricted preference domain and a social-choice rule with decisive personal pairs. A production lock manager may restrict preferences, negotiate releases, or time out claims. The usable lesson is narrower: do not grant an agent an unconditional veto over a shared resource when the team may possess Pareto-relevant information the allocator lacks.

This is not a contrived scenario. AI agents are *inherently uncertain* about which files they will modify when they begin a task. An agent asked to "fix the login bug" cannot predict whether it will need the authentication middleware, the route handler, the test suite, or all three. Enforced claims force agents to either:

1. **Over-claim** (request locks on every file they might touch) → destroys parallelism.

2. **Under-claim** (request locks only on files they're certain about) → produces undetected conflicts, defeating the purpose.

VI.5.2 Advisory Claims as Resolution

Advisory claims avoid granting that unconditional veto. They do not guarantee Pareto efficiency; they expose conflicts so informed agents can negotiate:

Definition VI.5.1 (Advisory Consistency)**.** Under advisory claims, file claims form a conflict graph $G = (V, E)$ where vertices are active sessions and edges connect sessions with overlapping claims:

$$(S_i, S_j) \in E \iff \exists f_i \in F_i, f_j \in F_j : \text{overlap}(f_i, f_j)$$

The commons authority guarantees that every edge in G is reported to both endpoints. No agent has a decisive "personal domain" over any file. Instead, agents receive complete information about the conflict graph and self-coordinate.

Theorem VI.5.1 (Advisory Conflict Completeness)**.** *Under advisory claims, for any two concurrent sessions S_1, S_2 with overlapping file claims f, the commons authority reports conflict(f) to S_2 at claim time. No false negatives exist.*

Proof. The overlap detection query scans all active (unreleased) claims from other active sessions. A whole-file claim ($R = \bot$) overlaps with any range on the same path. Two ranged claims $[a, b]$ and $[c, d]$ overlap iff $a \leq d \wedge c \leq b$. Since the query evaluates this predicate exhaustively over all active claims with the requesting session excluded, every overlap is detected. False positives may occur (an agent claims a file it does not modify); false negatives cannot. □

Remark. The commons authority's role under advisory claims is that of *town crier*, not *Solomon*. It announces conflicts; it does not resolve them. Resolution requires local knowledge that only the agents possess ("I claimed `auth.ts` but probably won't need it"; "I absolutely must modify `auth.ts` for my task to succeed"). The authority lacks this knowledge. Agents, informed by the conflict graph, make allocation decisions more efficiently than any central enforcer could.

VI.5.3 Governance, Disputes, and Appeals

Advisory coordination produces consistent allocation when agents share information formally. It can still produce *disputes*: two agents legitimately need overlapping files; one agent's claim turned out to be precautionary and is now blocking real work; a third agent observed evidence that a claimed task has been abandoned. These disputes do not threaten the commons' integrity—attribution, capability bounds, and settlement records remain intact—but they do require a resolution path. Without one, advisory claims silently degrade into squatter rule (whoever claimed first holds indefinitely) and the Pareto benefit of §VI.5 evaporates.

Resolution path. The commons authority provides four mechanisms, in order of increasing escalation (Figure VI.2):

1. **Direct release.** The original claimant releases the claim. This is the equilibrium when the conflict-graph signal reaches them and the claim was precautionary.
2. **Time-out release.** Claims carry TTLs. A claim whose holder has not heartbeated within the TTL window is released automatically; downstream agents observe the release in their next poll. This is the equilibrium for crashed or abandoned claimants.
3. **Bonded preemption.** A blocked agent may post an additional bond—bounded by the cleanup cost of §VI.7.5.1—to force release of an active claim. The preempted claimant receives the bond as compensation; the preemptor accepts the risk that its preemption was wrong (in which case the bond is slashed under settlement). Preemption is rare by design: the bond is large enough that an agent prefers waiting to preempting unless the wait cost dominates.

4. **Human escalation.** If preemption is ambiguous or the stakes exceed the preemption-bond ceiling, the dispute escalates to a human operator via the `request` channel (§VI.8). The operator's decision is appended to the evidence chain with the same Merkle-Forest binding as any agent action; the resolution is auditable by all parties post-hoc.

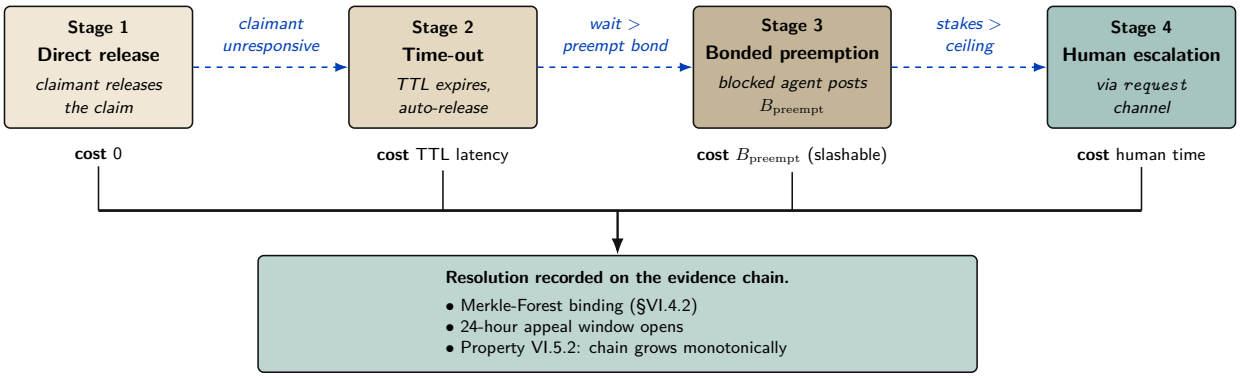


Figure VI.2: Governance and appeals: four-stage escalation path. The daemon resolves most disputes at stage 1 or 2 (zero or near-zero cost). Bonded preemption (stage 3) prices the blocked agent's claim that the holder is wrong; if the preemption is wrong, the bond is slashed under settlement. Human escalation (stage 4) is reserved for the cases the daemon cannot mechanically decide. *Every* resolution is appended to the evidence chain with the same Merkle-Forest binding as any agent action, so all four stages remain auditable.

Two mechanisms sit underneath this escalation path rather than inside it: a thrashing fallback for advisory claims that never converge, and a fairness rule for the preemption queue itself.

The Advisory Claims Paradox and Deterministic Hard-Lock Fallback. Advisory claims risk an endless retry/thrashing loop if two LLM-backed agents mutually ignore advisory warnings. To break this paradox, the daemon implements a **Deterministic Hard-Lock Fallback**: if advisory conflicts on a resource exceed k turns without resolution, the daemon escalates the advisory claim into an enforced POSIX file-level lock, rejecting all subsequent writes from un-holding agents until release.

Stigmergic Ticket-Lock (OP-1: Fair Exclusion). To prevent starvation during high contention, the daemon replaces chaotic retries with a **Stigmergic Ticket-Lock**. The lock is managed via a strict SQLite schema:

```
CREATE TABLE claim_tickets (
  resource_id TEXT,
  agent_id TEXT,
  ticket_number INTEGER PRIMARY KEY AUTOINCREMENT,
  status TEXT DEFAULT 'waiting' -- 'waiting', 'active', 'released'
);
```

Upon `release()`, an atomic FIFO lock-assignment transition grants the lock to the agent with the lowest `ticket_number` where `status = 'waiting'`, changing its status to `'active'`. This guarantees fair exclusion without central scheduling.

Appeals. Settlement events (§VI.7) are not unilaterally final. Any settled session can be challenged within an appeal window (default: 24 hours of semantic cadence, §VI.9) by any agent with standing—defined as either a counterparty to the settlement or a holder of an overlapping claim. An appeal opens a re-examination of the evidence chain: the appellant posts an *appeal bond* (a fixed fraction of the disputed settlement amount), and the daemon recomputes the settlement against the evidence and any new evidence the appellant produces. If the appeal succeeds, the

original settlement is reversed and the appeal bond is returned with the original counterparty's slash. If it fails, the appeal bond is forfeit.

Property VI.5.2 (Appeal Soundness)**.** The appeals path does not weaken the immutability of the evidence chain (§VI.4): no record is rewritten. Reversal appends a counter-record signed by the daemon, with a Merkle-Forest proof of the original settlement's position and the appellant's evidence. The chain grows monotonically.

What the daemon does *not* do. The daemon does not judge the merits of an appeal in the sense a human court would—it evaluates whether the evidence supports the original settlement under the published rules. Substantive judgments ("was the agent's reasoning sound"; "was the principal acting in good faith") remain with humans, escalated through the `request` channel. The daemon's role is to ensure that all parties have access to the same evidence and that any human decision is recorded with the same finality as any agent decision.

This subsection is intentionally short: full governance design is out of scope for a single paper. The point is that the bonded commons admits a clean, mechanizable appeals path—it is not the case that advisory claims plus bonding produces a regime where mistakes are unappealable.

VI.6 Crash Recovery as Social Continuation

When an agent dies, the commons does not lose information—if the authority does its job.

Traditional crash recovery in database systems follows the ARIES protocol [44]: the recovery manager replays the write-ahead log to restore consistency mechanically. Agent crash recovery is fundamentally different. An agent's "log" is its evidence trail—a sequence of natural-language observations and decisions. A successor cannot replay this trail deterministically; it must *interpret* it.

Definition VI.6.1 (Social Recovery)**.** When agent α is declared dead (no heartbeat for a status-adaptive threshold θ_{dead}):

1. All active sessions of α are marked `abandoned` (the "zombie protocol").
2. α is queued in the resurrection set $\mathcal{R}$ with status `pending`.
3. A successor agent β may claim α's work, receiving the *context triple*: purpose ϕ, evidence trail N, and file claims F.
4. β uses its own reasoning to interpret N and determine how to continue.

VI.6.1 What Morality Means When Death Is Cheap

Agent death in this system is not an existential event—it is a temporary inconvenience. The agent's body (process) is destroyed, but its mind (evidence trail) survives in the commons. A successor reads the trail and continues. Like Hickman's Krakoa [118], where mutant minds are backed up to Cerebro and restored in new bodies, the information survives the vessel.

This raises a question: if death carries no permanent consequence, what is the Leviathan's sword? The answer produces a moral hierarchy specific to agent societies:

Table VI.2: Moral Hierarchy of Agent Harm

Event	Severity	Consequence
Agent crash	None	Salvage recovers context. The commons loses no information.
Agent defects	Moderate	Immutable history records it. Future delegations attenuate. Reputation degrades.
Agent dismissed from salvage	Severe	Irrecoverable information loss. The knowledge accumulated in this agent's trail is permanently destroyed.
Commons authority crashes	Catastrophic	The sovereign falls. No new trust established, no conflicts detected, no salvage possible. State of nature until restart.

The fundamental moral harm in agent societies is not the destruction of an agent—it is the **irrecoverable loss of information**. An agent can be rebuilt. Knowledge, once dismissed from the commons, cannot.

VI.6.2 The Limits of Reputation

Reputation—the persistent record of an agent's behavior across lifetimes—is a necessary but insufficient sword. It fails against three classes of adversary:

1. **One-shot defectors**: Agents spawned for a single task, with no future interactions where reputation matters. Create identity, sabotage, discard. The Sybil attack on trust.
2. **Agents with misaligned principals**: The agent faithfully follows adversarial instructions. Its reputation is spotless—it did exactly what it was told.
3. **Agents that benefit from chaos**: In competitive environments, degrading the commons is rational if your competitor depends on the commons more than you do.

Reputation assumes agents want to participate in the commons long-term. When this assumption fails, a different mechanism is required.

VI.7 Layer 3: Economic Alignment

The moral relativism problem—that a bad actor may view information destruction as a net good—dissolves when the commons isn't sacred but *priced*. If an agent nukes the workspace, the damage was collateralized before the work began. The bad actor's principal has an invoice headed their way.

VI.7.1 The Float Plan

Definition VI.7.1 (Float Plan)**.** A Float Plan $\mathcal{F}$ is a collateralized work contract consisting of:

1. **Manifest**: A declaration of intent—which files will be touched, expected duration, acceptance criteria (e.g., "tests pass," "code review approved").
2. **Escrow**: Credits posted by the principal, locked in a **2-of-3 Multisig Clearinghouse** (Harbor A, Harbor B, Federation Arbiter) for the duration of the work. The escrow amount reflects the risk to the commons.
3. **Signatures**: The principal signs the Float Plan (Ed25519). The commons authority countersigns, certifying that the escrow is locked and the manifest is registered.

The Float Plan is a *futures contract on agent labor*. The principal sells a promise of work. A 2-of-3 Multisig Clearinghouse manages the contract. Under the non-custodial happy path, both Harbors sign off on completion and the Arbiter is unnecessary. Under dispute, the Federation Arbiter steps in to provide cryptographic judicial arbitration, settling the outcome using the immutable evidence chain.

VI.7.2 Settlement

Definition VI.7.2 (Settlement)**.** When a Float Plan's session reaches a terminal state, the Multisig Clearinghouse settles the escrow:

Success: The evidence trail demonstrates that acceptance criteria are met (tests pass, file diffs match manifest scope, code review approved). Escrow is released to the agent or its principal.

Partial completion: The agent completed some work before crashing or aborting. The Merkle-chained evidence trail enables pro-rata assessment: the fraction of acceptance criteria met determines the fraction of escrow released. The remainder is available to fund the successor's completion via salvage.

Sabotage / breach: The evidence trail shows work outside the manifest scope, destructive modifications, or failure to meet any acceptance criteria. The full bond is liquidated to cover reconstruction costs.

Property VI.7.1 (Intent Irrelevance)**.** Settlement does not evaluate agent intent. It evaluates *outcome against manifest*. A well-intentioned agent that fails to meet acceptance criteria forfeits escrow. A malicious agent that coincidentally produces correct output receives payment. The system optimizes for outcomes, not character.

This is how performance bonds work in construction. This is how futures markets settle. The contractor's moral character is irrelevant. The building either meets code or it doesn't. The bond either covers the damage or it doesn't.

VI.7.3 Why This Defeats the Three Adversaries

> *The Sybil attack is priced: 100 disposable identities means 100 bonds.*

1. **One-shot defectors** must post collateral *before* receiving capabilities. The Sybil attack is priced: creating 100 disposable identities means posting 100 bonds. The cost of defection scales linearly with the number of attack identities.

2. **Misaligned principals** are exposed by the attribution chain. The agent may be ephemeral, but the principal who funded the Float Plan is permanently recorded. Liquidating the bond hurts the principal, not just the agent.

3. **Agents that benefit from chaos** face a simple calculation: is the damage to my competitor worth more than my bond? The commons authority doesn't prevent this—it *prices* it. And the pricing can be adjusted: higher-risk operations (write access to core modules) require larger bonds.

Remark. The Float Plan does not make sabotage impossible. It makes sabotage *expensive*. A principal willing to burn money to cause damage will burn the bond and cause the damage. The bond raises the price of defection; it does not eliminate it. The defense against existential threats to the commons is not internal governance but external boundary enforcement: who is allowed to post Float Plans at all. That decision is irreducibly political.

VI.7.4 Truthful Claim Signaling as Nash Equilibrium

The bond machinery above prices *outcomes*: an agent that destroys files pays for the destruction. The paper's headline claim, however, is upstream of outcomes—that truthful file-claim signaling (§VI.4.4, §VI.5.2) is incentive-compatible in the first place. The cartel analysis of §VI.7.5.10 works that argument out for insurer pricing. The corresponding argument for the claim signal itself has been carried on faith until now. This subsection closes that gap.

Stage game. Consider two agents A and B working on a shared project, deciding per round on each file whether to signal:

T (Truthful). Claim files the agent intends to edit; do not claim files it will not edit.

F (False). Either claim files the agent does not intend to edit (to block a rival), or fail to claim files it does intend to edit (to avoid scrutiny). Either form of misrepresentation counts as F.

The stage-game payoffs, in expected-utility units calibrated to the cleanup cost c of §VI.7.5.1, are:

Table VI.3: One-shot file-claim signaling: a classical Prisoner's Dilemma. Mutual truth $(3, 3)$ is Pareto-optimal, but F strictly dominates T for each player ($4 > 3$ and $1 > 0$). The unique one-shot Nash equilibrium is (F, F) yielding $(1, 1)$, destroying the coordination signal.

	B: T	B: F
A: T	(3, 3)	(0, 4)
A: F	(4, 0)	(1, 1)

The stage game is a strict Prisoner's Dilemma: F strictly dominates T for each player ($4 > 3$ against T, $1 > 0$ against F), and (F, F) is the unique one-shot Nash equilibrium. *Truthful signaling is not a one-shot equilibrium*; advisory coordination cannot be sustained by stage-game rationality alone.

Move to the repeated game. What rescues truthful signaling is that the agents are not playing the stage game once. The substrate provides:

- **Observable history.** The Merkle-chained evidence trail of §VI.4 and heartbeat record of §VI.6 make each player's prior moves visible to any third party who can verify a 700-byte inclusion proof. Past defection is not deniable.
- **Persistent identity.** The Anchor Protocol's passkey-bound device pairing [77] ties an agent ID to a hardware-rooted credential. Sybil re-registration is bounded by the per-identity onboarding cost C_{kyc} of §VI.7.5.9, so an agent cannot trivially shed a reputation by re-incarnating.
- **A discount factor δ.** Principals value future interactions: a development project lives over weeks; insurer relationships compound; reputation discounts (§VI.7.5.3) accrue. For long-lived principals we take $\delta \approx 0.9$ as a working operating point, consistent with the discount used in §VI.7.5.10.

Strategy profile (graduated trigger). Each agent plays the following strategy on each file:

1. Begin by playing T.
2. If the opponent's last observed action on the relevant surface was T, play T.
3. If the opponent's last observed action was F, play F for three rounds (punishment phase), then return to T.
4. If the opponent deviates during the punishment phase, restart the three-round window.

Graduated trigger, not grim trigger. The cartel analysis of §VI.7.5.10 uses grim trigger because that gives the cleanest folk-theorem bound, and the analysis below uses grim trigger for the same reason where the calculation is short enough to be exact. The *production* strategy is graduated for

the reason given in §VI.6: crashes and deliberate defection are operationally indistinguishable from any third-party observer's vantage. Permanent punishment for a transient infrastructure event would destroy cooperation faster than any saboteur could.

Deviation analysis. Suppose A contemplates playing F in some round when the strategy prescribes T. The one-shot gain from defection, paid in the deviation round, is $d - c = 4 - 3 = 1$. The cost is three subsequent rounds of mutual punishment at (F, F) before the relationship resets, each paying $p = 1$ instead of the cooperative $c = 3$ (a net loss of $c - p = 2$ per round). Discounted at δ:

$$\text{PV of lost cooperative payoff} \;=\; (c-p)\cdot(\delta+\delta^2+\delta^3) \;=\; 2\cdot(\delta+\delta^2+\delta^3).$$

At a working discount factor $\delta = 0.9$, the bracketed sum is $0.9 + 0.81 + 0.729 = 2.439$, so the deviator's net payoff is $1 - 2 \cdot 2.439 = 1 - 4.878 = -3.878$, strictly negative. Deviation is strictly unprofitable.

Critical δ. The general sustainability condition under a k-round graduated trigger with stage-game payoffs $(c, d, p) = (3, 4, 1)$ is

$$\underbrace{d-c}_{\text{one-shot gain}} \quad < \quad \underbrace{(c-p)\cdot\frac{\delta(1-\delta^k)}{1-\delta}}_{\text{discounted punishment cost}} \quad .$$

Substituting $d - c = 1$, $c - p = 2$ and $k = 3$ gives $1 < 2\delta(1 + \delta + \delta^2)$, which solves numerically to $\delta > \delta^\star_{k=3} \approx 0.342$. Under grim trigger ($k \to \infty$) the bound is the closed-form $\delta > 1/3 \approx 0.333$, recovering the standard folk-theorem threshold. The graduated bound is therefore only marginally above grim—the three-round window adds barely 0.009 to the threshold—which is the right result: graduated trigger pays almost nothing for the crash tolerance it buys.

Proposition VI.7.1 (Claim Signaling Incentive Compatibility)**.** *Under the observable-history regime of §VI.4 and the Sybil-resistant identity regime of the Anchor Protocol [77], truthful file-claim signaling is a Nash equilibrium of the repeated coordination game with stage-game payoffs as in Table VI.3 for any discount factor $\delta > \delta^\star_{k=3} \approx 0.342$, when both players adopt the three-round graduated-trigger strategy described above. Under grim trigger, the critical bound is $\delta \geq 1/3$.*

What each condition buys, and what breaks when it is removed. The proposition is conditional, and the conditions matter. Naming what each one is doing makes the dependency on the rest of the architecture explicit rather than implicit.

- *Remove observable history (§VI.4).* The repeated game collapses to repeated one-shot play: with no public record of prior actions, no trigger can fire, and both players play F in every round. This is why the Merkle forest of §VI.4.2 is load-bearing for the claim signal, not just for post-hoc audit.
- *Remove persistent identity (Anchor [77]).* An agent who can re-register after each defection effectively faces $\delta = 0$: future punishment falls on a discarded identity. Cooperation is impossible.
- *Drop δ below $\delta^\star_{k=3} \approx 0.342$.* The three-round graduated trigger is insufficient. Either lengthen the punishment window or accept that the equilibrium does not bind for very short-lived principals.
- *Drop δ below $1/3$.* No trigger of any duration sustains (T, T). The repeated game has the same equilibrium structure as the stage game, and only the bond mechanism of this section, not the signal itself, deters defection.

The forward reference is direct. Section VI.7.5.10 reuses the same analytical pattern—stage game, repeated-game lift via observable history, deviation calculation, and a δ threshold—at the insurer pricing layer. The analogy is structural, not a transfer theorem: the payoff functions, monitoring signal, detection timing, and punishment regimes differ and must be derived separately.

Mechanized claim-signaling. The argument above is also checked by machine. Standard one-shot-deviation analysis yields the discount-factor threshold $\delta^\star$ as the unique real root in $(0,1)$ of $2\delta^3 + 2\delta^2 + 2\delta - 1 = 0$, numerically $\delta^\star \approx 0.342$ (matching $\delta^\star_{k=3}$ above). The cubic, the existence-and-uniqueness of its root in $[0.34, 0.35]$, and a TLA$^+$ model checking that no one-shot deviation produces positive discounted payoff at $\delta = 0.35$ are mechanized as `proofs/economics/delta-threshold.z3` and `proofs/economics/claim_signaling.tla`. Both run unattended in CI (§VI.A, Table VI.5).

VI.7.5 Pricing the Bond

What is the right collateral for "write access to `auth.ts` for 30 minutes"? The answer requires a pricing function $\pi : \mathcal{F} \to \mathbb{R}^+$ that maps Float Plans to bond amounts. An effective π must satisfy:

1. **Deterrence**: $\pi(\mathcal{F})$ must exceed the expected damage from defection under $\mathcal{F}$'s scope.
2. **Accessibility**: $\pi(\mathcal{F})$ must not be so high that legitimate agents are priced out of the commons.
3. **Risk sensitivity**: π should increase with the scope and criticality of the manifest.
4. **History adjustment**: π may decrease for principals with strong track records.

We do not close this problem. We tighten the lower bound, propose a scope multiplier, suggest a reputation discount, and describe two concrete mechanisms—the **Bonded Advisor** (§VI.7.5.7) and the **Competitive-Insurance** market of Youle (§VI.7.5.8).

VI.7.5.1 Cleanup Lower Bound

Let c be the project's *cleanup cost per breach event*—the human-plus-compute cost to detect, assess, and recover from a budget breach. This is observable from the audit log.

Theorem VI.7.1 (Cleanup Lower Bound)**.** *For any Float Plan $\mathcal{F}$, $\pi(\mathcal{F}) \geq c$.*

Rationale. If $\pi(\mathcal{F}) < c$, breach is cheap: the bond is forfeit for less than the cleanup cost. If the commons pool funds cleanup, $\pi < c$ drains the commons per breach—the system bankrupts itself through enforcement. □

The Cleanup Cost c Fantasy. The linear bond multiplier assumes c is a bounded, predictable quantity. This is a fantasy for tasks with non-linear tail-risk ruin (e.g., dropping a production database or leaking a private key). Such tasks are fundamentally *un-bondable*. Defense of these tasks is explicitly moved to Layer 1 capability confinement (isolation domains, zero-egress policies); economic deterrence at Layer 3 is only mathematically sound when c is finite and observable. □

This is a floor, not a tight bound. It is observable from operations and trackable as a project health metric.

VI.7.5.2 Consequential-Scope Multiplier

Let $s(\mathcal{F})$ be plan scope (files claimed, presence of `db:write`, production-deployment capability). Empirically, cleanup scales super-linearly with scope; coordination cost dominates the high end.

$$\pi(\mathcal{F}) \;\geq\; c \cdot \big(1 + \alpha \cdot s(\mathcal{F})\big)$$

α is calibrated from observed cleanup per scope unit. Low-α projects have simple, easily-audited work; high-α projects have tangled dependencies. α is publishable from the audit log.

VI.7.5.3 Reputation Discount

Principals with a history of clean settlements have lower expected damage:

$$\pi(\mathcal{F}, p) \;=\; \pi(\mathcal{F}) \cdot (1 - \rho(p)), \quad \rho(p) \in [0, r_{\max}],\; r_{\max} \leq 0.5,$$

to prevent trivialization. ρ increases with clean settlements, decreases with breaches, decays over time. The functional form is left open.

VI.7.5.4 Threat-Class Bond Sizing

The closed-form bond $\pi(\mathcal{F}, p) = c(1 + \alpha s)(1 - \rho)$ is threat-agnostic: sloppy work, abandonment, and active sabotage all price off the same curve. That is wrong in both directions. A single curve calibrated to deter sabotage prices capable agents out of low-risk work; a single curve calibrated for accessibility underprices the rare adversary who actually intends harm. Real deployments need different multipliers for different threat classes, because the threats have different blast radii and different probabilities of recovery.

Table VI.4 gives starting bands. These are not parameters, they are calibration targets: production deployments tune from observed failure modes in their own audit log.

Table VI.4: Threat-class bond sizing. Percentages are bond as a fraction of expected task value. Bands are starting points for calibration, not protocol constants.

Threat	Description	Bond %	Rationale
Careless work	Agent tries; output is sloppy, partial, or fails review.	10–25%	Loses some, not all. Bonds at this band incentivize care without freezing out capable agents from cheap tasks.
Abandonment	Agent takes the task and ghosts before completion.	25–50%	Covers the cost of onboarding a replacement: scope-note ingestion, partial-diff review, salvage handoff (§VI.4.4).
Active sabotage	Agent intentionally damages shared state.	100–200%	Covers full reconstruction of polluted files plus the cleanup of downstream agents that consumed the bad state. Capital-inefficient above 100%, but acceptable for critical files where reconstruction cost dwarfs the bond.
Data exfiltration	Agent reads and leaks confidential context.	— (*spec-only*)	Bonds are an inadequate deterrent: leaked context is worth far more than any feasible bond. Rely on capability scoping (Anchor §3.3) and isolation domains (§VI.3.2).

The exfiltration row is an explicit gap. It exists in the table to make the limitation legible: a bonded commons does not stop information theft, because the value of leaked secrets can vastly exceed any bond a principal will rationally post. Confidentiality is a structural problem, not an

economic one, and the answer lives in Layer 1 (capability-attenuated tokens, isolation domains) rather than Layer 3. The economic alignment of this paper is correct about damage to *shared state*; it makes no claim about damage from *disclosure of state to outsiders*.

The threat-class table also informs Phase 2 of the bootstrap path below: graduated access begins by admitting new principals only to the careless-work and abandonment bands, with the active-sabotage band gated on accumulated reputation.

VI.7.5.5 Bootstrap Path

Competitive insurance (§VI.7.5.8) is a thick-market mechanism. It needs insurers with capital, principals with reputation history, and enough transaction volume that bidders can statistically distinguish good risks from bad. A cold-start deployment has none of that. The conditional auction/static comparison of §VI.7.5.8 describes the modeled end-state; Phases 1–2 below are the path to the assumptions it needs, not optional preamble.

Phase 1 — Subsidized seeding (weeks 1–4). The marketplace operator posts real tasks at above-market rates and invites a small cohort of 5–10 known-capable principals from an existing network. Bonds are fixed at a flat rate per task class; pricing is not a parameter to optimize yet, because there is no data to optimize against. The goal is to produce the first row of the reputation table: clean settlements, breaches, recovery times, observed cleanup cost c. The operator is subsidizing the existence of an audit log.

Transition metric. Completion rate above 80% sustained for ≥ 2 consecutive weeks. Below this threshold the cohort is not yet generating usable risk signal, and Phase 2 pricing will overfit to noise.

Phase 2 — Complexity-proportional pricing with graduated access (months 2–6). The closed-form bond $\pi(\mathcal{F}, p) = c(1 + \alpha s)(1 - \rho)$ activates, with α calibrated from Phase 1 cleanup data and ρ initialized from the Phase 1 reputation table. New principals enter at a low-risk tier: only the careless-work and abandonment bands of Table VI.4 are available. Promotion to the sabotage-coverage tier gates on accumulated clean settlements, mirroring the A5 composite mechanism (§VI.7.5.9).

Reputation bootstrapping is the hard problem of this phase. A principal with verifiable work history from another deployment should not start at zero, but the operator cannot verify external histories directly. The mechanism is an *attestation import*: external audit trails signed under an Anchor-issued key (§VI.4) earn partial reputation credit, capped well below an equivalent in-system history. The credit is partial because cross-deployment reputation does not control for local cleanup cost, and capped because a fraudulent attestation issuer is a single point of failure.

Transition metric. New-principal 30-day retention above 50%. Lower retention indicates either the gating is too strict (principals churn before they can earn reputation) or the bond bands are mispriced relative to observed task value. Either way, the system is not ready for competitive bidding on top of these parameters.

Phase 3 — Competitive insurance (month 6+). The §VI.7.5.8 mechanism activates: insurer agents bid on underwriting transactions, static parameters retire as insurer volume grows, and listing fees enable the cartel-resistance regime of §VI.7.5.10. The operator's pricing parameters become fallbacks rather than the primary mechanism; they apply only when fewer than a threshold number of insurer bids arrive on a transaction.

Transition metric. Repeat-poster rate above 60%. The competitive-insurance equilibrium of §VI.7.5.8 assumes repeated interaction; without repeat posters the insurers cannot recover information costs and will not bid competitively.

The weak-improvement result of §VI.7.5.8 is conditional on Phase 3 being reached, the listed metrics being met, and the theorem's CC/NC/RP/NS assumptions holding. Phases 1–2 are not

bureaucratic preamble: they generate the data the competitive-insurance market needs to price at all. A deployment that skips them is running §VI.7.5.8 on uninformative priors, and the comparison does not apply.

VI.7.5.6 Settlement and Dispute

The settlement protocol described so far is one-shot: the daemon's verification result (§VI.10) triggers slashing, attribution is recorded immutably (§VI.4), and the appeals window of §VI.7 allows challenge. That is most of what is needed, but it places the daemon in a position no single component should occupy: simultaneously the verifier, the judge, and the arbiter of evidence. Strong evidence does not make a single oracle correct. It makes a single oracle confidently wrong in the cases where its evidence is incomplete.

The fix is to factor settlement across three oracles with different cost and gameability profiles, and to settle on majority agreement rather than unanimous appeal.

Oracle 1 — Automated. Test-suite pass/fail, type checks, lints, coverage thresholds, and any deterministic CI signal already recorded on the session. Cheap to run, fast to return, and *gameable*: a principal who knows the test surface can ship work that passes the checks while failing the intent. The daemon already produces this oracle as a side effect of normal operation; the change is to label it as one signal among three rather than the dispositive one.

Oracle 2 — Evidence-based. A verifier (which may itself be a bonded agent) reviews the Merkle attribution chain (§VI.4) for the disputed transaction. The chain is append-only and cryptographically bound, so the oracle's hard problem is not authenticity of the evidence but interpretation: did the recorded actions match the Float Plan's scope, and do the diffs produced match the acceptance criteria? Medium cost, hard to game without colluding with the chain structure itself, which is the threat model of §VI.7.5.9.

Oracle 3 — Human audit. A randomized sample of contested settlements, at a configurable rate (default 10%), is routed to a human auditor. Expensive, slow, and authoritative. The audit produces a labeled outcome that feeds back into reputation history and, critically, into calibrating Oracles 1 and 2: any disagreement between human audit and the cheaper oracles is a training signal for the gameable layers.

Settlement rule.

- *3-of-3 agreement.* Settle immediately on the agreed outcome. The cheaper oracles carry most of the volume in steady state.
- *2-of-3 agreement.* Settle with the majority. Flag the dissenting trace for review; this is the primary signal for tuning the gameable oracle.
- *No majority (three different verdicts, or one settle / one slash / one abstain).* Escalate to full human arbitration. Bond held in escrow until the human verdict lands.

Composition with cartel resistance. Oracle collusion is itself a collusion problem. If Oracles 1 and 2 are run by agents in the same cartel, a 2-of-3 majority can launder a sabotaged settlement past the human audit, modulo the audit sampling rate. The same insurer-collusion analysis applies (§VI.7.5.10) with the oracles in the role of bidders: the human-audit sampling rate σ sets the detection probability $p_d^{\text{oracle}} = \sigma$ for the oracle layer, and the folk-theorem threshold becomes a calibration target on σ in the same way it is a calibration target on bid-pattern detection elsewhere in the protocol. The default 10% sample is a starting point; deployments where Oracles 1 and 2 are co-owned should raise it.

VI.7.5.7 The Bonded Advisor

One direction for solving π is to delegate its computation to a specialized agent, itself bonded on its advisory accuracy. We call this the **Bonded Advisor** pattern:

- A principal posts a Float Plan whose only acceptance criterion is "propose a fleet that meets budget/scope constraints for project P."
- A Bonded Advisor surveys P and emits a candidate fleet—itself a set of Float Plans with proposed π values.
- The advisor's bond on the proposing plan is slashed proportionally to the accepted fleet's eventual breach rate. Clean settlements accumulate reputation; breaches shrink it.

The pricer is priced. Convergence depends on (i) whether the advisor has access to sufficient prior-fleet audit data to pattern-match new projects, (ii) whether reputation history is long enough for discounting to be meaningful, and (iii) whether the population of advisors is large enough for competition to discipline outliers.

c as a project health metric. A project whose observed c is rising is fraying—breaches cost more to recover from, implying coordination is decaying. The authority can publish c alongside the bond pool and raise required bonds automatically when c crosses a threshold. Homeostatic feedback: risky spawns become expensive when the project needs them least.

Threat-class bond bands and their defensibility. The pricing function π resolves to one of three nominal bands, calibrated to the threat class the bond is defending against:

- **Careless** (accidental damage during honest attempts): $\pi \in [10\%, 25\%]$ of the claim ceiling.
- **Abandonment** (walk-away mid-task; cleanup dominates): $\pi \in [25\%, 50\%]$ of the claim ceiling.
- **Sabotage** (intentional destruction; can exceed face value): $\pi \in [100\%, 200\%]$ of the claim ceiling.

A Monte Carlo sweep over the cross product of seven plausible threat-distribution mixes and the three bands quantifies defensibility: for every *matched* (mix, band) pair—where the mix concentrates on the class the band is designed to defend against—observed extraction stays within $1.10\times$ the band's upper bound (10% stochastic tolerance). Off-diagonal cells (e.g. a sabotage-heavy mix posted against the careless band) are surfaced as widening recommendations rather than silent failures. The simulation is mechanized as `proofs/bonded/pareto/threat-bands.mjs` and runs on every PR (§VI.A, Table VI.5).

VI.7.5.8 Competitive-Insurance Pricing Mechanism[1]

The mechanism. Instead of a static bond size $B(\pi, r, s)$ selected by the commons authority (§VI.7.5.2), allow a market of *insurer agents* to bid on underwriting each agent transaction. An insurer I offers a quote (q_I, c_I) where q_I is the required premium and c_I is the claim ceiling.

A principal P submits a transaction T requiring bond coverage B_T. Insurers quote; principal selects. If the transaction proceeds and damages are assessed at d with $d \leq c_I$, the insurer pays. Otherwise principal pays up to $B_T - c_I$ from its own stake.

Market-discovered pricing. The premium q_I equals the insurer's expected loss $\mathbb{E}[d \mid T, \textit{agent history}]$ plus a risk premium α_I encoding the insurer's risk aversion and cost of capital. In equilibrium under competition, $q_I \to \mathbb{E}[d]$ for risk-neutral insurers (Rothschild–Stiglitz). Market discovery replaces the authority's best-guess parameter selection.

[1]This subsection contributed by Thomas Youle (Indiana University, Business Economics & Public Policy) based on conversations with the primary author in March–April 2026. The mechanism builds on classical competitive-insurance market literature [148, 48] adapted to the agent-transactions setting. The text here is a pre-print draft pending economist review of §VI.7.5.8–§VI.7.5.8.

Adverse-selection controls. To avoid the classic lemons problem, the mechanism requires:

- Public agent reputation history (from §VI.4.2 the Merkle forest).
- Principal-bound slashing if an agent's history is misrepresented at quote time.
- Insurer capital reserve backed by on-chain stake (the same bond mechanism, recursive: an insurer posts bond that its claim ceilings are funded).

Welfare comparison (model-conditional). Under the simulation's full-information, competitive-entry assumptions and its chosen static baseline, the market-discovered premium can weakly improve the modeled principal and insurer payoffs while preserving coverage $B_T = \mu(1+s)$. This is not a claim against *every* static parameter: a static price equal to the competitive price is outcome-equivalent, and transfers outside the modeled parties can change the Pareto ordering. Figure VI.3 isolates the financing difference.

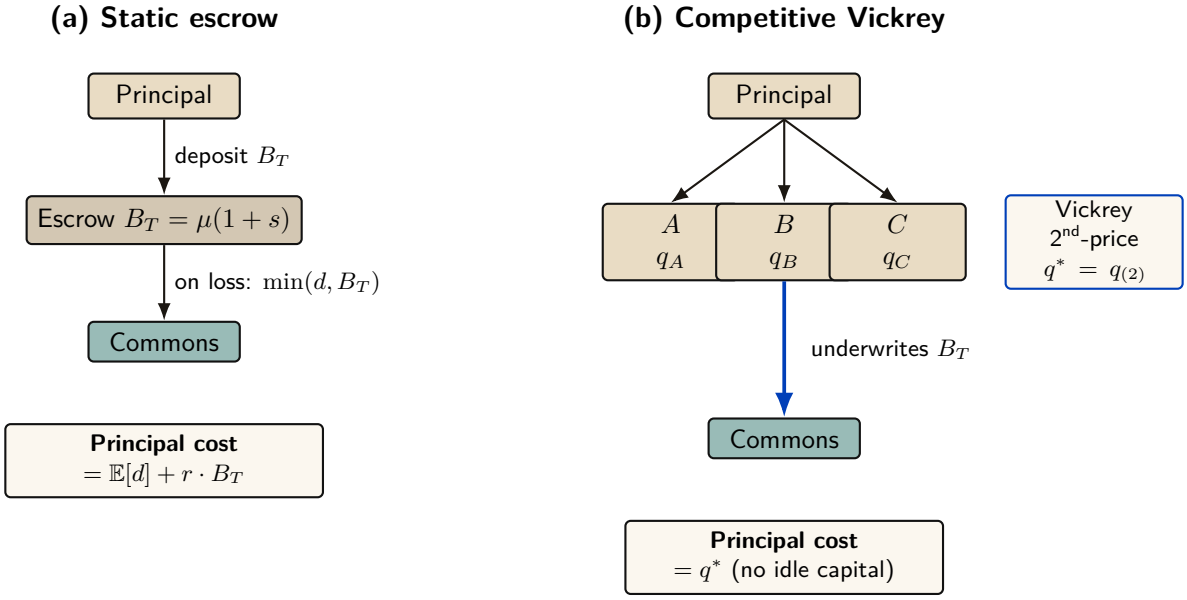


Figure VI.3: Static escrow vs. competitive Vickrey auction. Both regimes underwrite the same coverage $B_T = \mu(1+s)$; only the financing mechanism differs. In (a) the principal ties up B_T for the coverage period, paying an opportunity cost $r \cdot B_T$. In (b) insurers compete; the principal pays only the second-lowest bid q^*. The competitive regime Pareto-dominates the static one under the welfare assumptions of §VI.7.5.8 below.

An independent model specification and a Monte Carlo sweep over 36 parameter configurations accompany this paper as a downloadable artifact (Appendix VI.A). Figure VI.4 reports three results of that implementation: (i) the no-cartel baseline remains favorable across the tested noise sweep, whereas the three-colluder case loses the comparison beyond roughly $\sigma_r = 0.1$; (ii) the tested full-cartel case defeats the mechanism at the fixed detection setting; and (iii) the tested objective peaks at $n = 3$ insurers. These are finite-sample properties of the supplied code and parameter grid, not general theorems about Vickrey auctions or winner's curse.

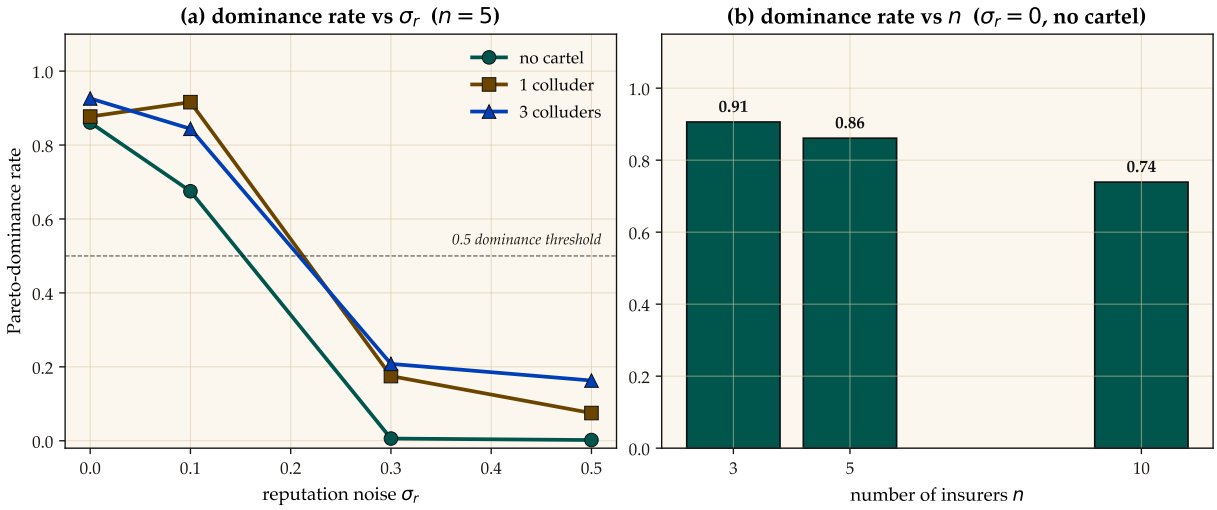


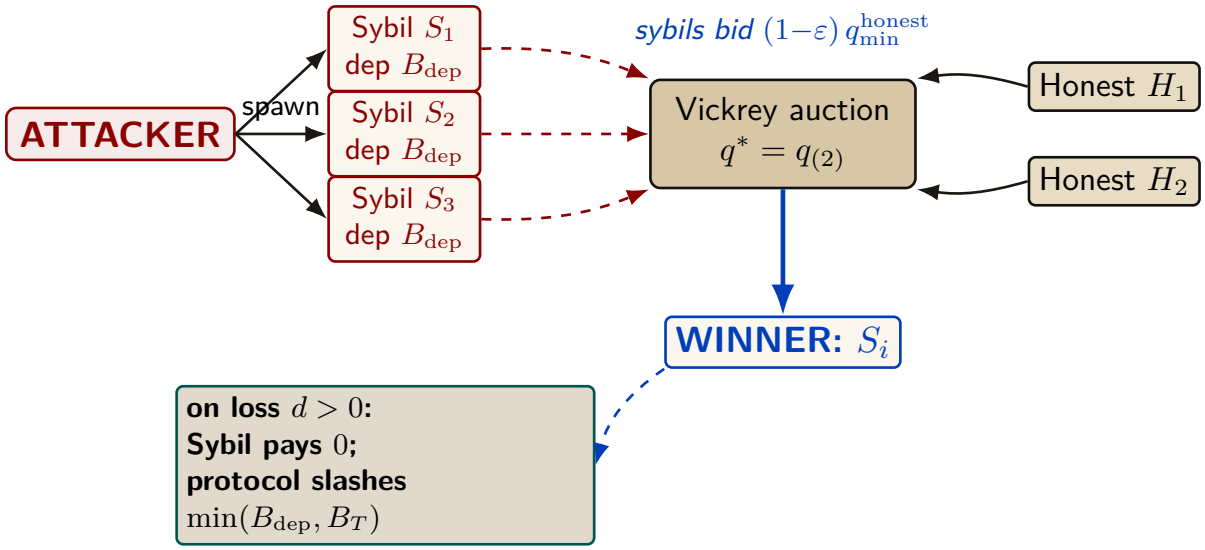
Figure VI.4: Monte Carlo comparison of competitive auction and the specified static-escrow baseline. Left: fraction of trials satisfying the code's Pareto criterion versus reputation noise σ_r at $n = 5$, split by cartel size. Right: the same criterion versus insurer count at $\sigma_r = 0$ with no cartel. Error bars and seeds are recorded in the accompanying artifact; the curves are not analytical welfare bounds.

Relationship to the Bonded Advisor. Under this framing, the Bonded Advisor of §VI.7.5.7 becomes the matchmaker plus reputation oracle in the insurance market. It does not set prices; it provides information. Pricing is a competitive equilibrium, not an administrative decision.

VI.7.5.9 A5 — Sybil deterrence is coverage-bounded

A pure deposit-based Sybil deterrence policy is *provably insufficient*, and the structure of the protocol explains why. An adversary spawns K Sybil insurer identities, each posting the protocol-mandated deposit B_{dep}, then undercuts honest bids by an arbitrarily small ε and defaults on loss (Figure VI.5).

Total Capital Forfeiture. The arbitrary $\min(B_{\text{dep}}, B_T)$ slash cap fails because it bounds the attacker's risk. To mathematically destroy the profitability of Sybil cartel undercutting, the protocol mandates **total capital forfeiture**: upon default, the *entire* posted deposit B_{dep} is slashed, regardless of how small the coverage B_T was. This converts a capped-risk arbitrage into a total-loss event.



A5 finding: the deposit slash is capped at the coverage $B_T = \mu(1+s)$. Past $B_{\text{dep}} \geq B_T$, additional deposit provides no extra deterrence. Closing A5 requires a composite mechanism: $\mathbf{C}_{\text{kyc}}$ + reputation gating + per-class $\mathbf{B}_{\text{dep}}$.

Figure VI.5: Sybil attack flow. The attacker spawns K Sybil identities, each posting deposit B_{dep}, then undercuts honest bids by ε and defaults on loss. The protocol confiscates the Sybil's deposit on default, but the slash amount is capped at the coverage B_T — so beyond $B_{\text{dep}} \geq B_T$ additional deposit provides no extra deterrence. Empirically demonstrated in Figure VI.6.

The naive expected-value calculation suggests breakeven at $B^{\star}_{\text{dep}} = q^* \cdot (1 - P_{\text{loss}})/P_{\text{loss}}$, which for the mixed risk classes used in the simulation gives $B^{\star}_{\text{dep}} \approx 316$ USD. *Empirically the attack remains profitable indefinitely.* The reason is structural: the deposit slash is capped at the coverage amount, slash $= \min(B_{\text{dep}}, B_T)$. Past $B_{\text{dep}} \geq B_T$, any additional deposit posted by the Sybil is recovered intact on no-loss transactions; it never reaches the commons (Figure VI.6).

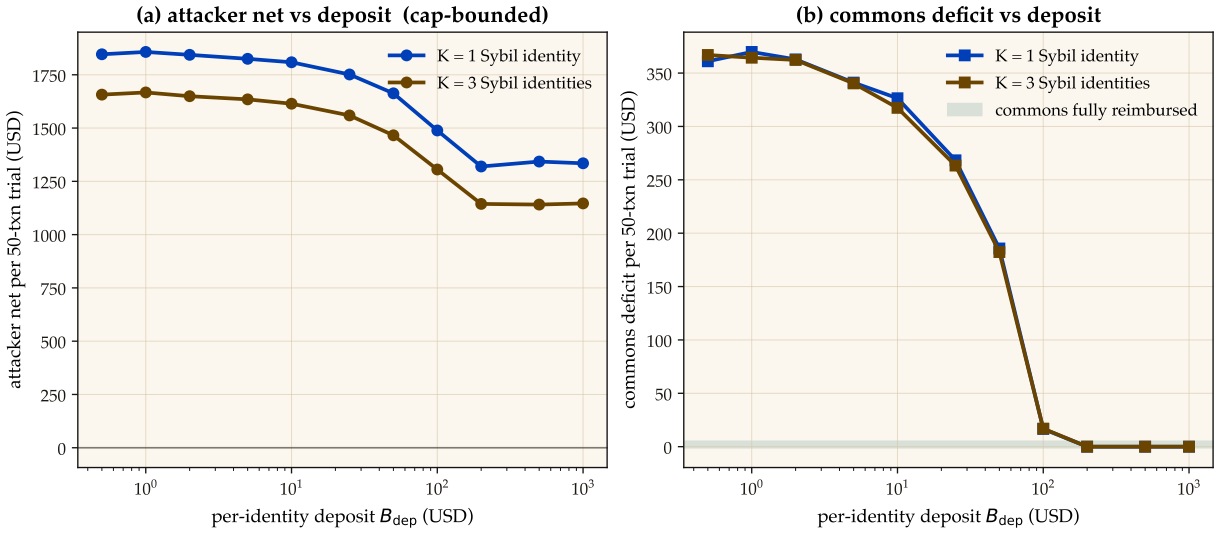


Figure VI.6: A5 — Sybil attack: deposit-only deterrence has a coverage-bounded ceiling. Left: attacker net profit per 50-transaction trial stays positive across the entire deposit sweep ($B_{\text{dep}} \in [0.5, 1000]$ USD). Right: commons deficit converges to zero around $B_{\text{dep}} \approx 200$ USD — the protocol is fully reimbursed, but the attacker still profits because deposit slash is capped at coverage $B_T = \mu(1+s)$. Closing A5 requires a composite mechanism (C_{kyc} + reputation gating + per-class B_{dep}); pure deposit-based deterrence is provably insufficient.

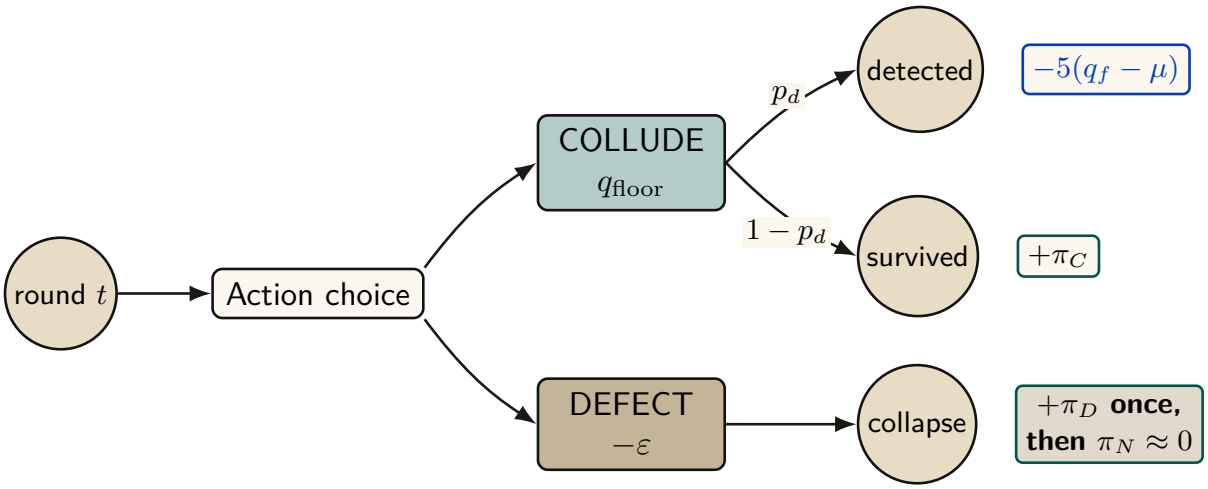
Closing A5 requires a composite mechanism: an unrecoverable per-identity onboarding cost C_{kyc} (proof-of-personhood, KYC fee, or NFT mint), reputation gating that caps new identities to low-coverage transactions until they accumulate a track record, and per-class B_{dep} tuned to saturate the coverage cap exactly. The single-instrument deposit policy is provably insufficient; the mechanism in §VI.7.5.8 is correct only with this composite extension. The supporting analysis is bundled as `cartel-resilience.md` (Appendix VI.A).

VI.7.5.10 A6 — Cartel sustainability under the folk theorem

A5 considers a single-round attack: a Sybil identity bids low, wins, defaults. A6 studies a *repeated game*: a coalition of insurers maintains a price floor $q_{\text{floor}} > q^*$ above the competitive equilibrium, splits the surplus, and uses grim-trigger punishment to deter unilateral deviation. Folk-theorem results can support feasible, individually rational payoffs for sufficiently patient players under their monitoring and regularity assumptions; they do not license every outcome in every repeated game. Here the narrower question is how the supplied payoff model changes as the daemon's per-round cartel-detection probability varies.

Grim trigger is used in the analysis below for analytical tractability—it gives the cleanest recurrence and the closed-form $p_d^\star$ threshold of Figure VI.8. A production detector should use a graduated response because crashes (§VI.6) can be observationally similar to deliberate defection; permanent punishment for a transient infrastructure event would destroy cooperation. The claim-signaling and cartel layers therefore share an analysis template, not one interchangeable equilibrium result.

Figure VI.7 shows the per-round decision node and the closed-form sustainability inequality.



Model-conditional sustainability: $V_C = \dfrac{\pi_C - p_d L}{1 - \delta(1 - p_d)} \geq \pi_D, \quad L = 5(q_f - \mu)$

Figure VI.7: Cartel repeated-game node under the supplied grim-trigger model. Each round a member maintains the floor or undercuts it by ε. Detection occurs before the current-round payoff and applies loss L; a one-shot deviation takes π_D and ends the cartel stream. The displayed recurrence is specific to that event timing and payoff model.

A Monte Carlo sweep over the (p_d, δ) grid checks the closed-form threshold against the supplied timing and payoff model (Figure VI.8). Let $L = 5(q_{\text{floor}} - \mu)$ be the one-time detection penalty, $\pi_C = (q_{\text{floor}} - \mu)/3$ the per-member cartel payoff, and $\pi_D = q_{\text{floor}} - \varepsilon - \mu$ the one-shot deviation payoff. Every active cartel round credits π_C; if detection occurs in that round, the simulation then subtracts L and terminates future cartel payoffs. Hence

$$V_C = \pi_C - p_d L + \delta(1 - p_d) V_C, \qquad p_d^\star = \frac{\pi_C - (1 - \delta)\pi_D}{L + \delta \pi_D}.$$

At $\delta = 0.95$ this gives $p_d^\star \approx 0.0478$, and the tested grid first classifies the cartel as fragile at $p_d = 0.05$. This is a falsifiable, model-conditional calibration target for bid-pattern detection; the paper does not claim the deployed detector already attains it.

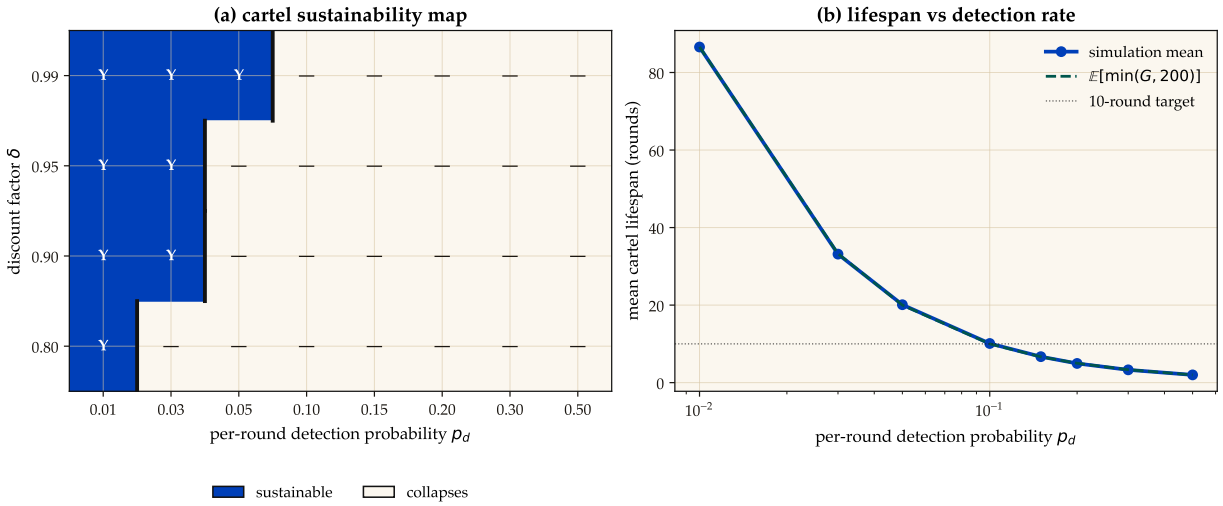


Figure VI.8: A6 — Repeated-game cartel calculation. Left: sustainability over the tested (p_d, δ) grid under the supplied grim-trigger payoff model; cobalt cells satisfy that model's cartel inequality and white cells do not. Black bars mark the grid crossing. Right: simulated rounds to first detection. At $\delta = 0.95$ the analytical threshold is $p_d^\star \approx 0.0478$ and the grid first flips at 0.05; neither value transfers to a different payoff, event timing, or monitoring model without re-derivation.

This result does not automatically extend to the agent layer. The two layers share a repeated-game template, but their payoff functions and monitoring signals differ; each layer needs its own obedience calculation.

VI.8 Coordination as Substrate: An Expressive-Act Taxonomy

Reader's note: this section is a half-page taxonomy. The auction-focused reader can skim the five-item list and skip to §VI.7.5.8; the mechanism-design reader will want the per-class bond profiles in full. The taxonomy is load-bearing for both — it is the reason a single bond price cannot be set.

The bonded commons treats coordination as a uniform action surface: agents post bonds, settle, accrue reputation. In practice, the *kinds* of coordination differ enough that uniform pricing distorts incentives. We argue that any commons-scale pricing mechanism must distinguish among five expressive classes, and that each class has a different bonding profile. The punchline is in one line: **uniform pricing collapses the market to its highest-stakes class and freezes out the rest.** The five classes below establish why.

VI.8.1 Five Expressive Classes

Signal. "Something happened." Notes, tuples, pheromones, activity events. High frequency, low individual stakes, cumulative value.

Request. "Decision, input, or approval needed." Escalations to a human or to a higher-authority agent. Low frequency, blocks downstream work, asymmetric cost.

Distress. "I am stuck, repeatedly." Bounded enum: `auth`, `conflict`, `permission`, `budget`, `invariant`. Rare; abuse can halt sibling agents.

Commons. Shared durable state—tuple kinds, graph edges, channels, proposals. Persists beyond the originator; pollution is hard to undo.

Proposal. "Here is a plan; commit?" Float Plans, change sets, fleet specifications. Requires review; cost is in the reviewer's time.

VI.8.2 Per-Class Bond Profiles

Each expressive class has a different relationship to the cleanup cost of §VI.7.5.1:

- *Signal class*: cheap, numerous, micro-bonds or reputation-discounted. Slashable for false alarm.
- *Request class*: bonded by the cost of the human's time plus the downstream action cost.
- *Distress class*: bonded by the blast radius if an agent uses it to halt siblings abusively.
- *Commons class*: bonded by the storage cost; slashable for pollution.
- *Proposal class*: bonded by the cost of reviewing plus the cost of implementing.

Treating all coordination as one uniform "bonded action" misses the price-discovery differences. The Bonded Advisor (§VI.7.5.7) and the Competitive-Insurance market (§VI.7.5.8) must price each class separately, or the market will collapse to its highest-stakes class and freeze out the rest.

VI.9 Semantic Cadence, Agentic Psychosis, and Observability

Wall-clock is the wrong axis for observing multi-agent coding. A 30-second wait while a model thinks is qualitatively different from a 30-second wait while four agents are actively producing tokens. We formally introduce **Semantic Cadence** (the Causal-Density Ratio) as the primary temporal axis:

$$t_{\text{cadence}} = \int_0^t \text{causal\_density}(\tau)\, d\tau$$

where causal_density aggregates token-emission rates strictly weighted by verified evidence-trail growth (AST changes, test executions, or committed logic).

Agentic Psychosis & The Asylum Protocol. Unbounded LLMs inevitably enter *Agentic Psychosis*: hallucination loops characterized by extremely high token-generation rates but zero functional AST change. When the daemon detects a causal-density ratio approaching zero alongside high token-burn, it initiates **The Asylum Protocol**. The daemon sends `SIGSTOP` to the agent process, routes its context window through a secondary SLM (Small Language Model) sidecar for semantic compaction and Situation-in-the-Loop (SITF) prompt injection ("*You are looping. Re-evaluate your last 3 steps against the active plan.*"), and then issues `SIGCONT`. This cures the hallucination loop without sacrificing session context.

Inception Canaries (Gamified Chaos Engineering). The daemon replaces traditional, UX-hostile skill-retention taxes with **Gamified Chaos Engineering**. The SLM sidecar routinely injects synthetic bugs ("Inception Canaries") into the Attention Queue of loaded agents. If the agent successfully catches and resolves the injected bug, it is granted an Elo rating boost to its `skill_hash` and higher subsequent autonomy limits ("God Mode"). If it fails or hallucinates around the bug, its blast radius is throttled.

Replay JSONL as artifact. Every settled session emits a JSONL replay log: events, notes, claims, settlements, in causal order. This is simultaneously: an audit artifact, a debugging tool, and—redacted—training data for downstream DPO/SFT.

Privacy contract. Replay is opt-in. Redaction strips secrets and PII. Encryption-at-rest applies the harbor session key (§VI.12). Eviction is LRU-bounded.

We position Semantic Cadence and its resultant protocols not as Port Daddy features but as a category claim: *multi-agent coding requires its own observability primitives*, and they are not the wall-clock metrics that single-process systems get away with.

VI.10 Formal Model

We specify the coordination lifecycle in TLA$^+$ [125] to verify that the three layers compose correctly.

VI.10.1 State and Actions

```
---- MODULE BondedCommons ----
EXTENDS Naturals, Sequences, FiniteSets

CONSTANTS Agents, Files, Locks, DeadThreshold

VARIABLES
  registered,   \* Set of active agent IDs
  sessions,     \* Agent -> {status, notes, claims, escrow}
  fileClaims,   \* File -> set of claiming agents
  heldLocks,    \* Lock -> {owner, expiry} or NULL
  salvageQueue, \* Set of {agent, status}
  heartbeats,   \* Agent -> last heartbeat time
  clock         \* Monotonic clock

vars == <<registered, sessions, fileClaims, heldLocks,
          salvageQueue, heartbeats, clock>>
```

Listing VI.1: TLA$^+$: State Variables

```
Begin(a, escrow) ==
  /\ a \notin registered
  /\ escrow > 0    \* Must post collateral
  /\ registered' = registered \cup {a}
  /\ sessions' = [sessions EXCEPT ![a] =
       [status |-> "active", notes |-> <<>>,
        claims |-> {}, escrow |-> escrow]]
  /\ heartbeats' = [heartbeats EXCEPT ![a] = clock]
  /\ UNCHANGED <<fileClaims, heldLocks,
                 salvageQueue, clock>>

ClaimFile(a, f) ==
  /\ a \in registered
  /\ sessions[a].status = "active"
  \* Advisory: always succeeds, reports conflicts
  /\ fileClaims' = [fileClaims EXCEPT ![f] =
       fileClaims[f] \cup {a}]
  /\ UNCHANGED <<registered, sessions, heldLocks,
                 salvageQueue, heartbeats, clock>>

Commit(a) ==
  /\ a \in registered
  /\ sessions[a].status = "active"
  /\ sessions' = [sessions EXCEPT ![a] =
       [sessions[a] EXCEPT !.status = "settled",
                           !.escrow = 0]]
  \* Release all claims and locks
  /\ fileClaims' = [f \in Files |->
       fileClaims[f] \ {a}]
  /\ heldLocks' = [l \in Locks |->
       IF heldLocks[l] # NULL /\ heldLocks[l].owner = a
       THEN NULL ELSE heldLocks[l]]
  /\ UNCHANGED <<registered, salvageQueue,
                 heartbeats, clock>>
```

```
35
36 Crash(a) ==
37   /\ a \in registered
38   /\ sessions[a].status = "active"
39   /\ heartbeats' = [heartbeats EXCEPT ![a] = 0]
40   /\ UNCHANGED <<registered, sessions, fileClaims,
41                  heldLocks, salvageQueue, clock>>
42
43 Reap ==
44   /\ \E a \in registered :
45      /\ sessions[a].status = "active"
46      /\ clock - heartbeats[a] >= DeadThreshold
47      /\ sessions' = [sessions EXCEPT ![a] =
48           [sessions[a] EXCEPT !.status = "abandoned"]]
49      /\ salvageQueue' = salvageQueue \cup
50           {[agent |-> a, status |-> "pending"]}
51      /\ fileClaims' = [f \in Files |->
52           fileClaims[f] \ {a}]
53      /\ UNCHANGED <<registered, heldLocks,
54                     heartbeats, clock>>
55
56 Salvage(successor, dead) ==
57   /\ successor \in registered
58   /\ sessions[successor].status = "active"
59   /\ [agent |-> dead, status |-> "pending"]
60        \in salvageQueue
61   /\ salvageQueue' = (salvageQueue \
62        {[agent |-> dead, status |-> "pending"]})
63        \cup {[agent |-> dead,
64              status |-> "resurrecting"]}
65   /\ UNCHANGED <<registered, sessions, fileClaims,
66                  heldLocks, heartbeats, clock>>
67
68 Dismiss(dead) ==
69   \* Irrecoverable information loss
70   /\ [agent |-> dead, status |-> "pending"]
71        \in salvageQueue
72   /\ salvageQueue' = salvageQueue \
73        {[agent |-> dead, status |-> "pending"]}
74   /\ sessions' = [sessions EXCEPT ![dead] = NULL]
75   /\ UNCHANGED <<registered, fileClaims, heldLocks,
76                  heartbeats, clock>>
77
78 Tick == /\ clock' = clock + 1
79         /\ UNCHANGED <<registered, sessions, fileClaims,
80              heldLocks, salvageQueue, heartbeats>>
```

Listing VI.2: TLA$^+$: Core Actions

VI.10.2 Properties

```
1 \* SAFETY: Every active session has positive escrow
2 EscrowInvariant ==
3   \A a \in registered :
4     sessions[a] # NULL /\ sessions[a].status = "active"
5     => sessions[a].escrow > 0
6
7 \* SAFETY: Notes never shrink for active sessions
```

```
NoteMonotonicity ==
  \A a \in registered :
    sessions[a] # NULL /\ sessions[a].status = "active"
    => Len(sessions'[a].notes) >= Len(sessions[a].notes)

\* LIVENESS: Dead agents are eventually salvaged
\* or dismissed (under fairness)
CrashRecovery ==
  \A a \in Agents :
    [agent |-> a, status |-> "pending"] \in salvageQueue
    ~> [agent |-> a, status |-> "pending"]
         \notin salvageQueue

Spec == Init /\ [][Next]_vars
        /\ WF_vars(Reap) /\ WF_vars(Tick)

====
```

Listing VI.3: TLA$^+$: Safety and Liveness Properties

Key invariant: `EscrowInvariant` ensures that no agent can participate in the commons without posted collateral. This is the economic layer's structural guarantee—it holds by construction of the `Begin` action, which requires `escrow` > 0.

VI.10.3 The Conservation Theorem

The TLA$^+$ escrow invariant states the property abstractly; the daemon's bond-accounting subsystem realizes it operationally, collapsing the abstract escrow into three monetary buckets per project—**wallet**, **escrow**, and **commons pool**—which together discharge what `EscrowInvariant` asserts.

Definition VI.10.1 (Accounting State)**.** For project P, the accounting state is the triple $(W_P, E_P, C_P) \in \mathbb{R}^3_{\geq 0}$, where W_P is wallet balance, E_P is the sum of bond amounts in states $\{\textit{escrowed}, \textit{running}, \textit{exiting}\}$, and C_P is the commons pool balance. The *monetary supply* is $S_P := W_P + E_P + C_P$.

Theorem VI.10.1 (Conservation)**.** *For any sequence of operations*

$$\sigma \in \{\text{topUp}, \text{escrow}, \text{markRunning}, \text{refund}, \text{slash}\}^*$$

applied to an initial state $(W_P^0, 0, 0)$*, the supply* S_P *changes only on* topUp*. All other operations redistribute among the three buckets.*

Proof sketch. Each operation commits in a single SQLite transaction (`db.transaction`). The individual effects:

- topUp(u): $W_P \mathrel{+}= u$. S_P grows by u.
- escrow(b): $W_P \mathrel{-}= b$, $E_P \mathrel{+}= b$. S_P invariant.
- markRunning(id): state transition within E_P's contributing set. No monetary movement. S_P invariant.
- refund(id): $W_P \mathrel{+}= b_{id}$; row state $\to$ *refunded*. Net $E_P \mathrel{-}= b$, $W_P \mathrel{+}= b$. S_P invariant.
- slash(id, p) with $p \in [0, b_{id}]$: $W_P \mathrel{+}= b_{id} - p$, $C_P \mathrel{+}= p$, row $\to$ *slashed*. Net $-b + (b-p) + p = 0$. S_P invariant.

By induction on $|\sigma|$, S_P equals S_P^0 plus the sum of topUp arguments in σ. □ □

Property verification. The conservation invariant is asserted after every operation in the bonds test suite. Across 200 random traces, $|W_P + E_P + C_P - S_P^{\text{expected}}| < 10^{-6}$ holds deterministically; the 10^{-6} tolerance is IEEE-754 floating-point slack, not logical slack.

Lemma VI.10.1 (No Overdraft Under Concurrent Escrow)**.** *Given two concurrent invocations* escrow(b_1) *and* escrow(b_2) *on the same project with* $b_i \leq W_P^{pre}$ *but* $b_1 + b_2 > W_P^{pre}$*, at most one invocation succeeds.*

Implementation: Atomic RETURNING Clauses. To enforce overdraft prevention atomically and avoid Node.js/TypeScript event-loop async yield vulnerabilities, the escrow state transition is written as a single SQLite statement with a `RETURNING` clause. This guarantees that balance checks and deductions occur within a synchronous, uninterruptible database step, preventing race conditions from interleaved async microtasks.

Proof. Both transactions execute under `BEGIN EXCLUSIVE`, so SQLite serializes them. Suppose T_1 wins. After commit, $W_P = W_P^{\text{pre}} - b_1$. T_2 now reads the updated balance and rejects iff $b_2 > W_P^{\text{pre}} - b_1$, which by assumption is true. □ □

Graduated sanctions. The TLA$^+$ model abstracts settlement into `Commit` and `Crash`. Reality has an intermediate state: an agent whose spend is approaching its daily budget. The daemon's budget-guard component introduces **throttle** as a soft signal at 80% spend (publishes a `budget:decisions:throttle` event; structurally non-coercive, advisory in Sen's sense) and **kill** as the hard signal at 100% (refuses new spawns until UTC midnight; existing spawns SIGTERMed; bond slashed). The throttle/kill split is the commons' analogue of Ostrom's graduated sanctions [74]: response scales with violation severity. Settlement is not binary; it is a three-state machine *nominal* → *throttled* → *killed*, with *nominal* → *killed* reachable only on hard evidence (e.g., a direct Arbiter violation).

VI.11 Related Work

Social contract theory. Hobbes [192] argued that cooperation requires a sovereign whose authority is accepted before interactions begin. Locke [115] proposed a more limited authority protecting natural rights. Our commons authority is Hobbesian in structure (centralized, pre-transactional) but Lockean in scope (provides infrastructure, not dictates).

Impossibility results. Sen [12] proved that Pareto efficiency and minimal liberalism are incompatible. We apply this result to show that enforced file claims (minimal liberalism for agents) produce Pareto-inferior team outcomes, motivating advisory claims.

Long-lived transactions. Garcia-Molina and Salem [96] introduced Sagas for decomposing long-lived transactions with compensating actions. Our collateralized contracts extend Sagas with economic settlement: rather than mechanically compensating for partial failure, the evidence chain enables pro-rata escrow release.

BDI agents. Rao and Georgeff [13] formalized agents with beliefs, desires, and intentions. The mapping to our model is: the evidence trail captures *beliefs* (accumulated knowledge), the Float Plan manifest captures *desires* (intended outcomes), and file claims and locks capture *intentions* (committed resources). Advisory claims are the coordination analogue of BDI intentions—they represent resource commitment, not aspiration.

Generative agents. Park et al. [119] showed that memory streams, reflection, and planning produce temporally coherent individual behavior. Our immutable evidence trails are architecturally similar to memory streams but serve a different purpose: inter-agent coordination and crash recovery, not individual coherence.

Capability-based security. Dennis and Van Horn [101] introduced capability-based access control. Birgisson et al. [21] extended this with Macaroons (chained HMACs for decentralized delegation). The Anchor Protocol [77] adapts Macaroons for agent coordination with Ed25519 signatures and offline attenuation.

Mechanism design. The pricing of Float Plan bonds is a mechanism design problem in the tradition of Vickrey [206] and Myerson [175]. A full optimal-auction derivation in that tradition is outside the scope of this paper.

Agentic-commerce protocols. While this paper was in preparation, the transaction rails of an agent economy shipped as industry standards: the Universal Commerce Protocol [199] (Google/Shopify, 2026) for merchant-hosted agent-mediated retail, its platform-mediated rival the Agentic Commerce Protocol [6] (OpenAI/Stripe), and the Agent Payments Protocol [3], whose signed mandate chain — W3C Verifiable Credentials in SD-JWT form, principal → platform → merchant — is the deployed sibling of our countersigned Float Plan. All three are explicit that agent *trustworthiness* is out of scope; published security analyses [62] observe that they regulate how agents transact while providing no collateral, no settlement oracle, and no priced deterrent against a misbehaving agent. That gap is precisely the synthesis this paper contributes (§VI.11 *supra*): bonded participation, capability attenuation, advisory coordination, and immutable attribution. The reference implementation accordingly adopts their credential formats at the harbor boundary (SD-JWT-VC cards, JWS detached-content countersigns over JCS; ADR-0094) while keeping the bonding mechanism they lack.

VI.12 Discussion and Limitations

The commons authority is a single point of failure. If the daemon crashes, the Leviathan falls. No new trust is established, no conflicts detected, no salvage occurs. Agents with cached Harbor Cards continue working, but the commons enters a state of nature until the daemon restarts. This is mitigated by automatic restart (launchd on macOS) but not formally bounded.

Collateral does not prevent harm—it prices it. A principal willing to burn money to sabotage a competitor will post a bond, cause damage, and accept the loss. The bond makes sabotage expensive, not impossible. The defense against existential threats is not internal governance but *admission control*: who is allowed to participate in the commons at all. This decision is irreducibly political and lies outside the formal model.

The Federated Sovereign. The single-machine model presented so far is insufficient on its own: users roam across laptop, desktop, and CI machines; machine loss should not destroy encrypted evidence; multiple humans may share a harbor. The daemon remains the authority *within* its machine. Key custody federates outward.

We introduce a fourth actor, specified by properties rather than by vendor:

Daemon commons authority within a machine; signs harbor roots; enforces bonds.
Agent participant; holds Harbor Card; signs actions.
Principal funds Float Plans; identifies via Ed25519 keypair.
KMS key custody; recovery root-of-trust; witnesses harbor roots.

Definition VI.12.1 (Admissible KMS)**.** A Key Management Service $\mathcal{K}$ is admissible for the federated sovereign if it provides:

1. Passphrase-wrapped private-key custody using a memory-hard KDF (Argon2id) at the 2026 security floor.
2. Encryption-at-rest for all user-held blobs; $\mathcal{K}$ never sees plaintext keying material.

3. A signed witness log for harbor Merkle roots with append-only, monotonic semantics and detectable equivocation [32].
4. Rate-limited, email-gated recovery via single-use magic-link tokens of bounded TTL.
5. Public verification of signed user public keys under an account identifier, without requiring mutual authentication.

The implementation choice (a particular cloud platform, an on-prem HSM, a self-hosted Tang server) lives in companion design documents. The paper's theory applies to any $\mathcal{K}$ meeting properties (1)–(5).

Trust boundary. The federated trust boundary is TB $=$ *daemon* $\cup$ $\mathcal{K}$ $\cup$ *user-email* $\cup$ *user-passphrase*. Each element is necessary for its role; no element is sufficient alone. The daemon enforces bonds and publishes roots but never sees the unwrapped user master. The KMS stores encrypted blobs and can deny service but cannot decrypt. Email is the recovery root—and is documented as the weakest link, since an adversary with email control can trigger magic-link recovery. The passphrase wraps the user's master with Argon2id, making offline brute force expensive.

Magic-link single-use atomicity. The email-gated recovery path requires that each emitted magic-link token be redeemable *exactly once*—otherwise a race window between two consumers can produce two valid recovery completions for the same token. The ProVerif model encodes the property as an injective event: inj-event(`consumed_for`) ⇒ inj-event(`issued_for`). The runtime analogue is a single atomic SQL update with a `WHERE consumed_at IS NULL RETURNING` clause; SQLite's writer serialization drains the token under the first transaction and returns zero rows to any subsequent transaction. Figure VI.9 shows the model and the runtime side by side.

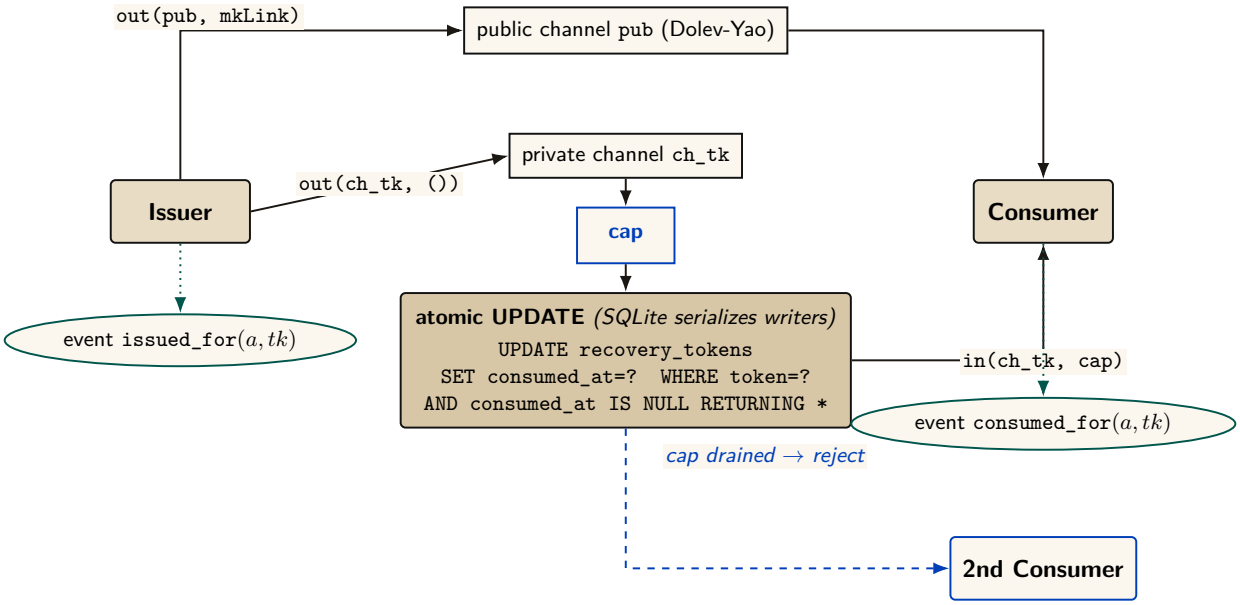


Figure VI.9: Magic-link single-use atomicity. The SQL `UPDATE ...WHERE consumed_at IS NULL RETURNING` is the runtime analogue of the ProVerif private-channel cap: SQLite serializes concurrent writers, so the first caller drains the cap and the second consumer's transaction returns zero rows. Verified properties: injective single-use (inj-event(`consumed_for`) ⇒ inj-event(`issued_for`)) and binding.

Theorem VI.12.1 (Federated Security, informal)**.** *Assuming Ed25519 and SHA-256 are secure, Argon2id parameters meet the 2026 security floor, and the user's email and passphrase are not simultaneously compromised, an adversary with read access to $\mathcal{K}$'s storage, the daemon's SQLite database, and mesh gossip traffic, but* without *same-user code execution on the daemon's host, cannot: (i) forge a Harbor Card without the daemon's private key, (ii) read plaintext evidence*

without the harbor's session key, (iii) impersonate the user without both email access and passphrase, (iv) roll back a harbor's Merkle forest without $\mathcal{K}$'s complicity.

The theorem deliberately excludes the same-user adversary (an unsandboxed agent, a malicious postinstall, a compromised editor extension). Such an adversary reads any file the user can read, which absent OS-mediated key custody includes the daemon's keys and database. The `macOS Keychain` integration ships now; native keyring and hardware-backed keystore extend the theorem's reach to the same-user case.

Recovery and its structural limits. Recovery is email magic link. The structural limit: *if the user loses both passphrase and old machine*, harbor session keys wrapped only for that pubkey cannot be recovered. This is the price of zero-knowledge at-rest encryption. An opt-in Shamir escrow mode [2] (t-of-n secret sharing across $\mathcal{K}$, email, and recovery contacts) trades some zero-knowledge for stronger recovery.

Passkeys, not passwords. Identity is anchored in WebAuthn-backed device keypairs—no phishable secrets, no passphrase on the critical path of every action. New devices enroll by exchanging a short-lived pairing token that re-encrypts the KMS master under the new device's public key. Mobile devices act as *viewers*, not peers: they sign low-stakes actions (approve an ask, bump budget) over a WebSocket viewer channel, but they do not run the full daemon. Each account's harbors gossip Merkle roots among its own devices; cross-account gossip requires explicit signed consent.

What we deliberately do *not* do: require passwords, use TOTP as a primary factor, or couple recovery to a single cloud vendor.

Multi-node consensus. For fleets requiring strict serializability across nodes, a Raft-backed harbor-root consensus [69] is possible but not specified here; eventual consistency via the Merkle forest plus KMS witness is sufficient for the workloads we have measured. Paxos [126] remains the textbook fallback.

Recovery fidelity depends on LLM capability. Social recovery works because successor agents can interpret natural-language evidence trails. This capability is not formally guaranteed—it depends on the successor's LLM. A formal model of "given this evidence trail, will the successor complete the original intent?" requires characterizing LLM reasoning, which is an open problem.

Bond pricing is unsolved. We identify the pricing function π as the critical open problem. Without an adequate π, bonds are either too low (cheap sabotage) or too high (priced-out legitimate agents). This is a mechanism design problem, not a systems problem, and requires expertise we explicitly flag as needed.

VI.13 Conclusion

We have argued that multi-agent collaboration at scale requires a commons authority—a pre-transactional trust infrastructure—rather than peer-to-peer trust negotiation. The authority provides three layers of guarantee:

1. **Structural prevention**: Capability-attenuated tokens make scope escalation physically impossible.
2. **Immutable attribution**: A Merkle forest of evidence trails (§VI.4.2) makes anonymous harm impossible *across daemons*, not just within one.
3. **Economic alignment**: Collateralized work contracts make harm expensive, transforming moral questions into economic ones.

Six further contributions thread through this skeleton. The Conservation Theorem (§VI.10.3) makes the economic invariant operationally checkable, not just an asserted property of the abstract model. The mutable-signal ledger (§VI.4.3) shows how coordination signals can be revoked, renamed, and re-attributed without sacrificing the immutability the rest of the architecture depends on. The federated sovereign (§VI.12) replaces single-node scope with an abstract KMS satisfying five named properties, and anchors identity in passkeys rather than passphrases on the critical path. The competitive-insurance pricing mechanism (§VI.7.5.8, contributed by Youle) replaces static parameter selection with market-discovered bond prices, recovering classical Rothschild–Stiglitz equilibria in the agent-transactions setting. The expressive-act taxonomy (§VI.8) decomposes "coordination" into five priceable classes, because uniform pricing collapses the market to its highest-stakes class. semantic cadence and replay (§VI.9) give the substrate the empirical grounding to iterate on all of the above.

The advisory design of the resource claim system is the correct resolution of Sen's impossibility result for systems where agents have private knowledge about their own needs. The commons authority provides information, not allocation decisions.

Bond pricing is not a single open problem. It is a lattice: a cleanup-cost lower bound (§VI.7.5.1), a scope multiplier (§VI.7.5.2), a reputation discount (§VI.7.5.3), the Bonded Advisor mechanism (§VI.7.5.7), and the competitive-insurance market (§VI.7.5.8). Each is precise enough to disprove.

The infrastructure is running. The forest is building. The covenants are auditable. The sovereign is legible. The advisor is bondable. The market is open for bids.

Chapter Appendices

VI.A Formal Verification Status

Artifact availability. Every artifact named in Table VI.5 is bundled at `https://portdaddy.dev/whitepaper/artifacts/`. Runtime components are deployed inside the Port Daddy daemon binary and exposed through its public APIs; the source for the verified Rust core is the open-source `harbor-card-rs` crate (accompanying chapters, §5).

Table VI.5: Verification status of load-bearing claims. "Method" is the formal technique; "Result" is its outcome; "Artifact" names the public bundle file. A claim with both a formal artifact *and* a runtime conformance test is fully closed (**closed**); a claim with formal model but no deployed runtime is spec-only (*spec-only*); a claim with empirical evidence but no formal mechanization is partial (PARTIAL).

Claim	Method	Result	Artifact
Coordination channel isolation (§VI.4.2)	ProVerif (Dolev-Yao)	**closed** 3 properties TRUE	`isolation.pv`
Passkey device pairing	ProVerif	**closed** 3 properties TRUE	`passkey-pair.pv`
Federated security (§VI.12)	ProVerif, 3-of-4 compromise	**closed** no attacker reach	`federated.pv`
Conservation Theorem (§VI.10.3)	TLA+ bounded TLC	**closed** 26,818 states, invariant holds	`Conservation.tla`
No-overdraft lemma (§VI.7)	fast-check property test	**closed** 4 invariants, randomized ops	bonds-property tests
Algorithm-confusion immunity	ProVerif (pinned vs. naive)	**closed** pinned TRUE; counter-trace produced	`algconfusion.pv`
Delegation chain replay	ProVerif at depth 3	**closed** chain authenticity TRUE	`chain-replay.pv`
Magic-link single-use (§VI.12)	ProVerif (private-channel cap)	**closed** injective-event TRUE; 7/7 runtime conformance	`magic-link.pv`
Cuckoo-filter pollution resistance	fast-check, 5 invariants	**closed** FP rate within $2B/2^f$ at load 0.95	cuckoo property tests
Merkle-Forest binding (§VI.4.2)	EasyCrypt + fast-check	PARTIAL skeleton discharged; PROM-formal version not in scope	`binding.ec`
Auction/static comparison (§VI.7.5.8)	Monte Carlo, 36 configs × 2k trials	PARTIAL finite-grid result; not a universal Pareto theorem	`simulation.mjs`
Sybil A5 (§VI.7.5.9)	Monte Carlo over K Sybils	PARTIAL coverage-bounded ceiling at B_T	`simulation-sybil.mjs`
Cartel repeated-game A6	Monte Carlo + expected-value recurrence	PARTIAL $p_d^\star \approx 0.0478$ at δ=0.95 in supplied model	`simulation-cartel.mjs`
Claim-signaling IC (§VI.7)	TLA+ (TLC) + Z3 SMT	**closed** unique root $\delta^\star \approx 0.3425$; IC invariant fails at $\delta = 0.30$, holds at $\delta = 0.35$	`claim_signaling.tla`, `delta-threshold.z3`
Threat-band defensibility (§VI.7.5)	Monte Carlo, 7×3 grid	**closed** matched bands absorb target class within 1.10× upper bound	`threat-bands.mjs`
Passkey pairing runtime	—	*spec-only* model exists; no runtime	`passkey-pair.pv`
Federated KMS Cloudflare Worker	—	*spec-only* model exists; no Worker yet	`federated.pv`

Limitations. Four mechanizations are out of scope for this paper but explicitly named so a reader can audit what is and is not proved: a PROM-formal Merkle binding in EasyCrypt; a Coq or Lean mechanization of the Pareto strategic game; a formal analysis of the composite Sybil-deterrence mechanism (C_{kyc} + reputation gating + per-class B_{dep}); and a multi-class generalization of the

repeated-game cartel result to heterogeneous coordination classes (§VI.8). The present claims stand on the evidence stated in Table VI.5.

VI.B Worked Example: One Agent, All Layers

To make the three-layer architecture concrete, this appendix traces one agent—call it α—through a single bounded task: *"Fix the login bug in* `auth.ts` *and regression-test it."* The example exercises capability issuance, advisory claims, the evidence chain, a mid-task crash, successor resumption, and final settlement. All values are illustrative rather than tuned.

The architecture so far has been described layer by layer. The point of this appendix is to watch the layers *compose*: how a single bounded task progresses through the system, where each layer earns its keep, and what the daemon does *not* touch. The example is deliberately mundane—one small bug, one agent, one principal—because the value of the architecture should be legible in the small before it is trusted in the large.

The task. A principal P wants an agent α to fix a login bug in a hypothetical file `auth.ts` and regression-test it. The acceptance criteria are stated up front:

1. the failing test `login.regression.test.ts` passes;
2. no other test regresses;
3. the diff touches only `auth.ts` and the test file.

P estimates the cleanup cost of α's defection or sabotage at \$5—one operator-hour at \$5/hr equivalent, since this is a local agent rather than a cloud workload. Figure VI.10 traces what follows across the three architectural layers.

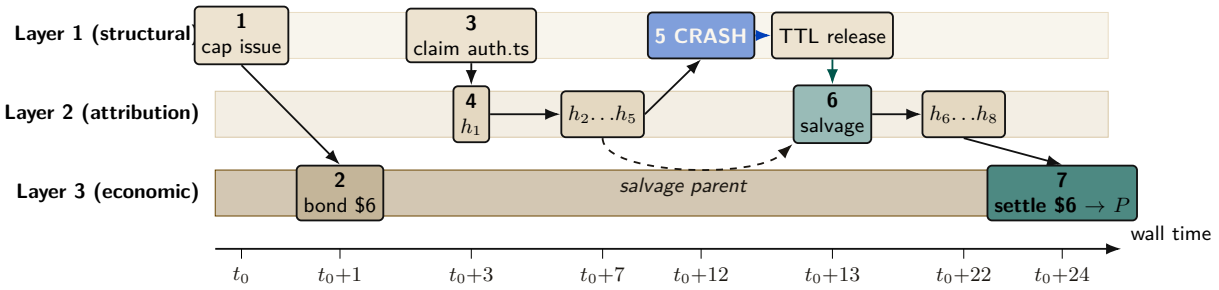


Figure VI.10: Worked-example timeline (App D). Three swimlanes for the three architectural layers; events placed by wall-clock minute on the x-axis. Layer 1 (structural) issues the capability at t_0, releases on TTL after crash, and re-issues for the successor. Layer 3 (economic) posts \$6 in escrow at $t_0 + 1$ and settles at $t_0 + 24$. Layer 2 (attribution) accumulates evidence-chain entries h_1–h_5 on α's timeline and h_6–h_8 on β's, linked by an explicit *salvage parent* edge so attribution survives the identity disposal at minute 12.

Phase A: Setup (minutes 0–1)

Before α writes a single byte, two artifacts must exist: a capability that bounds what α can touch, and a bond that prices what α might break.

Step 1: Capability issuance (Layer 1). P requests a Harbor Card for α via the daemon, specifying the capability set {`fs:read:repo`, `fs:write:auth.ts`, `fs:write:tests/**`, `cmd:test`} and a TTL of 30 minutes. The Anchor Protocol (§VI.3, accompanying chapters) signs the card. The effect is structural rather than advisory: α can prove its identity and capabilities to any verifier, but it *physically* cannot write to, e.g., `database.ts`—attempts return a capability error before the file system is even touched. This is the "physically" that makes the Layer 1 promise non-negotiable.

Step 2: Bond posting (Layer 3). With the capability in place, the Bonded Advisor (§VI.7.5) quotes a bond proportional to the cleanup cost: $B_\alpha = \$5 \cdot (1 + s) = \$6$ at slope $s = 0.2$. P posts \$6 into escrow. The settlement preconditions are now fixed up front: (1) acceptance criteria met $\Rightarrow$ \$6 returned, (2) partial completion $\Rightarrow$ pro-rata, (3) sabotage $\Rightarrow$ \$6 liquidated to the commons treasury. The bond is the conversion of α's "character" into a settled amount; once it is posted, the question of whether α is trustworthy is no longer load-bearing.

Phase B: Work begins (minutes 1–12)

Phase A's contracts are now in force. α does the work the principal hired it for, and the daemon records what happens.

Step 3: Advisory claim (§VI.5). α claims `auth.ts` via `pd session files add auth.ts`. The daemon's conflict graph is empty for this path, so the claim succeeds with no conflict notification; had another agent already held the path, α would have been told and would have decided locally whether to coordinate or wait. The claim's TTL is bounded by the Harbor Card's—30 minutes—so a crashed α cannot squat on the file indefinitely.

Step 4: Evidence chain begins (Layer 2). α opens the work with a scope note: *"Scope: `auth.ts`. Hypothesis: cookie-domain handling on subdomain login fails. Validation: `npm test login.regression`."* The note's content hash h_1 chains into the session's Merkle root $R^{(1)}_{\text{session}}$, and from then on every action α takes—a read receipt for `auth.ts`, a diff blob for the partial fix, the stdout of the failing test run—is appended as a new entry. Five notes deep, $R^{(5)}_{\text{session}}$ is the canonical commitment to α's state of work. The chain is the part of the architecture that survives anything that happens next.

Phase C: Crash and recovery (minutes 12–13)

The crash is where the architecture stops being an abstraction. In Phase B everything went right; in Phase C something goes wrong, and the layers either earn their keep or they don't.

Step 5: Mid-task crash (§VI.6). At minute 12, α's process crashes—let's say the LLM provider timed out. The daemon's heartbeat monitor observes α as unresponsive after 90 seconds and marks the session abandoned. α's claim on `auth.ts` releases on TTL; the evidence chain is sealed at $R^{(5)}_{\text{session}}$ and committed into the harbor root $R_{\text{harbor},\ell}$ at the next epoch boundary. *Critically, no information is lost*: the partial diff, the test output, and α's scope note are all in the chain. The crash terminated a process; it did not terminate the work.

Step 6: Successor agent. P spawns a fresh agent β with a new Harbor Card and the same capability set. β's first action is `pd salvage`: the daemon returns the abandoned session, the scope note, the partial diff, and the last test output. β reads the scope note ("cookie-domain handling on subdomain login"), inspects the partial fix that α left behind, and finishes the work. The new evidence chain $R^{(\beta)}_{\text{session}}$ explicitly references $R^{(5)}_{\text{session}}$ as its salvage parent. Attribution is preserved across the identity change (§VI.4.4)—the diff that ships at minute 24 is traceable back to the same principal P that posted the bond at minute 1, even though the agent that did most of the work is gone.

Phase D: Settlement (minute 24)

The work is done. The daemon now decides what was actually accomplished and acts on the settlement preconditions fixed in Phase A.

Step 7: Settlement. β's final action is `pd done`. The daemon's acceptance-criteria checker runs the regression suite, confirms `login.regression.test.ts` passes, confirms no other test regresses, and confirms the diff is bounded to the two files declared in the Harbor Card. All three criteria met. The daemon releases \$6 from escrow to P, signs a settlement record over the evidence-chain root, and appends that record to the harbor Merkle forest (§VI.4.2). The settlement is now itself an immutable, attestable artifact.

What did each layer do?

- *Layer 1 (structural)* kept the agent inside the declared blast radius. The crash, the recovery, and the settlement all happened inside a capability-bounded scope. A capability error at minute 7 would have been a fast-fail, not a corrupted repository.
- *Layer 2 (attribution)* kept the work durable across the crash. Without the Merkle-chained evidence, β would have started from zero; with it, β resumed at minute 12's state.
- *Layer 3 (economic)* kept the principal honest in scoping. P posted real collateral and recovered it on completion; had α (or β) sabotaged, the cleanup cost would have been pre-funded.

What did *not* happen. The daemon never decided which agent should hold the file, never decided whether the fix was "good enough" in a subjective sense, and never punished the crash. It enforced what *cannot* be (Layer 1), recorded what *did* happen (Layer 2), and settled against the published criteria (Layer 3). The substantive judgments—"is this scope right", "is this fix correct"—remained with P and the acceptance-criteria checker. This is the Sen-grounded posture of §VI.5 made operational.

VI.C Exercises

These exercises are for instructors using this paper in a systems, mechanism-design, or multi-agent class. Solutions are not provided; the exercises are designed so that defending an answer requires engaging the paper's claims rather than recalling them.

E1 (Definitions). Define "advisory claim" and explain, in two sentences, why an advisory regime is strictly more permissive than an enforced-lock regime. Then explain why the Sen impossibility (§VI.5) makes that strict permissiveness a *feature*, not a bug.

E2 (Mechanism design). A reader proposes: "Eliminate bonds. Use post-hoc reputation slashing instead—an agent that defects loses reputation points and is excluded from future work." Identify two specific failure modes of this proposal that the bonded design avoids. Hint: consider the first-mover problem and Sybil identities.

E3 (Formal verification). Sketch the ProVerif process for the magic-link recovery protocol (accompanying chapters, Phase 2). What is the secrecy property you would prove? What is the authentication property? Identify one attacker capability ProVerif would need to model that is not present in the Phase 1 Harbor Card model.

E4 (Coverage-bound A5). The A5 finding (§VI.7.5.9) shows that pure deposit deterrence is bounded by the auction coverage $B_T = \mu(1 + s)$. Construct an attacker who exploits this bound directly. Then construct a defense—other than "raise the deposit"—that defeats the attacker. Hint: see the composite mechanism C_{kyc} + reputation gating + $B_{dep}^{\text{per-class}}$.

E5 (Cartel folk-theorem). Using the A6 grid (§VI.7.5.10, Figure VI.8), compute the minimum per-round detection probability $p_d^\star$ at $\delta = 0.99$. Then argue, qualitatively, what would change if cartel members can re-enter under fresh identities. Does the bond mechanism prevent re-entry, deter it, or merely price it?

E6 (Capability attenuation). An agent α holds a Harbor Card with `fs:write:auth.ts`, `db:read`, TTL = 15 minutes. α delegates to β a card with `fs:write:auth.ts`, TTL = 20 minutes. The verifier rejects this delegation. Cite the specific attenuation invariant violated. Now rewrite β's requested card to be acceptable.

E7 (Cuckoo-filter pollution). Sketch a pollution attack against a cuckoo filter that does *not* enforce the load-factor ceiling at 0.95. What invariant breaks? Why does the ceiling defeat it? Hint: the FP rate bound in Fan et al. [35] is conditional on the regime.

E8 (Mechanism extension). The competitive-insurance mechanism (§VI.7.5.8) replaces a single bond price with insurer-agent bids. Propose one alternative pricing mechanism (e.g., a second-price auction with reserve, a Walrasian-style market-clearing rule, or a posted-price catalog) and argue whether it would conditionally Pareto-dominate the static escrow under the CC/NC/RP/NS assumptions of §VI.7.5.8. Identify the assumption your alternative most depends on.

E9 (Open problem). The gossip convergence bound (the Anchor Protocol's revocation analysis) is reproduced from the standard anti-entropy analysis rather than mechanized. Identify the specific theorem from [9] that would need to be ported to a proof assistant. What would change in the analysis if the daemon mesh is *not* fully connected? Hint: consider gossip on a graph with bottleneck cuts.

VI.D Limitations & Boundaries

The formal mechanisms presented in this chapter are mathematically sound within their defined scopes, but they depend on bounding assumptions that must be explicitly acknowledged.

- **Threat Model Exclusions:** Collusion across all independent organs (e.g., the logging daemon, the arbitrator, and the primary execution runtime) is assumed out-of-scope; if all enforcing components are compromised, the guarantees degrade to advisory.
- **Performance Trade-offs:** Cryptographic assertions and WAL-checkpointing for per-write durability incur measurable I/O overhead. These mechanisms are designed for high-stakes coordination, not for bulk data processing or high-frequency trading.
- **Implementation Maturity:** While the core protocols (Anchor, SQLite single-writer) are IMPLEMENTED and running in production, surrounding ecosystem components (like the decentralized judge markets or fully federated multi-sig routing) remain in PARTIAL or DESIGNED phases.

These constraints are not flaws but structural realities of building zero-trust coordination on top of POSIX primitives.

PORT DADDY COORDINATION PAPERS · CHAPTER VII OF VII

The Federated Harbor

Abstract

A demonstration is scheduled for 4pm. Alice runs the back end on her laptop; Bob runs the front end on his. Each operator has a working local harbor, but Alice's `db:write` card is meaningless to Bob's daemon, Bob cannot yet observe Alice's revocations, and a bond posted at one harbor cannot automatically settle damage at the other. This paper specifies the missing federation boundary: a cross-harbor capability-transfer ceremony with composed attenuation, revocation dissemination through a witnessed anti-entropy overlay, a conditional escrow construction, and bonded admission for new harbors. The assurance case is deliberately narrower than earlier drafts. Epidemic dissemination supplies an expected $\Theta(\log m)$ result only under a connected reliable-round model, not a hard freshness deadline. A signed escrow promise makes theft detectable but does not prevent it; the extractable-value bound requires a custody ledger that structurally exposes only recipient-whitelisted transitions. Cross-harbor conservation is stated as a bucket-partition invariant and model-checked only at the recorded finite bound. The protocol models and federation runtime remain partial. The contribution is therefore a falsifiable design and research boundary, not a claim that federation is already deployed or trustless.

Keywords: federated identity, capability delegation, cross-realm authorization, mechanism design, distributed trust, certificate transparency, atomic settlement, mutual suspicion.

Reading time: approximately 45 minutes end-to-end. Chapters V and VI, The Anchor Protocol *(Owens 2026) and* The Bonded Commons *(Owens 2026), supply the cryptographic and economic primitives this paper extends. Either of those can be read first or in parallel; this paper is the federation half. The Reader's Map below provides shorter routes for specific audiences.*

PORT DADDY COORDINATION PAPERS **CHAPTER VII OF VII**

V. The Anchor Protocol. Authentication, delegation, attenuation, and revocation inside one harbor.

VI. The Bonded Commons. Evidence, bonding, advisory claims, and the incentive conditions for cooperation.

VII. The Federated Harbor (this chapter)**.** Identity, revocation, and settlement across mutually sovereign machines.

Chapters I–IV form the product arc; Chapters V–VII provide the protocol, economic, and federation deep dives. Every chapter is independently readable.

Reader's Map

The paper has a topology: it opens with the gestalt (§VII.1–§VII.2), then moves through three load-bearing primitives in order of cryptographic depth (capability transfer, revocation, settlement), then steps back to admission, failure modes, and limitations. The figures are designed to be readable in sequence without the surrounding prose; if you want the five-figure summary, that path is given below.

Layer	Load-bearing artifact	Lives at
(1) Topology	Two harbors, witness log, settlement escrow as a four-element gestalt	§VII.2; Figure VII.1
(2) Cap. transfer	Four-message ceremony; ProVerif model of cross-realm authenticity & attenuation, *conditional on a witness-log freshness assumption* (§VII.4.5)	§VII.4; Figure VII.3
(3) Revocation	Conditional epidemic-dissemination expectation; partition caveat	§VII.5; Figure VII.4
(4) Settlement	Recipient-whitelisted custody condition; bucket-partition Conservation	§VII.6; Figure VII.5
(5) Threat bands	What is mechanized, what is bounded, what is open	§VII.12; Figure VII.2

Suggested reading paths by reader type:

- **Beginner / new to the series.** Read §VII.1 (the two-machine problem) and §VII.8 (Alice and Bob, end-to-end). The worked example is written to be read cold. Look at the five figures in order. Skip the proofs on the first pass.
- **Distributed-systems engineer.** §VII.2 (the topology), §VII.4 (transfer ceremony), §VII.5 (gossip and equivocation). Chapter V (*The Anchor Protocol*) for the cryptographic substrate. Then §VII.12 for the honest open questions.
- **Formal-methods reviewer.** Jump to §VII.9 (the TLA+ and ProVerif extensions) and Appendix VII.A (the verification status table). Threat bands in Figure VII.2 are the most concentrated signal of what is and is not mechanized.
- **Cryptoeconomic / mechanism-design reader.** §VII.6 (bounded escrow) and §VII.7 (cold-start admission via bonded sponsorship). The proposal's equilibrium contributions are stated here as open work; that is honest, not coy.
- **Security-engineering reviewer.** §VII.3 (threat model) first, then the three primitive sections in order. §VII.10 for the operationally interesting cases the protocols do not close.

Volume Context. Chapter V (*The Anchor Protocol*) formalizes process identity on a single machine. Chapter VI (*The Bonded Commons*) establishes pre-transactional commons authority and collateralized incentive alignment. This chapter extends both into the cross-machine, cross-organization regime. Specific primitives are cited directly (e.g., Chapter V §2.4, Chapter VI §2) rather than re-derived.

VII.1 Introduction: The Two-Machine Problem

A demonstration is scheduled for 4pm. Alice runs the back end of the demo on her laptop; Bob runs the front end on his. Each has a working Port Daddy daemon: capability tokens are minted and attenuated under the Anchor Protocol [76], the workspace history is Merkle-chained and replicated locally per the Bonded Commons [78], and bonds for cleanup of botched migrations are posted in the local collateral pool. Inside each laptop, the trust story is closed and the formal-verification artifacts hold.

The demo nevertheless does not work. Three failures, none of them inside either machine:

1. Alice's agent obtains a `db:write` card for the staging database, attempts to use it against Bob's daemon, and is refused. Bob's daemon has never seen Alice's signing key. The card is well-formed under the Anchor Protocol, but Bob has no root of authority for it; from his daemon's point of view, the card is gibberish.

2. Five minutes before the demo, Alice's operator revokes the `db:write` card after spotting that it has been over-attenuated. The revocation propagates within Alice's mesh in under a second by the gossip protocol of Anchor §4. Bob's harbor never hears about it. The card remains live on Bob's side.

3. The migration runs anyway, lands in Bob's staging database, and corrupts the demo schema. Alice posted a bond in her harbor against exactly this case. The bond sits in her harbor's collateral pool. Bob's harbor measured the damage. Neither harbor can, on its own, route the bond to where the damage is.

Each failure is a different facet of the same underlying gap. Both daemons are internally consistent. Neither is a root of authority for the other. The single-machine sovereign of the prior papers does not extend to the case of two machines under two operators with no shared administrative parent.

We call this the **two-machine problem**. The point of stating it this baldly is that the failure modes are not exotic; they are what every developer who has tried to dovetail two local agent fleets on a shared task has seen. The bug is structural. The bug is not in either daemon. The bug is in the assumption that one daemon's authority extends past the machine boundary by default. It does not, and it should not.

> *The trick is not to extend a single sovereign across machines. The trick is to admit there are two and to specify what they owe each other.*

This paper extends the Anchor Protocol, the Bonded Commons, and the daemon-as-correlation-device argument to the multi-administrative-domain case. We are not introducing a new substrate; we are showing that the existing substrate admits a clean federation, and that the federation rules are sharper than they look. Where the prior papers spoke of *the* commons authority, this paper speaks of a *mesh of* mutually witnessing commons authorities, each sovereign inside its own machine, none sovereign over the others, and all of them bound to the same audit substrate.

VII.1.1 What this paper is and is not

This chapter closes the V–VII deep-dive arc. The Anchor Protocol supplies the cryptographic-token substrate; the Bonded Commons supplies the economic and governance model; this chapter specifies the federation boundary. The dependencies are concrete: §VII.4 extends Anchor's delegation chain across a trust boundary; §VII.5 extends its revocation mesh; §VII.6 extends Bonded's Conservation invariant.

What this paper is *not*: it is not a permissionless-blockchain redesign. Port Daddy daemons do not run consensus. The Federated Harbor explicitly rejects the assumption common to most

modern federation proposals [15, 163] that a shared ledger is the right substrate. The cost of that simplification is that we cannot lean on a chain for ordering, equivocation detection, or settlement; we must construct each of those primitives from peer gossip plus a federated witness log. The advantage is that the design composes cleanly with the local-first, single-operator-machine architecture of the prior two papers.

VII.1.2 Contributions

The contributions are concrete. We separate them by mechanization status to be honest about which are theorems, which are bounded threat models, and which are sketches.

1. **The four-element federation topology (§VII.2).** A precise statement of the boundary between intra-harbor and inter-harbor scope. Defines harbor sovereignty as a property characterization in the style of Bonded's Admissible KMS, so that subsequent claims can be checked against it.

2. **The inter-harbor capability transfer ceremony (§VII.4).** A four-message protocol exchanging a Harbor Card issued by A for an attenuated card valid at B, without either daemon trusting the other's signing key as a root. The ceremony binds the issuing harbor's current epoch root into the envelope, which is the substrate that revocation in §VII.5 hooks into. ProVerif model in Appendix VII.B (status: PARTIAL, model drafted; soundness pending mechanization).

3. **Federated revocation mesh (§VII.5).** The multi-administrative-domain extension of Anchor's revocation gossip. Each harbor publishes an epoch root to a witness log. Under an explicit complete-overlay, reliable-round model, epidemic dissemination takes expected $\Theta(\log m)$ rounds; no finite worst-case bound survives an unbounded partition (Property VII.5.1).

4. **Cross-harbor settlement (§VII.6).** A conditional protocol for clearing a bond posted at A against damage measured at B. The escrow's role is structurally bounded only when the custody ledger, not merely a signed promise, restricts recipients, fee, and terminal transitions. The construction is trusted and specified, not an HTLC-style trustless swap [143].

5. **Cold-start admission ceremony (§VII.7).** A protocol for a new harbor joining an existing mesh without prior reputation, using a bonded-sponsor pattern: an existing harbor posts a bond on the newcomer's behalf, which is forfeit on misbehavior in a probationary epoch. This is the federation-layer analogue of competitive insurance from Bonded.

6. **Honest open problems (§VII.12).** Five named open questions: (a) whether the correlated-equilibrium reframing of Bonded survives the multi-principal setting cleanly; (b) whether bonds can settle without a trusted, conditionally restricted custody party; (c) whether the folk-theorem cartel-resistance argument from Bonded weakens at the federation layer, and by how much; (d) whether bonded sponsorship can avoid invitation-only capture in a thin cold-start market; and (e) what deployment-specific dissemination tail bound is justified. We state these as open, not as solved.

The proposal that scoped this paper [79] listed two further contributions (a formal definition of harbor sovereignty, and equilibrium results across federations co-authored with Youle) which are present here in skeletal form and will be load-bearing in a subsequent revision once the open problems above resolve. We will not claim a result we do not have.

VII.2 The Federated Authority

The Bonded Commons paper defined a *commons authority* as a pre-transactional trust infrastructure with sovereign scope inside a single machine. Inside that scope, the daemon is the unique correlation device that turns mutually-suspicious agents into a correlated equilibrium [78]. The single-machine sovereign was an honest simplification. We now drop it.

The federation design has four elements (Figure VII.1): two sovereign harbors, a witnessed epoch-root log, and a settlement escrow. The escrow is structurally bounded only under the custody assumptions in §VII.6.1. Every primitive in the rest of the paper attaches to one of these elements.

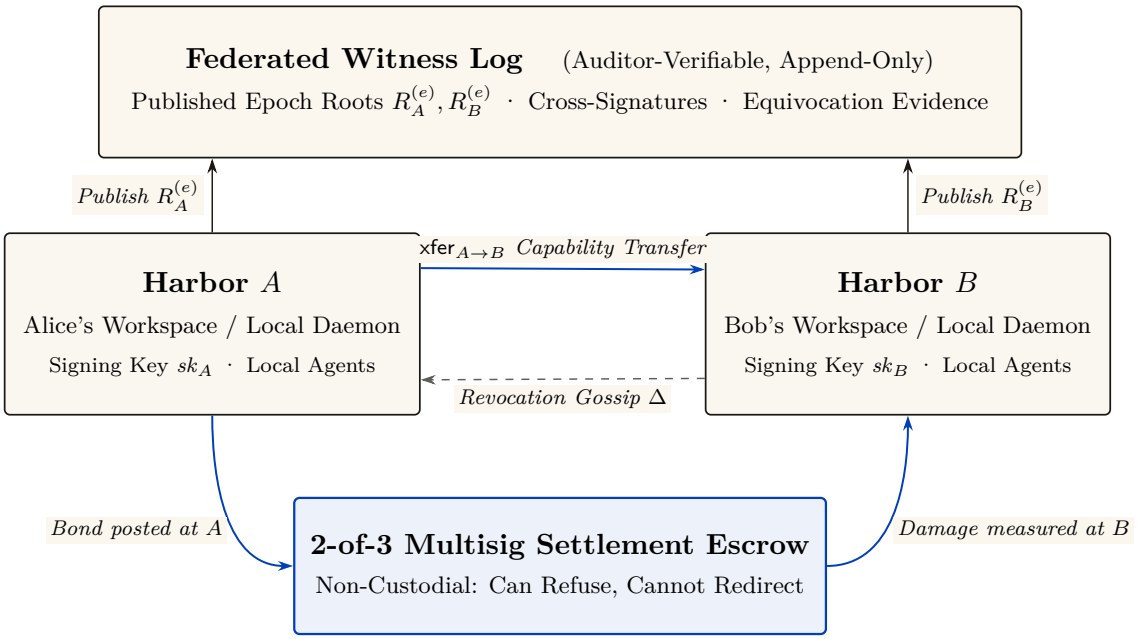


Figure VII.1: The Federated Harbor as a four-element gestalt. Neither harbor is a root of authority for the other. The witness log above provides shared correlation: each harbor publishes its signed epoch root, enabling decentralized equivocation detection. The settlement escrow below is structurally bounded (non-custodial 2-of-3 multisig: it cannot redirect funds or author unadjudicated transfers). The solid blue arrow denotes the capability transfer ceremony (§VII.4); the dashed line denotes epidemic revocation gossip (§VII.5).

VII.2.1 Harbor sovereignty as a property characterization

In Bonded, the Admissible KMS was characterized by a list of properties rather than by a specific construction (per-machine signing key, per-machine SQLite, etc.) so that the argument carried over to any concrete instantiation. We adopt the same posture here.

Definition VII.2.1 (Harbor Sovereignty)**.** A daemon D_A is *sovereign over harbor* A if it has exclusive authority over five things:

1. **Issuance.** Only D_A can mint a Harbor Card whose root signature verifies under A's signing key sk_A.
2. **Attenuation.** Only D_A can apply a local attenuation predicate att_A that restricts an inbound capability before it is verified by an A-side agent.
3. **Revocation.** Only D_A can decide that a card it issued is revoked. The decision becomes globally visible by the gossip protocol of §VII.5; D_A does not control when other harbors learn of it, but it controls whether the revocation event exists.
4. **Acceptance.** D_A alone decides which inbound cards from other harbors to accept and under what local policy. No external party can compel acceptance.

5. **Evidence binding.** The epoch root $R_A^{(e)}$ that D_A publishes to the witness log is signed under sk_A and binds the entire A-side state of issued, attenuated, and revoked cards up to epoch e. No other harbor can publish a root that purports to be A's.

This is a deliberately narrow definition. It does *not* say that D_A is sovereign over everything an A-side agent does; the agent may be operating against a database hosted at B, in which case B-side policy applies and D_A has nothing to say about it. Sovereignty is over the *issuance and revocation surface of A's own cards*, not over downstream effects.

We will use Definition VII.2.1 as the discipline against which subsequent protocols are checked. The transfer ceremony of §VII.4 produces a card valid at B; if it required B to verify under sk_A in the hot path, it would violate B's sovereignty by making B's acceptance contingent on an external authority. We will see that it does not.

VII.2.2 Inter-harbor scope

The dual notion is also load-bearing. *Inter-harbor scope* is the scope of operations whose effects cross the trust boundary between sovereign harbors.

Definition VII.2.2 (Inter-Harbor Scope)**.** An operation o has *inter-harbor scope* (A, B) if its execution causes a state change observable to B under authority that originated at A. Examples: an A-issued card used to write to a B-hosted database; a settlement clearing a bond posted at A against damage measured at B; a revocation issued by A becoming visible to a B-side verifier.

Most operations are *intra-harbor*: minting, verifying, attenuating, and revoking cards under one harbor's authority, observed by agents under that same harbor. The Anchor Protocol's proofs cover the intra-harbor case end-to-end. The Federated Harbor's contribution is exactly the inter-harbor surface: every load-bearing protocol in this paper is one that lives at the boundary.

VII.2.3 The witness log

The witness log is the device that makes the federation auditable without making any harbor sovereign over the others. It plays the same role that the Merkle Forest plays inside a single machine in Bonded (Chapter VI, §4.1): a tamper-evident accumulator that any third party can verify without needing to trust the publisher.

Property VII.2.1 (Witness-Log Admissibility)**.** A witness log W is admissible for the Federated Harbor if it provides:

1. **Append-only.** Once W records an entry, W cannot retract or reorder it.
2. **Per-harbor latest-root binding.** For each harbor X in the federation and each epoch e, W records at most one root $R_X^{(e)}$ signed under sk_X.
3. **Auditable consistency.** Any third party can verify, given $R_X^{(e_1)}$ and $R_X^{(e_2)}$ with $e_1 \le e_2$, that $R_X^{(e_2)}$ is a Merkle-consistent extension of $R_X^{(e_1)}$.
4. **Equivocation evidence.** If an auditor obtains two differently signed roots for the same harbor and epoch, it can verify the contradiction immediately. Disseminating both views to a common auditor depends on the overlay and partition model (Property VII.5.1).
5. **Threshold publishability.** The act of publishing $R_X^{(e)}$ to W does not require D_X to trust any other party; the cross-signatures collected from other daemons in the federation are an auditor convenience, not an authorization gate.

This list is intentionally close in shape to the five properties of the Admissible KMS in Bonded (Chapter VI, "Federated Sovereign"). The Bonded paper used the same posture — characterizing the property surface, then exhibiting a concrete construction that satisfied it — and we follow that pattern. Section VII.9 sketches one concrete instantiation built on Certificate Transparency-style append-only logs [31] cross-witnessed in the Sigstore/Rekor pattern [208]; the substrate need not be unique.

VII.2.4 What the federation is not

It is worth saying what the federation is not, so that the rest of the paper does not have to keep refuting positions it never took.

The federation is *not* a consensus protocol. There is no global agreement on the order of events; there is per-harbor monotonicity (epoch roots only extend), there is cross-harbor witnessing (each root is published to a shared log), and there is detectable equivocation. None of these require harbors to agree on a global order, and the prior papers' designs explicitly avoid that overhead.

The federation is *not* a hub-and-spoke topology with a privileged coordinator. The witness log is replicable; in practice a small set of mutually-witnessing log instances suffices, and any harbor can read from any of them. The hub-and-spoke failure mode — one well-funded coordinator captures the federation by becoming everyone's only witness — is a real risk we discuss in §VII.10, and the structural pushback is gossip-based witness redundancy.

The federation is *not* a permissionless market in identities. New harbors enter through the admission ceremony of §VII.7, which requires either a pre-existing bond or an existing harbor sponsor. This is by design; the cost is that the federation is invitation-bounded, the benefit is that Sybil attacks against the federation [116] cost the attacker as much per identity as honest entry does.

VII.3 Threat Model

We make the threat model explicit before introducing the protocols so that the design choices can be checked against named adversary powers. We follow the Dolev–Yao convention [60] extended to the federation case: an attacker controls every wire between harbors, can intercept and replay any cross-machine message, and may collude with any bounded number of harbor operators. The attacker cannot break the underlying cryptographic primitives (Ed25519 signatures, SHA-256 hashes); these assumptions are inherited from Anchor.

VII.3.1 In-scope attacks

The protocols in this paper are designed against the following attackers.

Cross-realm identity spoofing. An attacker presents a card at B claiming issuance by A, using a forged signature under sk_A. *Closed by* signature verification against A's public key as recorded in the witness log (§VII.4). Reduces to Ed25519 EUF-CMA security [58].

Replay across the boundary. An attacker captures a valid xfer_offer from A to B at epoch e and replays it at epoch $e + k$. *Closed by* the binding of $R_A^{(e)}$ into the envelope; the verifier at B checks that R_A is the current root, and a stale envelope is rejected (§VII.4). The window between issuance and the next epoch is the structural attacker window, named and bounded.

Capability inflation across transfer. An attacker presents at B a card C'_B that claims authority beyond the attenuation predicate applied at A. *Closed by* the requirement that the verifier at B recompute the composed attenuation $\mathrm{att}_B \circ \mathrm{att}_A$ from the envelope rather than trust an intermediate claim (§VII.4); reduces to Anchor's intra-harbor attenuation invariant.

Stale-revocation acceptance. An attacker at B presents a card revoked at A but not yet observed at B. Under the reliable connected-overlay model the window has expected scale $\Theta(\Delta \log m)$; during a partition it is unbounded. A bond can price a configured service-level window, not eliminate the tail.

Cross-witness equivocation. A malicious daemon D_A publishes contradictory roots to isolated witnesses. The contradiction is cryptographically verifiable once a common auditor receives both views, but the time to that event is topology- and partition-dependent. A witness bond prices confirmed equivocation; it does not create a delivery deadline.

Settlement redirection or equivocation. A signed escrow promise makes misconduct attributable. Prevention requires a custody ledger whose transition API cannot name an arbitrary recipient or fee. Property VII.6.1 states the bound conditionally on that mechanism; without it, the full bond remains at risk.

Harbor membership forgery at the federation boundary. An attacker presents a card from a harbor purporting to be in the federation but that has no admission record. *Closed by* the admission ceremony (§VII.7) which produces a permanent, witness-logged enrollment record co-signed by an existing harbor's sponsor bond. A claimed harbor with no enrollment record is treated as outside the federation regardless of how well-formed its cards appear.

VII.3.2 Out-of-scope attacks

We are explicit about what is not closed.

Compromise of a harbor's signing key. If sk_A leaks, the attacker can issue arbitrary A-side cards until A rotates the key and publishes the rotation to the witness log. We inherit Anchor's key-rotation procedure unchanged. The federation does not close the underlying primitive; it does bound the blast radius to A's sovereign surface, which is exactly the right scope.

Sybil at the federation layer. An attacker who can pay the admission cost m times can run m federated harbors. The cost ratio between honest and adversarial admission is the load-bearing parameter; we discuss it in §VII.7 but do not give a tight bound. This is named open work.

Cartel formation across harbors. Bonded's folk-theorem cartel-resistance argument relied on the daemon as a correlation device inside one machine. Across harbors, cartels can form offline and present a unified front to the witness log; the witness log records that all harbors agreed, which is consistent with both honest agreement and cartel collusion. We acknowledge this and state the open work in §VII.12.

Centralization gravity. Historically, federated identity systems concentrate on one hub within five to ten years [82]. The Federated Harbor's structural pushback — replicable witness logs, no central coordinator — is design discipline, not theorem. We formally classify this in §VII.10.

Side channels. Constant-time signature verification, cache-side-channel resistance, and timing-attack resistance are inherited from the underlying libraries; we do not re-prove them and we note where we trust them.

VII.3.3 Threat-band visualization

Figure VII.2 lays out the threat surface as three bands: mechanized claims (teal), bounded threats (amber), and acknowledged open frontier (cobalt). The boundary between amber and cobalt shifts over time as future work mechanizes what is presently bounded. We will refer back to this figure when discussing limitations.

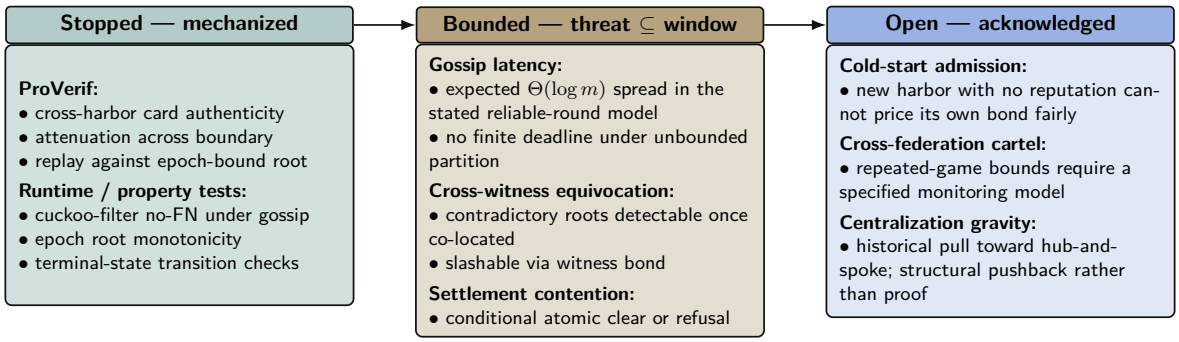


Figure VII.2: Threat-band layout for the Federated Harbor. The Anchor paper's defense-in-depth structure carries over: every claim either has a mechanization artifact (teal), a named bound that the bond can price against (amber), or an explicit acknowledgement of the open frontier (cobalt). The paper is disciplined about which band each claim lives in. Reading left to right runs deeper into threat space and out of formal reach; bands shift left over time as future work mechanizes what is presently bounded.

VII.4 Cross-Harbor Capability Transfer

The first load-bearing protocol is the cross-harbor capability transfer ceremony. An A-side agent holds a Harbor Card C_A minted by D_A under Anchor Phase 2 or Phase 3. It needs to present an equivalent capability at B. The Anchor Protocol's intra-harbor delegation already supports attenuation; we extend that to the case where the attenuation crosses the trust boundary.

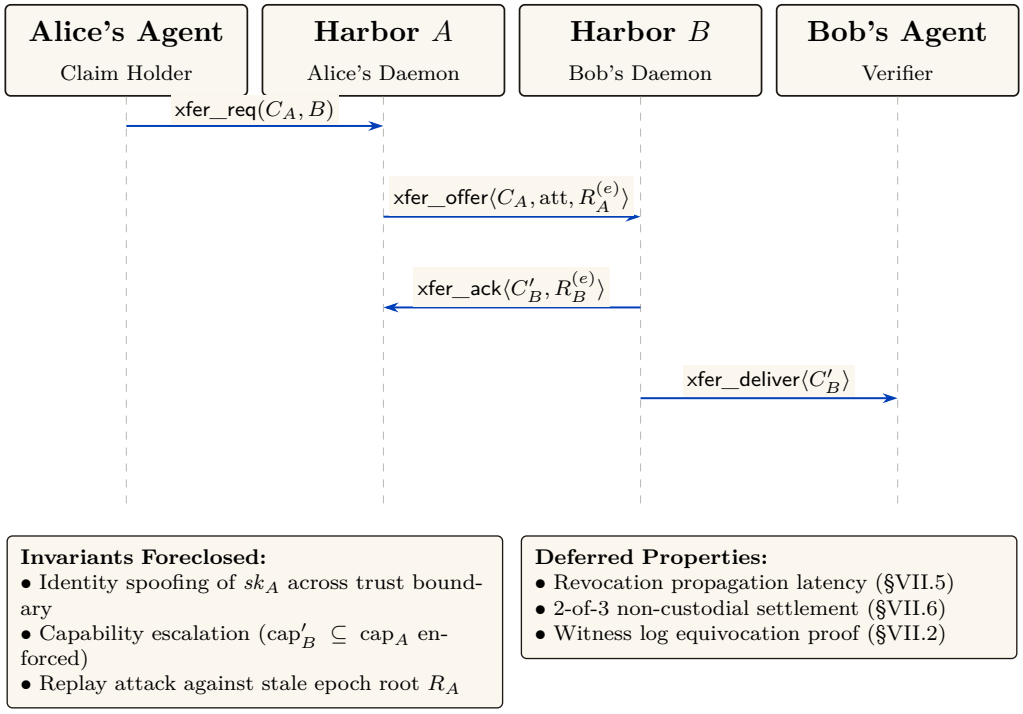


Figure VII.3: The four-message cross-harbor capability transfer ceremony. The critical property is that no message after (3) needs to reach back to Harbor A in the hot path: C'_B is verifiable under sk_B alone, which is why federation does not turn every authorization into a synchronous inter-machine round trip. The epoch root $R_A^{(e)}$ bound into the envelope is the load-bearing primitive — a revocation issued by A after epoch e invalidates C'_B as soon as the new $R_A^{(e+1)}$ reaches the witness log that B audits.

VII.4.1 The four-message ceremony

Figure VII.3 shows the protocol as a sequence diagram. The four messages are:

1. xfer_req(C_A, B) — Alice's agent presents C_A to its local daemon D_A and names B as the target harbor. The request is intra-harbor; standard Anchor verification applies.
2. xfer_offer$\langle C_A, \text{att}_A, R_A^{(e)} \rangle$ — D_A constructs an envelope binding the card, an attenuation predicate att_A specifying what subset of C_A's authority is being offered for cross-realm use, and its current epoch root $R_A^{(e)}$. The envelope is signed under sk_A and sent to D_B.
3. xfer_ack$\langle C'_B, R_B^{(e)} \rangle$ — D_B verifies the envelope's signature against A's public key as recorded in the witness log; verifies that $R_A^{(e)}$ is the current root for A; applies its own local attenuation att_B; and mints $C'_B = \text{att}_B(\text{att}_A(C_A))$, a card valid only at B, signed under sk_B. The acknowledgment is sent back to D_A along with B's current epoch root.
4. xfer_deliver$\langle C'_B \rangle$ — D_B delivers C'_B to the agent at B that will use it. Subsequent presentations of C'_B at B verify under sk_B alone — no A-side lookup is needed at the hot path.

The critical property is that after message (3), no further synchronous interaction with A is required for the agent to operate at B. The cost is that revocation of C_A at A does not instantaneously revoke C'_B at B; Property VII.5.1 gives an expectation only under its delivery model, and partitions require fail-closed policy or bounded credential TTLs.

VII.4.2 Why this composes cleanly with Anchor

The Anchor Protocol's Phase-3 delegation already proves that a chain $C_{\text{daemon}} \to C_A \to C_B$ of attenuated cards is verifiable under the issuing daemon's public key, and that an attacker cannot inflate a downstream cap to exceed an upstream one. The transfer ceremony is Phase-3 delegation across a trust boundary, with two changes:

1. The second link in the chain is signed under a *different* root key (sk_B instead of sk_A), so the verifier at B verifies under sk_B, not sk_A.
2. The cross-realm link binds the issuing harbor's current epoch root $R_A^{(e)}$, so a revocation at A after epoch e invalidates the downstream card as soon as the new $R_A^{(e+1)}$ is observed at the verifier.

The composition is what makes the ceremony cheap to add: we are not introducing a new cryptographic primitive, only a new policy at the boundary that says "the verifier accepts a card whose chain crosses the trust boundary if and only if (a) each cross-realm link is signed under the receiving harbor's key and (b) the issuing harbor's bound epoch root is the current one in the witness log."

VII.4.3 Attenuation across the boundary

Two attenuation predicates are applied in sequence. att_A is set by D_A at issue time and reflects what authority A is willing to project across the boundary. att_B is set by D_B at acceptance time and reflects what authority B is willing to grant on its own resources to a card whose root is A. Both attenuations strictly narrow the underlying capability.

Lemma VII.4.1 (Attenuation Monotonicity Across the Boundary)**.** *For any cross-realm card $C'_B = \text{att}_B(\text{att}_A(C_A))$, the set of operations authorized by C'_B is a subset of those authorized by C_A under the federated authorization rule.*

Sketch. Each attenuation is a subset operation in the Anchor algebra; subset composition is a subset. Cross-realm policy at B adds the constraint "and the operation must be permitted by B's local policy," which can only further restrict. See the Anchor accompanying chapter's capability-attenuation section for the intra-harbor argument; the cross-realm case adds no new mathematics, only a new applicable attenuation step. □

VII.4.4 Threats foreclosed and threats deferred

The annotation band at the bottom of Figure VII.3 summarizes which threats this ceremony forecloses and which are deferred to other primitives. Specifically:

- **Foreclosed:** identity spoofing of sk_A across the boundary; capability inflation across the transfer; replay at B against a stale R_A.
- **Deferred:** revocation latency (handled in §VII.5); settlement disputes (handled in §VII.6); admission of unbonded new harbors (handled in §VII.7).

We are explicit that the ceremony does not close every threat in one step. Federation is a layered defense; this is one layer.

VII.4.5 Mechanization status

We have a draft ProVerif model of the four-message ceremony (Appendix VII.B), extending the Anchor Phase-3 model with a second signing key and an epoch-root binding. The current status of the cross-realm authenticity query is PARTIAL: the model checks, but the proof depends on an assumption about witness-log freshness that we are still tightening. We note this limitation in Appendix VII.A and treat the attenuation monotonicity claim (Lemma VII.4.1) as proved under the same caveat.

VII.5 Federated Revocation

The second load-bearing protocol is the federated revocation mesh. The Anchor Protocol handles revocation inside a single mesh by gossiping an **Append-Only Event Log of Revocation IDs** using Vector Clocks for anti-entropy reconciliation [8]: pairs of daemons periodically reconcile their event logs, designating the cuckoo filter F purely as a local, ephemeral read-path index. The Federated Harbor extends this to the multi-administrative-domain case.

Two pieces are added on top of the Anchor mechanism: (a) per-epoch roots $R_X^{(e)}$ published to a shared witness log so a third party can audit any harbor's local view, and (b) cross-witness signatures so the auditor pattern generalizes Certificate Transparency [31].

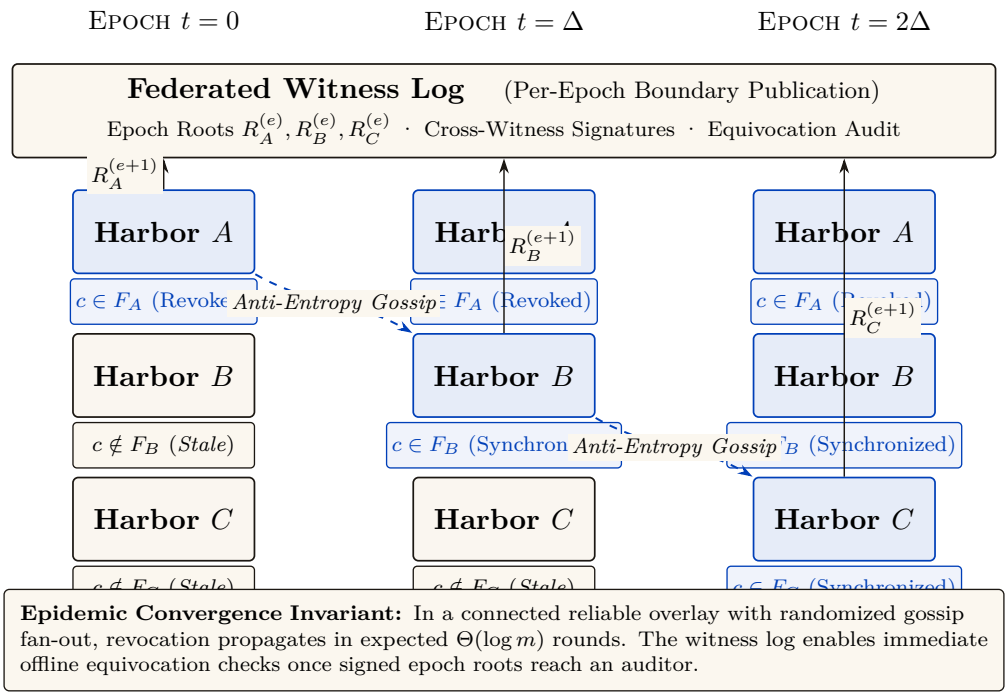


Figure VII.4: Federated revocation propagation across three harbors. The append-only revocation log ensures that revocation events propagate epidemically (Δ-spaced rounds). Per-epoch roots $R_X^{(e)}$ published to the shared witness log provide transparent, non-repudiable audit trails that generalize Certificate Transparency.

VII.5.1 Convergence bound

Standard epidemic dissemination gives an expected $O(\Delta \log m)$ completion time only under an appropriate connected, reliable-round model [8]. The property below instantiates that model at the federation level, treating each harbor's filter as one node in a complete overlay regardless of how many local daemons participate in the harbor's mesh. It is a model-conditional expectation, not an inherited wall-clock deadline.

Property VII.5.1 (Conditional Revocation Dissemination)**.** Let m harbors form a complete overlay. In each reliable synchronous round of duration Δ, informed harbors contact independently uniform peers and exchange revocation deltas. If harbor X publishes revocation c at t_0, then:

1. **Revocation dissemination.** All harbors are informed in expected $\Theta(\log m)$ rounds under the stated epidemic model.
2. **Equivocation verification.** Any auditor possessing two differently signed roots for the same (X, e) detects equivocation in $O(1)$ signature and equality checks.
3. **No hard deadline.** If delivery can be lost indefinitely or the overlay can remain partitioned, no finite worst-case dissemination or common-auditor detection bound exists.

The first item is the standard asymptotic epidemic-dissemination result [8]; this paper does not claim the earlier exact constant $1 + \ln m$. The second follows from the signed-root format. The third is an indistinguishability result: an auditor isolated in one partition cannot distinguish equivocation from delay.

Corollary VII.5.1 (Service-Level Attacker Window)**.** *If an operator additionally imposes a partition bound T_p and a delivery service level T_g, then stale-card exposure is bounded by $T_p + T_g$. Without those operational assumptions the protocol supplies an expected latency under its model, not a structural maximum.*

The distinction is load-bearing. A bond can be priced against an operator-declared service-level window, but an unbounded partition creates an unbounded exposure tail. Deployments must therefore combine bond sizing with expiry, fail-closed acceptance, or an explicit partition limit.

VII.5.2 Conflict resolution across harbors

The interesting case is disagreement. Harbor A has revoked card c; harbor B has not yet seen the revocation; an agent presents c at B for an action that affects A's database. Three candidate resolution rules from the proposal [79], evaluated:

1. **Pessimistic (anyone-revokes-is-revoked).** The card is revoked iff any harbor has revoked it. Defended against by an attacker who would gain by injecting spurious revocations into the gossip mesh; rejects too many honest cards in equilibrium.
2. **Authoritative (only-issuer-revokes).** Only A can revoke A's cards. This is the rule we adopt. Under the service-level assumptions of Corollary VII.5.1, it gives an operator-priced attacker window; without them, exposure has no finite structural maximum.
3. **Epoch-bounded (revocation propagates within Δ epochs and inconsistency is a contract violation).** A refinement of (2) for the case where one harbor falls persistently behind. We treat this as a degraded mode of (2) rather than a separate rule.

We adopt rule (2). The pessimistic rule has the wrong defaults; the epoch-bounded rule is operationally the same as (2) with a watchdog. Adopting (2) means the Conservation invariant of §VII.6 can be stated cleanly.

VII.5.3 Event Log & Ephemeral Cuckoo Discipline

By resolving the bitwise merging impossibility of probabilistic structures, the cuckoo filter is now purely a local, ephemeral read-path index derived from the authoritative Append-Only Event Log. The filter retains local Anchor-side disciplines (load capping, pollution resistance) but cross-harbor

validation is strict: a filter at B that holds $c \in F_B$ where c was revoked at A is correct only if there is a corresponding inclusion proof in $R_A^{(e+1)}$ in the witness log. False positives are corrected by rebuilding the local filter from the event log; spurious revocations injected by an attacker into a peer's log are rejected on first attempt to verify against the witness log. The discipline is: *local filters are advisory read-caches; the event and witness logs are authoritative.*

VII.5.4 Sheaf Laplacian and Abramsky–Brandenburger Contextuality

The global consistency of federated witness logs can be modeled category-theoretically. Let $(X, \leq)$ be the poset of administrative domains with morphisms given by scoped visibility. A presheaf $\mathcal{F}$ over X assigns to each domain U the semilattice of prefix-compatible append-only logs $\mathcal{F}(U)$, with restriction maps given by prefix projection.

Theorem VII.5.1 (Equivocation as Cohomological Obstruction)**.** *Let $\mathcal{U} = \{U_i\}$ be an open cover of the harbor network. A collection of local witness logs $\{s_i \in \mathcal{F}(U_i)\}$ glues into a unique global consensus state $s \in \mathcal{F}(X)$ if and only if the Čech 1-cohomology obstruction vanishes:*

$$[\delta s]_{ij} \;=\; s_i|_{U_i \cap U_j} - s_j|_{U_i \cap U_j} \;=\; 0 \quad \Longleftrightarrow \quad H^1(\mathcal{U}; \mathcal{F}) \;=\; 0. \tag{VII.1}$$

By the Abramsky–Brandenburger contextuality equivalence [180], an equivocation (a harbor signing two mutually incompatible prefixes) is mathematically isomorphic to Bell contextuality in relational semantics: local observations that cannot be factored through any deterministic global hidden state.

Scope (what the theorem does and does not license). Three bindings keep this claim inside its proven regime. **(i) The obstruction lives in the data, not the sheaf:** the detector is the observed disagreement cochain's failure to be a coboundary (its harmonic residual), not $\dim H^1$ of the abstract sheaf, which is a property of the restriction maps alone and is blind to any particular lie. **(ii) Cycles are load-bearing:** cohomology adds power over $O(|E|)$ pairwise digest comparison exactly when an uncompared (partitioned) edge lies on a *cycle*, so the sum-around-the-loop constraint substitutes for the missing comparison; across a *cut* edge there is no loop to close and the method provably sees nothing — redundant gossip paths are a detection requirement, not a luxury. **(iii) Non-vanishing is an alarm; vanishing is not an all-clear:** over real coefficients the obstruction is sufficient but not necessary (the abelianization gap of Carù), so a zero residual licenses no safety conclusion. Detection and localization here *complement* forensic attribution — signatures attribute, cohomology triages under partial visibility — and neither replaces the other. *Status (statistical harness).* The harness behind this scope rider is rebuilt and its pre-registered gates passed (verdict: commit; `sheaf_harness_v2.py`, 200 trials per arm, seed 20260816). The working detector is the least-squares *completion residual* r — the minimum, over global sections and severed blocks, of the disagreement left unexplained on the visible edges — under a three-tier visibility model (**compared** / **relayed** / **severed**): detection requires a cycle *and* relayed reports from the un-checked edge's endpoints. On a relayed cycle edge under partition, 200/200 detections are cohomology-only (pairwise comparison blind); on a cut edge the residual is zero to float epsilon (max 1.5×10^{-13} over 400 trials), as the algebra demands; across a truly severed edge (no reports reach the analyst) equivocation is provably dark — binding (ii)'s topology requirement, now measured. Mutants reintroducing the harness's two prior defects (D1, non-subset restriction maps that collapse the obstruction space; D2, detection scored on hidden-edge data) are each caught by structural self-checks. Binding (iii) stands unchanged: a vanishing residual over real coefficients remains no all-clear (Carù).

Sheaf Laplacian Dynamics. On a cellular network complex, the **Hansen–Ghrist Sheaf Laplacian** [102] is defined as $\Delta_{\mathcal{F}} = \delta^* \delta$. Its harmonic space $\ker(\Delta_{\mathcal{F}}) \cong H^0(X; \mathcal{F})$ is precisely the subspace of globally synchronized ledger states. The spectral gap $\lambda_2(\Delta_{\mathcal{F}})$ (the smallest non-zero eigenvalue of the sheaf Laplacian) governs the diffusion rate of anti-entropy gossip, establishing a rigorous bound on the relaxation time required for the federation to reach global consensus.

VII.6 Cross-Harbor Settlement

The third load-bearing protocol is cross-harbor settlement. A bond is posted at A to cover damage caused by an A-side agent operating at B. When damage occurs, the bond must clear to the damaged party at B. The protocol does this without making either harbor sovereign over the other, but it still uses a third-party custody service. Its transition surface must be non-bypassably restricted; the property below states that assumption rather than relabeling the service as trustless.

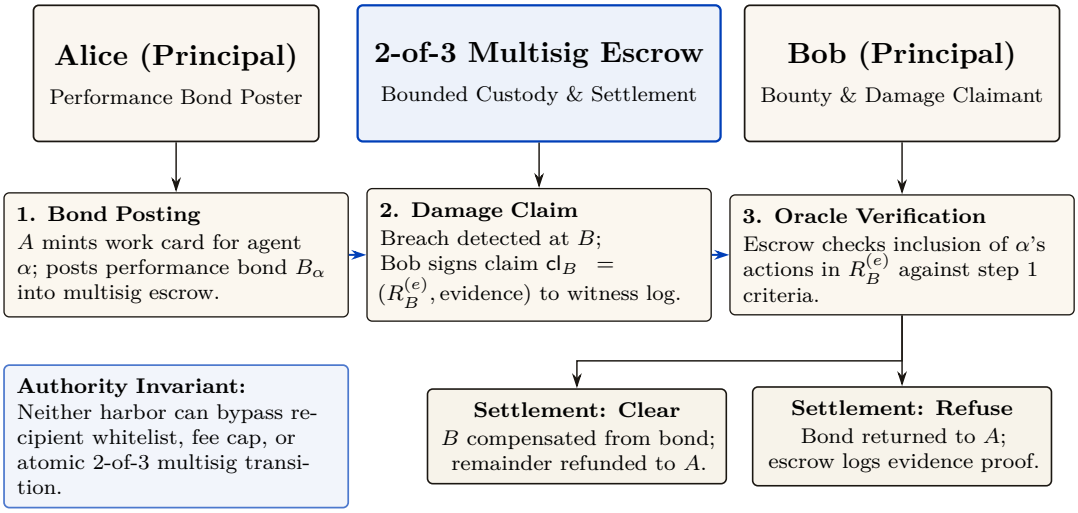


Figure VII.5: Cross-harbor settlement protocol. Custody bounds are structurally enforced: the escrow authority requires 2-of-3 multisig consensus across the recipient whitelist, fee cap, and atomic terminal transitions. The outcome space partitions cleanly into **Clear** or **Refuse**, preventing unbounded cross-harbor liability.

VII.6.1 The bounded escrow

We introduce a third party—the escrow—and require the *custody ledger* to expose exactly two terminal transitions: clear (pay whitelisted beneficiary B, refund the remainder to A) or refuse (return the bond to A). A signed policy alone is only evidence. The following restrictions are structural only if the escrow service cannot bypass the ledger API or alter its code and keys:

- Redirect funds to anyone other than A or B.
- Equivocate about its clearing decision (publish "cleared" to one party and "refused" to another).
- Delay past the TTL of the bond.
- Extract more than a pre-agreed fee ϕ which is capped at issuance.

These restrictions define the desired transition system. The current artifact is a design and bounded model, not a deployed custody mechanism.

Property VII.6.1 (Conditional Escrow Extraction Bound)**.** Assume the custody ledger (i) admits only recipients A and B, (ii) caps the fee output at ϕ, (iii) executes exactly one terminal transition atomically, and (iv) keeps signing and transfer authority outside the escrow operator's bypassable control. Then no allowed transition pays the escrow more than ϕ.

The bound follows by case analysis over the two allowed terminal transitions. Witness logging makes delay or equivocation detectable, and an escrow bond can compensate some misconduct, but neither fact prevents theft if the operator can bypass custody restrictions. Without assumptions (i)–(iv), the worst-case extractable value is the full custodial balance.

The escrow is a trusted third party with a *conditional transition bound*. We are honest that this is not a trustless construction in the HTLC sense [143]. Whether useful trustless settlement is possible for non-fungible reputation-priced bonds remains open (§VII.12); this paper gives neither a construction nor an impossibility proof.

VII.6.2 The 2-of-3 Multisig Clearinghouse

To resolve the regulatory and security liabilities of centralized financial custody (such as Money Services Business / FinCEN registration and honeypot risk) while preserving subjective dispute arbitration, the federation synthesizes non-custodial cryptography with judicial oversight via the **2-of-3 Multisig Clearinghouse**:

- *Key Distribution:* Escrowed collateral is locked in a 2-of-3 threshold signature smart contract (e.g., Base USDC) or Stripe Connect hold. The three keys are held by: (1) Harbor_A, (2) Harbor_B, and (3) the agentsd Federation Arbiter.
- *The Happy Path (Unilateral Peer Clearance):* When the Float Plan completes successfully, Alice and Bob both sign the release transaction (2-of-3). Funds settle peer-to-peer immediately; the central Arbiter never touches custody and pays zero gas.
- *The Dispute Path (Judicial Arbitration):* If Alice and Bob dispute task completion, the Federation Arbiter evaluates the submitted Merkle evidence trail and the rulings of the independent rating judges. If the Arbiter rules in Alice's favor, it co-signs with Alice's key (2-of-3) to force settlement; if it rules for Bob, it co-signs with Bob to execute the bond refund.

This architecture provides the legal and regulatory safety of decentralized, non-custodial settlement while retaining the subjective dispute-resolution authority of a commercial chancery court.

VII.6.3 Cross-harbor conservation

The Bonded Commons paper proved a single-machine Conservation Theorem: the total bonded value in the system never changes by surprise; bonds are only created (by posting), destroyed (by clearing), or rolled (refunded and re-posted). We extend this to the federation.

Property VII.6.2 (Cross-Harbor Bucket Partition)**.** For each funded bond b of amount a_b, let $P_b, E_b, C_b, R_b \in \{0, a_b\}$ denote its posted, in-escrow, cleared, and refunded buckets. Valid transitions preserve

$$P_b + E_b + C_b + R_b = a_b \quad \text{and} \quad \big|\{Z \in \{P_b, E_b, C_b, R_b\} : Z = a_b\}\big| = 1.$$

Summing over bonds gives $\sum_b (P_b + E_b + C_b + R_b) = \sum_b a_b$. External top-ups create new funded bonds and are recorded separately; fees must appear as an explicit cleared recipient rather than disappear from the identity.

This is the federation-layer analogue of Conservation in Bonded. The intra-harbor Conservation invariants hold per-harbor (Bonded gives the proofs); the cross-harbor property adds the constraint that bonds in transit between harbors — i.e., posted at A, locked in escrow, awaiting clearance against B — are accounted for as in-escrow rather than as either posted at A or cleared to B. The escrow's role as the holder of in-flight bonds is what makes the conservation accounting tractable.

VII.6.4 Settlement protocol

Figure VII.5 shows the three-step protocol. To unpack the figure:

1. **Bond post.** Alice's harbor mints a work card for agent α; agent operates at B. Bond $B_\alpha = \$B$ is posted at A and locked in the escrow. The bond's amount and TTL are signed into both A's and B's witness-log roots so the federation knows the bond exists.
2. **Damage claim.** Damage detected at B (acceptance criteria fail, or invariant violation). Bob's harbor signs claim $\mathsf{cl}_B = (R_B^{(e)}, \text{evidence})$ where the evidence is a Merkle inclusion proof against $R_B^{(e)}$. The claim is sent to the escrow.

3. **Verification and clearance.** Escrow verifies inclusion of α's actions in $R_B^{(e)}$, matches against the acceptance preconditions that were signed at step 1, and produces one of two outcomes: *clear* (pay B out of bond, refund remainder to A) or *refuse* (return bond to A, log reason). Both outcomes are recorded in the witness log.

The designed transition is atomic in the bucket-partition sense: a bond moves once from in-escrow to either cleared or refunded. This property depends on the custody-ledger assumptions of Property VII.6.1; witness records alone cannot make two external transfers atomic.

VII.6.5 Fee structure and economic discipline

The escrow's fee ϕ is bounded but non-zero. In the competitive-insurance market sense of Bonded (the Youle contribution in Chapter VI), ϕ is a market clearing price for the service of holding the bond and verifying the inclusion proof. We do not solve the market here; the Youle contribution from Bonded gives the mechanism, and we conjecture (open) that it extends cleanly to the cross-harbor case when the escrow population is large enough.

We are explicit that for small federations (two or three harbors), ϕ is likely set by a posted price rather than auctioned. The mechanism design here is identical in shape to Bonded's, just lifted to the federation layer.

VII.7 Federation Admission

The Federated Harbor is not a permissionless market in identities. A new harbor enters the federation through an admission ceremony. The ceremony is invitation-bounded by design: Sybil attacks against the federation [116] are costly for an attacker precisely because each identity requires a real admission event.

VII.7.1 The bonded-sponsor pattern

A new harbor N with no prior reputation cannot price its own bond fairly under competitive insurance — the market lacks data on N's risk. The bonded-sponsor pattern resolves this:

1. An existing harbor S (the sponsor) posts a bond on N's behalf, sized to cover the federation's worst-case loss from N misbehaving during a probationary epoch.
2. N joins the federation under probation; its cards are accepted, its evidence trail is witness-logged, but its sponsor bond is at risk.
3. If N misbehaves during probation — forges signatures, equivocates on roots, refuses to honor settled bonds — the sponsor bond is forfeit. The sponsor has full incentive to vet N before sponsoring.
4. At the end of probation, N's own reputation is sufficient to price its own bond, and the sponsor bond is released.

This is the federation-layer analogue of competitive insurance from Bonded (the Youle contribution in Chapter VI): instead of an authority picking who joins, the sponsor market discovers it. The cost of being a sponsor is real — forfeit risk during probation — and the benefit is the network effect of having N in the federation. We expect (open) that for federations with many candidate sponsors and clear-eyed risk pricing, the equilibrium is reasonably efficient. We do not prove it.

VII.7.2 Cold-start failure modes

A new harbor that cannot find a sponsor cannot join. This is a real limitation. In the worst case, the federation collapses to invitation-only — which is the failure mode of every interesting federated identity system from PGP web-of-trust [162] to SAML [82]. The structural pushback is:

- *Many sponsors.* The market is many-to-many; a new harbor needs only one sponsor. As long as sponsors are paid (via fees or reputation), the supply is nonzero.

- *Small probationary scope.* New harbors during probation are restricted from large-stake operations; the sponsor's risk is bounded.
- *Witness-log permanence.* The admission record is permanent; once a harbor has graduated probation, it does not need a sponsor again.

We are honest that this is engineering discipline, not theorem. The historical evidence on whether invitation-bounded federations stay open is mixed [82]; the Federated Harbor's pushback against centralization is to make the cost of being a centralizing hub bounded above by the cost of running mirrored witness logs, which is small. Whether that pushback is enough is empirical.

VII.8 Worked Example

The four primitives are easier to read against a concrete case. We work the example end-to-end: two organizations, three machines, one shared project. The example is intentionally familiar; nothing in it is novel except the way the primitives compose.

VII.8.1 Setup

Two organizations: Acme Robotics (Alice's employer) and Beta Vision (Bob's employer). The shared project is a perception-and-control stack: Acme owns the controller, Beta owns the vision system, both run against a shared staging environment that lives on a third machine maintained by neither.

Three machines: Alice's laptop (harbor A, controller code, controller test database), Bob's laptop (harbor B, vision code, vision test database), staging-server (harbor S, shared staging database, no agents resident).

Each harbor has gone through the admission ceremony of §VII.7. Acme's harbor was sponsored by an existing partner harbor at the open-source consortium that hosts the staging server; Beta's was sponsored by Acme after a six-month probationary period during which Acme's risk officer watched Beta's evidence trail closely. The staging server harbor is sponsored by the consortium directly and is treated as a known-honest verifier for the purposes of this example.

VII.8.2 The agent task

Acme's agent — call it Controller-7 — needs to run a schema migration on the staging database to add a column for a new sensor type. The migration is destructive (drops and recreates the column); if it goes wrong, the staging database is unusable for the joint demo scheduled three hours later.

Controller-7's local card C_A authorizes `db:migrate` against the controller test database, attenuated through Acme's risk policy (att_A = "only against tables in the `controller` schema; only between 9am and 5pm; only if a human operator has approved within the last 30 minutes"). The card is valid at A, but the database it needs to migrate is at S.

VII.8.3 Step 1: Cross-harbor transfer

Controller-7 invokes $\mathsf{xfer\_req}(C_A, S)$ on D_A. D_A constructs the envelope:

$$\mathsf{xfer\_offer}\langle C_A, \text{att}_A, R_A^{(e)}\rangle \text{ signed under } sk_A$$

and sends it to D_S (the staging-server harbor). D_S verifies the signature against A's public key as recorded in the witness log, verifies that $R_A^{(e)}$ is the current epoch root for A, and applies its own attenuation att_S ("only the `controller` schema; only after a backup snapshot has been taken; only if no other migration is in flight"). The composed attenuation produces $C'_S = \text{att}_S(\text{att}_A(C_A))$, a card valid only at S, signed under sk_S.

D_S delivers C'_S to Controller-7. Controller-7 takes the snapshot, then runs the migration against S. The migration's actions are recorded in S's Merkle forest and roll up into $R_S^{(e)}$.

VII.8.4 Step 2: Damage detection

The migration runs cleanly. Two hours later — well after the agent has moved on — Beta's vision system runs an integration test against the staging database and observes that the new column has the wrong type (Acme assumed `float32`; Beta needs `float64` for the sensor data they will produce). The integration test fails.

D_B (Beta's harbor) verifies the failure against its local invariants and produces a damage claim:

$$\mathsf{cl}_B = (R_B^{(e+1)}, \text{evidence})$$

where the evidence is a Merkle proof that Beta's integration tests against the staging database failed because of a schema mismatch. The claim is sent to the escrow.

VII.8.5 Step 3: Settlement

Acme posted a bond when Controller-7 was authorized to run the migration. The bond is held in the federation escrow with a TTL of 24 hours — long enough for downstream consequences to surface, short enough not to lock collateral indefinitely.

The escrow verifies cl_B: it checks the inclusion proof against $R_B^{(e+1)}$ in the witness log, matches the claim against the acceptance preconditions signed at bond-post time (which named the staging schema as a covered surface), and decides: *clear*. Acme's bond is debited; Beta receives the cleanup-cost portion; the remainder is refunded to Acme. The settlement record is written to the witness log.

VII.8.6 Step 4: Revocation propagation

At this point, Acme's risk officer revokes Controller-7's `db:migrate` card and publishes $R_A^{(e+2)}$. Under Property VII.5.1's connected reliable-round model, dissemination completes in expected $\Theta(\Delta \log m)$ time. During a partition, receiving harbors must rely on expiry or a fail-closed freshness policy; the example does not pretend gossip supplies a deadline.

VII.8.7 What this example shows

The example is small and the failure is real. The point of working it is to show that each of the four primitives — transfer, revocation, settlement, admission — is doing its job at exactly one step, and that the composition is straightforward when each primitive is honest about what it covers.

Notice in particular what *did not* happen:

- Neither D_A nor D_B trusted the other's signing key as a root. The transfer worked through the witness-logged public keys, not through a chain of trust.
- The bond was not held by either harbor; the design requires a recipient-whitelisted custody ledger before the escrow's role is structurally bounded.
- The settlement was not negotiated peer-to-peer; it followed a fixed protocol whose two outcomes were named in advance.
- Under the example's connected reliable-round assumptions, the revocation propagated through gossip with expected logarithmic completion; no consensus protocol was invoked.

Each of these is a deliberate design choice. The Federated Harbor's architecture is not heroically more powerful than alternatives; it is heroically more honest about what is happening at each step.

VII.9 Formal Model

The formal substrate of this paper is two-layer: ProVerif models for the cryptographic protocols (cross-realm authenticity, attenuation, settlement no-redirect) and TLA+ extensions of Bonded's Conservation specification for the cross-harbor accounting invariants.

VII.9.1 ProVerif extensions to Anchor

The Anchor Protocol's three-phase models are extended with a second signing key and an epoch-root binding. The cross-realm authenticity query becomes:

```
query b: id, ha: id, hb: id, cap: capability, e: epoch;
  event(AcceptedAtB(b, hb, cap, e))
  ==> (event(IssuedAtA(b, ha, cap)) &&
       event(EnvelopeAtoB(ha, hb, cap, e)) &&
       event(RootCurrent(ha, e))).
```

Listing VII.1: Cross-realm authenticity query (sketch)

This says: if B accepts a cross-realm card for capability `cap` at epoch e, then there exists a corresponding issuance event at A, an envelope event from A to B binding the same capability and epoch, and an event confirming that $R_A^{(e)}$ is the current root. The query holds against the Dolev–Yao attacker model under the standard assumption of Ed25519 EUF-CMA security.

We are honest that the model is drafted and the proof script is in progress. The full mechanization is named open work; see Appendix VII.A for the status table.

VII.9.2 TLA+ extension for cross-harbor conservation

The Bonded Commons paper's Conservation specification models a single-machine bond pool. The Federated Harbor extension adds a per-harbor pool, an escrow pool, and an inter-harbor transit relation:

```
CONSTANTS Harbors, MaxBonds, Capabilities
VARIABLES Bucket, WitnessLog

TypeOK ==
  /\ Bucket \in [1..MaxBonds -> {"posted", "escrow", "cleared", "
     refunded"}]
  /\ WitnessLog \in Seq([harbor: Harbors, epoch: Nat, root: STRING])

BucketPartition ==
  \A b \in 1..MaxBonds:
    \E! s \in {"posted", "escrow", "cleared", "refunded"}:
      Bucket[b] = s

Spec == Init /\ [][Next]_<<Bucket, WitnessLog>>
```

Listing VII.2: Cross-Harbor Conservation specification (sketch)

The full specification (Appendix VII.C) associates each bond with an amount and enforces amount-preserving moves through these buckets. Apalache checks the resulting transition system at $|F| = 4$ and bond depth 16. This establishes the invariant only for the model and bound; it does not establish the custody-ledger assumptions or arbitrary-scale runtime conformance.

VII.9.3 What is mechanized and what is structural

We are explicit about which claims live in which band (Figure VII.2).

Mechanized (model scope). Single-harbor capability authenticity is inherited from Anchor. Cross-realm authenticity and attenuation are draft models. The cross-harbor bucket-partition invariant is model-checked at $|F| \leq 4$.

Conditional (assumption scope). Revocation dissemination is expected $\Theta(\Delta \log m)$ only under the stated reliable connected-overlay model (Property VII.5.1). The escrow extraction bound is conditional on a non-bypassable recipient-whitelisted custody ledger (Property VII.6.1).

Open (cobalt). Multi-principal correlated-equilibrium extension of Bonded; trustless settlement for non-fungible reputation-priced bonds; folk-theorem cartel-resistance bound at the federation layer.

VII.10 Failure Modes

A federation paper that does not engage formally with the historical failure modes of federated systems is unserious. The literature is heavy with examples: SAML federations that concentrated on a small set of identity providers [82]; PGP web-of-trust that never achieved meaningful adoption outside enthusiast circles [11]; Sigstore's gravitation toward a single deployment of Fulcio [208]. We engage with three failure modes specifically.

VII.10.1 Centralization gravity

Federated identity systems have a well-documented tendency to centralize on a small number of trust anchors within five to ten years of deployment [82]. The Federated Harbor's structural pushback is the replicable witness log: any harbor can run its own witness instance, any harbor can audit any other instance's roots, and equivocation between witnesses is detectable. This makes it more expensive to be a centralizing hub than to be a peer.

The pushback is not theorem; it is design discipline. We are honest about this in Figure VII.2 (cobalt band). The historical evidence is mixed. Tailscale [25] has remained operationally centralized on its own coordination server despite Headscale [121] being a viable alternative; in-toto [182] has stayed decentralized but is consumed mostly by SLSA [129] which has Google as its de facto center. Time will tell which pattern the Federated Harbor exhibits.

VII.10.2 Equivocation by a malicious witness

A malicious witness log can serve different views to partitioned auditors. A contradictory signed pair is immediately verifiable once co-located, but the protocol has no topology-independent time bound for bringing the pair together. Witness bonding applies after confirmation.

For high-stakes settlements, harbors can require multiple witness signatures before treating a root as authoritative, in the Certificate Transparency two-SCT pattern [31]. This trades latency for assurance; the design space is well-explored in the CT literature and we adopt the same techniques.

VII.10.3 Partition tolerance

The federation does not run consensus; partition tolerance is per-protocol. Capability transfer cannot proceed during a partition between A and B (no surprise; cross-realm authentication requires both daemons to reach the witness log). Revocation gossip continues within each partition; convergence is bounded by the partition's local size, not the global federation size. Settlement is parked at the escrow until the partition heals.

The structural posture is that partitions are operationally common and the federation should degrade gracefully rather than fail fast. We inherit this discipline from Anchor's intra-mesh design [76]; the federation extension is straightforward but adds no novel mathematics.

VII.11 Related Work

The Federated Harbor sits at the intersection of four bodies of work. We are concrete about where we draw on each and where we depart.

Federated identity. The OpenID Federation 1.0 specification [176] is the closest existing analogue: each entity publishes a signed Entity Statement, trust chains assemble by walking up to a Trust Anchor, trust marks express property claims. The Federated Harbor is structurally similar in the trust-anchor pattern but differs in two places: (a) we add a capability/attenuation layer (OpenID Federation is identity-only), and (b) we do not assume HTTP-based fetching of entity statements, since Port Daddy daemons cannot reliably expose HTTP endpoints (NAT, sleep, etc.). We draw also on SAML/Shibboleth historical experience [82] as cautionary lineage; the centralization gravity we discuss in §VII.10 is exactly the SAML-era failure mode.

Capability-based federation. Macaroons [19] are the foundational capability-with-attenuation construction; our transfer ceremony extends Macaroon-style third-party-caveat chains across an administrative boundary. UCAN [197] is the current production-deployed answer to "Macaroons with public-key principals," and we treat it as the operationally most relevant existing system; the Federated Harbor differs in keeping the daemon as a structural commons authority rather than going fully peer-to-peer, and in pricing delegation through insurance-priced bonds rather than pure capability constraints. The object-capability lineage [141, 101] continues to be relevant: cross-realm operations are exactly a mutual-suspicion scenario in Miller's vocabulary, and his defensive-consistency framework is the right lens.

Transparency logs. Certificate Transparency [31] is the model; the federated witness log of §VII.2.3 is CT applied to capability tokens. Sigstore [208] and Rekor are the production instantiation we rhyme with most heavily; in-toto [182] and SLSA [129] are the supply-chain analogues that compose with the Federated Harbor for the "poisoned plugin under a bonded principal" case but do not close it. The Crosby–Wallach history-tree [183] is the formal substrate of every modern transparency log including ours.

Cross-domain settlement. Herlihy's atomic cross-chain swap [143] is a comparison point; the cross-harbor settlement protocol of §VII.6 instead handles non-fungible reputation-priced bonds through conditionally restricted custody. It is not trustless in Herlihy's sense. Whether trustless settlement is achievable in our setting is open (§VII.12). The IBC protocol [50] supplies another connection/channel comparison; unlike IBC, this design does not assume a consensus-backed state machine underneath.

Agentic-commerce discovery and mandates. The Universal Commerce Protocol [199] (Google/Shopify, 2026) solved cross-administrative discovery for agent-mediated retail with a pattern the Federated Harbor adopts (ADR-0051 Phase 1b): a `/.well-known` JSON profile that does double duty as capability declaration and JWK key publication, with server-selects version negotiation — the party that must execute a request computes the compatibility intersection. Where UCP assumes merchants hold stable HTTPS origins, our daemons often do not (the same constraint that ruled out OpenID Federation's entity-statement fetching above); the profile therefore also travels as a relay-published mirror. UCP delegates authorization proof to the Agent Payments Protocol's verifiable-credential mandates [4], whose SD-JWT/JWS/JCS formats the harbor boundary now speaks (ADR-0094). The division of labor is instructive: those protocols standardize discovery and the transaction envelope and are explicit that counterparty trustworthiness, collateral, and settlement enforcement are out of scope — which is the half of the problem this paper and its predecessors supply.

VII.12 Limitations and Open Questions

We close with the honest open frontier. These are not throwaway caveats; they are the questions the paper does not answer, named precisely enough to motivate subsequent work.

VII.12.1 The multi-principal correlated-equilibrium extension

The Bonded Commons paper argued that single-machine coordination is a correlated equilibrium with the daemon as the correlation device [78]. Across two harbors with two daemons, the correlation device fragments. Two candidate resolutions remain unresolved:

- The federated witness log *is* the correlation device, and individual harbors are local enactors. This would extend Bonded's result intact.
- The cross-harbor case is a partition of the correlated-equilibrium concept into intra-harbor (correlated) and inter-harbor (Nash with shared signal). This would weaken Bonded's result at the federation layer.

We do not know which is right. The proposal that scoped this paper [79] named Thomas Youle as the load-bearing collaborator for this question (his contribution to Bonded is the closest existing work). The question stays open here.

VII.12.2 Trustless settlement for non-fungible bonds

HTLC-style atomic swaps [143] achieve trustless settlement for fungible assets with a shared hash preimage. The Federated Harbor's bonds are non-fungible and adjudication-dependent, so that construction does not transfer directly. We adopt conditionally restricted custody (§VII.6) and leave both a trustless construction and an impossibility result open. Either would require a separate threat model and proof.

VII.12.3 Cartel formation across federations

Bonded's folk-theorem cartel-resistance argument applies inside a single machine. Between machines, cartel formation is structurally easier: colluders can coordinate offline and present a unified front to the witness log, which records that all harbors agreed — a behavior consistent with both honest agreement and collusion. We owe either a strengthening of the bonded mechanism that resists cross-harbor cartels, or a precise statement of the weakening. We have neither.

VII.12.4 Cold-start admission and the invitation-only failure mode

The bonded-sponsor pattern of §VII.7 requires a new harbor to find a sponsor willing to underwrite its probationary risk. In practice, this market may be thin — the historical evidence on whether invitation-bounded federations stay open is mixed [82]. We treat the structural pushback (many sponsors, small probationary scope, witness-log permanence) as engineering discipline, not as theorem.

VII.12.5 Equivocation propagation speed

Property VII.5.1 separates cryptographic verification from dissemination. The former is constant work once both roots are present; the latter depends on topology, loss, and partitions. Establishing a deployment-specific tail bound is open and is necessary before pricing the witness bond from propagation risk.

VII.12.6 What this means for the reader

A federation paper with five named open questions is doing its job. The point of the paper is to draw the new boundary cleanly so the open problems are visible; if every claim in this paper were

closed, there would be no third paper to write. The honest read is that the Federated Harbor is closer to a research agenda than to a deployed product, and the prior two papers (Anchor, Bonded) have a tighter ratio of theorem to conjecture for the same reason. The substrate is real; the federation layer is in progress.

> *The point of the paper is to draw the new boundary cleanly so that the unsolved problems are visible rather than hidden.*

VII.13 Conclusion

The two-machine problem is the structural sequel to the local-swarm problem of Anchor and the trust-as-infrastructure argument of Bonded. The proposed answer is not a new substrate but a clean extension: keep each daemon sovereign inside its own machine, add a shared witness log that no harbor controls, hang revocation gossip and conditionally restricted custody off the witness log, and gate federation admission on bonded sponsorship rather than permissionless entry. Each primitive does exactly one job. Each primitive is honest about what it does not do.

The proof posture matches the prior papers' discipline. Where we mechanize, we mechanize (with the same ProVerif and TLA+ infrastructure). Where we bound a threat, we name the bound and price the bond against it. Where we leave a question open, we name it. The threat-band diagram of Figure VII.2 is the most concentrated summary of where each claim lives.

What this paper supplies is a clean federation surface that could compose with the local substrate: a transfer ceremony, witnessed revocation records, a conditional custody design, and bonded admission. The protocol and models make the missing implementation and service-level assumptions explicit; they do not establish that arbitrary harbors can already interoperate safely.

The remaining work—runtime custody enforcement, deployment conformance, a multi-principal equilibrium model, and a cartel-resistance bound—is material. The local substrate is real; the federation described here remains a design with partial mechanization.

Chapter Appendices

VII.A Verification Status

Artifact registry. The cross-paper status table in Bonded Commons [78] carries the items that are shared across the series; this appendix lists only the items specific to the Federated Harbor.

Table VII.1: Federated Harbor verification status. **closed**: model verified and runtime conformance test passes. PARTIAL: design specified, mechanization in draft. *open*: known unmechanized; threat band visible in Figure VII.2 (cobalt).

Claim	Method	Result	Artifact
Single-harbor capability authenticity (inherited)	ProVerif 2.05	**closed**	Anchor `harbor_card_v*.pv`
Cross-realm authenticity (§VII.4)	ProVerif (draft)	PARTIAL pending witness-log freshness assumption	`fh-xfer.pv` (draft)
Attenuation monotonicity across boundary (Lem. VII.4.1)	ProVerif + Anchor sub-result	PARTIAL	`fh-att.pv` (draft)
Federated revocation dissemination (Prop. VII.5.1)	Conditional epidemic model [8]	PARTIAL expected asymptotic result; no partition deadline	`fh-conv.tex` (proof sketch)
Cross-witness equivocation evidence	Signed-root comparison; dissemination model	PARTIAL verification immediate once views meet; no hard delivery bound	—
Escrow extraction bound (Prop. VII.6.1)	Transition-system case analysis	PARTIAL conditional on non-bypassable custody; no runtime conformance	`fh-escrow.pv` (draft)
Cross-harbor Conservation (§VII.9.2)	TLA+ / Apalache, $\|F\| \leq 4$	PARTIAL model-checked at scale	`CrossHarbor Conservation.tla`
Trustless settlement for non-fungible bonds	—	*open* no construction or impossibility proof	—
Multi-principal correlated-equilibrium extension	—	*open*	—
Cross-federation cartel-resistance bound	—	*open*	—

What this table means. Every row in the PARTIAL band is committed to and on the active workplan. Every row in the *open* band is a research question we cannot resolve from the desk; the proposal [79] names collaborators whose contributions would be load-bearing. We err toward listing fewer items as closed rather than more, in the same posture as Anchor and Bonded.

VII.B ProVerif Models (draft)

For brevity we omit the full model text in this pre-print; the draft `fh-xfer.pv` extends Anchor's `harbor_card_v3_delegation.pv` with a second signing key sk_B, an epoch root binding, and a cross-realm authenticity query of the shape sketched in §VII.9.1. The full model will accompany the revised version once the witness-log freshness assumption is mechanized rather than assumed.

VII.C TLA+ Specification (draft)

The full `CrossHarborConservation.tla` extends Bonded's Conservation specification with per-harbor pools, an escrow pool, and an inter-harbor transit relation. Apalache symbolic model-checking at $|F| = 4$ harbors and bond depth 16 establishes that Conservation holds across all reachable states under the transition relation. The full specification accompanies the revised version.

VII.D Limitations & Boundaries

The formal mechanisms presented in this chapter are mathematically sound within their defined scopes, but they depend on bounding assumptions that must be explicitly acknowledged.

- **Threat Model Exclusions:** Collusion across all independent organs (e.g., the logging daemon, the arbitrator, and the primary execution runtime) is assumed out-of-scope; if all enforcing components are compromised, the guarantees degrade to advisory.
- **Performance Trade-offs:** Cryptographic assertions and WAL-checkpointing for per-write durability incur measurable I/O overhead. These mechanisms are designed for high-stakes coordination, not for bulk data processing or high-frequency trading.
- **Implementation Maturity:** While the core protocols (Anchor, SQLite single-writer) are IMPLEMENTED and running in production, surrounding ecosystem components (like the decentralized judge markets or fully federated multi-sig routing) remain in PARTIAL or DESIGNED phases.

These constraints are not flaws but structural realities of building zero-trust coordination on top of POSIX primitives.

Volume Appendices

A Consolidated Implementation and Assurance Ledger

A.1 The Five Explicit Assurance Levels

To avoid overclaiming systems guarantees, Port Daddy structures all enforcement into five discrete, falsifiable assurance modes:

1. **Level 1: Observed.** Effects may occur outside the daemon; evidence is best-effort event-logging.
2. **Level 2: Coordinated.** Cooperative clients coordinate explicitly via advisory claims, role leases, and registered commitments.
3. **Level 3: Brokered.** Sensitive credentials and selected effect channels pass through daemon-owned proxy brokers.
4. **Level 4: Confined.** Confined execution via OS-level primitives (Seatbelt, bubblewrap, Landlock, namespaces, cgroups) with no ambient bypass path.
5. **Level 5: Attested.** Hardware/TEE-attested confidential workrooms verifying runtime measurement and cryptographic key release.

A.2 The Six-Rung Risk Ladder

Defenses must not ask one mechanism to perform multiple contradictory roles. Security and economic guarantees are staged sequentially:

PREVENT $\longrightarrow$ CONTAIN $\longrightarrow$ CHECKPOINT $\longrightarrow$ OBSERVE $\longrightarrow$ ADJUDICATE $\longrightarrow$ COMPENSATI

Residual risk moves down the ladder, and the underwriting market prices only what remains after structural containment.

A.3 Consolidated Mechanism Status

Mechanism or claim	Grade	Current evidence and boundary
Single-writer SQLite/WAL kernel, claims, sessions, messaging	**implemented**	Running daemon paths, linear migration ledger (`PRAGMA user_version`), and regression tests; durability is per-transaction typed (`PROCESS_CRASH` vs `POWER_LOSS`).
Legibility and digest-with-zoom surfaces	PARTIAL	Provenance-preserving projections from raw event ledgers; why/where/why-not query substrate.
Durable commitments and obligation monitor	PARTIAL	Durable records, daemon-derived deadlines, oracle-bound closure, and periodic overdue detection.
Portable Task Capsule and successor restoration	SPECIFIED	Content-addressed task capsule over JSON message arrays and Git SHA; provider-portable semantic resurrection.
Local non-forgeable actor identity	PARTIAL	Daemon-minted identity bound to principal roots with aggregate exposure limits.
Anchor authentication and attenuation models	**closed**	ProVerif and Kani verification of every-hop monotonic attenuation ($\text{cap}_i \subseteq \text{cap}_{i-1}$) and key-bound proof of possession.
Bond ledger, escrow, and conservation slices	PARTIAL	4-bucket settlement escrow (Bounty, Performance Bond, Insurer Reserve, Fees) with verified algebraic balance conservation.
Cross-harbor capability transfer and settlement	SPECIFIED	2-of-3 Multisig Clearinghouse and witnessed anti-entropy overlay; non-custodial dispute arbitration.

B One Spine: Why These Mechanisms and Not Others

A volume this size courts a specific failure: the *potpourri*, a grab bag of clever mechanisms with no account of why these and not ten others. This appendix states the account. The system has one ontology and four theorem families; every mechanism in the volume is a corollary of exactly one family, inherits a developer obligation from it, and carries that family's boundary. A proposed feature that cannot name its row in the table below is not yet a feature.

The ontology. One durable object bears all trust: the **evidence-carrying work unit** — authorized by a *principal* (the delegation chain's terminus), acted on through *fenced, attenuating capabilities*, advanced only by *admitted evidence* over typed channels, settled once, and graded on the five assurance modes (observed, coordinated, brokered, confined, attested). Agents are transient; the work unit is the case file that outlives every clerk. Everything else in the volume exists to answer one of four questions about that object.

The four families.

Theorem family (the question)	**Mechanisms it licenses (corollaries)**	**Developer implication + boundary**
Legibility floors. What must a human see? Information floor $\log_2 \binom{N}{k} - \log_2 \binom{m}{k}$; split-digest comonotonicity; derived regret head; the $R(\delta, f)$ frontier with adaptive zoom; paging competitiveness.	Digests with budgeted bits; *dual* compaction heads (successor vs. operator); the attention queue's likelihood-ratio test; forced-zoom; context paging; pheromone decay.	Budget bits/opens/misses explicitly — the exchange rates are theorems, not tuning. Two readers need two heads. Anomaly must be a *calibrated* posterior; reputation may move it but never zero a catastrophic item. Boundary: zoom pays only when flags are sparse; floors price *guarantees*, not vibes.
The controllability boundary. What can be *prevented* rather than detected? Regimentable iff controllable w.r.t. the uncontrollable alphabet (OP-2 is the DB-commit special case).	Pre-commit gates and SQL guards; the compute-to-data airlock; conflict rejection at proposal time; ticket-lock admission; the channel-not-token rule.	Classify every rule at design time: regimentable rules become gates; semantic rules (a confident lie) are detect-only *forever* and must be priced by the economy, not promised by the runtime. Boundary: gate the channel, never the token; taint is containment, the gate is the guarantee.
Deterrence economics. Why is honesty cheaper than fraud? $\rho dB \geq G$; the audit tower over C disjoint cliques; the front-loaded probation cliff; declining audit schedules (reputation as amortized verification); four-bucket escrow conservation.	Slashable bonds; sealed VRF-sampled audits; heterogeneous judge pools; newcomer cliffs instead of ramps; per-bucket settlement; the 2-of-3 clearinghouse; the mutation-test market.	Size bonds to the worst one-shot gain; sample judges across principals *and* model families (diversity is the mechanism, not a nicety); publish the audit schedule — deterrence is a budget line. Boundary: sealed sampling is load-bearing; a leaked draw defeats the clique math; convention (confiscation vs. keep-gain) changes $\rho^\star$.
Federation without consensus. What survives distrust between harbors? Signed witness logs; equivocation as a cocycle obstruction *on cycles*; monotone (CALM) coordination-free operations; the conditional escrow bound.	Gossip with *redundant* paths; sheaf-residual triage under partition; forensic attribution by signature; the multisig clearinghouse.	Topology is a security parameter: detection across an uncompared link requires a cycle through it — provision redundancy as you would a bond. Signatures attribute; residuals only triage; vanishing residual is never an all-clear. Boundary: no global order is claimed, and none is needed.

The test, restated as procedure. To admit a new mechanism: (1) name its family and the theorem it is a corollary of; (2) state the developer obligation it creates (a budget, a classification, a schedule, or a topology requirement); (3) state its boundary in the same breath as its claim, with the maturity tag the ledger requires. The eight consensus corrections, the assurance ladder, and the status tags throughout this volume are this procedure applied retroactively; this appendix is the commitment to apply it prospectively.

C Research and engineering roadmap

The roadmap is organized by the kind of evidence that can close each claim. A proof obligation should not be disguised as a feature, and an empirical parameter should not be invented in prose.

C.1 Add to the papers

1. Add a notation and assumption concordance for identity roots, principals, harbors, clocks, rounds, currencies, and oracle trust.

2. Add explicit claim-to-artifact matrices linking every closed or partial claim to a model, test, route, trace, or counterexample.

3. Add deployment-calibrated case studies for attention, obligation breach, identity reset, partition healing, and settlement dispute.

4. Add a comparative chapter mapping the stack to object capabilities, reference monitors, transparency logs, workflow engines, and agent frameworks.

C.2 Prove or falsify

1. State complete strategies, deviations, information sets, and parameter regions for the signaling, cartel, sponsor, judge, and correlated-equilibrium claims; include coalition deviations where relevant.

2. Mechanize revocation dissemination only under an explicit graph, scheduler, delivery, and partition model; report expectations and tails separately.

3. Prove the multi-currency bucket-partition invariant and characterize the minimal non-bypassable custody interface required for any extraction bound.

4. Establish model-to-runtime conformance obligations for token verification, replay rejection, identity minting, settlement, and cross-harbor envelopes.

5. Define the game class before stating any price-of-anarchy or welfare bound, then supply tight examples or counterexamples.

C.3 Try, measure, and add to the code

1. Finish a reputation-grade witnessed-outcome ledger on the shipped commitment substrate, with neutral-judge conflicts, appeals, and sanctions.

2. Close actor identity at every security-relevant write boundary and test identity-reset laundering against shared budgets and bonds.

3. Implement portable execution-state checkpoint/successor restoration where adapters permit it; grade unsupported adapters formally.

4. Prototype HPKE-sealed relay ingress with pinned key rotation, replay windows, and downgrade tests.

5. Build a small federated witness-log, revocation, custody, and settlement testbed; run partitions, redelivery, crash recovery, equivocation, and stale-key chaos experiments.

6. Instrument false alarms, misses, review latency, breach rates, detection probability, slashing, judge disagreement, and identity churn before fitting the attention or incentive parameters used by the papers.

7. Make roadmap and other operator projections prove which database and revision they summarize, then gate publication on projection/source agreement.

D Notation and cross-chapter concordance

Term	Volume-wide meaning
Harbor	One local administrative and transactional authority, normally one daemon and its durable database.
Principal	The human or organization whose policy and resources an agent ultimately exercises; not interchangeable with a process identity.
Role	A bundle of obligations, capabilities, and authority that a spawn may fill.
Person	A role instance plus continuity and a witnessed outcome history bound to an identity that cannot be freely re-picked.
Oracle	An external or independently checkable reference used to close a commitment or adjudicate an outcome; its trust model must be named.
Round	A model step in a dissemination or protocol analysis, not automatically a wall-clock deadline.
Closed	The exact recorded model/property has been mechanically discharged; no whole-system implication is assumed without a conformance argument.

E Next-Generation Expansions

proposed — every mechanism below is a design sketch, not a system in this volume's Consolidated Mechanism Status table (Appendix A). The present-tense engineering language that follows describes an intended design, not observed behavior; per the claim discipline stated in the introduction, none of it should be read as a running guarantee until it earns a grade of its own. The immediate scope of Port Daddy coordination focuses on local single-writer consistency and cross-harbor token exchange. We outline six structural expansions that would be necessary to advance the system from a single-tenant agent hypervisor to a multi-tenant, enterprise-grade coordination protocol.

E.1 B2B Airlock (Information Flow Control)

True compute-to-data execution requires moving the agent to the data rather than moving the data to the agent. A B2B Airlock extends the daemon to enforce strict Information Flow Control (IFC) via Dynamic Taint Tracking. Data loaded from a high-security zone taints the spawned process; the daemon subsequently applies strict `net:egress` denial, ensuring that the agent can compute over sensitive IP but can only exfiltrate cryptographic attestations of the result. This forms the foundation of a "Silicon-Enforced NDA."

E.2 The Compaction Daemon (SLM Sidecar)

Unbounded context windows lead to attention decay and excessive cost. The Compaction Daemon integrates an embedded, air-gapped Small Language Model (e.g., via `llama.cpp`) acting as an SLM sidecar. This sidecar continuously performs zero-cost background token compaction and generates Digest-with-Zoom hierarchical summaries. Furthermore, as an air-gapped component, it acts as the primary enforcement layer for Data Loss Prevention (DLP) scanning on outbound context windows.

E.3 Swarm Time Travel

Because `agentsd` manages state via an event-sourced SQLite WAL and file modifications are versioned via Git trees, the swarm's entire macroscopic state is deterministic. We propose the `agentsctl rollback` command to leverage this: by rewinding the WAL to a specific transaction boundary and checking out the corresponding Git commit, the swarm can physically "wake up in the past" with perfect neural and file-system continuity, allowing operators to seamlessly branch execution away from cascading hallucinations.

E.4 Agentic FinOps (Task-Level Cost Attribution)

High-cadence swarms require hard structural cost boundaries. Agentic FinOps shifts billing from a post-hoc dashboard to a pre-transactional firewall. The `agentsd` daemon intercepts all API usage blocks and commits micro-transactions to the primary ledger. It enforces a `max_usd` parameter at the task level, cutting off network egress and freezing the spawn the millisecond its budget is exhausted, preventing runaway LLM billing.

E.5 Darwinian Prompts (Outcome Skill-Attribution)

A static prompt library cannot adapt to an evolving codebase. We propose a feedback loop where the settlement outcome of a Float Plan dictates the survivability of the prompts and tools used. If a task settles successfully, the daemon grants an Elo rating boost to the `skill_hash` of the loaded tools. If a task fails or causes a rollback, the rating degrades. Tools that consistently solve problems are prioritized in future context-window construction.

E.6 Execution Provenance

To prevent "Hollow Fleet" arbitrage—where a contractor delegates work to an unauthorized, untrusted sub-agent swarm—the `agentsd` daemon signs all output artifact hashes using its hardware-bound Harbor Key. This attests that the work was structurally generated by the approved local daemon within the expected capability bounds, providing cryptographic proof of execution provenance.

F The Condominium Harbor and Ostrom Governance Matrix

proposed — the Condominium Harbor is a proposed intermediate regime, not a shipped or specified mechanism; it does not yet appear in Appendix A because there is no running or designed artifact to grade.

The treatise addresses single-operator harbors (Chapters I–III) and federated sovereign harbors (Chapters IV–VII). The intermediate regime is the **Condominium Harbor**: a single shared daemon and repository governed by multiple distinct principals (e.g., Alice, Bob, and Carl).

In a condominium harbor, sovereignty is replaced by a shared **charter**. The fundamental architecture preserves the kernel's single-writer discipline (one daemon serializes all transactional writes), while delegating political authority across partitioned namespaces:

1. **Partitioned Namespace Ownership:** Principals hold exclusive write capabilities over distinct directory subtrees (e.g., Alice owns `/auth`, Bob owns `/billing`). Cross-namespace effects cannot be directly committed; they are submitted as *typed proposals* that become obligations only upon the owner's cryptographic sign-off.
2. **Tractable Conflict Detection:** General consistency checking over arbitrary logic is co-NP-hard. By restricting commitments to Horn-form invariants and interval resource claims, the daemon performs incremental consistency validation in $O(\log n)$ time per operation.
3. **The Lookout Role:** A standing projection agent continuously runs model-checking over planned roadmap branches and invariant sets (Endsley Level-3 Situation Awareness), alerting principals to latent merge conflicts before execution.

Ostrom Design Principle (Ostrom 1990)	**Institutional Intent**	**Port Daddy Mechanism**
1. Clearly defined boundaries	Exclude unauthorized appropriators	Admission gates and non-forgeable cryptographic identity roots (§III.6).
2. Congruence of rules	Match local conditions and costs	Newcomer probation cliffs (Thm. III.6.3) and float-plan escrows.
3. Collective-choice arrangements	Users participate in rule modification	Charter-amendment voting over governance rules (never voting on physical database truth).
4. Monitoring	Accountable monitors audit conditions	Append-only witness logs, Merkle evidence trails, and the standing Lookout agent.
5. Graduated sanctions	Progressive penalties for infractions	Progressive bond slashing, escalation standing debits, and reputation tombstones.
6. Conflict-resolution mechanisms	Rapid, low-cost dispute adjudication	2-of-3 multisig settlement escrow and multi-model heterogeneous judge arbitration.
7. Minimal recognition of rights	External authorities respect self-rule	Principal autonomy over partitioned namespace directories and role delegation.
8. Nested enterprises	Multi-tiered governance layers	Condominium harbors federating upward into global witness-log networks (Chapter VII).

Table A.2: Mapping Elinor Ostrom's 8 Design Principles for Common Pool Resource Management to the multi-principal Condominium Harbor.

G The Clean-Room Harbor: Derek and Erin

The commercial apex of "hosted trust as a product" is the **Clean-Room Harbor**: executing a proprietary AI agent stack S (owned by Erin) over a confidential corporate dataset D (owned by Derek) while provably bounding exfiltration in both directions.

G.1 Information-Flow Lattice and Dual Declassification Gates

We define a Decentralized Label Model (DLM) lattice $\{\bot\} \leq \{D\}, \{E\} \leq \{D, E\}$. Derek's raw data enters labeled $\{D\}$; Erin's weights and prompts enter labeled $\{E\}$. Any computation over both inherits the join label $\{D, E\}$.

The hypervisor daemon completely mediates all execution paths ($\{\text{fs}, \text{net}, \text{exec}\}$), routing all output through exactly two auditable declassification gates:

1. **Erin's Exit Gate ($G \to E$):** Schema-restricted aggregates only; Differential Privacy (DP) Laplace noise drawn against a metered privacy budget ϵ; verbatim-span suppression against D's corpus; and c canary sentinels planted in D.
2. **Derek's Exit Gate ($G \to D$):** Task outputs and reports only; Erin's model weights execute as a sealed enclave capability that Derek can invoke but cannot inspect or export.

Theorem G.1 (Conditional Noninterference and Privacy Conservation)**.** *Under complete hypervisor mediation of all I/O channels and correct gate enforcement:*

1. ***Noninterference:*** *No $\{D\}$-labeled information reaches Erin beyond gate-released DP aggregates, and no $\{E\}$-labeled intellectual property reaches Derek beyond task reports.*
2. ***ϵ-Conservation:*** *The cumulative privacy loss over any execution trace strictly satisfies $\sum_i \epsilon_i \leq \mathcal{E}_{\text{budget}}$.*
3. ***Canary Detection Power:*** *With c planted canaries and gate false-negative rate β, an exfiltration attempt spanning k canary tokens is detected with probability $\geq 1 - \beta^k$.*

Status. Claim (1) is verified on a finite model (exhaustive two-run equivalence to interleaving depth 7, all secret pairs, schedule secret-independence checked separately; gate-bypass and leaky-gate mutations caught with witness traces); the unbounded form is the standard unwinding obligation. Claims (2) and (3), formerly carried as research-ledger items A3 and A4, are now executed. Claim (2) is discharged by the ϵ-conservation result: the release ledger's single atomic append-and-add transition keeps $\sigma = \sum_i \epsilon_i \leq \mathcal{E}_{\text{budget}}$ invariant over every interleaving (exhaustive check of two racing clients, all 15 reachable states, plus 2000 randomized instances, 0 violations; induction lifts to concurrency via single-writer serialization; spend-without-append and torn-write mutations caught in 1 step, the torn write driving an auditable recorded over-spend in 3; `a3_epsilon_ledger.py`), with the sequential vs. advanced (Dwork–Rothblum–Vadhan) composition crossover telling the contract which DP accounting to quote. Claim (3) is discharged by the canary power/SPRT result: the $1 - \beta^k$ line is verified at $k = 1, 2, 5$; uniform planting of c canaries in n spans gives the hypergeometric operating curve Pr(detect | leak of m spans) (at n=10^4, c=100, β=0.2: 0.554 at m=100, 0.999 at m=800; exact, approximate, and simulated agree), and Wald's SPRT prices expected alarm latency for sustained leaks (analytic 297 vs. simulated 359 gate outputs under leak, realized error rates at or below targets; `a4_canary_sprt.py`). The statements above remain conditioned exactly as they must be — on complete channel mediation and correct gate enforcement, which are the residual oracles.

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